

Cooperative Multilateral Tariff Negotiations: A Quantitative Analysis Using Shapley Value

Draft v2 (August 21, 2026)

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August 21, 2026

Abstract

We replace the estimated bilateral bargaining weights in [Bagwell et al. \(2021\)](#) (BSY) with a weight-free normative benchmark — utilitarian-optimal coalition surplus divided by the [Shapley \(1953\)](#) value — on BSY’s own six-country 1990 tariff-negotiation panel and calibrated primitives. We replicate their equilibrium map and key counterfactuals (their Table III to 0.024pp) but not their $\hat{\zeta}$ estimator; the benchmark itself carries no bargaining weight, and a symmetric-weight placebo shows the fairness gap Γ measures the asymmetry BSY’s estimation found, not a structural distance from Shapley. Γ is positive for the United States and Canada and negative for Japan under all four efficient, individually-rational allocation rules tested (magnitudes are rule-dependent), a pattern largely anticipated by a one-parameter rescaling of the efficient policy’s own incidence. The Shapley allocation lies in the transferable-utility core but is self-enforcing only if transfers are feasible: without them, even the efficient policy violates Australia’s participation constraint, and implementation requires gross transfers of 13.5% of grand-coalition surplus. BSY’s cached bargained outcome falls short of the full-schedule line-by-line optimum by 79.5% of that surplus, decomposing into agenda coverage beyond BSY’s 151-line estimation sample (62.2%), transferability (1.2%), and a within-agenda bargaining gap under 1990 participation constraints (16.1%) — an aggregate transferable-utility gap, not a demonstrated Pareto improvement. The model’s optimal cuts track observed 1990–2000 liberalization no better than simple tariff persistence.

Keywords: Multilateral bargaining, tariff negotiations, Shapley value, cooperative game theory, normative benchmark, replication, MPEC.

JEL codes: F13, F17, C71, C68.

1 Introduction

Multilateral tariff negotiations offer significant potential for global welfare improvements but are hindered by the complexity of coordinating diverse national interests. In a multilateral setting, countries’ decisions generate spillover effects on nations not directly involved, complicating the bargaining process. These spillovers—changes in international prices, trade flows, and terms of

trade—make balanced agreements harder to achieve (Ossa, 2014). Addressing these challenges requires a framework that explicitly accounts for spillover effects and heterogeneity in bargaining power across countries.

This paper extends the quantitative analysis of tariff negotiations from a bilateral to a multilateral setting, building directly on Bagwell et al. (2021) (BSY). Their work analyzes bilateral negotiations under the GATT/WTO framework, where the most-favored-nation (MFN) clause ensures non-discriminatory tariffs, and estimates the bilateral Nash bargaining weight that best rationalizes the tariff concessions of their six-country 1990 negotiating panel. We proceed in three steps on that same panel and on those same calibrated primitives: we replicate BSY’s equilibrium map and counterfactuals; we derive a multilateral cooperative-bargaining model that requires no estimated bargaining weight; and we apply the derived model to BSY’s own 1990 economy, comparing the result to their own estimated outcome. As a robustness check, we then apply the identical machinery to a second, structurally distinct economy built on Caliendo and Parro (2015).

Contribution 1: A full replication of Bagwell et al. (2021), with an honest account of what does and does not reproduce. We replicate BSY’s six-country 1990 negotiating panel end to end. The calibrated primitives load against their original `.mat` inputs to a maximum difference of 2.5×10^{-14} (16 arrays plus 3 scalars, `phase1_loader_acceptance.json`). Their Nash-in-Nash equilibrium map reproduces from a cold start plus box-level feasibility search to 0.40pp mean tariff deviation at their own point estimate $\hat{\zeta}$ (`phase4_parity_rerun.md`), and their Table III counterfactual reproduces to a maximum absolute deviation of 0.024pp (`phase6a_table3.json`). We do *not* reproduce their Table V $\hat{\zeta}$ estimator: our re-estimation fails to converge, with the Levenberg–Marquardt damping parameter diverging and the finite-difference Jacobian degenerating, the same class of numerical difficulty their own coarse `DiffMinChange` setting works around (`phase5_verdict.md` §6). We report this as a disclosed negative result, not a caveat buried in an appendix (§4.1). Reading off BSY’s own published estimates—not our failed re-estimation—we document two features of their bargaining equilibrium that motivate the rest of the paper: three of their seven estimated $\hat{\zeta}$ parameters sit exactly on the optimizer’s box bounds, and the United States is a zero-surplus Nash-in-Nash participant, with its individual-rationality constraint binding to $\sim 10^{-7}$. Both observations are read directly off BSY’s own reported solutions and motivate asking whether a bargaining-weight-free normative benchmark assigns the United States a comparably small share of the cooperative surplus.

Contribution 2: A coalition-value characteristic function with no bargaining-weight parameter. We replace the bilateral Nash bargaining solution of Bagwell et al. (2021) with a cooperative game in which every coalition S of countries has a well-defined dollar-denominated value $v(S)$: the utilitarian-optimal general-equilibrium surplus attainable by S ’s members acting jointly, holding outsiders at their status-quo tariffs (eq. (7), “D1”). Unlike a weighted Nash bargain, this characteristic function carries no bargaining-weight parameter of any kind—the optimal policy s_S^* is a straightforward maximization, solved once per coalition, entirely independently of how the resulting surplus is later divided among members. We compute $v(S)$ on BSY’s own six-country panel—the paper’s primary application—and, as a robustness check, on

a second economy built on the [Caliendo and Parro \(2015\)](#) 31-country dataset (Section 4.2).

Contribution 3: A normative Shapley imputation applied to BSY’s own economy, and a fairness-gap bridge to their estimated outcome. We divide the cooperative surplus $v(N)$ using the Shapley value ϕ , the unique allocation rule satisfying efficiency, symmetry, the dummy property, and additivity ([Shapley, 1953](#); [Hart and Mas-Colell, 1989](#)). We frame this explicitly as a *normative efficiency benchmark*, not as an estimate of real-world bargaining power. Because the optimal policy s_N^* that generates $v(S)$ is computed independently of which allocation rule is applied afterward, the choice of Shapley over any alternative imputation (nucleolus, Kalai–Smorodinsky, egalitarian) changes nothing about predicted tariffs and therefore cannot be identified from tariff-level data—see “What we do not claim” below. What the framework does deliver is a fully pinned-down allocation: given our maintained assumptions (A1–A9, `paper/theory/00_assumptions.md`) and the four Shapley axioms, there is no residual free parameter left to calibrate, unlike [Bagwell et al. \(2021\)](#)’s own bilateral weights, which are estimated by GMM from observed RTA tariff changes.

Among cooperative solution concepts, the Shapley value is the natural choice for our setting on three grounds. First, when restricted to two players the Shapley value coincides with the symmetric Nash bargaining solution under a fixed disagreement point (Proposition 1), so adopting ϕ as the imputation rule is the natural extension of [Bagwell et al.](#)’s bilateral Nash bargain to $N > 2$ that preserves the underlying axioms of the two-player benchmark. Second, the Shapley value pays each country its expected marginal contribution to a uniformly random coalition. This formalizes a long-standing intuition in the [Bagwell and Staiger \(2002\)](#) reciprocity literature—that a country’s negotiating leverage scales with the externality its tariffs impose on partners—and operationalizes it inside a structural GE model. Third, [Shapley \(1953\)](#) and [Hart and Mas-Colell \(1989\)](#) show that ϕ is the unique allocation satisfying efficiency, symmetry, the dummy property, and additivity (or, equivalently, balanced contributions); competing concepts such as the nucleolus or the kernel either lack closed-form computability or are set-valued, both of which are awkward in a quantitative model that must be re-solved at every coalition. [Agnew \(2023, 2025\)](#) earlier proposed the Shapley value as an allocation rule for cooperative gains from international trade and illustrated it on a stylized $N \leq 3$ USMCA example. Our contribution relative to that work is threefold: (i) we replace the hand-stipulated $v(S)$ with a coalitional value derived from a structural GE model, solving the full hat-algebra equilibrium under each coalition’s optimized tariff schedule, on BSY’s own economy and, as a second economy, on [Caliendo and Parro \(2015\)](#)’s; (ii) we scale to full 63-coalition exact enumeration on BSY’s six-country panel and to $N = 7$ exact enumeration on our second economy; and (iii) our second economy inherits the multi-sector input–output linkages of [Caliendo and Parro \(2015\)](#), which are absent from the single-sector framework in [Agnew](#)’s analysis. To the best of our knowledge, this is the first GE-based structural quantification of Shapley-value allocations in international trade.

We then compare the Shapley benchmark to BSY’s own estimated bargaining outcome through a *fairness-gap* statistic $\Gamma_i^{\text{raw}} = \phi_i(v)/v(N) - s_i^{HW}(\hat{\zeta})$ (§2.6), used descriptively—as a partial-identification comparison between the normative benchmark and BSY’s estimated Horn–Wolinsky equilibrium—never as a channel for identifying bargaining weights from data.

On BSY’s own panel, $\Gamma_{US} > 0$, $\Gamma_{Japan} < 0$, and $\Gamma_{Canada} > 0$ are sign-robust across four efficient, individually-rational allocation rules we test (Shapley, equal split, 1990 revenue share, 1990 income share); magnitudes are strongly rule-dependent (Γ_{US} spans +3.3 to +23.2pp across rules), and Γ_{EU} , $\Gamma_{Australia}$, $\Gamma_{South\ Korea}$ are sign-unstable (Section 4.1). We read this as evidence that the Shapley benchmark redistributes along a dimension—marginal contribution—that BSY’s observed bargaining outcome does not track, in a direction that favors the US and Canada and disfavors Japan robustly, not as evidence that BSY’s bargaining outcome is specifically unfair to the United States.

A companion recomputation restricting the coalition-value lever to BSY’s own 151-line negotiated set, solved on a genuine, weight-corrected line-by-line lever (predictions P-31/P-33, $F_{line}^{v2} = 0.436840$, $F_{line}^{IR} = 0.428519$), together with the identical instrument extended to the full 294-line schedule ($D = 0.885046$, prediction P-34), closes an *exact* identity against BSY’s aggregate bargained surplus ($\Sigma_k g_k^{HW} = 0.312441$): $D - \Sigma_k g_k^{HW} = 79.51\%$ of $v(\mathcal{N}_6)$, splitting additively into three matched-instrument terms (§4): the first two are non-negative by construction of the nested optimization problems that produce them; the third is not — it would require BSY’s own cached tariff schedule to be itself a feasible point of the constraint set that generates it, and a direct check finds it is not (5 of 151 masked lines, 3.3%, all below zero and legal under BSY’s own solver bound $lb = -0.25$), so its non-negativity is confirmed empirically in this run’s numbers, not guaranteed by construction.

The three terms, each measured consistently against the TI-5-corrected frontier $F^{IR,corr} = 0.428073$ (§4): 62.24% from extending coverage to lines outside BSY’s own bilateral negotiating structure — an estimation-sample construct, not the Uruguay Round’s actual agenda — to the full schedule, at a fixed line-by-line instrument; 1.22% from the transferable-utility/ individually-rational wedge on the 151-line set itself; and 16.06% from a within-agenda shortfall restricted to points no member finds worse than the 1990 threat point, $3\times$ the pre-registered materiality band and essentially unchanged by that restriction (the three two-decimal components sum to 79.52% by independent rounding; the exact total, from the underlying unrounded values, is 79.51%, above). This 16.06% figure is the TI-5-correct point estimate, using each member’s own line-by-line unilateral floor rather than a uniform zero (§4); a uniform-zero floor instead gives a 16.12% upper bound, only 0.06 percentage points higher, because the one large floor (Japan’s) is never binding at the optimum while the small floors that do bind (EU, Australia, and the US) cost little.

A stress test using BSY’s actual bargained outcome, rather than 1990, as every member’s floor confirms the shortfall is not an artifact of that choice: 85% of the 1990-IR-constrained shortfall survives as a genuine, no-transfer-feasible, *weak* Pareto improvement over what the six negotiators actually achieved — four of six members are exactly indifferent between the two outcomes, and the entire net gain accrues to Japan and Canada (§4). The feasibility axis is therefore not the source of the shortfall. Coverage is worth **3.88** times the within-agenda shortfall on a regime-matched instrument, now a fully floor-corrected point estimate on both operands (the 294-line individual-rationality floor correction on the coverage side has been executed, moving the ratio down slightly from a previously reported $3.91\times$, as expected since tightening a floor can only lower a constrained maximum): widening what is negotiable, not extracting more from

a fixed bilateral architecture, is this panel’s larger source of unrealized surplus — consistent with an *architectural* shortfall in what the GATT principal-supplier mechanism puts on the table, though cut-depth on the shared negotiated lines (axis (a)) and coalition-structure effects (axis (d)) remain jointly embedded in this reading and have not been separately measured (§4), so this is a reading, not a clean decomposition.

D1’s own optimal line-by-line cuts on this agenda correlate positively with the actual 1990–2000 tariff change (Spearman $\rho = +0.64$, clearing the pre-registered $\rho \geq 0.3$ band; prediction P-37, §4) — the paper’s first external-data comparison, though a paired bootstrap shows a model-free uniform-cut placebo scores higher on the identical pooled metric, resolved ($\rho = +0.71$; the 95% CI on the difference excludes zero and favors the placebo).

On BSY’s own cached $\hat{\zeta}$ schedule, 86 of these 151 negotiated lines (57%) are left at exactly BSY’s own 10^{-6} solver tolerance — effectively uncut, though the data show these same lines were, on net, cut over 1990–2000 — while D1’s efficient frontier cuts 118 of them (78%), including all but 27 of the lines that carried a positive 1990 tariff. A mid-rank-corrected rank comparison on the same metric favors D1 over BSY’s own fitted model on the pooled sample ($\rho_{BSY} = +0.32$ against $\rho_{D1} \in [+0.64, +0.67]$; prediction P-42, §4), a margin that survives a direct test against the tie-count mechanism but not against coverage: BSY structurally cannot lose rank ties on lines it never negotiated, and on the same-coverage 65-line subsample — the caliber this paper’s own registry treats as the fair, terminal comparison — none of the three pairwise comparisons (D1 vs. BSY, D1 vs. placebo, BSY vs. placebo) is resolved (prediction P-45, §4), so the official disposition is **intermediate**, not a settled D1 victory; part of the pooled-sample gap reflects D1’s own closer resemblance to naive 1990-tariff persistence than BSY’s fitted schedule shows. The two are close on the level (Pearson) face, and neither carries content beyond own-tariff persistence once rank is partialled on τ^{1990} .

The extensive-margin contrast — BSY’s own model leaving a majority of negotiated lines essentially untouched, D1 cutting a large majority of them — is the more defensible reading: on BSY’s own cached $\hat{\zeta}$ schedule it establishes directional agreement with the data and a pooled-sample rank advantage over BSY’s own estimated model that does not survive a coverage-matched test, plus a documented coverage difference from it, but not incremental agreement beyond simple persistence, nor an advantage over a naive persistence rule.

The 151-line set itself is BSY’s own bilateral-negotiation estimation sample, not the Uruguay Round’s actual agenda—agriculture and tobacco are excluded by that sample’s own construction. Agriculture’s exclusion reflects a genuine structural difference, not an omitted liberalization channel: the Round’s Agreement on Agriculture brought agricultural protection under *effective* GATT discipline for the first time—GATT 1947 formally covered agriculture from the outset, but waivers and Article XI exceptions had hollowed that coverage out in practice (Hoda, 2001)—via *tariffication*: converting non-tariff barriers into bound tariff-equivalents subject to a separate reduction formula, rather than through the bilateral principal-supplier channel BSY estimate and this lever restricts to (Ingco, 1996; Hoda, 2001); this project’s own applied-tariff data shows measured applied agricultural protection instead *rose*, on average, over 1990–2000 in this sample (five of six Track A countries rising on both the mean and the within-country median; pooled median across all 54 country-sector cells +1.45pp, 32 of 54 cells rising), consis-

tent with a change in the protective instrument itself—from non-tariff barriers largely invisible to a tariff series, to explicit, in many cases inflated (“dirty”), bound tariff-equivalents—rather than with a genuine increase in the level of protection; whatever the AoA’s formula-cut track accomplished on bound rates, it is not visible in, or commensurable with, the applied-MFN channel either model’s lever restricts to (§4).

What we do not claim (identification scope). Following Manski (1995), identification requires that distinct parameter values produce distinct distributions of *observed* outcomes. In our setting, only $v(N)$ —the grand-coalition value—has a plausible data analog (the panel’s own partial liberalization, itself a calibration target); the remaining $2^N - 2$ sub-coalition values $v(S)$ are model-output counterfactuals. Sharper than the generic Manski point: under D1 the optimal policy that generates every $v(S)$ is computed *before*, and independently of, any choice of allocation rule, so different solution concepts (Shapley, nucleolus, Kalai–Smorodinsky, the γ -core of Chander (2007)) produce different imputations ϕ on the identical $v(S)$ and hence identical predicted tariffs at every coalition. We therefore explicitly *do not* claim that (i) the framework identifies real-world bargaining power; (ii) the allocation rule is identifiable from tariff moments—no observed tariff outcome could in principle discriminate the Shapley value from any competing efficiency-preserving imputation rule, because none of them moves the predicted policy (§2.3); or (iii) the Shapley ranking constitutes evidence about strategic positions. The applied tariff schedule that anchors our threat point is itself an equilibrium outcome of decades of prior GATT bargaining and is thus not exogenous to historical bargaining power; alternative threat points (zero-tariff, prohibitive-tariff à la Ossa (2014), Limão (2007) 1947 Smoot-Hawley) are a natural robustness path that we leave to subsequent work. We treat these limitations as the price of the imputation framing rather than as deficiencies relative to a bargaining-power identification claim that the data cannot support.

Contribution 4: Core membership is self-enforcing only given transfers GATT does not provide. Two further results establish the framework’s internal coherence on BSY’s own economy, each with a fixed outsider convention and each directly verified rather than merely inferred from convexity: the game is not convex (superadditivity fails outright on 20/301 disjoint pairs, worst case the US joining {EU, Canada}), yet the Shapley imputation is nonetheless a member of the transferable-utility core, checked directly on all 62 proper-coalition constraints (0/62 violations, minimum slack 2.59% of grand-coalition surplus, status-quo-outsider convention). Core membership is a property of the *transferable-utility* game: it shows the Shapley-fair allocation is self-enforcing *given* feasible transfers, not that the underlying efficient policy is implementable without them. On this economy that distinction has teeth: absent transfers, the efficiency-maximizing policy itself is blocked outright by Australia’s participation constraint, and reaching the Shapley allocation from the efficient policy requires a budget-balanced gross transfer of 13.51% of grand-coalition surplus (the US pays 2.49%, Australia receives 7.0%; §4.1). GATT/WTO is, by design, a no-transfer institution, operating through issue linkage and reciprocal concessions rather than side payments, so this is a first-order implementability caveat on the efficiency benchmark, not a technical footnote.

Contribution 5: A single-step MPEC formulation that jointly optimizes tariffs and the trade equilibrium. Bagwell et al. (2021) and much of the quantitative bargaining literature adopt a two-step procedure in which the trade equilibrium is solved separately at each candidate tariff vector and the bargaining objective is optimized over the resulting reduced-form welfare surface. We instead cast the coalition-value problem (7) as a single nonlinear program in which the full hat-algebra trade equilibrium (1)–(5) enters as equilibrium constraints solved jointly with the outer objective—the Mathematical Programming with Equilibrium Constraints (MPEC) representation of Luo et al. (1996), eq. (11). On our second economy (Section 4.2), a controlled comparison on the NAFTA grand coalition illustrates that separating the trade equilibrium from the outer optimization can both leave surplus on the table and silently select a policy that fails individual rationality once general-equilibrium feedback is accounted for (Appendix C); that comparison predates the current D1 characteristic function and its magnitude is superseded pending a D1 recompute, so we illustrate the mechanism there without quoting the specific number here. The trade-off is computational: the single-step solve is orders of magnitude slower per coalition than the two-step heuristic, because each outer iteration triggers a full GE fixed-point solve. Tractability of the single-step approach therefore relies on formulating the equilibrium as smooth complementarity constraints; a full Pyomo+IPOPT implementation that scales to a larger country panel with sectoral tariff heterogeneity is the natural next step and is left to an extended version of this paper.

Contribution 6: Robustness on a second, structurally distinct economy. As a check that our results are not an artifact of BSY’s particular six-country panel, we apply the identical D1/Shapley machinery to a second economy built on the Caliendo and Parro (2015) 31-country, 40-sector 1993 dataset, exactly enumerated on coalitions of size $N = 3$ (NAFTA) through $N = 7$ (NAFTA + China, India, Brazil, Korea). The same raw-share tilt toward high-tariff, high-headroom economies documented on Track A (Section 4.1) emerges here too, together with informative contrasts. Splitting each country’s raw share into a unilateral (own-tariff) and genuinely cooperative component reverses the Track B ranking—once a high-tariff outsider enters ($N \geq 4$), only 13.1–16.1% of grand-coalition surplus is genuinely cooperative, and this cooperative surplus accrues disproportionately to the United States (35.5–45.2% of $w(N)$), the largest contributor at every $N \geq 4$ (`ir_verification.md`). Both figures diverge sharply from Track A: the cooperative share there is 94.05% ($\kappa_A^{s \leq 1}$, Section 4.1), and its largest recipient is Japan (42.9%), with the United States third (13.3%); the raw-versus-cooperative reversal documented here has no analog on Track A, where $\phi_i(v) = \phi_i(w)$ for four of six players by construction ($v(\{i\}) = 0$). The raw imputation is close to empirically indistinguishable from a simple tariff-revenue share on this economy (Pearson 0.84–0.87 across $N = 4$ –7; markedly weaker at $N = 3$, +0.29)—an adverse fact for any claim that the Shapley structure does discriminating work beyond a largely mechanical headroom calculation, reported honestly rather than as something the exercise establishes. The Shapley imputation is also directly verified to lie in this economy’s core (0/126 constraints violated at $N = 7$, minimum slack 0.37% of $v(N)$). Full detail, tables, and figures are in Section 4.2 and Appendix C.

Roadmap. Section 2 sets up the multilateral bargaining model, the D1 characteristic function, and the Shapley imputation. Section 3 describes BSY’s own 1990 calibration, our primary dataset, and the [Caliendo and Parro \(2015\)](#) dataset used for the second-economy robustness check (with explicit treatment of the Rest-of-World residual, Appendix B). Section 4 presents the primary results on BSY’s six-country panel—the replication credentials, coalition values, the fairness-gap bridge, and the core/transfer result—followed by a condensed robustness section on the second economy, with full legacy detail in Appendix C. Section 5 discusses scope and policy implications.

2 The Model

Notation: D1 supersedes the Nash-product characteristic function (ADR-1, Round 8). Earlier drafts of this section defined $v(S \mid \sigma)$ as the dollar value attained at the argmax of a σ -weighted Nash product (eq. (8) below, kept for historical reference only). That definition was retired: it is not the value function of any single program (the Nash-product objective and the dollar payoff are different objects), degenerates to $v(S) = 0$ on 13% of coalitions at $N = 7$ when the individual-rationality-feasible set admits no strict Pareto improvement, and introduces a bargaining weight σ into the characteristic function itself, forcing the self-referential fixed-point apparatus discussed (and retired) in §2.4 below. The current, canonical characteristic function is **D1**, defined in eq. (7) after the scalar-lever paragraph: an unconstrained utilitarian maximization with no bargaining weight and no IR constraint of any kind. See `paper/theory/00_model_blueprint.md` §3 (ADR-1) for the full argument and disclosed costs.

Notation convention. Throughout the model, σ denotes the *retired* bargaining weight of the Nash-product characteristic function (§2.2, superseded by D1, eq. (7)); it appears only in historical discussion of what was removed and never in a live definition. We never use σ for the within-sector CES elasticity of substitution—when the latter is referenced, we follow the rename η_j (Caliendo–Parro 2015 §3.1 use σ_j for the same object, but the bargaining-literature σ collides). Trade elasticity is the Fréchet shape parameter θ^j (CP 2015 convention), which satisfies $\theta^j > \eta_j - 1$ for the gravity-Gamma integral to converge. Real income / welfare of country n is denoted W_n , never π_n (reserved for trade shares).

2.1 Trade Equilibrium: Eaton–Kortum with Intermediates

We adopt the multi-sector Eaton–Kortum production structure of [Caliendo and Parro \(2015\)](#). The world consists of N countries (indexed $n, i, h \in \{1, \dots, N\}$) and J sectors (indexed j, k). In each country, sector j produces a continuum of varieties using labor and intermediate goods from all sectors. Variety-level productivities are drawn from a Fréchet distribution with shape parameter θ^j (sector-specific) and country-sector location λ_n^j . Bilateral trade incurs an iceberg cost $d_{ni}^j \geq 1$ and an ad-valorem tariff $\tau_{ni}^j \geq 0$, so that the effective cost of delivering sector- j varieties from i to n scales by $\kappa_{ni}^j \equiv (1 + \tau_{ni}^j) d_{ni}^j$. Final consumers have Cobb–Douglas preferences with sector shares α_n^j summing to one; intermediate use is described by an input–output matrix $\gamma_n^{k,j}$ (share of sector- k inputs used in sector- j production) and a labor share γ_n^j , with constant

returns ($\gamma_n^j + \sum_k \gamma_n^{k,j} = 1$).

Level equilibrium. Given fundamentals $(L_n, \lambda_n^j, d_{ni}^j, \tau_{ni}^j, \alpha_n^j, \gamma_n^j, \gamma_n^{k,j})$ and trade deficits D_n , an equilibrium consists of wages w_n , sectoral prices P_n^j , input-bundle costs c_n^j , trade shares π_{ni}^j , sectoral expenditures X_n^j , and total incomes $I_n = w_n L_n + R_n$ (labor income plus tariff revenue R_n) satisfying the system of unit-cost, price-index, gravity, expenditure-balance, and trade-balance conditions of [Caliendo and Parro \(2015\)](#).

Hat-algebra transformation. Define $\hat{x} \equiv x'/x$ for any variable x where x is the observed base-year value and x' is the counterfactual value under a new tariff vector τ' . Assuming technology (λ_n^j) and iceberg costs (d_{ni}^j) are time-invariant, the level equilibrium transforms into a system in changes that depends only on *observable* base-year shares $\{\pi_{ni}^j, X_n^j\}$, the elasticity parameters θ^j , and the tariff change $\hat{\kappa}_{ni}^j = (1 + \tau_{ni}^{j'})/(1 + \tau_{ni}^j)$. The five equilibrium conditions in changes are:

$$\hat{c}_n^j = \hat{w}_n^{\gamma_n^j} \prod_{k=1}^J (\hat{P}_n^k)^{\gamma_n^{k,j}}, \quad (1)$$

$$\hat{P}_n^j = \left[\sum_{i=1}^N \pi_{ni}^j (\hat{\kappa}_{ni}^j \hat{c}_i^j)^{-\theta^j} \right]^{-1/\theta^j}, \quad (2)$$

$$\hat{\pi}_{ni}^j = \left(\frac{\hat{c}_i^j \hat{\kappa}_{ni}^j}{\hat{P}_n^j} \right)^{-\theta^j}, \quad (3)$$

$$X_n^{j'} = \sum_{k=1}^J \gamma_n^{j,k} \sum_{i=1}^N \frac{\pi_{in}^{k'} X_i^{k'}}{1 + \tau_{in}^{k'}} + \alpha_n^j I_n', \quad (4)$$

$$\sum_{j,i} \frac{\pi_{ni}^{j'} X_n^{j'}}{1 + \tau_{ni}^{j'}} - D_n = \sum_{j,i} \frac{\pi_{in}^{j'} X_i^{j'}}{1 + \tau_{in}^{j'}}, \quad (5)$$

with $\pi_{ni}^{j'} = \pi_{ni}^j \hat{\pi}_{ni}^j$ throughout. Equations (1)–(3) are written in changes, while (4)–(5) are in levels of the counterfactual; they form an exactly-determined nonlinear system in $(\hat{w}_n, \hat{P}_n^j, \hat{c}_n^j, \hat{\pi}_{ni}^j, X_n^{j'})$ once a wage numeraire is fixed.

Why hat-algebra. The unobserved fundamentals λ_n^j and d_{ni}^j cancel symmetrically between x' and x in (1)–(3), leaving only base-year trade shares and the tariff change. This eliminates the need to estimate productivity and iceberg costs—the most demanding identification step in level-form calibrations ([Eaton and Kortum, 2002](#); [Caliendo and Parro, 2015](#)). We use this property heavily: every coalitional value $v(S)$ in Section 2.2 is computed from the same base-year $\{\pi_{ni}^j, X_n^j\}$ and a coalition-specific $\hat{\kappa}$.

Welfare measure. For each country n , real income is $W_n = I_n / \prod_j (P_n^j)^{\alpha_n^j}$, so the welfare change is

$$\hat{W}_n = \frac{\hat{I}_n}{\prod_j (\hat{P}_n^j)^{\alpha_n^j}}, \quad (6)$$

which by the ACR aggregation result (Arkolakis et al., 2012) captures the full general-equilibrium welfare effect from the tariff change.

2.2 Cooperative Bargaining over a Coalition

Coalitional value function: D1 (canonical, ADR-1). Fix a coalition $S \subseteq N$. The characteristic function is an unconstrained utilitarian maximization over the coalition’s scalar policy levers (defined below, eq. (10)):

$$v(S) = \max_{s_S \in [0,1]^{|S|}} \sum_{n \in S} I_n^{\text{base}} \cdot \left(\widehat{W}_n(s_S, \mathbf{1}_{-S}) - 1 \right), \quad v(\emptyset) = 0. \quad (7)$$

There is *no* bargaining weight and *no* individual-rationality (IR) constraint inside this program. Non-negativity is constructive: the status-quo lever $s_S = \mathbf{1}$ is always feasible and gives a value of 0, so $v(S) \geq 0$ for every S . Outsiders $n \notin S$ maintain their status-quo tariffs ($s_n = 1$); their equilibrium response affects coalition members through general-equilibrium spillovers (the same status-quo outsider convention as before, now stated without an IR constraint attached to it).

Why D1, not a weighted Nash product (ADR-1). An earlier draft of this section defined $v(S \mid \sigma)$ as the dollar value attained at the argmax of a σ -weighted Nash product,

$$\max_{\tau_S} \prod_{n \in S} \left(\widehat{W}_n - 1 \right)^{\sigma_n} \quad \text{s.t. trade equilibrium and } \widehat{W}_n \geq 1 \text{ for all } n \in S, \quad (8)$$

with $\sigma_S = (\sigma_n)_{n \in S}$ non-negative weights summing to 1 and the IR constraint $\widehat{W}_n \geq 1$ enforced for $n \in S$. This definition is **retired**: the Nash-product objective and the dollar payoff (7) are different functions, so the "value" it produced was not the value function of any single optimization program; on coalitions where the IR-feasible set admits no strict Pareto improvement the Nash product is identically 0 on the whole feasible set, so the numerical argmax reports the *minimal* dollar element of that face rather than a well-defined value (13% of coalitions at $N = 7$); and it introduced the bargaining weight σ into the characteristic function itself, forcing a self-referential fixed-point layer that turned out to carry zero data-side identifying variation across three review rounds. Under D1, all three problems vanish: $v(S)$ is a linear (additive) functional of $\widehat{W}$, so any two maximizers of the compact-box program attain the identical value; existence follows directly from Weierstrass on the compact box $[0, 1]^{|S|}$ (no ϵ -buffer needed — see `paper/theory/03_cooperative_bargaining.md` Proposition 3.4 for the full statement); and no bargaining weight enters the characteristic function at all. But this does not mean policy selection is distribution-free. The dollar-utilitarian objective (7) that *selects* s_N^* implicitly embeds **income-proportional (Negishi-type) weights**: it maximizes an unweighted dollar sum, which is equivalent to weighting each country’s utility by the inverse of its marginal utility of income, and is not neutral across policies that redistribute differently across countries (the $[0, 1, 0, 0, 0, 0]$ best *vertex* on Track A, in which the EU alone cuts and captures 130% of that vertex’s surplus while the USA loses in absolute terms, is the sharpest illustration — though we flag that this vertex is *not* the efficiency optimum, which is interior and strongly asymmetric on five of six coordinates — **Korea sits exactly at the zero boundary**,

$s_{\text{Korea}}^* = 0$; the registered strong form of prediction P-19 failed on that coordinate, and “interior” here is the post-hoc weak form (corrected 2026-08-14, TEST-13-02) — $s^* = [0.533, 0.472, 0.027, 0.326, 0.111, 0.000]$ with $g_{US} > 0$; disclosed in full below, §2.6). This has direct precedent: Bagwell et al. (2021)’s own Table III reports a world-income-maximizing row as one of their comparison benchmarks. What *is* true, and is the correct narrower claim, is that the Shapley *imputation* layer (§2.4) re-divides the resulting surplus independently of which allocation rule is chosen — alternative imputation rules (Kalai–Smorodinsky, nucleolus, egalitarian) all operate on the identical $v(S)$ and produce identical predicted tariffs (§2.3). The imputation choice is distribution-free with respect to *policy*; the policy choice itself is not distribution-free with respect to *countries*. Eq. (8) is kept here only so the historical asymmetric-Nash-bargaining motivation for the model (§2.2 title) remains legible; it does not define $v(S)$ anywhere below.

IR relocated to the imputation layer. D1 has no IR constraint of any kind inside $v(S)$. The economic content of “no member does worse than its outside option” is checked instead at the *imputation* layer: $\phi_n(v) \geq v(\{n\})$ (equivalently $\phi_n(w) \geq 0$), which we verify directly on the Shapley allocation (§2.5). This is a relocation, not a removal. Non-members $n \notin S$ remain unprotected — $v(S)$ is not required to be a Pareto improvement over the status quo for $n \notin S$ — consistent with modeling *plurilateral liberalization with free-riding* (importer $n \in S$ cuts proportionally on all origins, including non-members, who receive the market-access externality without a reciprocal cut) rather than a discriminatory-PTA exclusion mechanism; see Chander (2007) on γ -cores with externalities for the closest formal treatment. A robustness re-run with outsiders playing γ -response best replies (rather than passive status quo) is delegated to the empirics section.

Cooperative normalization. A singleton value $v(\{n\})$ is the welfare gain country n obtains by *unilaterally* optimizing its own tariff (the status-quo $s_n = 1$ is always feasible, so $v(\{n\}) \geq 0$; it is strictly positive for high-tariff economies, e.g. $v(\{\text{China}\}) = \$119.681\text{B}$, the exact D1 optimum — `paper/replication/results/extended_shapley/N7_D1.json`). A country’s unilateral gain is not cooperative surplus, so we define the *cooperative game* by zero-normalizing the singletons,

$$w(S) \equiv v(S) - \sum_{n \in S} v(\{n\}), \quad w(\{n\}) = 0 \text{ by construction}, \quad (9)$$

and decompose the grand-coalition value into a unilateral and a cooperative component, $v(N) = \sum_n v(\{n\}) + w(N)$. The Shapley imputation reported as the paper’s *primary* object is computed on the cooperative game w ; the raw- v Shapley is reported both as a total-attribution transparency benchmark *and* as the primary operand of the positive–normative bridge (§2.6), since it is the object directly comparable to Bagwell et al. (2021)’s threat point (§2.6 explains why). By Shapley linearity these decompose exactly, $\phi_n(v) = v(\{n\}) + \phi_n(w)$ (Section 4).

Operational simplification: scalar tariff lever. For tractability we restrict the policy space to a per-importer scalar *uniform proportional cut*:

$$\tau_{ni}^{j,\text{new}} = s_n \tau_{ni}^{j,0}, \quad s_n \in [0, 1] \quad \forall n \in S, \quad s_n = 1 \quad \forall n \notin S, \quad (10)$$

so that importer n chooses a single s_n that scales every applied tariff on its imports proportionally; non-members keep their status-quo schedule. This is a stylized policy lever chosen for tractability, not a representation of an actual WTO bargaining unit. Real RTA negotiations (NAFTA/USMCA, CETA, RCEP, the Doha Swiss formula $\tau' = a\tau/(a + \tau)$) are sectoral and frequently non-linear in τ^0 . We adopt (10) so that exact Shapley enumeration over 2^N coalitions remains computationally feasible at $N \leq 7$ (per-coalition decision space drops from $\sim N^2 J$ to $|S|$). Sectoral, asymmetric-pair, and Swiss-formula extensions are left to future work. A direct consequence: for low-tariff countries (Canada, Korea, USA), the feasible window $s_n \in [0, 1]$ is mechanically narrow, capping their marginal contributions to any coalition's D1 value—an issue we revisit in Section 2.5.

Single-step MPEC formulation (D1 objective, no IR row). Combining the D1 objective (7), the scalar lever (10), and the hat-algebra equilibrium (1)–(5) yields a Mathematical Program with Equilibrium Constraints (Luo et al., 1996) in which equilibrium variables are decision variables rather than nested-iteration outputs:

$$\begin{aligned} & \max_{\{s_n\}_{n \in S}, \widehat{w}, \widehat{P}, \widehat{c}, \widehat{\pi}, X'} \sum_{i \in S} I_i^{\text{base}} (\widehat{W}_i - 1) \\ \text{s.t. } & \widehat{c}_n^j = \widehat{w}_n^j \prod_k (\widehat{P}_n^k)^{\gamma_n^{k,j}} \quad \forall n, j \\ & \widehat{P}_n^j = [\sum_i \pi_{ni}^j (\widehat{\kappa}_{ni}^j(s_n) \widehat{c}_i^j)^{-\theta^j}]^{-1/\theta^j} \quad \forall n, j \\ & \widehat{\pi}_{ni}^j = (\widehat{c}_i^j \widehat{\kappa}_{ni}^j(s_n) / \widehat{P}_n^j)^{-\theta^j} \quad \forall n, i, j \\ & X_n^{j'} = \sum_k \gamma_n^{j,k} \sum_i \frac{\pi_{in}^{k'} X_i^{k'}}{1 + \tau_{in}^{k'}} + \alpha_n^j I_n' \quad \forall n, j \\ & \sum_{j,i} \frac{\pi_{ni}^{j'} X_n^{j'}}{1 + \tau_{ni}^{j'}} - D_n = \sum_{j,i} \frac{\pi_{in}^{j'} X_i^{j'}}{1 + \tau_{in}^{j'}} \quad \forall n \\ & \widehat{\kappa}_{ni}^j(s_n) = (1 + s_n \tau_{ni}^{j,0}) / (1 + \tau_{ni}^{j,0}) \quad \forall n, i, j \\ & s_n \in [0, 1] \quad \forall n \in S; \quad s_n = 1 \quad \forall n \notin S. \end{aligned} \quad (11)$$

The IR row present in earlier drafts ($\widehat{W}_i - 1 \geq \epsilon$) is **removed**: it is neither needed for existence (the whole box $[0, 1]^{|S|}$ is the feasible set, Weierstrass applies directly) nor economically meaningful at this layer under ADR-1 (IR now lives at the imputation layer, above). The decision count is $|S|$ outer scalars plus the $\sim N(1 + 3J + NJ)$ equilibrium variables; one trade-balance constraint is redundant by Walras' law and is dropped at solve time. Our current implementation uses a two-step nested loop (outer SLSQP/L-BFGS-B over $\{s_n\}$, inner CP fixed-point); the D1 refactor requires deleting the IR filter and penalty terms and widening the search grid to include the endpoints $\{0, 1\}$ (`scripts/python/cp2015_value.py`); all headline numbers under the retired Nash-product objective (e.g. the two-step-vs-MPEC benchmark, $v = \$5.49\text{B}$ vs.

\$5.89B, Section C.5) are superseded pending the D1 recompute pass. A full MPEC implementation in IPOPT/CONOPT, feasible at $N \leq 7$ and required for an $N = 31$ extension, is left to future work.

2.3 Identification Scope

Following Manski (1995), identification requires that distinct parameter values produce distinct distributions over *observed* outcomes. Our framework does not meet this bar, and we state the limit explicitly rather than relegate it to a footnote. **Under D1 this is sharper than a generic caveat:** the optimal lever s_N^* is computed independently of any imputation rule, so the choice of Shapley over alternatives literally cannot move any predicted tariff — see §2.6’s positioning paragraph, stated once there and not repeated in full here.

Of the 2^N coalitional values, only $v(N)$ —the grand-coalition payoff under realized RTAs—has a conceivable data analog (e.g., NAFTA 1993–2005 partial liberalization, itself used by Caliendo and Parro (2015) to calibrate the underlying GE primitives, introducing circularity). The remaining $2^N - 2$ sub-coalition values $\{v(S)\}_{S \neq \emptyset, N}$ are pure model output, jointly determined by four modeling choices:

1. **Imputation rule**—Shapley value on the cooperative game w (§2.4). Under D1 this choice lives *entirely outside* $v(S)$ — alternatives (Kalai–Smorodinsky, nucleolus, γ -core, egalitarian) yield different ϕ_i ’s on the identical $v(S)$ and hence identical optimal tariffs; the choice is not pinned down by any tariff data (a stronger, structural version of the generic Manski point, not merely an instance of it).
2. **Tariff parameterization**—scalar s_n uniform across sectors j and trading partners i . Real RTAs are sectoral; this restriction is a tractability choice (see (10)).
3. **Disagreement payoff**— W_n^0 at the 1990/1993 applied schedule τ^0 (A7). The applied tariff is itself the equilibrium outcome of 1947–1990s GATT bargaining (Geneva, Kennedy, Tokyo, Uruguay rounds), not exogenous to the object the model studies (Limão, 2007).
4. **Non-member treatment**—outsiders are not IR-protected under D1 either (§2.2); this is now read as a plurilateral-with-free-riding institutional choice, not a discriminatory-PTA exclusion mechanism.
5. **Equilibrium selection (A2)**—added 2026-08-09, IDENT-09-03: once A2 is honestly restated as “the branch selected by the damped fixed-point iteration” rather than a claim of global uniqueness (§2, Assumption A2), $v(S)$ is a counterfactual defined *relative to that solver-selected branch*, not relative to “the” equilibrium in any stronger sense (Marschak’s Maxim, in the spirit of Heckman, 2010). The live monitors are multi-damping agreement (2.5×10^{-5} relative) and independent re-solve agreement (127/127 bitwise), both reported in `paper/replication/results/noise_floor_eps_B.md`. This modeling choice is disclosed *symmetrically*: the branch index b that qualifies $\Gamma_i(\zeta, b)$ on the BSY/comparator side (§2.6) has an exact structural counterpart on our own side, and both must be reported together, not only on the comparator’s.

Two implications follow. First, different imputation rules produce different Shapley-analogue

shares ϕ_i but, under D1, *identical* $v(S)$ and identical predicted tariffs at every coalition — so our choice of Shapley over alternative imputation rules is not merely “not identified by data” in the generic partial-identification sense, but structurally decoupled from any tariff prediction whatsoever. Second, there is no longer a fixed-point weight to interpret (§2.4 retires that apparatus): the Shapley shares reported in Section 4 are a *computed structural imputation under stated D1/A1–A9 assumptions*, not an estimate of any empirical bargaining-power parameter.

The contribution of this paper is therefore methodological and normative: we propose (i) the [Caliendo and Parro \(2015\)](#) GE for a canonical (D1) characteristic function $v(\cdot)$, (ii) the Shapley value for normative allocation under efficiency, symmetry, dummy, and additivity, and (iii) a level-set bridge (§2.6) that compares this normative benchmark to [Bagwell et al. \(2021\)](#)’s estimated bargaining outcome without claiming to identify bargaining weights from data. We explicitly do *not* claim that the framework identifies real-world bargaining power, that any object in §2.6 is an estimate of an unknown structural parameter, or that the cross-country pattern of ϕ constitutes evidence about strategic positions.

2.4 Shapley Imputation

Definition 1 (Shapley value). For a TU game $v : 2^N \rightarrow \mathbb{R}$, the Shapley value of player i is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(N - |S| - 1)!}{N!} (v(S \cup \{i\}) - v(S)). \quad (12)$$

No self-referential weight map under D1 (retired, Round 8). Earlier drafts of this subsection defined a fixed-point map $\Phi(\sigma) \equiv \phi(v(\cdot \mid \sigma)) / \sum_n \phi_n(v(\cdot \mid \sigma))$ and iterated it to a self-consistent weight σ^* , because the retired Nash-product characteristic function (8) depended on σ . Under D1 (eq. (7)), $v(S)$ does not depend on any weight, so this entire apparatus—the iteration Φ , its existence conjecture, and the Brouwer-condition list that depended on properties of v_σ —is **vacuous** and is retired (full text preserved for the record in `paper/theory/05_iterative_weight_update.md`, marked ARCHIVED). This is a net simplification, not a loss: the identification review found across three rounds that σ^* carried zero data-side identifying variation, and the fixed point’s existence gap — condition (i) below in the retired list, continuity of the *argmax* rather than the value function — is exactly the Berge-gap problem that D1 also eliminates for $v(S)$ itself. The academic content of “a weighted imputation” is carried forward, if needed, as a robustness fork using weighted Shapley value with *exogenous* weights ([Kalai and Samet, 1987](#)), not a self-referential fixed point.

Superadditivity/monotonicity status. Single-player w -monotonicity ($w(S \cup \{i\}) \geq w(S)$, distinct from the general superadditivity property $v(S \cup T) \geq v(S) + v(T)$ for arbitrary disjoint S, T) is a checkable question about D1’s characteristic function. **On Track B**, the D1 recompute sweep confirms it holds on 100% of tested pairs (2059/2059 at $N = 7$, 3019/3019 pooled $N = 3$ – 7 , `paper/replication/results/superadditivity_sweep_D1.md`), and imputation IR is verified directly at every $N \in \{3, \dots, 7\}$ (`paper/replication/results/ir_verification.md`). This is strong empirical evidence and *not* a proof — see `paper/theory/03_cooperative_bargaining.md` (the R6 replication-lower-bound lemma, a sufficient-condition result that does not close this

question for high-tariff countries) and `paper/theory/04_shapley_value.md` Proposition 4.3, both of which remain unable to *prove* the property analytically. Any downstream claim resting on this monotonicity (imputation IR, the Brouwer-style codomain condition above, or Proposition D3.2 in `paper/theory/derivations/03_zeta_corner_vs_shapley.md`) is stated as conditional on the maintained hypothesis `hyp:wmono` (`appendix_proofs.tex`). **On Track A the hypothesis is *refuted*, not merely untested:** 17/192 single-player additions violate it and general superadditivity fails 20/301, with the failure already present at the pairwise level ($w(\{US, CAN\}) = -0.0056$); Canada is present in 16/17 of the violations — see [Proposition A.2\(iii'\)](#) and `paper/replication/results/bsy_replication/phase7_full_primary.md`. The downstream conclusions named above survive on Track A by *direct verification* (IR at every player, min slack 2.59% of $v(N_6)$; core membership 0/62), not via `hyp:wmono`; “empirically well supported” describes Track B alone.

$N = 2$ limit: symmetric Nash bargain only. We close this subsection with a sanity check that anchors our framework to [Nash \(1950\)](#) and [Bagwell et al. \(2021\)](#) (BSY).

Proposition 1 (Symmetric Nash bargain in $N = 2$). *Let $N = \{i, j\}$ and compute the Shapley imputation on the cooperative game w (9), for which $w(\{i\}) = w(\{j\}) = 0$ by construction. Then $\phi_i(w) = \phi_j(w) = \frac{1}{2}w(N)$, and this allocation coincides with the symmetric Nash bargaining solution of a two-player TU game with disagreement point $d_i = v(\{i\})$, $d_j = v(\{j\})$ (i.e. each country’s unilateral optimum) — not with [Bagwell et al. \(2021\)](#)’s disagreement point, which is the 1990 status quo. On the raw game v , the analogous symmetric-Nash-bargain coincidence (disagreement point = 1990 status quo, matching BSY exactly) holds if and only if $v(\{i\}) = v(\{j\})$; see the companion propositions B-1a/B-1b and R8 in `paper/theory/derivations/03_zeta_corner_vs_shapley.md` §2.5 for both directions and their economic interpretation.*

Proof sketch. With two players and $w(\emptyset) = 0$, the Shapley formula (12) applied to w gives $\phi_i(w) = \frac{1}{2}w(\{i\}) + \frac{1}{2}[w(\{i, j\}) - w(\{j\})]$. Since $w(\{i\}) = w(\{j\}) = 0$ by the cooperative normalization, this collapses to $\phi_i(w) = \phi_j(w) = \frac{1}{2}w(\{i, j\}) = \frac{1}{2}w(N)$. This is precisely the symmetric Nash bargaining solution of the two-player TU game with feasible set $\{x_i + x_j \leq w(N)\}$ and disagreement point $(0, 0)$ in w -space — which, translated back to v -space via $x_i = v(\{i\}) + (\text{share of } w)$, is the disagreement point $d_i = v(\{i\})$. $\square$

The raw-game split $\phi_i(v) = v(\{i\}) + \frac{1}{2}w(\{i, j\})$ therefore need *not* be 50–50: it equals $\phi_j(v)$ only when $v(\{i\}) = v(\{j\})$ (Proposition 1’s “if and only if” clause). For two low-tariff economies $v(\{i\}) \approx v(\{j\}) \approx 0$, so raw and cooperative coincide and both reduce to the symmetric Nash bargain; for an asymmetric pair (e.g. a high-tariff and a low-tariff economy) they diverge, and it is the *raw* split that shares BSY’s disagreement point.

The proposition shows that our framework nests [Bagwell et al. \(2021\)](#) *only when their estimated $\mu_i = \mu_j = 1/2$* ; the asymmetric case ($\mu_i \neq 1/2$, GMM-estimated from observed RTAs) is *not* nested by Shapley alone—reproducing it requires the weighted Shapley value of [Kalai and Samet \(1987\)](#), with exogenous weights w_i that are not derived from the game. We flag this as an open extension and emphasize that, at $N = 2$, our framework is observationally equivalent

to the symmetric Nash bargain regardless of underlying country characteristics; the marginal value-add of Shapley appears only at $N \geq 3$ through cross-coalition heterogeneity in $v(S)$.

2.5 Game-Theoretic Properties of $v(\cdot)$

The Shapley value is uniquely characterized by efficiency, symmetry, dummy, and additivity. Whether the resulting allocation is also *stable*—i.e. lies in the core—depends on additional structural properties of v that we now examine.

Singletons. By convention $v(\emptyset) = 0$. For any n , the singleton problem $\max_{s_n \in [0,1]} I_n^{\text{base}}(\widehat{W}_n - 1)$ (D1, no bargaining weight, no IR constraint) admits the status-quo point $s_n = 1$ (no change, $\widehat{W}_n = 1$) as a feasible point achieving value 0, so the maximum satisfies $v(\{n\}) \geq 0$ constructively. Whether $v(\{n\})$ is strictly positive depends on the magnitude of the unilateral consumer-surplus gain from liberalizing imports vs. the loss of tariff revenue. In our 7-player panel, $v(\{n\})$ scales monotonically with applied tariff $\times$ trade exposure:

$$\begin{aligned} v(\{\text{USA}\}) = v(\{\text{Canada}\}) = v(\{\text{Korea}\}) &= \$0.000\text{B (exactly),} & (\text{applied} \leq \text{unilateral optimum}) \\ v(\{\text{Brazil}\}) &= \$1.669\text{B,} & (\text{small/moderate}) \\ v(\{\text{Mexico}\}) &= \$1.941\text{B,} & (\text{moderate}) \\ v(\{\text{India}\}) &= \$26.411\text{B,} & (\text{high tariff} \times \text{scale}) \\ v(\{\text{China}\}) &= \$119.681\text{B,} & (\text{high tariff} \times \text{scale}). \end{aligned}$$

Strict positivity of singletons does *not* preclude near-dummy Shapley shares for these players: the Shapley value averages marginal contributions across coalition orderings, and a player whose marginal contributions are uniformly small (whether in the singleton or larger coalitions) still receives a small share. Conversely, a player with a *large* singleton (China, India) can carry a large *raw* Shapley share that is mostly its own unilateral gain; the cooperative game w (9) strips this out, and we report both raw and cooperative shares in Section 4.

Pair coalitions. Under D1, a pair $\{i, j\}$'s value $v(\{i, j\}) = \max_{s_i, s_j \in [0,1]} I_i^{\text{base}}(\widehat{W}_i - 1) + I_j^{\text{base}}(\widehat{W}_j - 1)$ is bounded below by $v(\{i\}) + v(\{j\}) \geq 0$ (the singleton-lever point $(s_i^*, s_j^* = 1)$ and its mirror are both feasible), so $v(\{i, j\}) \geq 0$ constructively — there is no longer a Pareto-improving-cut condition to check. Whether the coalition adds value *beyond* the two singletons (i.e. whether $w(\{i, j\}) > 0$) is **no longer open**: the D1 recompute sweep discharged it empirically (single-player w -monotonicity 100% pooled at $N = 3\text{--}7$, `superadditivity_sweep_D1.md`; imputation IR 5/5 N , `ir_verification.md`) — see §4 for the per- N numbers. This paragraph's earlier numbers (\$20–23B for China pairs, $v = 0$ for low-tariff pairs) were computed under the retired Nash-product value and are superseded.

Superadditivity and the R6 replication-lower-bound lemma. We say v is superadditive if $v(S \cup T) \geq v(S) + v(T)$ for any disjoint $S, T \subseteq N$; single-player w -monotonicity ($w(S \cup \{i\}) \geq w(S)$, equivalently $v(S \cup \{i\}) \geq v(S) + v(\{i\})$) is the special (singleton- T) case needed for imputation IR, and is *not* equivalent to general superadditivity.

Under D1, the available result is weaker but unconditional: for any coalition S and country $i \notin S$, replicating S 's optimal lever and adding i passively at the status quo is feasible for $S \cup \{i\}$, so

$$\Delta_i^v(S) \equiv v(S \cup \{i\}) - v(S) \geq I_i^{\text{base}}[\widehat{W}_i(s_S^*, s_i=1) - 1] \equiv \text{outsider_gain}_i(S), \quad (13)$$

a one-line construction (*sufficient*, not “if and only if”). If $\text{outsider_gain}_i(S) \geq 0$ for all (S, i) , this lemma implies v is monotone; the converse need not hold. Subtracting $v(\{i\})$ gives the w -analogue $\Delta_i^w(S) \geq \text{outsider_gain}_i(S) - v(\{i\})$, which is **generically non-binding** for high-tariff countries (whose $v(\{i\})$ reaches \$119.681B for China, \$26.411B for India, dwarfing a passive spillover term) — so this lemma *cannot, on its own logical content*, establish single-player w -monotonicity, which as a *theoretical* matter remains open (the R6 lemma is a sufficient-condition argument that does not bind for high-tariff countries, and no other proof is offered).

Empirically, on Track B, single-player w -monotonicity holds on 100% of tested pairs pooled across $N = 3\text{--}7$ (`superadditivity_sweep_D1.md`), and imputation IR is verified directly at every N (`ir_verification.md`) — see §4. **On Track A this is false, not merely open:** single-player w -monotonicity is violated 17/192 times on Track A, with the failure already present at the pairwise level (see the Track qualifier at §2.5 above and [Proposition A.2\(iii'\)](#) in the appendix). On Track B, the R6 lemma's inability to *prove* w -monotonicity is an open theoretical question about *why* the property holds; on Track A, by contrast, *whether* it holds has been answered, and the answer is no — there the R6 lemma's insufficiency is not merely a proof gap, it correctly anticipates a property that fails. See `paper/theory/03_cooperative_bargaining.md` (“R6”) for the full statement.

Convexity: convex to within numerical resolution (Track B only). Track qualifier. Everything in this paragraph is Track B (the seven-country Caliendo–Parro economy); Track A's supermodularity is not measured here (only single-player w -monotonicity and general superadditivity are, both above and in [Proposition A.2](#)), and Track A's general superadditivity outright fails 20/301 times (§2.5 above) — a fact this paragraph's “convex to within numerical resolution” language describes on a different track and must not be read to contradict. v is convex (supermodular) if $v(S \cup \{i\}) - v(S) \leq v(T \cup \{i\}) - v(T)$ for all $S \subseteq T \subseteq N \setminus \{i\}$, i.e. marginal contributions weakly grow with coalition size. By [Shapley \(1971\)](#), convexity implies the Shapley value lies in the (non-empty) core. On the D1 recompute, the game is convex to within numerical resolution: supermodularity fails on only 31 of 5103 valid $S \subseteq T$ triples at $N = 7$ (0.61%) and 40 of 7101 pooled across $N = 3\text{--}7$; the violating-triple gaps (\$0.016–\$0.046B) are within a few multiples of the cross-configuration sensitivity (\$0.003–\$0.006B) and are not evidence of genuine non-convexity (`paper/replication/results/superadditivity_sweep_D1.md`). This is nonetheless a **pre-registered FAIL** of the non-convexity half of prediction P-12 (threshold: failure rate $\geq 10\%$; measured 0.61%), reported as a failure and not reframed, while the companion w -monotonicity half of P-12 **PASSES** at 100% — the joint verdict is a FAIL of the conjunctive hypothesis, and the pre-registered Branch B fires: core non-empty (to within the same numerical resolution), Shapley value $\in$ core, so the fairness benchmark is also self-enforcing given feasible transfers (A5; TI-16(b)) — not unconditionally — and the free-riding/non-convexity

thread is demoted to a supporting result rather than the paper’s headline (`predictions.md` P-12 Discrepancies). A natural conjecture — that a coalition’s liberalization gives non-members a positive externality, so joining a large coalition contributes less at the margin than joining a small one, hence non-convexity — is refuted quantitatively: on the v3 numbers **tariff complementarity dominates** free-riding at every tested N , which is the opposite direction from what that mechanism predicts. Critically, convexity is a robustness property, not load-bearing: the Shapley value remains uniquely characterized by efficiency, symmetry, dummy, and additivity (Shapley, 1953) regardless of convexity.

Implications for stability. Core membership is verified directly rather than inferred from the approximate-convexity argument above: Track B’s core is directly checked and holds 0/126 times across the full non-convex-superadditivity placebo sweep (P-28, `p28_trackB_core.md`), and Track A’s core is directly checked and holds 0/62 times (P-23, `phase7_full_primary.md`). The residual 0.61% supermodularity gap above is numerical, not structural, and on that gap alone the fairness benchmark could only be *inferred* to be self-enforcing given feasible transfers (A5; TI-16(b)) — never unconditionally — since Shapley (1971)’s theorem requires exact, not approximate, convexity; the direct core verification above makes that inference chain unnecessary going forward. Bargaining IR in the sense of earlier drafts ($\widehat{W}_i \geq 1$ enforced inside the per-coalition program) no longer exists as a separate object under D1 — there is no constraint of that kind inside $v(S)$ (§2.2). The entire IR content of the model is now *imputation IR*: $\phi_n(v) \geq v(\{n\})$, equivalently $\phi_n(w) \geq 0$, which is verified directly on the Shapley allocation at every $N \in \{3, \dots, 7\}$ (`ir_verification.md`), discharging the "pending D1 recompute" status of earlier drafts (TI-5, `theory-invariants.md`).

2.6 Positive–Normative Bridge: Fairness Gap and Shapley-Consistent Weights

The Shapley imputation $\phi(w)$ computed above is a purely *normative* object, derived from the Caliendo and Parro (2015) primitives under the efficiency/symmetry/dummy/additivity axioms. Bagwell et al. (2021) instead *estimate* an exogenous pairwise bargaining weight ζ from observed RTA tariff changes via a Horn–Wolinsky (HW) bilateral bargaining network (their eq. 7, no-transfers/NTU). This subsection defines the objects that connect the normative benchmark to BSY’s estimated positive object—the *fairness gap* (in two variants) and its *inverse map*, and states, up front, exactly what these objects can and cannot establish. Throughout, ζ denotes exclusively BSY’s own pairwise weight parameter.

Positioning (E3, stated before the definitions, not after). This bridge is a **normative and efficiency benchmark** — *efficiency defined over the member group $\mathcal{N}$ under D1’s utilitarian objective* (eq. (7); A1’s member-only-objective scope clause, `00_assumptions.md`), **not** world efficiency inclusive of non-member (NES) welfare, and **not** efficiency relative to any alternative cooperative solution concept: this scope qualifier must travel with every use of the bare word “efficiency” below, since the term carries at least three mutually incompatible senses elsewhere in this paper (member-group utilitarian optimum; world welfare inclusive of NES spillovers, A1; Pareto-efficiency of a specific policy incidence, §4.1’s IR/core discussion) — not

a channel for identifying bargaining weights from data. Under D1 (eq. (7)), the optimal lever s_N^* is a pure utilitarian argmax, computed independently of which imputation rule is later applied to divide the resulting surplus — so the allocation rule (Shapley vs. any ζ -weighted alternative) produces *no distinguishable prediction about equilibrium tariffs*, and cannot be identified from tariff moments. The only data that could in principle discriminate between allocation rules are observed international transfers, which do not feature in GATT-era plurilateral liberalization (§2.2, TU discussion). We do *not* claim, anywhere below, that observed tariff outcomes identify a bargaining-weight parameter; the object identified (in the ordinary sense) is the deterministic exact level-set Z_0^* of ζ -vectors that *exactly* reproduce the Shapley-fair benchmark (§2.6 below, Definition 3) — a *normative-consistency* object with no sampling error of its own, not a discovered structural parameter. A numerical-tolerance neighborhood Z_δ^* is carried alongside it for solver-precision reasons only, and is explicitly not a confidence set; all statistical content is carried separately, by the P-09 Wald statistic comparing $\hat{\zeta}$ to Z_0^* .

Why the raw game v , not the cooperative game w , is the bridge’s primary operand. $s_i^{HW}(\zeta)$ ’s underlying gains $g_i(\zeta)$ are measured from the **1990 status quo** with nothing subtracted; $\phi_i(w)/w(N)$ has already netted out $v(\{i\})$. The two objects are not on the same baseline — on the CP-calibrated v3 evidence, raw USA share is 36.50% at $N = 3$ (no China present) and 5.25–6.06% at $N = 4–7$ (China present, and accounting for 67–85% of $v(N)$), while cooperative USA share is 35.5–45.2% at $N = 4–7$ — a gap of roughly 6–8 $\times$ at $N = 4–7$, wider than earlier (pre-D1) estimates suggested, driven by player-set composition (China’s presence or absence) rather than by calibration choice. Only $\phi_i(v)/v(N)$ shares BSY’s disagreement point (both are welfare gains relative to 1990 status quo, consistent with assumption A7 in the theory notes), so it is the **primary** bridge operand; $\phi_i(w)/w(N)$ remains a well-defined, independently interesting **secondary** object that answers a different question — see the companion propositions (Prop. B-1a/B-1b and “R8” in `paper/theory/derivations/03_zeta_corner_vs_shapley.md` §2.5): zero-normalizing w and taking its Shapley value is, in the two-player TU case, exactly equivalent to re-defining the disagreement point from “1990 tariffs frozen” to “every country at its own unilateral optimum.” That re-defined disagreement point is not behaviorally credible on its own terms (if China could unilaterally capture \$119.681B (corrected from an earlier draft’s placeholder “\$20–\$50B”), why did it not do so?), a further reason the raw game is primary.

Definition 2 (Fairness gap, raw (primary) and cooperative (secondary)). Let $\hat{\zeta}$ denote the Bagwell et al. (2021) GMM point estimate and let $s_i^{HW}(\zeta)$ denote country i ’s dollar-denominated surplus share in the resulting Horn–Wolinsky equilibrium, $s_i^{HW}(\zeta) = g_i(\zeta) / \sum_{k \in \mathcal{N}} g_k(\zeta)$, where $g_i(\zeta)$ aggregates i ’s welfare gains across every bilateral pair it negotiates in the BSY network and $\mathcal{N}$ is the full set of 6 BSY negotiating countries (not a 5-country in-pair subset: even the country absent from every pair has a non-zero general-equilibrium welfare gain, verified numerically). The **primary** fairness gap is

$$\Gamma_i^{\text{raw}}(\zeta, b) \equiv \frac{\phi_i(v)}{v(N)} - s_i^{HW}(\zeta, b), \quad (14)$$

where b indexes the HW equilibrium branch (§2.6 inherits the within-branch continuity caveat

below). The **secondary**, MC-16-aligned cooperative gap uses a comparator netted by the *same* unilateral term that normalizes the numerator,

$$\Gamma_i^{\text{coop}}(\zeta, b) \equiv \frac{\phi_i(w)}{w(N)} - \frac{g_i(\zeta, b) - v(\{i\})}{\sum_{k \in \mathcal{N}} [g_k(\zeta, b) - v(\{k\})]}. \quad (15)$$

Comparator provenance, and a symmetric disclosure (author adjudication, 2026-08-16, Round 14 item 3; IDENT-13-05). Every reported $\Gamma_i^{\text{raw}}(\hat{\zeta})$, $s_{US}^{HW}(\hat{\zeta}) = 10.04\%$ (MC-17), and every other headline number keyed to $\hat{\zeta}$ takes $s_i^{HW}(\hat{\zeta})$ from **BSY’s own cached equilibrium** (`track_bargaining_closest.mat`’s terminal solution), **not** from a re-solve by our own HW pair solver. This is a deliberate choice, not an oversight: our independent re-solve pipeline (`run_zstar.py`, §4) does not reproduce it exactly — at $\hat{\zeta}$, our solver lands 26 of 151 negotiated bilateral lines (across BSY’s $K_{\zeta} = 7$ bilateral *pairs*, not 151 pairs) in a different (no-deal) equilibrium branch than BSY’s cache, giving $\Sigma g^{HW} = 0.1487$ against the cached $\Sigma g^{HW} = 0.3124$, a $2.1\times$ discrepancy (Round 14 math/identification review, MATH-66/IDENT-14-01). We therefore treat the **cached** equilibrium as authoritative for every already-published Γ , and record the 26/151 branch discrepancy as a **known limitation** of our own re-solve infrastructure, not as evidence against the cached numbers. This choice has a symmetric disclosure obligation the paper must also carry: the test center $\hat{\zeta}$ itself is likewise *borrowed*, not independently reproduced — our own GMM re-estimation of $\hat{\zeta}$ (Phase 5, `phase5_verdict.md`) did not converge to BSY’s published point (§2 of that verdict: SSR fell only 5% before the solver stalled). Both the bargaining-weight point $\hat{\zeta}$ and the equilibrium branch $s^{HW}(\hat{\zeta})$ evaluated at it are therefore taken on trust from BSY’s own published/cached objects, not independently verified end-to-end. This strengthens, rather than undermines, this paper’s C2 finding (BSY’s own estimation and equilibrium-selection machinery is numerically fragile): we are reporting that fragility, not papering over it by silently substituting a different, self-computed $\hat{\zeta}$ or equilibrium branch that happens to be easier to reproduce.

The two gaps satisfy an exact dollar identity (a one-line algebraic check) that separates a *distributional* term from a *level* term,

$$\phi_i(v) - g_i(\zeta, b) = v(N) \Gamma_i^{\text{raw}}(\zeta, b) + s_i^{HW}(\zeta, b) [v(N) - \sum_k g_k(\zeta, b)], \quad (16)$$

whose second term is exactly the object compared in the cooperative-efficiency benchmark of Section 4 (BSY’s own equilibrium total surplus vs. the D1 grand-coalition optimum): this identity is what lets us say a stated Γ_i^{raw} finding is a distributional claim and not silently a re-statement of an efficiency comparison.

Granularity-consistency discipline for the level term. The level term $s_i^{HW}(\zeta, b) [v(N) - \sum_k g_k(\zeta, b)]$ above is, in every reported instantiation, a comparison between quantities that can differ along three separate axes: *instrument granularity* (our country-scalar lever s_n versus BSY’s own line-by-line negotiated outcome), *agenda coverage* (which tariff lines are on the table at all), and *transferability regime* (TU vs. NTU, §2.6 above; TI-13(d’)). A subtraction that moves across more than one of these axes at once (e.g. $v(N)$, a full-coverage scalar-instrument quantity, minus a restricted-coverage, line-instrument comparator) conflates them, and the re-

sult is, at best, a **lower bound** on the single axis it is nominally meant to isolate — *not* that axis’s value itself. This is not a hypothetical risk: it is the precise failure mode that produced this project’s original, retracted two-way coverage/shortfall split (P-31, MATH-83/84/85) and, independently, a front-matter passage that mis-summed three non-additive, differently-instrumented quantities as if they were parts of one decomposition (MATH-90/91). Any reading of this level term as a clean single-axis quantity must therefore first hold instrument granularity fixed on both sides of the comparison (line-matched vs. line-matched, or scalar vs. scalar) and hold the transferability regime fixed on both sides (or carry it as an explicit separate term, per clause (d’)); where a fixed-granularity, fixed-regime comparison cannot be constructed, the resulting number must be reported and labelled as a lower bound, never silently upgraded to an exact reading. This paragraph is stated at the point where the level term is first defined so that any future instantiation of (16) inherits the discipline by construction — the failure mode above has already recurred twice in this project’s history, and this pin exists to foreclose a third instance.

HW comparator is not itself box-feasible: nesting claims for multi-term dollar identities are empirical, not constructive. A natural strengthening of the granularity-consistency discipline above would claim that a multi-term additive breakdown of a level-type gap (e.g. $D - F_{\text{line}}^{v2}, F_{\text{line}}^{v2} - F_{\text{line}}^{IR}, F_{\text{line}}^{IR} - \sum_k g_k(\hat{\zeta}, b)$) has every term non-negative *by construction*, on the reasoning that the three terms’ underlying feasible sets nest and BSY’s own cached bargained-equilibrium point is a member of the innermost one. This reasoning fails at its final step: restricted to the 151 masked (negotiated) lines, 5 of 151 (3.3%) of BSY’s own cached $\hat{\zeta}$ tariff schedule lies genuinely *outside* this paper’s cuts-only $[0, \text{mfn90}]$ box — all five below zero (legal under BSY’s own lower bound $t \geq -0.25$, outside our $s \geq 0$ floor; the known lower-bound asymmetry of clause (d)(v)) — so BSY’s cached point is not a feasible point of our own restricted polytopes. BSY’s comparator is otherwise confirmed cuts-only at its own source (its own MATLAB solver’s upper bound is `ub_t=mfn_tariffs1990+1e-6, est_barg_power_all_lsqrnonlin.m:217-219`), so clause (d)’s “same truncation” comparability defence stands. The multi-term identity itself is unaffected (it is exact real-number subtraction, true regardless of feasibility), but each term’s non-negativity, wherever reported, is a fact about the specific computed numbers on record, verified independently term-by-term, not a geometric guarantee inherited from set inclusion. This mirrors, and should be read together with, the granularity-consistency discipline immediately above: both paragraphs exist to stop a plausible-sounding “the algebra guarantees this” reading from substituting for the actual verification.

Axes (a) and (d) remain jointly embedded in $\sum_k g_k$. TI-13(d)(vi) names five axes that must be disclosed when the level term above is read as evidence about any one of them: (a) cut-depth on the shared negotiated lines, (b) agenda-scope, (c) TU-vs-NTU (clause (d’)), (d) coalition structure (the grand coalition analyzed here vs. BSY’s actual seven-pair bilateral architecture), and (e) instrument granularity (the paragraph immediately above). Only (b), (c), and (e) have, as of this draft, been separately measured. Axes (a) and (d) are **not** separately isolated by the current empirical program: both are embedded jointly inside $\sum_k g_k(\hat{\zeta}, b)$, BSY’s own actual bilaterally-negotiated outcome, which mixes “how deep the agreed cuts are on the

negotiated lines” with “what a seven-pair bilateral architecture can deliver relative to a single grand-coalition optimizer” in one number. Any residual gap attributed to “within-agenda bargaining shortfall” after axes (b)/(c)/(e) are accounted for is therefore still a *joint* reading of (a)+(d), not a pure bargaining-inefficiency measure net of coalition structure; separating them would require re-solving BSY’s actual bilateral-pair architecture under the weight-corrected D1 objective, which was attempted (P-40, 2026-08-19, `run_p40_f_pairs_eq.py`) and returned a boundary/no-verdict disposition, not a clean separation of (a) from (d): the fixed point of the *local* best-response map over BSY’s seven bilateral pairs — each pair’s per-round sub-problem is re-solved from a single warm start, so the object is, by construction, at most a fixed point of a *local*, not a certified global, best-response correspondence (00_assumptions.md A10’s Excludes clause, rewritten 2026-08-20, MATH-117/IDENT-21-03) — $F_{\text{pairs}}^{\text{eq}} = 0.305867$ (00_assumptions.md A10: existence stipulated; three independent trajectories — forward from the BR1 assembly, reverse-order from the same start, and forward from the joint 151-line planner’s own optimum F_{line}^{v2} ’s argmax — converge to this identical fixed point, dispersion 7.22×10^{-8} in objective value and 1.50×10^{-6} per-line L^∞ ; A10’s multi-start monitor, discharged 2026-08-19), sits within the pre-registered boundary band of $\Sigma g^{\text{HW}} = 0.312441$ (margin 0.006574, the same order of magnitude as the fixed point’s own *within-trajectory* convergence displacement — the round-1 $\rightarrow$ eq step, 0.003624, and the round-1 best-response residual, 0.003879), and the object’s primary driver is a single member’s individual-rationality violation at that equilibrium (EU, $g_{\text{EU}} = -0.1383$, a margin roughly 21 $\times$ the boundary gap and measured entirely within this project’s own pipeline), not a coalition-structure effect that would isolate axis (d) — see `paper/replication/results/bsy_replication/f_pairs_member_aggregate.md`. **A0 comparator-branch caveat (added 2026-08-20, IDENT-21-01).** $\Sigma g^{\text{HW}} = 0.312441$ is BSY’s own *cached* estimated equilibrium, not observed data; this project’s independent resolve at the identical $\hat{\zeta}$ does not reproduce it (`zstar_gate1.md` GATE 1 ABORT: 26 of 151 lines register no-deal, resolved objective roughly double the cached one) — precisely the branch-mismatch the A0 doctrine above ($L(\hat{\zeta})$ cached-vs-resolved) requires disclosing whenever Σg^{HW} is cited as a comparator. The 0.006574 margin is therefore, read on its own, a comparison *across* two different pipelines’ branches, not purely a within-pipeline threshold test. This does not change the disposition: the **operative** reason this section reports *no verdict* is EU’s individual-rationality violation above, measured entirely within this project’s own pipeline and immune to the comparator-branch issue; the level comparison against Σg^{HW} is downgraded to a disclosed diagnostic — **not adjudicable at this precision** — rather than the operative reason for the disposition. This boundary reading is therefore **marginal relative to Σg^{HW}** , but it is **robust to starting point and update order** (the three-trajectory discharge above); the qualifier “**conditional on equilibrium selection**” (A10) accordingly no longer applies to this margin. What was open — the *local*-vs-global qualifier on each pair’s sub-problem — has been discharged to *spot-checked global* (2026-08-20, MATH-119/IDENT-22-04, Round 22 canonical remediation; MATH-117 cold-start spot check, bundled per author-approved adjudication item 3, `paper/replication/results/bsy_replication/p43_pair_ir_probe.md` §C): all 7 pairs were cold-started from free trade with the other 6 pairs frozen at the reported $F_{\text{pairs}}^{\text{eq}}$ fixed point, and every pair re-converged to the *identical* per-pair value, with

every displacement in the 1.5×10^{-12} – 5.6×10^{-9} range — the numerical noise floor, not a real improvement. The “local” qualifier is accordingly upgraded to “**spot-checked global (one alternative start per pair; not a full global-optimality certificate)**” (00_assumptions.md A10 Excludes (i), same date). If a pair’s true global best response were to dominate the local one found here, $F_{\text{pairs}}^{\text{eq}}$ would be *underestimated*, which — consistent with the bias-direction disclosure in that artifact’s §5 (IDENT-19-01(a)) — pushes in the same direction as the boundary branch’s trigger (understating $F_{\text{pairs}}^{\text{eq}}$ raises its probability of falling below Σg^{HW}); the spot check now finds this hypothetical did not materialize at any of the 7 pairs, so this directional argument is retained only for completeness. **Scope correction (math minor 8, IDENT-22-02(d), 2026-08-20):** this directional argument is a statement about *one pair’s own re-optimization, holding every other pair fixed* — it does not extend to, and must not be read as absorbing, either (a) the distinct fixed-point-multiplicity risk (00_assumptions.md A10 Excludes (ii)), which the 7-pair spot check does not address (it holds six of seven pairs fixed and cannot detect a jointly different trajectory), or (b) a separate risk, now diagnosed and scoped (added 2026-08-20, exit-plan canonical sync, model-refiner): the same family’s related P-43 EU-IR probes were found to exhibit a **37–113× discrepancy** between the SLSQP-internal objective read-off and an independent GE re-evaluation at the identical terminal tariff vector. This was *initially* diagnosed as an MPEC equilibrium-branch mismatch — the same general branch-selection phenomenon A2 already names for this paper’s other GE solver (§2, 00_assumptions.md A2: “branch selection is therefore in principle damping-dependent”) — because both wage vectors independently satisfied their own market-clearing residual to machine precision (10^{-16} – 10^{-14} ; p43_resid_checkpoint.md), which rules out an unconverged solve and is consistent with, but does not establish, branch multiplicity. A dedicated discriminant test (math124_discriminant.md) has since run the check that distinguishes the two hypotheses — evaluating each wage vector against the *other* route’s own market-clearing condition, not only its own — and finds the true cause is a **trade-deficit (D) convention mismatch**: the in-solve MPEC constraint uses the raw, nonzero observed D (make_static(p)), while the robust GE re-evaluation forces $D = 0$ (welfare_case(..., D_zero=True)) — two different economies, not two branches of one system. **Corrected cross-reference:** this is not a further instance of the A2 phenomenon — it was initially misread as an A2-type branch problem on the evidence then available, and is, on discriminant testing, confirmed to be a D -convention implementation inconsistency instead, a single-economy code-level mismatch rather than genuine multi-root branch selection on one shared system. **Scope, corrected:** of the 13 originally flagged scripts, 3 (run_f_line_family.py and its two now-superseded predecessors) are grep false positives, confirmed safe to machine precision (10^{-13} – 10^{-16} vs. a $\sim 3.09 \times 10^{-2}$ -order positive control); the genuinely-affected scope is 10 scripts in the F_{pairs} family (the P-40/P-41/P-43 probe scripts). The F_{line} headline family ($F_{\text{line}}^{v2} = 0.436840$, $F_{\text{line}}^{IR} = 0.428519$, $F_{\text{line}}^{IR,corr} = 0.428073$, and the H2 shortfall built on them) is therefore **confirmed safe, argmax-certified, and stands exactly as previously reported**. $F_{\text{pairs}}^{\text{eq}} = 0.305867$ is **not similarly cleared**, correcting the claim in an earlier draft of this paragraph that every banked $F_{\text{pairs}}/F_{\text{line}}$ number was already clear of this risk: it is built by run_p40_f_pairs_eq.py, which is on the confirmed-affected 10-script list and has not been re-verified or re-optimized under the corrected D -convention

— its level number and every per-country g_i built on it are **provisional pending R&R re-optimization** (00_assumptions.md A10 Excludes (iv)). This does not retroactively flip $F_{\text{pairs}}^{\text{eq}}$ ’s own already-banked disposition (the boundary/no-verdict framework built on it, §4, which turns on EU’s individual-rationality violation, measured independently of this risk), but any prose citing $F_{\text{pairs}}^{\text{eq}}$ ’s specific level as settled must carry this provisional flag. Axes (a) and (d) therefore remain jointly embedded, now for a re-diagnosed, re-scoped reason. This scope limitation is stated here, at the level of the model definition, so that it is visible independently of any specific empirical write-up’s framing.

Discharge (added 2026-08-21, paper-writer, source p40_dconvention_rerun_result.md/.json): the provisional flag above is lifted, and the paragraph’s specific numerical claims are superseded, several of them by a sign reversal rather than a magnitude revision. $F_{\text{pairs}}^{\text{BR1}}$ and $F_{\text{pairs}}^{\text{eq}}$ were both re-run with $p[\text{“D”}] = 0$ set before `make_static(p)`, matching the F_{line} family’s confirmed-safe pattern, and independently verified safe to machine precision on both legs (in-solve residual at the robust wage $\leq 3.3 \times 10^{-16}$; positive control on the raw, unzeroed- D economy $\approx 3.1 \times 10^{-2}$, the same order the discriminant test used to diagnose the original bug). Both headline numbers move by more than the pre-registered 1% threshold: $F_{\text{pairs}}^{\text{BR1}}$: $0.302243 \rightarrow 0.310033$ (+2.58%); $F_{\text{pairs}}^{\text{eq}}$: $0.305867 \rightarrow 0.311364$ (+1.80%). **The paragraph’s stated primary driver is retracted, not merely revised:** the claim above that “the object’s primary driver is a single member’s individual- rationality violation at that equilibrium (EU, $g_{\text{EU}} = -0.1383$, ... measured entirely within this project’s own pipeline and immune to the comparator-branch issue)” does not survive the corrected D -convention — at the new fixed point the EU *satisfies* its own 1990 floor ($g_{\text{EU}} = +0.0155$), so this is a sign flip in the paper’s own explanatory mechanism, not a magnitude update, and the $21\times$ multiple built on $g_{\text{EU}} = -0.1383$ no longer describes anything real. The margin against $\Sigma g^{\text{HW}} = 0.312441$ itself shrinks from 0.006574 (0.913% of $v(N)$) to 0.001077 (0.149% of $v(N)$), an 83.6% relative contraction — the NO-VERDICT-BANKED disposition is therefore *weaker*, not stronger, than this paragraph originally reported, though the categorical bucket is unchanged. The within-trajectory displacement diagnostics cited above (0.003624 , 0.003879) and the cross-trajectory/cold-start spot checks (7.2×10^{-8} , 5.6×10^{-9}) were not themselves part of this re-optimization batch and are **pre- D -fix figures, recomputation pending**, not silently retained as current. The provisional-flag policy this section states — that a level number built on a confirmed-affected script must be flagged provisional until re-optimized (00_assumptions.md A10 Excludes (iv)) — remains the correct general rule and is unchanged; only its application to $F_{\text{pairs}}^{\text{eq}}/F_{\text{pairs}}^{\text{BR1}}$ specifically is **since discharged by the re-run (2026-08-21)**. Full numbers: p40_dconvention_rerun_result.md/.json; narrative detail in §4.

ζ sensitivity is now decisive for the P-40 boundary disposition, and a scope limitation this draft cannot lift. The boundary disposition above is itself a function of which realized-outcome comparator is used: at the paper’s own $\zeta = 0.5$ robustness point, $\Sigma g^{\text{HW}}(0.5) = 0.297081 < F_{\text{pairs}}^{\text{eq}} = 0.311364$ (re-optimized under the corrected D -convention, superseding the pre-fix 0.305867 ; see the discharge note above) $< \Sigma g^{\text{HW}}(\hat{\zeta}) = 0.312441$, so the boundary branch does *not* fire at $\zeta = 0.5$ (§4). This is the first point in the paper where ζ ’s own estimation uncer-

tainty (Z_δ^* below; BSY’s Table V standard errors on $\hat{\zeta}$) is decisive for a banked disposition, not merely a robustness caveat; mapping BSY’s own $\hat{\Sigma}$ into an interval for Σg^{HW} and re-checking whether the boundary condition holds throughout that interval is registered future work, not attempted here. Separately: this section’s “architecture ceiling” language is deliberately never used for $F_{\text{pairs}}^{\text{eq}}$ (00_assumptions.md A10’s own Consequence clause forbids that reading for a single computed fixed-point set), and that restriction cannot be lifted within this submission — doing so would require a genuinely different bilateral-pair construction (a full six-objective, IR-consistent Nash-in-Nash pair equilibrium, $F_{\text{pairs}}^{\text{eq},6\text{obj}}$) that is scoped to R&R, not computed here.

Reporting protocol: not a point value, not frozen. We do not report $\Gamma_i^{\text{raw}}(\hat{\zeta})$ as a single number, but the object we report is now split into two, deliberately, because they answer different questions and only one of them is a statistical object. The **primary** object is the *deterministic exact level set*

$$Z_0^* \equiv \{\zeta \in \mathcal{Z} : s^{HW}(\zeta) = \phi(v)/v(N)\} = \{\zeta \in \mathcal{Z} : L(\zeta) = 0\}, \quad (17)$$

the (possibly empty) set of ζ -vectors that *exactly* reproduce the Shapley-fair raw share — a purely algebraic object, with no sampling error attached to the criterion $L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(N)\|_2^2$ itself ($\phi(v)/v(N)$ is a computed structural quantity, not an estimate with a standard error, and $s^{HW}(\zeta)$ is model output, not data). All statistical content is carried by a single object built *on top of* Z_0^* : the pre-registered Wald statistic (prediction **P-09**), $\min_{\zeta \in Z_0^*} (\zeta - \hat{\zeta})' \hat{\Sigma}^{-1} (\zeta - \hat{\zeta})$ compared against its χ^2 critical value, which asks whether BSY’s own estimation uncertainty around $\hat{\zeta}$ (Table V standard errors) can statistically distinguish $\hat{\zeta}$ from the Shapley-consistent set. This is where Andrews (1999)’s warning about boundary parameters is operative — three of $\hat{\zeta}$ ’s seven coordinates sit on the optimizer’s corner bounds with standard errors as large as 0.13–0.47, so delta-method confidence intervals around $\hat{\zeta}$ itself are invalid, and the Wald statistic must be built from a boundary-robust (S-set / residual-bootstrap) construction, not a naive delta-method ellipsoid.

The **secondary** object, Z_δ^* , is retained purely as a *numerical-tolerance* level set — **explicitly not a confidence set** — because $L(\zeta^*) = 0$ may fail to be attained exactly by the solver even when it is attained in principle (finite-precision optimization), and because Z_0^* itself may be empty on the data while a small- δ neighborhood is not. δ is set to the numerical noise of the HW solve, but L is built in *share* space ($s^{HW}(\zeta) - \phi(v)/v(N)$, both unitless shares of the pie), not in the native dollar-denominated $g_i(\zeta)$ units that `eq_tol` is measured in — **these are different units and the naive equation $\delta = \text{eq_tol}$ silently skipped the conversion (corrected 2026-08-13, IDENT-11-13)**. The correct chain: BSY’s own bargaining-equilibrium outer loop terminates at `eq_tol` = 10^{-4} in native $g_i(\zeta)$ units (paper/literature/bsy2021_replication_code_notes.md §2.2); converting to the share space L is built on divides by the total surplus $\Sigma g^{HW} = \sum_k g_k(\hat{\zeta}) \approx 0.312$ (sanity_assertions.md), giving a per-coordinate share noise of $10^{-4}/\Sigma g^{HW} \approx 3.2 \times 10^{-4}$; squaring and summing over the $N_6 = 6$ coordinates of L ’s ℓ_2^2 norm (the six BSY negotiating countries’ shares — **not** the seven estimated bargaining pairs K_ζ that index ζ itself; these are two different dimensions,

kept notationally distinct from here on, corrected 2026-08-14, MATH-64/IDENT-13-06) gives a worst-case $\|dL\| \approx 6 \times (3.2 \times 10^{-4})^2 \approx 6.2 \times 10^{-7}$. **The volume-ratio exponent uses the wrong dimension count (corrected 2026-08-16, math minor 3):** the six coordinates entering L 's ℓ_2^2 norm are *shares* constrained by $\sum_i s_i^{HW} = \sum_i \phi_i(v)/v(N) = 1$, so only $N_6 - 1 = 5$ of them are free; the correct exponent is $160^{5/2}$ (radius ratio $\sqrt{160} \approx 12.7$ raised to the free-dimension count 5, $\approx 8.1 \times 10^4$), not $160^{N_6/2} = 160^3$, which double-counts the adding-up-constrained coordinate.

δ re-anchored to a direct noise-floor measurement, not the propagation calculation above (corrected 2026-08-16, IDENT-14-02(c)). The chain above ($\text{eq_tol} = 10^{-4}$ native units $\rightarrow$ share-space propagation $\rightarrow \|dL\| \approx 6.2 \times 10^{-7}$) is a *theoretical* error-propagation estimate, not a measurement, and the claim that followed it in an earlier draft — “ $\delta = 10^{-6} \gtrsim$ the measured noise floor” — is **false as stated**: it was never actually measured against a noise floor, only against its own propagation calculation. A direct measurement now exists (`run_zstar.py`'s two independent re-solves of the same ζ point, Round 14 identification review IDENT-14-02, provenance to be consolidated in the forthcoming `noise_floor_eps_L.md`, the L -space analogue of `noise_floor_eps_B.md`): the same ζ resolved twice gives L -values differing by $\varepsilon_L \approx 4.1 \times 10^{-4}$ — **about 660 $\times$ larger** than the theoretical propagation estimate of 6.2×10^{-7} above (the gap is consistent with additional noise sources the propagation calculation does not model, e.g. equilibrium-branch selection at corner ζ , (Andrews, 1999), and the inner-loop HW solver's own tolerance, not just BSY's outer-loop `eq_tol`). **We therefore re-anchor δ to the measured floor, not the propagation estimate**: following this paper's own 10 $\times$ -noise-floor convention used elsewhere for ζ^* -solver tolerances (`tol_g` = $10\varepsilon_\zeta$, `paper/theory/derivations/04_shapley_consistent_weights.md` §3.1(a)3), the operative δ moves to $\delta \approx 10 \times 4.1 \times 10^{-4} \approx 4 \times 10^{-3}$, order 10^{-3} , **not** 10^{-6} — three orders of magnitude looser than the value used to adjudicate the 2026-08-15 run (§4). This is a *provisional* re-anchoring pending the dedicated `noise_floor_eps_L.md` measurement (parallel empirics-runner batch, Round 14); any Z_δ^* adjudication computed under the old $\delta = 10^{-6}$ must be re-read against this corrected order of magnitude, and **no $Z_\delta^* = \emptyset$ finding is valid until δ is re-anchored this way** (see the empty-set adjudication rule below, revised 2026-08-16 to require this as a precondition). The original propagation calculation is retained above only as a lower bound on the noise a correctly specified δ must exceed, not as the calibration itself. It is a *numerical* tolerance either way, and **Chernozhukov–Hong–Tamer-style (Chernozhukov et al., 2007) calibration does not apply to it**, because CHT calibrates a confidence set to a criterion function's *sampling* variation, and $L(\zeta)$ here has none — both $s^{HW}(\zeta)$ and $\phi(v)/v(N)$ are deterministic model outputs at any fixed ζ . **Resolving power of the re-anchored δ (added 2026-08-18, IDENT-15, missing-moments item #5).** The re-anchoring above has a cost worth stating in the same share-space units used throughout this section: $\sqrt{\delta} = \sqrt{4 \times 10^{-3}} \approx 6.3\text{pp}$ against $\sqrt{L(\hat{\zeta})} = \sqrt{0.0196} \approx 14.0\text{pp}$ — a discriminating ratio of only $\approx 2.2\times$. Re-anchoring δ to the measured noise floor buys *constructibility* of a non-degenerate Z_δ^* (rather than a δ set so tight nothing could ever qualify), at the cost of a resolving power that is itself a substantial fraction of the very distance (Γ , s^{HW} share gaps) it is meant to discriminate among candidate ζ values for. This is the most condensed available statement of why the ζ^* layer is not identified at this

numerical precision: the tolerance required to make the object computable is comparable in magnitude to the quantity it is supposed to measure.

Both registered branches of this pre-registered directional claim have fired. The claim was pre-registered as a two-sided test precisely because the observed $s_{US}^{HW}(\hat{\zeta}) \approx 10\%$ put a naive +5pp lower bound at meaningful risk. Phase 7 has since computed $\phi_{US}(v)/v(N) = 13.33\%$ (§4), so the *point estimate* $\Gamma_{US}^{\text{raw}}(\hat{\zeta}) = +3.29\%$ is positive but below the +5pp secondary threshold: **P-07'' (magnitude, $\geq +5\text{pp}$) closed FAIL, 2026-08-12.** The **primary** object, P-07', is defined not on the point estimate but on the sign of the $\hat{\Sigma}$ -confidence-set interval's lower bound (IDENT-11-01 object correction); that interval could not be constructed — the Z_δ^* -based search stalled on all 8 pre-registered starting points (§3.1(g) of `derivations/04_shapley_consistent_weights.md`) — so **P-07' closed INCONCLUSIVE, 2026-08-15:** this does *not* close under the row's own pre-registered stalled-search branch rule (IDENT-11-04(c)) — that rule's antecedent never obtained, since the $\hat{\Sigma}$ -confidence-set interval was never constructed for the rule to be evaluated against; closure instead cites the plain Status-legend INCONCLUSIVE definition directly: the primary object could not be constructed on the evidence available, full stop, with no branch-rule machinery invoked. Both dispositions are final, not pending.

Definition 3 (Shapley-consistent bargaining weight sets). The **primary** identified object is the deterministic exact level set Z_0^* of (17). The **secondary**, numerical-tolerance level set is

$$Z_\delta^* \equiv \{\zeta \in \mathcal{Z} : L(\zeta) \leq \delta\}, \quad L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(N)\|_2^2, \quad \delta \approx 4 \times 10^{-3} \text{ (order } 10^{-3}; \text{ re-anchored to } (18))$$

with $\mathcal{Z} = [0.005, 0.995]^{K_\zeta}$ BSY's own optimizer box ($K_\zeta = 7$ the number of estimated bilateral pairs — kept notationally distinct from $N_6 = 6$, the number of BSY negotiating countries indexing the target vector $\phi(v)/v(N)$; corrected 2026-08-14, MATH-64/IDENT-13-06); Z_δ^* is a numerical-tolerance neighborhood of Z_0^* , not a confidence set.

Empty-set adjudication rule (author adjudication, 2026-08-14, IDENT-13-02(b); amended 2026-08-16, MATH-70/IDENT-14-05/TEST-14-04 with three power pre-conditions after the rule as originally stated proved unreachable and was correctly declined by the author on 2026-08-15 — see the closing paragraph of this note). Z_δ^* , and *a fortiori* Z_0^* , may be empty: nothing above guarantees L attains a value at or below δ anywhere on $\mathcal{Z}$. If $Z_\delta^* = \emptyset$ under the registered **absolute** definition (18), predictions P-08 and P-09 — both defined as objects *on* Z_δ^* — are adjudicated **STRUCTURAL-REJECT**, not INCONCLUSIVE: an empty level set means no admissible bargaining-weight vector in BSY's own optimizer box reproduces the Shapley-fair share to within numerical tolerance, which is a stronger, more informative finding than a stalled or inconclusive search, and is pre-registered as such precisely to remove any temptation to substitute a *relative* object (e.g. $\{\zeta : L(\zeta) \leq L_{\min} + \delta\}$, which is never empty by construction) for the registered absolute one.

Notation (introduced 2026-08-16, matching MATH-70's requirement): $L_{\min} \equiv \min_{\zeta \in \mathcal{Z}} L(\zeta)$ denotes the *true* minimum over the entire box — generally uncomputable, since $\mathcal{Z}$ is a continuum — while $\hat{L}_{\min} \equiv \min_{\zeta \in \mathcal{D}} L(\zeta)$ denotes the *sample* minimum over whatever finite design $\mathcal{D} \subset \mathcal{Z}$ was actually evaluated (e.g. the ≤ 8 pre-registered starting points of §3.1(g)).

Any numerical run reports $\hat{L}_{\min}$, and $\hat{L}_{\min} \geq L_{\min}$ always: the sample minimum is an **upper bound** on the true minimum, never a lower bound and never the true minimum itself except by coincidence.

Three preconditions for a legitimate STRUCTURAL-REJECT (amended 2026-08-16, closing MATH-70/IDENT-14-05/TEST-14-04). The rule as stated above is, read literally, **unreachable by any finite search**: certifying $Z_{\delta}^* = \emptyset$ requires certifying $L_{\min} > \delta$ over a continuum, and a finite design can only ever report $\hat{L}_{\min}$, an upper bound. `run_zstar.py`’s 2026-08-15 run mechanically fired STRUCTURAL-REJECT on four starting points anyway; the author correctly declined to bank it (reported in §4’s discussion of the ζ^* search), on exactly this ground — **an upper bound cannot certify a lower bound**, the same error this note’s own P-29/TW-max discussion (Round 14 econ review, ECON-60) warns against in the opposite direction. That declined event is the operative precedent for the three conditions below; absent all three, any apparently-empty Z_{δ}^* must be adjudicated INCONCLUSIVE, never STRUCTURAL-REJECT:

- (i) **δ at or above the measured noise floor.** δ must be calibrated to a *direct* measurement of L ’s solver noise (see the re-anchoring above, IDENT-14-02(c)), not to a theoretical propagation estimate alone — a δ set below the true noise floor makes every point’s $L(\zeta)$ indistinguishable from δ by construction, and an “empty” set under such a δ is an artifact of the threshold, not a finding about s^{HW} .
- (ii) **Design coverage and success rate meet a pre-registered threshold.** The evaluated design $\mathcal{D}$ must satisfy §3.1(g)’s ≥ 5 -non-stalled-starting-point gate *and* the realized success rate (evaluations that actually returned a finite L , divided by evaluations attempted) must be disclosed alongside the verdict — a design that nominally proposes 8 or 10 starting points but successfully evaluates only 2 (as in the 2026-08-15 run) has not met this condition regardless of how many points were *proposed*.
- (iii) **The empty evaluated set must be a genuine $\{\zeta \in \mathcal{D} : L(\zeta) > \delta\}$ over a finite design that satisfies (ii), not an artifact of evaluation failure.** A starting point that times out, returns NaN/Inf, or otherwise fails to produce a finite $L(\zeta)$ contributes **zero** evidence toward emptiness — it is a missing observation, not an observation of $L(\zeta) > \delta$. An “empty set” assembled by reclassifying non-finite evaluations as exceeding δ is not the registered object; **an evaluated set being empty is not the same claim as the true set being empty** (the distinction the 2026-08-15 adjudication turned on).

In either a legitimate STRUCTURAL-REJECT or the non-empty case, $\hat{L}_{\min}$ — the closest *evaluated* HW weight vector comes to the Shapley-fair share, an upper bound on $L_{\min}$ — is promoted to a **primary reported quantity** in its own right (not merely a diagnostic), and must be reported alongside $L(\hat{\zeta}) = 0.0196$ on the same line, explicitly labelled as an upper bound: it is interpretable without any statistical apparatus as “how close the Horn–Wolinsky architecture can get to the Shapley benchmark *on the points we managed to evaluate*,” independent of whether the registered δ -tolerant set happens to be empty. **A0 precondition on the comparison itself (added 2026-08-18, IDENT-15-04(c)).** $L(\hat{\zeta}) = 0.0196$ is BSY’s own *cached* equilibrium branch. This project’s own independent re-solve at the identical $\hat{\zeta}$ lands on a different branch, $L(\hat{\zeta})_{\text{resolved}} = 0.0381$ (roughly double the cached value, and itself well above the frozen $\delta =$

4×10^{-3} below — §4), so any ratio built from $\hat{L}_{\min}$ against $L(\hat{\zeta}) = 0.0196$ (the “ $2.4\times$ closer” headline) is a *cross-pipeline* comparison that presupposes the A0 comparator-reproduction gate has confirmed the re-solve reproduces the cached branch. That gate has been attempted and aborted at its own acceptance condition without confirming agreement (`zstar_run.json` GATE 1 addendum). Absent that confirmation, the cross-pipeline “ $2.4\times$ ” reading must always be reported alongside the *same-pipeline* reading — this run’s own $\hat{L}_{\min}$ against this run’s own re-solved $L(\hat{\zeta})_{\text{resolved}} = 0.0381$, giving “ $3.3\times$,” not “ $2.4\times$ ” — never the more favourable number alone. See `paper/theory/derivations/04_shapley_consistent_weights.md` §3.1(i) for the matching protocol clause (kept synchronized with this note; both amended together, 2026-08-16, and again 2026-08-18 for this A0 precondition). A regularized point $\zeta_{\text{reg}}^* = \arg \min_{\zeta \in Z_0^*} (\zeta - \hat{\zeta})' \hat{\Sigma}^{-1} (\zeta - \hat{\zeta})$ (Mahalanobis selection using BSY’s own Table V standard errors) is reported for reference, not as the headline object; the headline statistical statement is the P-09 Wald statistic defined above, not a claim that Z_0^* or Z_{δ}^* is itself a confidence region.

Existence of a minimizer of $L(\zeta)$ follows from the same compactness-plus-continuity (Weierstrass) argument used to establish that $v(S)$ is well-defined in the first place: *given* the standing continuity assumption A2 (`paper/theory/03_cooperative_bargaining.md` Assumption A3.2), s^{HW} is continuous on the compact box, so a best-fit minimizer always exists. **Fixed-branch qualification (added 2026-08-18, MATH-75).** This existence claim, precisely stated, is limited to a *fixed* equilibrium-selection branch: s^{HW} ’s continuity is *within-branch* only (A2/A-D4.1), and this project’s own re-solve infrastructure has since observed four independent instances of the discrete deal/no-deal branch selection changing at a fixed ζ under nothing more than a different solver warm start — the 26/151-line branch mismatch at $\hat{\zeta}$, a $\Delta L = 4.10 \times 10^{-4}$ gap between two re-solves of $\zeta = 0.5^7$ differing only in inner warm-start perturbation, and (added 2026-08-21, isolated single-anchor re-run, `zstar_gate1_half_isolated.md`) a fourth, decisive instance: an isolated $\zeta = 0.5^7$ re-run given a 3600s budget and dedicated resources converges fully, in 1018s, to machine precision ($\max |\Delta \text{mfn}| = 5.085 \times 10^{-7}$) — yet the converged point still misses BSY’s cached comparator by 10.0% on $\Sigma_k g_k$ and 5.08pp on the share vector L_2 distance, both roughly an order of magnitude past the GATE 1 tolerances. This confirms the branch mismatch at $\zeta = 0.5^7$ is determinate — a genuinely different stable fixed point, not an artifact of an incompletely converged solve — alongside the 94% larger $L(\hat{\zeta})_{\text{resolved}} = 0.0381$ versus BSY’s cached 0.0196 at the identical $\hat{\zeta}$ (§4 below documents $L(\zeta)$ as “better described as a path-dependent step function than as a smooth objective” for exactly this reason). Consequently the Weierstrass argument here establishes existence of a minimizer of L restricted to each fixed branch cell, not existence of a minimizer of the branch-dependent, warm-start-selected $L(\zeta)$ a solver actually realizes — whose very well-definedness as a function of ζ alone is not established by the three observations above. See `paper/theory/derivations/04_shapley_consistent_weights.md` Proposition D4.3 (revised 2026-08-18) for the full statement and what it does and does not prove. *Exact* solvability ($L = 0$) and uniqueness of that minimizer are not guaranteed in general—at $N = 2$ under a symmetric Pareto frontier, the unique solution is $\zeta = 1/2$ exactly, consistent with Proposition 1; for a general asymmetric bilateral pair, the solution is constructed by matching the Shapley *ratio* to a point on the Pareto frontier (not by substituting Shapley dollar values directly into

an on-frontier tangency condition, which is invalid off the frontier) and is otherwise corrected by the local slope of the Pareto frontier at that point; for $N \geq 3$ with more estimated pairs ($K_\zeta = 7$) than target dimensions ($N_6 - 1 = 5$, BSY’s actual case), the inverse map is generically underdetermined and uniqueness within Z_0^* is an open numerical question. **Rank-condition statement (added 2026-08-14, IDENT-13-06, closing an open leg of IDENT-11-12; domain restricted 2026-08-18, IDENT-15-05).** In Rothenberg (1971)’s local-identification sense, identifying an isolated ζ^* requires $\text{rank}(\partial s^{HW}/\partial \zeta) = K_\zeta = 7$ at the solution, which is impossible against a target of effective dimension $N_6 - 1 = 5$ (six country shares subject to one adding-up constraint): the Jacobian can have rank at most $\min(K_\zeta, N_6 - 1) = 5 < 7$, so Z_0^* is *generically* a $(K_\zeta - (N_6 - 1)) = 2$ -dimensional manifold rather than an isolated point, and only the *set* Z_0^* — not a point ζ^* — is identified in the ordinary sense of (Rothenberg, 1971; Manski, 1995). **Domain of this statement.** The rank condition presupposes $\partial s^{HW}/\partial \zeta$ *exists* at the solution, which is a strictly stronger regularity condition than the within-branch *continuity* A2 actually supplies (Assumption A3.2 grants existence and continuity of the selected equilibrium, not differentiability); this statement is therefore local not just in the usual Rothenberg sense but also in the branch-cell sense of the fixed-branch qualification above — it applies to ζ^* in the interior of a single branch cell $\overline{Z_b}$, on the maintained (unproved) supplementary assumption that s^{HW} restricted to that cell is differentiable there. At a branch boundary, where the four MATH-75 observations above show s^{HW} need not even be continuous, neither the rank condition nor the resulting 2-manifold statement is meaningful. This upgrades the informal “more pairs than target dimensions, underdetermined” observation above from a numerical symptom to a structural local-identification statement. Full statements and proofs are given in the companion derivation note (theory derivation 04, “Shapley-Consistent Bargaining Weights,” Propositions D4.1–D4.3), which also formalizes existence, finiteness, and the $v(\emptyset) = 0$ convention for $v(S)$ itself (theory note 03, Proposition 3.4, *given* A2) and the comparison between the status-quo and γ -response outsider conventions (theory note 03, Remark 3.5, downgraded from Proposition per MATH-18) underlying the IR-scope discussion above.

What implementing the benchmark would require. The point $s = 0.25^6$ used in an earlier estimate of the required transfer was a uniform diagnostic probe, not the optimum: the certified continuous optimum of the Track A grand-coalition program (`phase7_full_primary.json`; L-BFGS-B with exact implicit-function-theorem gradients, all 63 coalitions converged) is strongly asymmetric, $s^* = [0.533, 0.472, 0.027, 0.326, 0.111, 0.000]$, and attains $v(\mathcal{N}) = 0.7202$, 11.7% above the probe’s value. At s^* the United States *gains*, $g_{US} = +0.114$ (+15.8% of $v(\mathcal{N})$) — indeed more than its Shapley share $\phi_{US}(v)/v(\mathcal{N}) = 13.3\%$ — and the only member left worse off in level terms is **Australia**, $g_{AUS} = -0.029$ (−4.1% of $v(\mathcal{N})$). Since $\sum_i g_i(s^*) = v(\mathcal{N}) = \sum_i \phi_i(v)$, the side-payment vector $T_i \equiv \phi_i(v) - g_i(s^*)$ required to move from the efficient policy’s physical incidence to the Shapley allocation is budget-balanced and fully computable: the US pays 2.5% of $v(\mathcal{N})$, the EU 4.1%, Japan 6.9%; Canada receives 0.5%, South Korea 6.0%, and Australia — the one member whose participation constraint the efficient policy violates outright, $g_{AUS} < v(\{AUS\}) = 0$ — receives 7.0%. The gross one-way volume is 13.5% of $v(\mathcal{N})$. (This does not contradict $\Gamma_{US} > 0$: Γ_{US} compares Shapley with the *bargained* HW outcome, T_{US}

compares Shapley with the *efficient policy’s own* incidence; the US is under-compensated relative to the former and over-compensated relative to the latter, and both statements are true.) Imputation IR on Track A is verified *directly* — $\phi_i(v) \geq v(\{i\})$ holds at every player with minimum slack 0.0186 (2.59% of $v(\mathcal{N})$), and $w(\mathcal{N}) = 0.677 > 0$ (phase7_full_primary.md §2); the sufficient-condition route via Proposition A.1 is *not available* on Track A, where that hypothesis is refuted (17/192 single-player violations; §2, TI-12). The best *vertex* $s = [0, 1, 0, 0, 0, 0]$ (only the EU cuts; $g_{US} = -0.494$, the EU alone capturing 130% of that vertex’s surplus) is retained here purely as a diagnostic contrast — it is *not* the efficiency optimum, which is interior and strongly asymmetric on five of six coordinates — **Korea sits exactly at the zero boundary, $s_{\text{Korea}}^* = 0$: the registered strong form of prediction P-19 failed on that coordinate, and “interior” here is the post-hoc weak form.** GATT/WTO provides no transfer instrument: it is a no-transfers institution by design, operating through issue linkage and reciprocal concessions rather than side payments (§2.2, TU discussion). This is the concrete empirical content of Assumption A5’s “as-if transferable-utility” scope restriction (paper/theory/00_assumptions.md) — treating $v(S)$ as a coherent dollar-denominated characteristic function is defensible only because we are willing to abstract from the absence of a real transfer instrument — and it remains this paper’s sharpest policy conclusion: *at the certified optimum* the efficient policy’s own incidence leaves one member strictly worse off, so a transfer-free institution cannot implement the efficiency benchmark *at that point* — and we have not located any maximizer of the coalition objective that is individually rational, though non-existence of one is not established (the optimum is certified local-with-multi-start, not global; quantifier qualified per MATH-53) — let alone the Shapley reallocation on top of it, and the required gross side-payment volume (13.5% of the coalition surplus) is far outside anything GATT’s reciprocity mechanism can replicate.

2.7 Computational Strategy

Two-step vs. single-step solves of (11). The MPEC (11) can be solved in two styles:

1. *Two-step (BSY-style)*—approximate the equilibrium response using base-year shares and re-solve only the outer bargaining problem; fast but loses GE feedback and may silently violate IR (see Section C.5).
2. *Single-step*—solve the full hat-algebra system at every outer iteration so that all GE spillovers are captured in the bargaining objective; the limit case is the IPOPT/CONOPT solve of (11) in which equilibrium variables and tariff scalars are all decision variables.

Our current implementation is a Python two-step nest: the outer optimization uses `scipy.optimize.minimize` with SLSQP/L-BFGS-B over the box-constrained D1 objective (no Lagrangian IR handling needed under D1, §2.2); the inner equilibrium uses a hand-coded hat-algebra fixed-point iteration (`toy_model.solve_hat_iterative`) validated against the Caliendo and Parro (2015) NAFTA replication. The D1 refactor of `scripts/python/cp2015_value.py` (deleting the IR filter/penalty and widening the search grid to include $\{0, 1\}$) is a prerequisite for any headline number reported under D1; numbers computed before that refactor are from the retired Nash-product objective and are superseded. A full MPEC realization of (11) in Pyomo+IPOPT,

feasible at $N \leq 7$ and required for an $N = 31$ extension, is left to future work.

Shapley computation: exact vs Monte-Carlo. The Shapley value (12) requires evaluating $v(S)$ on all 2^N coalitions. For $N \leq 7$ exact enumeration is feasible (with cross-coalition warm-starts: each $v(S)$ takes ~ 10 – 100 s, total runtime ≤ 30 minutes). For larger N we adopt the random-permutation sampler of Castro et al. (2009), which yields an unbiased estimator of ϕ with standard error $O(1/\sqrt{M})$ in M permutations. Stratification by coalition size (Castro et al. §4) reduces variance an additional 30–50%.

3 Data

3.1 Primary Dataset: BSY’s Own 1990 Negotiating Panel

We use Bagwell et al. (2021)’s own calibrated dataset for their six-country 1990 tariff-negotiation panel (USA, EU, Japan, Canada, Australia, South Korea). Following their own construction, five “not-elsewhere-specified” (NES) composite regions fill out an 11-region $\times$ 67-sector world economy; the six named countries are the negotiating players, and the five NES regions are non-strategic outsiders held fixed at the data, for the same reason Caliendo and Parro (2015)’s Rest-of-World residual is excluded from the second economy’s player set below (Appendix B). Primitives include bilateral iceberg trade costs, Fréchet trade elasticities, expenditure shares, sectoral input–output (I–O) linkages, factor endowments, and the 1990 applied MFN tariff schedule that anchors our threat point (A7).

Loader fidelity. We load BSY’s original .mat inputs (`current_estimates_cp1.mat`, `aggregated_data.mat`, `gravity_variables.mat`, `gamma_transforms.mat`, `pre_estimation.mat`) directly, reproducing every parameter array their own `counterfactual_preliminaries_main.m` constructs: the Fréchet shape parameter θ , productivity shifters z , expenditure shares α (row-normalized), iceberg costs d (their sinh functional form plus six bilateral dummies, non-traded $d = 200$, diagonal 1), wages, and trade deficits D , including their Australia tariff correction (sectors 8/11/12, 1990, divided by $(1 - 0.4739)$) and PTA matrix. Against a MATLAB-exported ground truth from their own script, all 16 arrays plus 3 scalars match to a maximum absolute difference of 2.5×10^{-14} (worst array `d`, shape $11 \times 11 \times 67$; `phase1_loader_acceptance.json`), well within the pre-registered 10^{-12} acceptance threshold.

Players. The six negotiating countries are the Shapley players; the five NES regions are held fixed at their 1990 data throughout, never entering any coalition, consistent with the treatment of the Rest-of-World residual in our second economy (Appendix B, TI-7). All 63 non-empty coalitions of the six players are solved exactly (Section 4.1).

Comparator source ($s_i^{HW}(\hat{\zeta})$, author decision 2026-08-16). Every $\hat{\zeta}$ -dependent quantity we report — $s_i^{HW}(\hat{\zeta})$, Γ_i^{raw} , and the $D4'$.0 level pool (§4.1) — is evaluated at BSY’s own *cached* Horn–Wolinsky equilibrium (their own solved bargaining outcome at $\hat{\zeta}$), not at an equilibrium

re-solved by our own Nash-in-Nash pipeline. This is a deliberate, disclosed choice: a cold-started re-solve of the identical $\hat{\zeta}$ point, using the same pairwise bargaining code, reproduces BSY’s equilibrium tariff map to 0.40pp mean deviation overall (§4.1) but disagrees on the discrete deal/no-deal outcome for 26 of the 151 negotiated lines — BSY’s own cached solution records a deal on every line, while our re-solve lands 26 lines at the no-deal (1990-baseline) corner (`phase4c_diagnostic.md`, `phase4_parity_run.log`). This branch-level disagreement is a known limitation of the pairwise Nash-in-Nash fixed point’s sensitivity to starting values, further documented in §4.1.5, not evidence against the aggregate replication above. Using BSY’s own cached solution as the comparator avoids importing this branch-sensitivity into every Γ_i we report, and means the comparator side of the bridge is, by construction, exactly BSY’s own published bargaining outcome rather than a re-derived approximation of it.

3.2 Second Economy: The [Caliendo and Parro \(2015\)](#) 1993 Panel (Robustness)

As a robustness check on a second, structurally distinct economy (Section 4.2), we use the calibrated dataset constructed by [Caliendo and Parro \(2015\)](#), which provides, for the year 1993, country-level data on bilateral trade flows, sectoral value-added, gross output, and tariffs across 30 countries plus a Rest-of-World residual, and 40 sectors (20 tradeable, 20 non-tradeable). The choice of 1993 as the baseline year reflects [Caliendo and Parro](#)’s focus on the pre-NAFTA-1994 environment; we inherit this calibration directly.

Tariffs. Bilateral applied MFN ad-valorem tariffs are sourced from UNCTAD TRAINS, restricted to manufacturing sectors. Non-tariff barriers and anti-dumping duties are excluded; this is consistent with [Bagwell et al. \(2021\)](#) but implies our welfare estimates are *conservative* for jurisdictions whose true protection rates exceed applied tariffs (e.g., the EU, Japan).

Tariff revenue. For the Appendix C horse race (Section C.7), we construct each country’s base-year tariff revenue as $R_n = \sum_{i \neq n} \sum_j \tau_{ni}^{j,0} M_{ni}^j$, where $\tau_{ni}^{j,0}$ is the 1993 applied MFN ad-valorem tariff (UNCTAD TRAINS) on sector- j imports from source i , and M_{ni}^j is the corresponding base-year bilateral import value (in CIF dollars) from the [Caliendo and Parro \(2015\)](#) calibration. Equivalently, $R_n = \sum_j \tau_n^j M_n^j p_n^j$ once bilateral imports are aggregated to a sector-level value-weighted tariff τ_n^j . We use R_n only as a benchmark for the cooperative imputation; it does not enter the equilibrium solver, which uses the full bilateral $\tau_{ni}^{j,0}$ tensor directly.

Rest-of-World aggregation (R5 revision). The ROW residual is an accounting aggregate of the ~ 130 countries outside the sample. Trade, value-added, and gross output for ROW are computed as world total – sum of sample 30. Input-output coefficients for ROW are imputed as the median of the sample-30 I-O matrix (see Appendix B for empirical correlation 0.99 between sample-30 and ROW industry shares). This implies that ROW is *not* a strategic agent: in our cooperative game ROW is treated as an outsider whose tariffs and preferences are fixed at the data, and never as a coalition member. Including ROW as a player would conflate accounting aggregation with strategic value and inflate coalition surplus mechanically.

Player selection for Shapley computation. Because exact enumeration scales as 2^N , we restrict attention to coalitions of moderate size and select players based on (i) inclusion in the original [Caliendo and Parro](#) sample, (ii) economic relevance for multilateral negotiations, and (iii) heterogeneity in tariff levels and sizes. Concretely, we report results for the following nested sequences:

- $N = 3$: USA, Mexico, Canada (NAFTA);
- $N = 4$: + China;
- $N = 5$: + India;
- $N = 6$: + Brazil;
- $N = 7$: + Korea.

Beyond $N = 7$ we deploy Monte-Carlo Shapley sampling ([Appendix C](#), [Section C.4](#)).

Data summary statistics. [Table 1](#) reports key calibration moments for each of the seven players used in our reported coalition sequence. Two patterns are immediately relevant for the Shapley computation in [Section 4.2](#): (i) average applied tariffs vary by an order of magnitude (USA at 3.4% vs. India at 34.5%), implying very heterogeneous disagreement payoffs across players, and (ii) the four NAFTA-and-OECD members (USA, Mexico, Canada, Korea) have low-to-moderate tariffs, while China and India have substantially higher protection. This heterogeneity is precisely the source of the Shapley asymmetry documented in [Section 4.2](#): low applied tariffs imply small unilateral gains from cooperation, hence small marginal contributions, hence small Shapley weights.

Table 1: Calibration moments for the seven players in our second economy’s coalition sequence (1993 baseline, source: [Caliendo and Parro \(2015\)](#)).

Country	GDP (\$B)	Imports/GDP	Exports/GDP	Avg. applied tariff (%)
USA	6,546	0.09	0.07	3.4
Mexico	390	0.15	0.12	13.0
Canada	565	0.22	0.24	3.6
China	601	0.16	0.21	31.9
India	264	0.07	0.09	34.5
Brazil	444	0.06	0.09	13.1
Korea	357	0.22	0.22	8.7

Notes: GDP is total value added across the 40 sectors. Trade ratios are computed from gross bilateral flows (excluding tariff revenue). Average applied tariff is the bilateral-import-weighted MFN ad-valorem rate across all foreign source countries and tradable sectors. Computed by `cp2015_loader.py`. The [Appendix C](#) GDP horse race ([Section C.6](#)) uses base-year household income I_n^{base} in place of GDP; in the CP-style calibration $I_n^{\text{base}} = \text{VA}_n + \text{tariff revenue}_n + \text{trade deficit}_n$, so the two measures coincide up to a tariff-revenue and deficit term ($\leq 5\%$ for the seven sample players) and the correlation patterns in [Table 13](#) are robust to this distinction.

4 Results

This section applies the D1 characteristic function and Shapley imputation of [Section 2](#) to two economies. [Section 4.1](#) is the paper’s primary application: a direct replication of [Bagwell et al. \(2021\)](#)’s own six-country 1990 negotiating panel, on which we (i) reproduce their equilibrium

map and counterfactuals, disclosing what we could not replicate; (ii) replace their estimated bilateral bargaining weight $\hat{\zeta}$ with the derived Shapley benchmark; and (iii) evaluate the resulting fairness gap against their own estimated bargaining outcome. Section 4.2 reports, as a robustness check, the parallel exercise on a second, structurally distinct economy — the $N = 3$ –7 Caliendo–Parro coalitions that were this paper’s original empirical core — in condensed form, with full detail in Appendix C.

4.1 BSY’s Six-Country 1990 Negotiating Panel (“Track A”): Replication, Coalition Values, and the Core

We replicate Bagwell et al. (2021)’s own six-country 1990 negotiating panel (USA, EU, Japan, Canada, Australia, South Korea) end to end. The primitives load with fidelity: 16 arrays plus 3 scalars against the original `.mat` inputs, maximum difference 2.5×10^{-14} (`phase1_loader_acceptance.json`). We reproduce BSY’s equilibrium map and key counterfactuals — our Nash-in-Nash fixed point from a cold start plus box-level feasibility search matches their solution to 0.40pp mean tariff deviation at their own point estimate $\hat{\zeta}$ (`phase4_parity_rerun.md`), and their Table III counterfactual to a maximum absolute deviation of 0.024pp (`phase6a_table3.json`) — but we do *not* reproduce their Table V $\hat{\zeta}$ estimator: our re-estimation fails to converge, with the Levenberg–Marquardt damping parameter diverging ($7.4 \times 10^{12} \rightarrow 2.9 \times 10^{18}$) and the finite-difference Jacobian degenerating, the same class of numerical difficulty their own coarse `DiffMinChange` setting works around (`phase5_verdict.md` §6). Nothing in this paper’s numbers depends on that failed re-estimation: every $\hat{\zeta}$ -dependent object below ($s_i^{HW}, \Gamma_i^{\text{raw}}$) is evaluated at BSY’s own published point estimate, fed into the equilibrium map just validated, and under D1 the characteristic function $v(S)$ itself carries no bargaining weight and is unaffected by $\hat{\zeta}$.

Having established the replication, we replace BSY’s estimated bilateral $\hat{\zeta}$ with a derived cooperative benchmark and apply it to their own economy. We solve all 63 coalitions of the D1 characteristic function under the primary cuts-only policy space $s_n \in [0, 1]$ (continuous L-BFGS-B with exact implicit-function-theorem gradients, multi-start; all coalitions converged, `phase7_full_primary.md`). This is also the panel on which the positive–normative bridge to BSY’s own estimated bargaining outcome (§2.6) is evaluated, since $\hat{\zeta}$ was estimated on exactly this panel — our second economy (Section 4.2) has no analogous estimated-weight comparator. Individual rationality and core membership are both verified directly: $\phi_i(v) \geq v(\{i\})$ at every player, minimum slack 2.59% of $v(\mathcal{N}_6)$ (binding at the singleton {Canada}); core 0/62 (all proper-coalition constraints hold), same minimum slack, status-quo-outsider (α -style) convention — despite superadditivity failing outright on 20/301 disjoint pairs, so the sufficient-condition route via `hyp:wmono` is unavailable here and both results survive only because they are checked directly. Restricting to coalitions of size 2 or 3 that are *not* singletons or the grand coalition — the genuinely-unseen constraints least favorable to a construction argument — the tightest slack is still 5.51% of $v(\mathcal{N}_6)$, at {Canada, Australia} (`p23_core_slacks.json`), an order of magnitude looser than the global 2.59% minimum. The analogous checks on our second economy (Section 4.2) confirm the same qualitative pattern.

A structural panel on the w -monotonicity violations (P-35). The 20/301 disjoint-pair superadditivity violations above have a companion, more granular object. Of the 192 single-player additions (S, i) , $i \notin S$, ranging S over every subset of the other five countries (including $S = \emptyset$), 17 violate single-player w -monotonicity ($w(S \cup \{i\}) < w(S)$), worst margin -0.026338 at the US joining $\{\text{Canada}, \text{EU}\}$ — a zero-new-GE arithmetic check on Phase 7’s own certified 63-coalition census that reproduces TI-12’s citation exactly (`p35_coalition_panel.md`). TI-12’s qualitative signature — “US and Canada in the same coalition without Japan or South Korea” — is a strong but imperfect predictor: conditional on the signature firing (12 of 192 additions), the violation rate is 75% (9 of 12), an $8.21\times$ lift over the base rate excluding the six by-construction-zero $S = \emptyset$ rows ($17/186 \approx 9.14\%$; the naive $17/192 \approx 8.9\%$ base rate, which includes those six rows, overstates the lift at $8.47\times$ and is superseded here), but the signature is neither necessary (8 of the 17 violations occur without it) nor sufficient (3 of the 12 signature-firing additions are non-negative). A **post-hoc, critic-initiated** exact permutation test over all $6! = 720$ country-role relabelings — holding the entire coalition-composition structure fixed and permuting only which country plays which role, the same convention already used for P-22’s $n = 6$ tests, but devised only after inspecting the 17 violations it evaluates, and not part of P-35’s original pre-registration — offers suggestive, not confirmatory, evidence that this association is not a small-sample artifact: the observed signature-match count among the 17 violations (9) has exact one-sided $p = 0.0111$ against the resulting null distribution (range 0–9, mean 1.40 over the 720 relabelings; `p35_coalition_panel.md` §3a). Three caveats attach, and none is in tension with the OLS’s own no-sampling-null statement later in this paragraph, which concerns a different statistic (the OLS standard errors on the full deterministic census), not this permutation p -value. First, a test designed by a critic after seeing the same 17 violations it is run against should be read as descriptive corroboration, not an independent confirmatory test. Second, the observed count (9) is exactly the maximum of the empirical null’s own range, which is itself a sign that the null’s discreteness at $n = 6! = 720$ makes $p = 0.0111$ an optimistic (too small), not conservative, estimate of the true tail probability. Third, the signature under test was itself induced from these same 17 violations, and no correction for that selection is applied here. A sharper, single-country characterization fits better: **Canada is present, as joiner or incumbent, in 16 of the 17 violations** — only the smallest-margin case (US joining $\{\text{Australia}, \text{EU}\}$, -0.000083) involves no Canada at all. An OLS regression of the margin on coalition size, $\tilde{P}_i^{\text{base}}$ (“economic size,” $= I_i^{\text{base}}/W_i^{\text{base}}$), own 1990 tariff, and the TI-12 signature indicator ($R^2 = 0.252$, $n = 192$) is a *descriptive* decomposition, not an inferential test: the 192 observations are the full, deterministic census of single-player additions on Phase 7’s 63-coalition solve, not a sample from a larger population, so no sampling-based null hypothesis or p -value attaches to the OLS’s own standard errors, and prediction P-35’s own registration explicitly rules out significance language for exactly this reason.¹ Read descriptively, holding

¹The pre-registered protocol instead specified a bivariate Spearman rank correlation on two legs (tariff level, revenue share), not the OLS above. An exact-permutation version of that correlation, over the identical 720 relabelings, gives $\rho = +0.104$ ($p = 0.736$) for the 1990 tariff level and $\rho = +0.372$ ($p = 0.164$) for the revenue share — neither reaches the registered $\rho \geq 0.5$ (“size/tariff-loaded”) threshold; only one of the two statistics — the revenue share ($\rho = +0.372$) — actually falls *within* the intermediate interval $[0.2, 0.5]$, since the tariff-level leg ($\rho = +0.104$) itself falls below the 0.2 threshold that would otherwise trigger the “composition effects dominate” verdict; because the protocol requires *both* legs below 0.2 for that verdict, and neither leg clears 0.5,

coalition size and $\tilde{P}_i^{\text{base}}$ and own tariff fixed, the signature indicator’s coefficient is negative and comparable in magnitude to the other three ($\hat{\beta} = -0.089$ against $+0.053$ for coalition size and -0.829 for own tariff, `p35_coalition_panel.md` §3): on the census we observe, the signature is not fully explained by size or tariff level alone, a claim confined to this census with no implied claim about a broader population. We read this as sharpening, not overturning, TI-12: the mechanism is a country’s unilaterally profitable cut degrading the terms of trade of a larger coalition partner by more than that country gains, and on this panel the mechanism is present whenever Canada sits on either side of the addition, not only in the joint US-and-Canada-without-Japan-or-Korea configuration originally documented.

4.1.1 The fairness-gap bridge

Table 2: Track A fairness-gap bridge. $\phi_i(v)$ and $\phi_i(w)$ are normalized so that $v(\mathcal{N}_6) = 0.720163$; “share” is $\phi_i(v)/v(\mathcal{N}_6)$; $s_i^{HW}(\hat{\zeta})$ is country i ’s dollar share of BSY’s own estimated Horn–Wolinsky equilibrium; $\Gamma_i^{\text{raw}} = \phi_i(v)/v(\mathcal{N}_6) - s_i^{HW}(\hat{\zeta})$ (Definition 2, eq. (14)). Source: `phase7_full_primary.md` §1.

Country	$\phi_i(v)$	share (%)	$\phi_i(w)$	$s_i^{HW}(\hat{\zeta})$ (%)	Γ_i^{raw} (pp)
USA	0.09599	13.33	0.09599	10.04	+3.29
EU	0.18015	25.02	0.18015	20.71	+4.30
Japan	0.30916	42.93	0.27192	55.47	−12.54
South Korea	0.08974	12.46	0.08974	10.33	+2.14
Canada	0.02422	3.36	0.01863	1.24	+2.12
Australia	0.02090	2.90	0.02090	2.21	+0.69

$$v(\mathcal{N}_6) = 0.720163, w(\mathcal{N}_6) = 0.677338, \kappa_A^{s \leq 1} = 94.05\%, \sum_i v(\{i\}) = 0.042825.$$

Japan is the only negative gap in the panel, and the largest in magnitude: BSY’s estimated Horn–Wolinsky bargain awards Japan 55.5% of the estimated bargained surplus where the Shapley benchmark assigns 42.9%; every other player is under-compensated relative to Shapley on this raw-share metric. This is the C3 finding, restated against the present evidence (see the closing discussion of this subsection for the full, adjudicated reading).

Comparator-branch sensitivity: an alternative Nash-in-Nash equilibrium branch at $\hat{\zeta}$. The comparator side of the bridge, $s_i^{HW}(\hat{\zeta})$, is itself the output of a Gauss–Seidel fixed point that can select different deal/no-deal branches across nominally identical re-solves (§4.1.5). Table 2 above uses BSY’s own *cached* equilibrium, which we select as the primary comparator throughout this paper because it is the branch BSY themselves published; we disclose the sensitivity to that choice explicitly rather than treat it as settled. An independently re-solved branch at the identical $\hat{\zeta}$ (`zstar_evals_2026-08-15_pre-r14.jsonl.bak`, eval #7, a normally-converged evaluation) returns a materially different share vector, $s^{HW}(\hat{\zeta}) =$

the decision rule still routes to the registered protocol’s **INTERMEDIATE band, no verdict language** — on this more precise reading, not because both statistics land in the band’s interior. This discharges leg (a) cleanly against its own pre-registered criterion, superseding the earlier disposition, which substituted the unregistered OLS above and recorded leg (a) as **INCONCLUSIVE (protocol deviation)** for lack of the pre-registered statistic (`predictions.md`, P-35; `p35_coalition_panel.md` §3a).

[7.521, 14.029, 56.709, 0.321, 3.812, 17.608]% for [US, EU, Japan, Canada, Australia, South Korea], and a correspondingly different Γ_i^{raw} :

Table 3: Fairness gap on the cached (published) branch used throughout this paper versus an alternative, independently re-solved branch at the identical $\hat{\zeta}$. Neither branch is privileged by the model.

	USA	EU	Japan	Canada	Australia	S. Korea
Γ_i^{raw} , cached branch (primary)	+3.29	+4.30	−12.54	+2.12	+0.69	+2.14
Γ_i^{raw} , alternative branch	+5.81	+10.99	−13.78	+3.04	−0.91	−5.15

Two consequences follow. First, Australia and South Korea flip sign across branches; both are already in the sign-unstable set on the four-rule check below (Table 6), so this is corroborating rather than new instability. Second, and more consequential: Γ_{US} on the alternative branch is +5.81pp, which *clears* prediction P-07''’s frozen +5pp threshold — reversing, on this branch alone, the certified FAIL reported below (§4.1.7) for the cached branch. We disclose this branch-conditionality explicitly rather than let the FAIL stand as if it were branch-independent: P-07''’s FAIL verdict holds on the primary (cached, published) branch this paper uses throughout, and is not robust to this comparator-side branch instability. On the constructive side, this sensitivity check is not purely a liability: the sign-robust core of this paper’s finding survives on *both* branches at zero additional compute cost — $\Gamma_{US} > 0$, $\Gamma_{\text{Japan}} < 0$, and $\Gamma_{\text{Canada}} > 0$ hold under the cached branch (Table 6) and under this alternative branch alike.

An agenda-mismatch caveat this bridge cannot resolve on its own (ECON-19, ECON-56). BSY’s $\hat{\zeta}$ is estimated from bilateral tariff *concessions* alone. Surplus the United States plausibly extracted through TRIPS, GATS, and agricultural market-access commitments in the Uruguay Round is invisible to that estimation strategy: none of it enters s_i^{HW} , which is built entirely from tariff-driven welfare gains $g_i(\zeta)$. If that omitted surplus is economically large, s_{US}^{HW} is understated by construction and $\Gamma_{US}^{\text{raw}} = \phi_{US}(v)/v(\mathcal{N}_6) - s_{US}^{HW}$ is correspondingly overstated. The *direction* of this agenda-truncation bias is unambiguous; its *magnitude* is not quantified here, and there is no structural reason to expect it is smaller than the +3.29pp point estimate it would inflate — an external estimate of Uruguay Round non-tariff (TRIPS/GATS/agricultural) concession magnitude is needed to bound it. Quantifying this bound is left to future work; candidate order-of-magnitude estimates exist in the CGE literature (e.g. Goldin, Knudsen & van der Mensbrugghe 1993) but have not been independently verified to citation-grade precision here, and are not cited formally for that reason.

The same mismatch cuts in the other direction too, and it is not merely the mirror image of the point above: it also runs through our own lever, not only through BSY’s estimator. Our s_n scales country n ’s *entire* MFN tariff row (eq. (7)), whereas BSY’s own 151-line negotiating set is the product of their *estimation-sample screen* (`structures.py`), not of the Uruguay Round’s own agenda: it excludes agriculture (sectors 1–9) and tobacco entirely by construction, and further restricts to mutual, non-developing, bidirectional principal-supplier pairs above a trade-share threshold, dropping any pair already covered by a preferential agreement. The clearest evidence this is a sampling construct rather than an institutional agenda is

Canada, which negotiates *zero* lines under it — its natural principal supplier, the United States, is screened out by the CUSFTA 1989 PTA exclusion — despite Canada plainly participating in, and making tariff concessions during, the Uruguay Round. Calling the 151-line set “the Uruguay Round agenda” therefore misstates what it is: it is BSY’s own bilateral request-offer sample, a proper and much narrower subset of what the Round actually did. Agriculture illustrates a related but distinct point, and this project’s own data corrects an earlier draft’s account of it. The Round’s Agreement on Agriculture (AoA) was the first GATT instrument to bring agricultural protection under *effective* multilateral discipline: GATT 1947 formally covered agriculture from the outset, but a web of waivers and Article XI exceptions had hollowed that coverage out in practice, so the AoA is properly read as the first instrument with real bite, not the first coverage of any kind (Hoda, 2001). It proceeded via *tariffication*: pre-existing non-tariff barriers (quotas, variable levies, state trading) were converted into bound tariff-equivalents, then cut by a separate formula (developed countries 36% average / 15% line-level minimum, 1995–2000; developing countries 24%/10%, 1995–2004) (Ingco, 1996; Hoda, 2001). This does *not* imply applied agricultural protection fell over 1990–2000: this paper’s own loaded tariff data shows measured applied protection instead *rose* on average, consistent with a change in the protective instrument itself — from opaque non-tariff barriers, largely invisible to a tariff series, to explicit (and in many cases “dirty,” i.e. inflated relative to the NTB they replaced) bound tariff-equivalents — rather than with a genuine increase in the level of protection. Comparing `mfnTariffs_1990` to `mfnTariffs_2000` sector by sector, average *applied* agricultural tariffs (sectors 1–9) *rose* for five of six Track A countries on both the mean and the **median** measure (only Australia falling on either). We report the within-country **median** as the primary magnitude descriptor. The within-country medians are: US +0.99pp, EU +11.77pp, Japan +0.62pp, Canada +5.43pp, Australia –0.31pp, South Korea +8.41pp; pooled across all 54 country-sector cells, the median change is +1.45pp, with 32 of 54 cells rising and 20 declining (`agriculture_mfn90_vs_mfn00_addendum.md`). Every country’s non-agricultural traded-goods tariffs fell over the identical window (–1.1pp to –6.1pp, the pattern ordinary MFN liberalization predicts, `agriculture_mfn90_vs_mfn00.md`).² Ingco’s own contemporaneous assessment of the tariffication exercise is consistent with this pattern: many countries set the new bound tariff-equivalent substantially above the protection the prior non-tariff barrier actually conveyed (“dirty” tariffication), so a rising *applied* rate is not inconsistent with formula cuts applied to an inflated baseline, alongside genuine new market-access commitments (minimum- and current-access tariff-rate quotas) that an applied-rate comparison alone cannot see (Ingco, 1996). Whatever reform agriculture underwent in this decade, it is not visible in, and not commensurable with, the applied-MFN-tariff channel this paper’s $v(S)$ construction and BSY’s

²The country-level *mean* range spans +7.35pp (US) to +35.91pp (South Korea); the upper end is sensitive to a small number of individual tariff lines that report an identical 1.500 value (150%) in `mfnTariffs_2000` — one line for Japan, two for South Korea — consistent with a data ceiling or cap rather than a distinct economic observation. Excluding those lines lowers Japan’s agriculture-sector *mean* change from +17.40pp to +9.5pp and South Korea’s from +35.91pp to +6.6pp; Canada’s +18.22pp mean is not cap-driven but is single-line-driven, with 12.45pp of the 18.22pp coming from a single line (leave-one-out mean +5.77pp). The qualitative finding — five of six countries rising on both mean and median, direction robust — does not depend on this sensitivity; the country-level *mean* magnitudes, especially South Korea’s, do; the *median* figures reported in the main text are little affected, since Japan’s and South Korea’s medians (+0.62pp, +8.41pp) are already well below their cap-inflated means (`agriculture_mfn90_vs_mfn00_addendum.json`, per-line arrays).

own s_i^{HW} both use. This reinforces, on cleaner evidence than an earlier draft of this paragraph offered, the same mechanism the ECON-19 point above establishes: agricultural surplus does not enter s_i^{HW} because agriculture sits structurally outside BSY’s bilateral principal-supplier negotiating mechanism (a formula-cut, not a principal-supplier, track, [Hoda, 2001](#)) — not because some other applied-tariff-cutting channel captured it instead; on the evidence above, no such channel operated in this sample period at all. $v(\mathcal{N}_6)$ and $\phi_i(v)$, by contrast, price surplus on the full tariff schedule — agricultural liberalization above all — that lies outside BSY’s bilateral negotiating structure regardless of channel, not by any shortfall in their bargaining outcome. The clearest symptom is Japan: the D1 optimum’s $s_{\text{Japan}}^* = 0.027$ cuts 97% across Japan’s *entire* tariff schedule, including rice and the other politically sensitive agricultural lines that sit outside BSY’s 151-line bilateral sample, and $\Gamma_{\text{Japan}} = -12.54\text{pp}$ — the largest gap in the panel — inherits that coverage difference along with everything else in its construction.

Both directions of this agenda mismatch push the bridge the same qualitative way — comparing a bargained outcome measured on a narrow, institutionally restricted agenda against a benchmark solved on a broader one mechanically inflates the apparent gap between them, regardless of which side of the bridge the broader agenda sits on. Consequently, Γ_{US} ’s *sign* claim rests on the four-rule robustness check below (Table 6, prediction P-27), not on the $\hat{\zeta}$ point value; its *magnitude* is not established as robust to either direction of this bias and should not be quoted as a precise number. The level-pool comparison immediately below inherits the coverage direction of this caveat and is qualified accordingly.

4.1.2 The $D4'.0$ decomposition and the level pool

Table 4: The $D4'.0$ three-term decomposition, $\phi_i(v) - g_i(s^*) = v(\mathcal{N}_6) \Gamma_i^{\text{raw}} + s_i^{HW}(\hat{\zeta})[v(\mathcal{N}_6) - \sum_k g_k^{HW}(\hat{\zeta})]$ (eq. (16)). $\Delta_i = g_i(s^*)/v(\mathcal{N}_6)$ is the efficient policy’s own physical incidence share; $R_i = \Gamma_i - \Delta_i$ is the residual rule component; $T_i = \phi_i(v) - g_i(s^*)$ is the transfer, in the same normalized units as $v(\mathcal{N}_6)$, required to move from the efficient policy’s own incidence to the Shapley allocation (positive = receives). DESCRIPTIVE (P-24): never cite as a predictive success. Source: `phase7_zero_ge_batch.md` §3.

Country	Γ_i (pp)	Δ_i (pp)	R_i (pp)	fairness	level	T_i
USA	+3.29	+5.78	-2.49	+0.0237	+0.0409	-0.0179
EU	+4.30	+8.43	-4.13	+0.0310	+0.0845	-0.0298
Japan	-12.54	-5.65	-6.90	-0.0903	+0.2262	-0.0497
Canada	+2.12	+1.60	+0.52	+0.0153	+0.0051	+0.0038
Australia	+0.69	-6.29	+6.98	+0.0050	+0.0090	+0.0503
South Korea	+2.14	-3.87	+6.01	+0.0154	+0.0421	+0.0433

Gross one-way transfer = 13.51% of $v(\mathcal{N}_6)$ (USA, EU, Japan pay; Canada, Australia, South Korea receive).

Column sums: fairness ≈ 0 , level = +0.4078.

The level pool is the largest object in this decomposition, and it dwarfs every Γ_i : $\sum_k g_k^{HW}(\hat{\zeta}) = 0.31244$ against $v(\mathcal{N}_6) = 0.720163$ (level pool 56.6% of $v(\mathcal{N}_6)$). Reading this gap as “BSY’s bargain falls well short of full efficiency” would attribute the entire 57% shortfall to bargaining failure. Three companion recomputations below (predictions P-31/P-33/P-34), correcting an earlier pass whose headline split mixed agenda-coverage and instrument-granularity effects

(ECON-74/MATH-84), instead anchor an *exact additive* decomposition on the full-coverage, line-by-line frontier $D = 0.885046$ (P-34): $D - \Sigma_k g_k^{HW} = 0.572605 = \mathbf{79.51\%}$ of $v(\mathcal{N}_6)$, which splits exactly into three matched-instrument terms, each varying one dimension — coverage, instrument fineness, or individual rationality — while holding the other two fixed: $D - F_{\text{line}}^{v2} = 0.448206 = \mathbf{62.24\%}$, extending coverage at a fixed line-by-line instrument from BSY’s 151-line negotiated set — itself an estimation-sample construct, not the Uruguay Round’s actual agenda — to the full 294-line schedule (P-34); $F_{\text{line}}^{v2} - F^{\text{IR,corr}} = 0.008767 = \mathbf{1.22\%}$, the transferable-utility/ individually-rational wedge on the identical 151-line set, measured against the TI-5-corrected frontier $F^{\text{IR,corr}} = 0.428073$ (P-33, discharged below; the 0-floor predecessor $F_{\text{line}}^{\text{IR}} = 0.428519$ gives an internally-consistent but now-superseded 1.15%, retained only where explicitly flagged as the uncorrected reading); and $F^{\text{IR,corr}} - \Sigma_k g_k^{HW} = 0.115632 = \mathbf{16.06\%}$, the within-agenda bargaining shortfall restricted to points no member finds worse than each member’s own line-by-line 1990 threat point (P-31/P-33). Two of these three inequalities hold by construction and one does not, and the distinction matters for how the identity should be read. $D \geq F_{\text{line}}^{v2}$ and $F_{\text{line}}^{v2} \geq F_{\text{line}}^{\text{IR}}$ are genuine nested-feasible-set arguments — extending the line set from 151 to 294 lines, and adding six IR inequality constraints to an otherwise identical problem, can each only weakly raise or weakly lower the maximized objective, independently of what value any particular candidate point takes. $F_{\text{line}}^{\text{IR}} \geq \Sigma_k g_k^{HW}$ is different in kind: it would hold by construction only if BSY’s own cached bargained-equilibrium tariff schedule at $\hat{\zeta}$ were itself a feasible point of $F_{\text{line}}^{\text{IR}}$ ’s constraint set, and a direct check finds it is not. Restricting BSY’s cached schedule to the 151 negotiated lines, 5 of 151 (3.3%) genuinely fall outside this paper’s own cuts-only box $[0, \tau^{90}]$, all below zero and legal under BSY’s own solver bound $lb = -0.25$ (TI-13(d)(v)’s known asymmetry, deepest at Japan, -2.6pp); the remaining 86 lines sit exactly at BSY’s own solver slack (excess $= 1.000 \times 10^{-6}$, identical to BSY’s own $ub = \tau^{90} + 10^{-6}$) — pinned at the uncut 1990 rate, not an economic increase; the true increase count is 0/151. The companion IR condition, $g_i(\hat{\zeta}) \geq 0$ for all six members, does hold (A8-c, verified empirically, `sanity_assertions.md` MC-15) (`check_hw_box_constraint.md`). BSY’s own equilibrium is therefore approximately, though not exactly, a point this paper’s own optimization problems could themselves have selected — the residual gap is confined to 5 lines, all below zero — and the third inequality — the one that produces $H2$ — is confirmed only *empirically*, in this run’s own computed numbers, not guaranteed by the structure of the optimization problems the way the first two are: $H2$ is “non-negative in this run’s numbers, independently verified,” not “non-negative by construction”; this changes nothing about $H2$ ’s magnitude, only the epistemic status of the claim that it must be non-negative. Separately, converting the line-by-line instrument back to $v(\mathcal{N}_6)$ ’s own country-wide scalar, $D - v(\mathcal{N}_6) = 0.164883 = \mathbf{22.90\%}$ — the pure value of instrument granularity, holding coverage fixed at full; this is a different total than the 79.51% identity above (its own base point is $v(\mathcal{N}_6)$, not $\Sigma_k g_k^{HW}$) and the two must not be added together. On a matched, unconstrained instrument, coverage is worth roughly $3.6\times$ the within-agenda shortfall ($F_{\text{line}}^{v2} - \Sigma_k g_k^{HW} = 17.27\%$, so $62.24/17.27 = 3.60$); comparing directly against the IR-constrained, TI-5-corrected 16.06% shortfall instead gives 3.88, an instrument-*mismatched* ratio (TU-regime coverage against IR-regime shortfall). The regime-matched version — D^{IR} , the IR-constrained analogue of D , against the identical IR-constrained

shortfall — has since been computed and confirmed (two of two planned starts converged on the value, cross-start F difference 3.1×10^{-10} ; the two starts' per-line argmax differs on a flat ridge, per-line $L_\infty = 0.1512$ apart, though F itself and the endpoint census agree exactly, so D^{IR} 's value is confirmed while its argmax is not point-identified): $D^{IR} = 0.879995$ (0-floor; retained only as a labelled historical upper bound, see below), only 0.70% of $v(\mathcal{N}_6)$ below D itself (0.57% of D itself), so the IR constraint costs almost nothing at full coverage either. **The per-country floor correction pending as of the previous draft has since been executed** (`d_ir_vline_line_member_aggregate.md`, R&R item 2): replacing the constant $g_i \geq 0$ floor with the instrument-matched singleton floor $v^{\text{line,full}}(\{i\})$ (the 294-line analogue of $v^{\text{line}}(\{i\})$, six values computed directly, monotonicity $v^{\text{line,full}}(\{i\}) \geq v^{\text{line}}(\{i\})$ verified at every country) gives $D^{IR,corr} = 0.877226$, a small downward correction (-0.38% of $v(\mathcal{N}_6)$, -0.31% of D^{IR} itself) driven by a single binding constraint (Australia's floor binds to 14 decimal places; the other five countries are strictly interior). Nesting holds exactly: $D^{IR,corr} \leq D^{IR} \leq D$. Measured against the TI-5-corrected frontier, $D^{IR,corr} - F^{\text{IR,corr}} = 62.37\%$ of $v(\mathcal{N}_6)$ (against the uncorrected-numerator reading $D^{IR} - F^{\text{IR,corr}} = 62.75\%$, and the pure 0-floor $D^{IR} - F_{\text{line}}^{\text{IR}} = 62.69\%$, both now superseded) — coverage-maximality is robust across all four cells of the $\{\text{TU}, \text{IR}\} \times \{\text{negotiated}, \text{full}\}$ grid. The regime-matched ratio, both operands now singleton-floor-corrected, is $62.37\%/16.06\% \approx 3.88\times$ (`d_ir_vline_line_member_aggregate.md`) in place of the mismatched $3.60/3.88$ readings above (that instrument-mismatched 3.88 figure numerically coincides with this paragraph's own fully-corrected $3.88\times$ but is a different quantity and the two should not be conflated) and the previously reported mixed reading, $3.91\times$. **This ratio is now a fully regime- and floor-matched point estimate, not an upper bound on either side:** $H2$'s denominator was already corrected to the TI-5-correct, line-by-line floor $v^{\text{line}}(\{i\})$ (16.06%); the numerator $D^{IR,corr}$ has now received the analogous full-294-line correction $v^{\text{line,full}}(\{i\})$, discharging the one-sided upper-bound caveat this paragraph previously carried. The correction moved the ratio down slightly, as expected (tightening a floor can only lower a constrained maximum): $3.91\times \rightarrow 3.88\times$, a $0.03\times$ ($\approx 0.8\%$ relative) change, consistent with the small magnitude of the underlying $D^{IR,corr}$ correction. **Value certified, argmax not:** as with D^{IR} itself, $D^{IR,corr}$'s value is certified to high precision (cross-start F difference 4.09×10^{-10}) while its argmax sits on a flat ridge (per-line $L_\infty = 0.1513$ between starts, essentially identical to D^{IR} 's own 0.1512) — no line-level or per-country cut composition claim should be drawn from this artifact; only the aggregate value and the ratio built from it are citable. (i) On the 151 lines both objects can be compared directly, a naive tariff-depth comparison suggests D1 cuts substantially deeper than BSY's own negotiated outcome — mean 5.79pp versus 2.91pp (`p17_tariff_fit.md`; this comparison should itself be read together with prediction P-17's placebo-adverse rider below, since an atheoretical uniform-cut baseline fits the actual line-level tariff change at least as well as either model) — but a depth comparison alone cannot settle whether the gap is bargaining shortfall or agenda coverage, which requires a value comparison on the same lines. (ii) $v(\mathcal{N}_6)$ also prices surplus on the several hundred lines — essentially all of agriculture and tobacco — that lie outside BSY's own bilateral negotiating structure, itself an estimation-sample construct rather than the Uruguay Round's actual agenda (the agenda-mismatch caveat above); P-29 isolates exactly this component and finds

it, on its own, material at more than ten times the pre-registered threshold. (iii) $v(\mathcal{N}_6)$ is a transferable-utility, grand-coalition object, while BSY’s estimated outcome is the fixed point of seven *bilateral*, non-transferable bargains (Section 2); prediction P-33 directly tests how much of the within-agenda shortfall this transferability wedge accounts for and finds it small (§4.1.2 below). (iv) $\Sigma_k g_k^{HW}$ nets out none of the coalition-structure gap this creates.

The agenda-coverage component, quantified (P-29, ECON-56(c)). We recompute the D1 planner’s optimum restricting the lever s_n to act only on the 151 lines in BSY’s own negotiating set — the identical sector-screen (agriculture, sectors 1–9, and tobacco excluded) and mutual, non-developing principal-supplier construction that builds their own `tariffs_under_negotiation` table — holding every other line at its 1990 applied rate. This restricted-lever value is $v^{\text{negotiated}}(\mathcal{N}_6) = 0.303512$, against $v^{\text{full}}(\mathcal{N}_6) = 0.720163$: a gap of 0.41665 (57.9% of $v(\mathcal{N}_6)$), clearing the pre-registered 5% materiality band by more than an order of magnitude (`p29_restricted_lever.json`, prediction P-29). This is a genuinely *consistent-instrument* measure of agenda coverage at the country-scalar lever: both sides (v^{full} and $v^{\text{negotiated}}$) retain D1’s single country-wide scalar s_n , differing only in which lines that scalar is allowed to touch, so 57.9% is not a coarse or conflated number — it answers a well-posed question (“how much does agenda width matter, holding the country-scalar instrument fixed”) precisely. It is simply a *different* question from the one a line-by-line instrument answers: prediction P-34 below repeats the identical coverage comparison at the finer, line-by-line lever both P-31 and BSY’s own negotiators use, and finds a *larger* figure, 62.2% of $v(\mathcal{N}_6)$ — a finer instrument extracts more value from the additional coverage too, not only from the lines already on the table, so the two coverage figures are not in tension: 57.9% is the country-scalar-instrument answer, 62.2% is the line-instrument answer, and both are internally consistent on their own terms (§4.1.2). A naive tariff-depth reading (component (i) above) that ignores coverage cannot be the whole story either way. Separately, and on a distinct evidentiary footing, BSY’s own estimated bargaining outcome, $\Sigma_k g_k^{HW}(\hat{\zeta}) = 0.312441$, is *not below* $v^{\text{negotiated}}(\mathcal{N}_6) = 0.303512$: on their own agenda, BSY’s bargained outcome meets a coarser, country-scalar planner’s constrained optimum evaluated on the identical 151 lines. **We read this as a bound, not as evidence of within-agenda efficiency, and the bound runs the wrong way to license that stronger reading.** The two objects use different policy instruments on identical lines — $v^{\text{negotiated}}$ retains D1’s single country-wide scalar s_n , while BSY’s negotiators set each of the 151 lines individually, a strictly finer instrument — and the resulting policy spaces are *not nested* in the direction that would make $v^{\text{negotiated}}$ a valid upper bound on $\Sigma_k g_k^{HW}$: this run’s own singleton table contains the direct counterexample, $v^{\text{negotiated}}(\{\text{Australia}\}) = 0.002060$ exceeding $v^{\text{full}}(\{\text{Australia}\}) \approx 0$ (`p29_restricted_lever.md` §4), which is possible only because the country-scalar and full-schedule optima are not nested inside one another. Consequently $\Sigma_k g_k^{HW} \geq v^{\text{negotiated}}$ establishes only that BSY’s bargained outcome is not below a coarser benchmark on the same lines; it does *not* establish that BSY’s bargaining is close to the true line-by-line efficient frontier on its own agenda, which requires a finer-instrument benchmark. The first attempt to construct that benchmark (P-30, immediately below) fails on its own terms; the second (P-31, §4.1.2 below) succeeds, and **resolves this question against the within-agenda-efficiency reading:**

BSY’s bargaining falls short of the true line-by-line frontier by a genuine, quantified shortfall, established most defensibly at 16.1% of $v(\mathcal{N}_6)$ once the comparison is both weight-corrected and restricted to individually-rational points (P-33, §4.1.2). What can be said once P-29, P-31, P-33, and P-34 are combined: the 56.6% level pool reflects agenda width, instrument granularity, and within-agenda bargaining performance jointly, in that order of magnitude — a real within-agenda bargaining shortfall survives, but it is the smallest of the three identified components. The one instrument available on this project’s own artifacts that operationalizes “finer than country-scalar” on the identical 151 lines — BSY’s own Table III TW-max counterfactual — is examined immediately below (prediction P-30); it does not resolve the question, for reasons stated there, which is why P-31 re-purposes the same MPEC machinery to solve the genuine line-by-line optimum directly.

BSY’s own Table III “TW-max” counterfactual, which we independently replicate (world-aggregate welfare gain +0.34% relative to the 1990 baseline, matching BSY’s own published value to a maximum deviation of 0.024pp across all reported cases, `phase6a_table3.json`; MPEC re-solve on the identical restricted line set, `phase6b_resolve.json`), is *not* directly usable as the finer-instrument benchmark P-29 needs, and an earlier draft’s characterization of it as a “weakly tighter coverage-matched upper bound” was a category error: TW-max maximizes *world* welfare over the full modelled tariff schedule — every sector, not only the 151 lines in BSY’s own negotiated set — and all eleven modelled regions, including the five non-strategic (NES) composites, not the sum of our six negotiating countries’ welfare restricted to a subset of lines.³ On the same 151-line set, TW-max’s own regional decomposition shows the EU (−3.06%) and Australia (−1.15%) *losing* while each NES region gains (+1.6 to +2.3%), so a naive six-country member sum read off TW-max is not even guaranteed to be positive, let alone a certified upper bound on $\sum_{i \in \mathcal{N}_6} g_i^{HW}$. The object that would resolve this — $\max_{s: 151 \text{ lines}} \sum_{i \in \mathcal{N}_6} I_i^{\text{base}} \Delta W_i(s)$, the six-country member sum evaluated at the line-by-line (not country-scalar, not world-welfare) optimum on the identical negotiated lines — is registered as prediction P-30. We have since computed a masked version of it: TW-max applied only on the 151-line mask, with every other line — including every NES region’s tariffs and the six countries’ own non-negotiated lines — held at its 1990 baseline. This masked point has the correct *support* for the six-country line-by-line problem (only negotiated lines move, everything else fixed at 1990), but a direct box-constraint check — whether every masked line’s tariff also lies in this paper’s cuts-only range $[0, \tau^{90}]$, since BSY’s own TW-max optimization permits tariffs outside that range ($lb_t = -0.5$, $ub_t = 1.0$, `run_phase6b.py`) — **fails**: 127 of the 151 masked lines violate the box (56 below zero, concentrated in the EU and Australia; 71 above the 1990 applied rate, concentrated in the US, Japan, and South Korea; Canada has zero negotiated lines and is unaffected either way, `p30_box_constraint_check.json`). The masked point is therefore *not* a feasible point of this paper’s cuts-only policy space, and its value is not a lower bound on the true frontier in either direction — it is descriptive context only. That value is −3.220370 — deeply negative,

³Raw TW-max changes lines outside the 151-line negotiated mask for five of six countries, but leaves every one of Canada’s tariffs, negotiated or not, at its 1990 level (`p30_tymax_member_aggregate.md` §2). We do not know whether this reflects a corner solution at Canada’s own optimum or a restriction in BSY’s own TW-max computation to some subset of countries; if the latter, BSY’s “world optimum” may itself not be a solution over the full policy space, a possibility this paper has not investigated.

driven almost entirely by the EU’s terms-of-trade loss ($g_{\text{EU}} = -7.87$) — because TW-max was never trying to make this six-country subgroup well off, on a cuts-only or any other constrained policy space: it maximizes *world* welfare without that restriction, and masking its footprint to the negotiated lines changes which point is evaluated without changing what objective produced it, or bringing it inside this paper’s box (p30_twmax_member_aggregate.md, 2026-08-18 box-constraint addendum). P-29’s own $v^{\text{negotiated}}(\mathcal{N}_6) = 0.303512$ remains a genuinely feasible lower bound on the same problem, since its country-scalar lever is constructed within $[0, 1]$ by definition: this exercise therefore does not improve on what P-29 already established, for two independent reasons — its value is far looser even taken at face value, and it fails the feasibility check a bound requires in the first place. The mechanical decision rule’s literal “established” verdict is deliberately not banked (author ruling, p30_twmax_member_aggregate.md §3, box-constraint addendum). **Neither “within-agenda efficiency” nor “essentially none of the gap is bargaining underperformance” is established by this exercise.** We record P-30 as **INCONCLUSIVE-BY-CONSTRUCTION** — a registered test that ran and produced a diagnostic, not an answer. The genuine 151-dimensional line-by-line maximization P-30 could not deliver is not left to future work: prediction P-31, immediately below, re-purposes the same MPEC machinery to solve it directly and settles the coverage-vs-bargaining-shortfall question.

The genuine line-by-line frontier, weight-corrected (P-31, F_{line}). Where P-30 could only evaluate an already-solved *world-welfare* optimum at a masked point, P-31 solves the six-country member-sum problem directly: $\max_{s \in [0,1]^{151}} \sum_{i \in \mathcal{N}_6} I_i^{\text{base}} \Delta W_i(s)$, over BSY’s own 151-line negotiated mask, holding every other line and every NES tariff at its 1990 baseline — the identical feasible set as $v^{\text{negotiated}}$ (P-29), but with a genuine line-by-line lever in place of a country-wide scalar. This re-purposes `run_phase6b.py`’s already-validated JAX-jit continuous-optimization machinery (the same code that reproduces BSY’s own Table III TW-max counterfactual to $4.8 \times 10^{-5}\%$), with the objective weight vector swapped to the $\mathcal{N}_6$ indicator and bounds restricted to the 151 negotiated lines. The original run used an *equal-weighted* percent-welfare objective rather than this paper’s own dollar-denominated $D1$ objective, whose implicit per-country weight on percent welfare change, $\tilde{P}_i^{\text{base}} = I_i^{\text{base}}/W_i^{\text{base}}$, ranges from 992.6 (South Korea) to 3292.7 (Canada) — a $3.3\times$ dispersion (`ptilde_dispersion_check.json`) — so its reported value, $F_{\text{line}} = 0.360516$, is a *feasible lower bound* on the registered objective, not its argmax (MATH-83/IDENT-16-01). A weight-corrected re-run, using $\tilde{P}_i^{\text{base}}$ as the per-country weight (`run_f_line_family.py -variant v2`), converges from both starting points — the 1990 baseline and the free-trade corner — to $F_{\text{line}}^{v2} = 0.436840$ (cross-start dispersion $< 10^{-5}$ relative, box constraints hold, TI-15 identity check passes, `p31_fline_v2.md`).

The pre-registered decision rule (prediction P-31, thresholds frozen before compute at $\Sigma_k g_k^{HW} + 5\%$ of $v(N) = 0.348449$) fires **refutation** on both the original and the weight-corrected run, more decisively at the corrected weight: the original run’s clearance margin was $F_{\text{line}} - 0.348449 = 0.012067$; the corrected run’s is $F_{\text{line}}^{v2} - 0.348449 = 0.088391$ — $7.3\times$ the original margin (not $2.5\times$, which compares against a different, $H2$ -ratio benchmark; corrected here). **Every citation of the original $F_{\text{line}} = 0.360516$ and its 49.94%/6.68%/88.2%/11.8% split is superseded** by the weight-corrected figures below and is retained in `predictions.md` only for

provenance. Country-level detail at F_{line}^{v2} : $g_{\text{US}} = -0.0935$, $g_{\text{EU}} = -0.0569$, $g_{\text{Japan}} = +0.5318$, $g_{\text{Canada}} = +0.0160$, $g_{\text{Australia}} = +0.0034$, $g_{\text{Korea}} = +0.0360$ (`p31_fline_v2.json`). Two countries (US, EU) remain below their 1990 threat point at this unconstrained optimum; Australia’s borderline violation at the original, mis-weighted optimum disappears once the weight is corrected. The corrected weight *shrinks* the EU’s loss by 84% relative to the original run ($-0.3654 \rightarrow -0.0569$) — exactly the mechanism the corrected weight was designed to produce, since a dollar-denominated objective penalizes EU’s disproportionately large *percent* loss less once weighted by $I_i^{\text{base}}/W_i^{\text{base}}$. Japan’s gain at F_{line}^{v2} is $\sim 3.1\times$ its dollar share of BSY’s cached bargained outcome ($g_{\text{Japan}} = 0.5318$ against $55.47\% \times \sum_k g_k^{HW} = 0.1733$) — the efficient within-agenda frontier’s incidence is, if anything, more concentrated in Japan than BSY’s own bargaining outcome already is, even as the Shapley imputation (§2.6) reallocates share away from Japan toward the US and Canada (ECON-77).

Holding coverage fixed at the 151-line agenda, the 56.6% level pool splits at this instrument into $H1^{v2} = v(\mathcal{N}_6) - F_{\text{line}}^{v2} = 39.34\%$ of $v(\mathcal{N}_6)$ (69.5% of the gap) and $H2^{v2} = F_{\text{line}}^{v2} - \sum_k g_k^{HW} = 17.27\%$ of $v(\mathcal{N}_6)$ (30.5% of the gap; $H1^{v2} + H2^{v2} = 56.61\%$ closes this two-term identity exactly, by construction, since the two terms partition $v(\mathcal{N}_6) - \sum_k g_k^{HW}$ additively). **$H1^{v2}$ is not itself a coverage-only figure:** $v(\mathcal{N}_6)$ retains D1’s country-wide scalar lever while F_{line}^{v2} uses a line-by-line lever restricted to 151 lines, so $H1^{v2}$ nets a coverage effect against an instrument-granularity effect that push in the same direction and cannot be separated within this two-term split alone — it is a *lower bound* on pure coverage (coverage net of the granularity dividend), not coverage itself. It is nonetheless numerically consistent with the matched-instrument decomposition below: $D - F_{\text{line}}^{v2}$ (62.24%, pure coverage) minus $D - v(\mathcal{N}_6)$ (22.90%, the granularity dividend) equals $62.24 - 22.90 = 39.34$, exactly $H1^{v2}$ — an internal consistency check across two independently solved MPEC families (the 151-line family here and the 294-line family below), not a coincidence. The within-agenda shortfall very nearly *triples* relative to the original, mis-weighted reading, and its share of the total gap rises from 12% to 31%: the original run’s coverage-dominant framing understated the within-agenda component by roughly $2.6\times$.

Is the within-agenda shortfall a TU/NTU artifact? (P-33, F_{line}^{IR}). $\sum_k g_k^{HW}$ is BSY’s cached bargained outcome, verified empirically to be individually rational for all six members (A8-c, `sanity_assertions.md` MC-15); F_{line}^{v2} ’s unconstrained optimum is not (US and EU both below their 1990 threat point, above). Comparing a transferable-utility (TU) unconstrained optimum against a non-transferable-utility (NTU)-feasible bargained outcome without further comment would violate TI-13(d)’s comparability protocol: some of $H2^{v2}$ could be a transferability-regime wedge — surplus reachable only via intra-coalition transfers BSY’s own no-transfer bargaining architecture cannot make — rather than a genuine bargaining shortfall on the feasible set BSY’s negotiators actually bargain over. Prediction P-33 tests this directly: re-solve the identical weight-corrected MPEC with six additional inequality constraints, $g_i(t, w) \geq 0$ for every $i \in \mathcal{N}_6$, restricting the comparison to the individually-rational-relative-to-1990 (NTU-*feasibility*-comparable) subset of the 151-line policy space. This constraint bears on the *individual-rationality* axis only; the objective maximized, $\sum_{i \in \mathcal{N}_6} I_i^{\text{base}} \Delta W_i(s)$, remains a transferable-utility dollar sum throughout, so F_{line}^{IR} is an upper bound on what any no-transfer

bargaining solution could jointly achieve on this feasible set, not itself the output of an NTU solution concept. Both starts converge to $F_{\text{line}}^{IR} = 0.428519$, with the US and EU individual-rationality constraints binding to machine precision (`f_line_ir_member_aggregate.md`). The IR restriction removes only 1.9% of the unconstrained objective ($F_{\text{line}}^{v2} - F_{\text{line}}^{IR} = 0.008321$): Japan absorbs almost the entire adjustment (g_{Japan} falls from 0.5318 to 0.3851), redistributing surplus to bring the US and EU exactly to their threat points rather than eliminating their gains from the optimum. **The within-agenda shortfall therefore survives the individual-rationality-relative-to-1990 restriction almost entirely:** $H2^{IR} = (F_{\text{line}}^{IR} - \sum_k g_k^{HW}) / v(\mathcal{N}_6) = \mathbf{16.12\%}$ of $v(\mathcal{N}_6)$, only 1.15pp below the unconstrained 17.27%, and still $3\times$ the pre-registered 5% materiality band.

This IR-feasibility check is narrower than a demonstrated Pareto improvement over BSY's actual outcome, and the two must not be conflated. At F_{line}^{IR} 's optimum, four of the six members are *worse off* than at BSY's own cached bargain: $g_{\text{US}} = 0$ (vs. 0.0314 at $\sum_k g_k^{HW}$), $g_{\text{EU}} = 0$ (vs. 0.0647), $g_{\text{Australia}} = 0.0035$ (vs. 0.0069), and $g_{\text{South Korea}} = 0.0260$ (vs. 0.0323) — together 14.7% of $v(\mathcal{N}_6)$ redistributed away from these four members, almost entirely toward Japan (+0.212) and the zero-negotiated-lines Canada (+0.010). $H2$ therefore measures a TU-aggregate efficiency gap relative to BSY's own 151-line agenda, individually rational only relative to the 1990 status quo, not a point every member would accept over their actual bargain: under BSY's own no-transfer architecture, four of six members would veto it. We therefore report the within-agenda bargaining shortfall at $H2 = 16.12\%$ of $v(\mathcal{N}_6)$ at this (0-floor) IR constraint; the following paragraph corrects this to the TI-5-correct point estimate, **H2 = 16.06%**, which is the paper's most defensible reading of the *aggregate* efficiency gap, and, per TI-13(d')'s disclosure requirement, state precisely which axis the TU-only qualifier attaches to: the *individual-rationality* axis no longer requires it (the transferability wedge P-33 isolates accounts for at most 1.15pp of the 17.27pp unconstrained reading, not for the shortfall itself); the *aggregation* axis still does, since F_{line}^{IR} remains a TU dollar-sum maximization and is an upper bound, not a demonstrated feasible point, for any no-transfer solution that must also respect every member's participation constraint *relative to the actual bargain*, not only relative to 1990. (A full λ -transfer/NTU-Shapley recomputation — the remaining, narrower obligation under TI-13(d')(iii) — is deferred to R&R.) As with every $\hat{\zeta}$ -dependent object in this paper, $\sum_k g_k^{HW}$ is evaluated on the primary, cached Nash-in-Nash comparator branch (Table 3); $H2$ is branch-conditional in the same sense Γ_{US} is above.

Is F_{line}^{IR} 's IR floor itself correct, and does correcting it move $H2$? (TEST-17-04, discharged). F_{line}^{IR} 's individual-rationality constraint, $g_i(t, w) \geq 0$, uses the 1990 status quo as every member's outside option. But the registered TI-5 baseline for individual rationality is $\phi_i(v) \geq v(\{i\})$, and on a line-by-line instrument the correspondingly correct unilateral outside option is $v^{\text{line}}(\{i\})$ — country i 's own best response using *only* its own negotiated lines, all other lines and countries held at 1990 — not the country-scalar $v(\{i\}) \approx 0$ this paper reports elsewhere. We compute $v^{\text{line}}(\{i\})$ directly for all six members (`p33_line_singletons.md`): it is *not* uniformly zero. Three of six are materially positive — Japan $v^{\text{line}}(\{\text{Japan}\}) = 0.071712$, Australia 0.005106, EU 0.003184 — against the US, South Korea, and Canada (no negotiated

lines) at or indistinguishable from zero. Since $g_i \geq v^{\text{line}}(\{i\})$ is a strictly tighter constraint than $g_i \geq 0$ whenever $v^{\text{line}}(\{i\}) > 0$, this flagged $F_{\text{line}}^{\text{IR}} = 0.428519/H2 = 16.12\%$ as an upper bound on the true within-agenda shortfall, pending a re-run with the tightened, per-country IR floor.

That re-run has since been executed (`f_line_ir_vline_member_aggregate.md`, two starting points, converged to 6+ significant figures, per-line constraint residual $\leq 6 \times 10^{-16}$): $F^{\text{IR,corr}} = 0.428073$, only 0.000446 (0.06pp of $v(\mathcal{N}_6)$) below $F_{\text{line}}^{\text{IR}} = 0.428519$. **The correction is small, not large:** the triple nesting $F^{\text{PD}} = 0.410870 \leq F^{\text{IR,corr}} = 0.428073 \leq F_{\text{line}}^{\text{IR}} = 0.428519$ holds exactly, but $F^{\text{IR,corr}}$ sits far closer to the loose (0-floor) end than to F^{PD} . The mechanism: Japan's own floor, though the largest of the three (0.0717), is never binding at the optimum — Japan's optimal gain (0.380–0.385) already exceeds it by a wide margin, so raising Japan's floor from 0 to 0.0717 costs nothing. What does bind, at machine precision, are the two *small* floors — EU (0.003184) and Australia (0.005106) — alongside the already-binding US floor (≈ 0); Canada and South Korea remain slack. **This yields the TI-5-correct point estimate, discharging the upper-bound flag:** $H2 = (F^{\text{IR,corr}} - \Sigma_k g_k^{\text{HW}})/v(\mathcal{N}_6) = 0.115632/0.720163 = 16.06\%$ of $v(\mathcal{N}_6)$, in place of the flagged 16.12% upper bound — a 0.06pp difference, two orders of magnitude below this paper's 5% materiality band, that does not affect the REFUTATION verdict below. Japan's own dollar gain at the corrected optimum, 0.3801, continues to exceed $v^{\text{line}}(\{\text{Japan}\}) = 0.0717$ by a wide margin, so the shortfall does not evaporate. As a fraction of the line-by-line instrument's own IR-constrained frontier, rather than the country-scalar $v(\mathcal{N}_6)$, the identical shortfall reads $H2/F^{\text{IR,corr}} = 0.115632/0.428073 \approx 27.0\%$: roughly a quarter of what the identical 151-line, line-by-line agenda could jointly attain remains unrealized. Both denominators are reported because $v(\mathcal{N}_6)$ (country-scalar) and $F^{\text{IR,corr}}$ (line-by-line) are different- granularity instruments, and the granularity-consistency protocol (TI-13(vi)) requires disclosing both rather than defaulting silently to one. Citations of $H2$ elsewhere in this paper, including the abstract and introduction, now report the 16.06% point estimate; the $g_i \geq 0$ -floor 16.12% figure is retained only as the (now discharged) upper bound this correction supersedes, cited only where an internally-consistent, uncorrected 0-floor comparison is explicitly flagged as such. The D^{IR} -based coverage ratio elsewhere in this section has been updated to use the corrected 16.06% denominator throughout, and, following the 294-line per-country floor correction to D^{IR} itself (`d_ir_vline_line_member_aggregate.md`, R&R item 2), is now a fully floor-matched point estimate, $62.37\%/16.06\% \approx 3.88\times$, superseding the previously reported mixed reading $\approx 3.91\times$ (above).

The full-coverage frontier and a matched-instrument ranking (P-34, D). P-29's 57.9% and the original, superseded 49.9% figure mixed two distinct axes — coverage and instrument fineness — inside a single number (ECON-74/MATH-84). Prediction P-34 separates them by solving a fourth point: the identical weight-corrected line-by-line MPEC extended to *all* 294 traded lines (49 sectors $\times$ six countries, agriculture and non-negotiated lines included), $D = 0.885046$ (`d_full_line_member_aggregate.md`) — the anchor point for the exact three-term identity: $D - \Sigma_k g_k^{\text{HW}} = 79.51\%$ of $v(\mathcal{N}_6)$, splitting as 62.24% coverage +1.22% TU–NTU wedge +16.06% within-agenda shortfall. All three terms are mea-

sured consistently against the TI-5-corrected frontier $F^{\text{IR},\text{corr}} = 0.428073$ ($D - F_{\text{line}}^{v^2} = 62.24\%$; $F_{\text{line}}^{v^2} - F^{\text{IR},\text{corr}} = 0.008767 = 1.22\%$; $F^{\text{IR},\text{corr}} - \Sigma_k g_k^{\text{HW}} = 0.115632 = 16.06\%$), summing to $0.572605 = 79.51\%$ of $v(\mathcal{N}_6)$ exactly. The internally-consistent, uncorrected 0-floor alternative ($62.24\% + 1.15\% + 16.12\%$, also 79.51% exactly on its own basis) is retained only as the pre-TEST-17-04-discharge reading. **The two bases must not be cross-mixed:** pairing the uncorrected 1.15% wedge with the corrected 16.06% shortfall gives 79.45% , not 79.51% , and does not close the identity; every citation of this identity elsewhere in this paper is corrected to the fully-corrected reading, $62.24\% + 1.22\% + 16.06\% = 79.51\%$ (the two-decimal components are rounded independently and sum to 79.52% ; the exact total, from the underlying unrounded values, is 79.51% , as reported above). D 's own optimum is *not* individually rational: at D , $g_{\text{Australia}} = -0.0323$, the sole IR violation among the six members at this point, unsurprising since no IR constraint was imposed on this unconstrained, larger-coverage variant. On a matched, unconstrained instrument, coverage is worth roughly $3.6\times$ the within-agenda shortfall ($D - F_{\text{line}}^{v^2} = 62.24\%$ against $F_{\text{line}}^{v^2} - \Sigma_k g_k^{\text{HW}} = 17.27\%$, so $62.24/17.27 = 3.60$). The regime-matched version of this ratio — $D^{\text{IR},\text{corr}}$ (0.877226 , the 294-line per-country floor-corrected value, superseding the 0-floor $D^{\text{IR}} = 0.879995$ retained only as a labelled historical upper bound) against the TI-5-corrected, IR-constrained shortfall on both sides — is $3.88\times$ ($D^{\text{IR},\text{corr}} - F^{\text{IR},\text{corr}} = 62.37\%$ against $H2 = 16.06\%$, above), essentially the same reading on a properly instrument- and regime-matched basis, which we cite as the headline ratio going forward in place of the mismatched $3.60/3.88$ figures above (that 3.88 is a different, instrument-mismatched quantity — TU-regime coverage against IR-regime shortfall — and its numerical coincidence with this paragraph's own fully-corrected $3.88\times$ is not to be conflated with it). **This ratio is now a fully regime- and floor-matched point estimate, not an upper bound:** both $H2$'s denominator and $D^{\text{IR},\text{corr}}$'s numerator use the TI-5-correct, instrument-matched singleton floor (`d_ir_vline_line_member_aggregate.md`, R&R item 2), discharging the one-sided upper-bound caveat this paragraph previously carried; the correction was small and in the expected direction ($3.91\times \rightarrow 3.88\times$, a $0.03\times$ move, since tightening a floor can only lower a constrained maximum). As with D^{IR} , $D^{\text{IR},\text{corr}}$'s value is certified (cross-start difference 4.09×10^{-10}) while its argmax is not (flat ridge, per-line $L_\infty = 0.1513$); the $3.88\times$ ratio, built from the certified value, is safely citable. Widening what is on the table — extending coverage beyond BSY's own estimation-sample agenda, an institutional restriction consistent with P-32's finding that applied agricultural protection *rose*, not fell, over 1990–2000 in this sample (§4 above) — is worth several times more than extracting additional surplus from the agenda already on the table.

We record P-31 as **REFUTATION (established, weight-corrected, IR-tested, and floor-corrected, author-adjudicated 2026-08-18)**: within-agenda bargaining leaves a real, quantified, IR-feasible 16.06% of $v(\mathcal{N}_6)$ (TI-5-correct point estimate; 16.12% under the uncorrected 0-floor) on the table, on top of — not instead of — the larger agenda-coverage (62.24%) and TU–NTU-wedge (1.22%) terms of the same additive identity (`p31_fline_v2.md`, `f_line_ir_member_aggregate.md`, `f_line_ir_vline_member_aggregate.md`, `d_full_line_member_aggreg`).

What does the within-agenda shortfall reflect? A natural candidate mechanism — that BSY’s estimated bargaining weight $\hat{\zeta}$ is asymmetric, and a more symmetric bargain would close more of the 16.06% gap — is tested with the one comparator this paper has computed, with no new compute: `sanity_assertions.md` already reports $\Sigma_k g_k^{HW}(\hat{\zeta}) = 0.31244$ against a symmetric-weight placebo $\Sigma_k g_k^{HW}(0.5) = 0.29708$ (the same placebo convention as §4.1’s Γ placebo below). Holding the corrected frontier $F^{\text{IR,corr}} = 0.428073$ fixed — it does not depend on $\hat{\zeta}$ — and substituting the symmetric-weight comparator gives $H2(0.5) = (F^{\text{IR,corr}} - \Sigma_k g_k^{HW}(0.5))/v(\mathcal{N}_6) = 18.19\%$: *larger*, not smaller, than $H2(\hat{\zeta}) = 16.06\%$. This is a *single* alternative weight point, not a scan over ζ , and $\Sigma_k g_k^{HW}(\zeta)$ has not been shown to be monotone or single-peaked in ζ ; the correct statement is that **the one available alternative weight point moves the gap the wrong way** for the disproportionate-weight story — symmetrizing $\hat{\zeta}$ should shrink the gap if it reflected a weight favoring Japan, and instead it widens by 2.1pp — which is evidence *against*, not a refutation of, the weight-asymmetry channel; fully excluding the channel would require $\max_{\zeta} \Sigma_k g_k^{HW}(\zeta)$, not computed here. This points instead to the two signatures the *w*-monotonicity panel above (P-35) already identifies: bilateral bargaining pairs cannot price the MFN spillovers a country’s own concession confers on parties outside that pair, and Canada — present in 16/17 of the panel’s violations and the only member of $\mathcal{N}_6$ absent from every bilateral pair, yet a genuine positive free-rider on others’ liberalization ($g_{\text{Canada}}(\hat{\zeta}) = 0.0039$, `sanity_assertions.md` MC-14) — carries both fingerprints at once. Read this way, 16.06% is **consistent with an architectural reading** — the price of using a bilateral solution concept for a problem whose externalities are inherently multilateral — but this is a *reading*, *not a decomposition*: as this paper’s own model discussion notes, cut-depth on the shared negotiated lines (axis (a)) and coalition structure (axis (d)) remain *jointly embedded* in $\Sigma_k g_k^{HW}$ and have not been separately measured (coverage, axis (b), has in fact been separately measured above at 62.24%, and this coverage match is now **verified**, not merely presumed: BSY’s own cached $\hat{\zeta}$ equilibrium moves zero tariff lines outside the 151-line negotiated mask, 0 of 251 off-mask cells checked across all six members, Canada’s full 67-line row included (`check_hw_offmask_displacement.md`); it is (a) and (d) that remain entangled, not (b) and (d)), so this evidence does not isolate the bilateral bargaining protocol itself from within-agenda cut-depth, and should not be cited as a residual measure of how hard any given bilateral pair bargained net of coalition structure. Prediction P-40 (`predictions.md`, registered before compute) was designed to separate axes (a) and (d) directly, via a pairwise-decomposition optimization; the result, reported next, does not resolve the entanglement — axes (a) and (d) remain *jointly embedded*, and this remains the paper’s operative description of *H2*.

Scope note: the F_{pairs} family has been re-optimized under the corrected *D*-convention (MATH-124 finding); the numbers below are superseded, not merely re-verified. The provisional flag this paragraph previously carried is **discharged**. `run_p40_f_pairs.py` (BR1) and `run_p40_f_pairs_eq.py` (the Gauss–Seidel fixed point) were both re-run with $p[\text{“D”}] = 0$ set before `make_static(p)`, matching the F_{line} family’s confirmed-safe pattern, and the fix was independently verified to machine precision on both legs: in-solve residual at the robust wage $\leq 3.3 \times 10^{-16}$, in-solve-vs-robust welfare difference $\leq 2.1 \times 10^{-16}$, posi-

tive control on the raw, unzeroed- D economy $\approx 3.1 \times 10^{-2}$ — the same order of magnitude the discriminant test used to diagnose the original bug, confirming the check has power, not merely code-pattern analogy (`check_p40_d_convention.md`). Both headline numbers move by more than the pre-registered 1% materiality threshold: F_{pairs}^{BR1} : $0.302243 \rightarrow 0.310033$ (+2.58%); F_{pairs}^{eq} : $0.305867 \rightarrow 0.311364$ (+1.80%). More consequentially, the re-optimization **reverses the sign of the paper’s previously banked explanatory mechanism**: the EU’s individual-rationality floor, previously reported violated by -0.1383 (-19.2% of $v(N)$) and named the boundary result’s “primary driver,” is in fact *satisfied* under the corrected economy ($g_{\text{EU}} = +0.0155$ at the new fixed point). The $21 \times / 30.9 \times$ EU-violation-magnitude readings, the “4 of 6 members better off than $\Sigma_k g_k^{HW}$ ” distributional claim, and the paragraph built on them (previously below) are therefore **retracted, not revised**, and replaced with the corrected account below. The categorical disposition itself — $F_{\text{pairs}}^{eq} < \Sigma_k g_k^{HW}$, **NO-VERDICT- BANKED** — is unchanged, but its margin shrinks from 0.006574 (0.913% of $v(N)$) to 0.001077 (0.149% of $v(N)$), an 83.6% relative contraction that leaves the boundary result *weaker*, not stronger, than previously reported: the new margin is smaller than the fixed point’s own within-trajectory convergence displacement (below), essentially indistinguishable from solver noise. The pair-externality decomposition and Δ_d^{TU} are updated in place below with the corrected numbers; where a sub-diagnostic (the cross-start dispersion check, the round-1-to-converged displacement, the $\zeta = 0.5$ comparator’s own internal decomposition) was not itself part of this re-optimization batch, it is marked *pre-fix figure, recomputation pending* rather than silently retained. Full numbers: `p40_dconvention_rerun_result.md/.json`.

Does P-40 separate axes (a) and (d)? A boundary result, no verdict banked (Round-20 remediation). P-40 constructs F_{pairs} : a **pair-decomposed welfare-sum equilibrium** of BSY’s own seven disjoint bilateral principal-supplier pairs, each pair maximizing the *dollar sum* of its own two members’ welfare on its own line subset among the 151 negotiated lines, taken to mutual best-response consistency. This is not a Nash bargain: there is no disagreement point and no Horn–Wolinsky-style weight, only a transferable-utility planner restricted to two members’ own lines (ECON-115). This describes what the bilateral architecture’s own equilibrium delivers under *this solution concept*, not an optimality property for the six-country group; it is not a ceiling on any six-country objective, and — because BSY’s own $\hat{\zeta}$ equilibrium is itself an individually-rational equilibrium of the identical seven-pair architecture, already solved, estimated, and cached (§3.1) — this construction’s failure to be individually rational (below) is a property of *this solution concept*, not evidence that the bilateral architecture cannot support an IR-consistent outcome at all. Two variants were computed against pre-registered, pre-result decision bands (`f_pairs_member_aggregate.md`). The one-shot simultaneous best response to the 1990 status quo gives $F_{\text{pairs}}^{BR1} = 0.310033$ (re-optimized under the corrected D -convention, superseding the pre-fix 0.302243 ; `p40_dconvention_rerun_result.md`); iterating each pair’s best response against the *current*, not 1990-conjectured, lines of the other six pairs (Gauss–Seidel) converges at round 4 ($\max |\Delta t| = 4.7 \times 10^{-7}$, pre-fix convergence diagnostic, recomputation pending) to the amended primary object, $F_{\text{pairs}}^{eq} = 0.311364$ (superseding the pre-fix 0.305867). This fixed point is not an artifact of the particular starting point or up-

date order: a reversed update order from the same start, and a forward pass initialized at F_{line}^{v2} 's own argmax (as far from the primary run's starting configuration as an economically meaningful point gets), both converge to the identical value to within 7.2×10^{-8} , per-line L_∞ distance 1.5×10^{-6} (`f_pairs_member_aggregate.md` §8, **pre- D -fix figure, recomputation pending** — this multi-start check was not part of the re-optimization batch); a further cold-start spot-check — each of the seven pairs individually restarted from free trade with the other six pairs held at the converged fixed point — returns to the identical value at every pair, $\Delta \leq 5.6 \times 10^{-9}$ (`p43_pair_ir_probe.md` §C, **pre- D -fix figure, recomputation pending**). The object is therefore best described as **spot-checked against a second start at every pair**, not merely a single-start local best-response fixed point; this is a spot-check, not a certificate — one alternative start per pair, holding all other pairs fixed at the reported solution, not a full joint multi-start sweep over the entire seven-pair system, which is what would be required to rule out an entirely different fixed point reachable from a fully different trajectory. Both values are **below** $\Sigma_k g_k^{HW} = 0.312441$ — BSY's own actual $1990 \rightarrow \hat{\zeta}$ outcome, measured, like F_{pairs}^{eq} , on the identical 151-line mask (verified below — the coverage-mismatch explanation offered in an earlier draft of this passage is withdrawn). This triggers the boundary branch pre-registered for exactly this contingency: neither dispositive branch — architectural constraint ESTABLISHED, nor non-architectural-residual- dominant — can be banked, on either the 1990-IR leg ($F^{\text{IR},\text{corr}} = 0.428073$) or the transfer-unconstrained leg ($F_{\text{line}}^{v2} = 0.436840$); both legs agree the disposition is boundary. The margin driving this is thinner than previously reported, and shrank further under the corrected D -convention: $\Sigma_k g_k^{HW} - F_{\text{pairs}}^{eq} = 0.001077$ (0.149% of $v(N)$), down 83.6% relative from the pre-fix margin, 0.006574 (0.913% of $v(N)$) (`p40_dconvention_rerun_result.md`). The pre-fix within-trajectory displacement diagnostics (round-1-to-converged step +0.003624; round-1 consistency residual -0.003879) have not themselves been recomputed against the new fixed point and are marked **pre- D -fix figures, recomputation pending**; qualitatively, the new margin is smaller than that displacement scale by construction (it shrank 83.6% while the fixed point itself moved +1.8%), so if anything the “margin is within noise” reading is *strengthened*, not weakened, by the correction — not, as an earlier draft stated, the fixed point's dispersion *across* starting points and update orders, which the spot-checks above instead measure at 7.2×10^{-8} (pre-fix figure), nearly five orders of magnitude smaller; the correct sensitivity this margin should be read against is *cross-convention* (a materially different way of solving the pair sub-problem), for which the D -convention fix itself is now the best available data point: a +1.80% move in F_{pairs}^{eq} from correcting one implementation convention dwarfs a 0.149%-of- $v(N)$ margin. Its firing is also a knife-edge function of the realized-outcome comparator used: at the paper's own $\zeta = 0.5$ robustness benchmark, $\Sigma_k g_k^{HW}(0.5) = 0.297081 < F_{\text{pairs}}^{eq} = 0.311364$, landing squarely inside branch (i)'s own pre-registered “architectural constraint ESTABLISHED” threshold (≤ 0.370257 on the IR-constrained leg, ≤ 0.374640 on the transfer-unconstrained leg) rather than merely failing to trigger the boundary — i.e. the identical construction, evaluated against a different but equally legitimate ζ point, would be banked as a positive architectural result. This is not cited to argue for banking that branch here: the paper's primary comparator is $\hat{\zeta}$, not 0.5, and the near-noise-level $\hat{\zeta}$ margin documented above (**corrected**: under the D -convention fix there is no longer

an EU individual-rationality violation “present at both anchor points” to invoke here — see the discharged scope note above — so the reason to distrust this margin is the margin’s own thinness and comparator-sensitivity, not a floor violation) is reason enough not to treat switching anchors as settling anything. It is cited because a margin this comparator-sensitive is, on its own terms, not robust evidence for either dispositive branch. An earlier, unamended version of the decision criteria would have misclassified this result as architectural-constraint ESTABLISHED, since $F_{\text{pairs}}^{\text{eq}} = 0.311364 \leq 0.370257$ satisfies branch (i)’s own pre-registered threshold in isolation; the pre-result amendment that added the boundary check (`predictions.md`, 2026-08-19) is what caught this before the result existed and converted a false positive into an honest “no verdict.”

P-40’s original question — whether $H2$ is genuinely architectural or an inseparable mixture with within-agenda cut-depth — therefore remains open, and this run’s boundary disposition should be read as marginal, not as a robust separation from $\Sigma_k g_k^{\text{HW}}$.

Is the individual-rationality failure below a genuine non-existence result, or a solver limitation? (Caveat, added with the D -convention fix: this paragraph’s motivating premise is now largely moot.) Under the corrected economy the EU does not violate its own individual-rationality floor at the $F_{\text{pairs}}^{\text{eq}}$ fixed point (above), so there is no “failure” left for the question below to resolve; `run_p43_pair_ir_probe.py` is itself one of the ten scripts confirmed on the D -convention-affected list and has not been re-run, so every number in this paragraph is **likely itself confounded by the same bug, re-run deferred to R&R**. The paragraph is retained as a diagnosed lower bound on the EU’s unilateral bargaining room under the pre-fix economy, not as evidence resolving a floor violation this batch’s fix shows does not survive. A regime-matched comparator, $F_{\text{pairs}}^{\text{eq,IR}}$, adding $g_i \geq 0$ to every pair’s own sub-problem, failed to converge (below); the frozen decision criterion this row pre-registered treats $\max g_{\text{EU}} < 0$ at *every* EU-containing pair as sufficient to conclude the EU’s best-response correspondence is empty. We test only that one-sided, per-pair floor: holding all other six pairs fixed at the $F_{\text{pairs}}^{\text{eq}}$ fixed point, does either EU-containing pair admit a strictly positive, unilaterally IR-feasible improvement for the EU on its own line subset, ignoring the partner’s own floor? Both do — US–EU, $\max g_{\text{EU}} = +0.130034$; EU–Japan, $\max g_{\text{EU}} = +0.043184$ (`p43_pair_ir_probe.md` §A) — so the one pre-registered route to an emptiness *proof* is closed: a feasible point with $g_{\text{EU}} > 0$ exists in each pair considered alone. This does **not** establish that the pair’s two-sided IR-floored sub-problem is jointly feasible — the partner’s own floor ($g_{\text{US}} \geq 0$ or $g_{\text{Japan}} \geq 0$) was never imposed alongside the EU’s — nor that a fixed point of the IR-floored map across all seven pairs simultaneously exists; only the simplest failure mode, EU unilateral infeasibility, is ruled out. **A second caveat, disclosed here:** every value in this paragraph and the placebo below is this paper’s own general-equilibrium re-evaluation at the solver’s reported argmax, not the solver’s raw in-solve objective at termination, which is unreliable across the board — it overstates the US–EU probe by $37\times$ (4.8967 at termination vs. 0.1300 re-evaluated) and the EU–Japan probe by $113\times$ (4.8774 vs. 0.0432), and inverts the sign entirely for the Japan and Australia single-country placebos (−5.0457 at termination vs. +0.3131 re-evaluated for Japan; sign-flipped for Australia as well, re-evaluated value +0.0085 as reported below) — all while every probe reports `success=True`.

This is not a vague or generic MPEC symptom: it has since been traced to a concrete mechanism. `run_p43_pair_ir_probe.py` is one of ten scripts, confirmed by direct code inspection, whose in-solve SLSQP constraint uses the raw, non-zeroed trade-deficit array D while every downstream welfare re-evaluation forces $D = 0$ — two different economies evaluated as if they were the same one (MATH-124, `math124_discriminant.md`); the 37–113 $\times$ gap and the sign inversions above are exactly this mismatch’s signature (a raw- D /robust-re-evaluation residual of $3.08\text{--}3.09 \times 10^{-2}$, reproduced as a positive control on this same probe family). A comparably-sized MPEC in the same code family, `run_f_line_family.py` (residual $10^{-15}\text{--}10^{-16}$), is *not* a counterexample of the same phenomenon under a different guise: it was directly, numerically confirmed safe — the identical D -zeroing step executes in place immediately before its solve, so its in-solve and reported economies agree to machine precision (`f_line_family_dfix_rerun.md`) — not merely observed to carry a smaller residual. Every number cited in this section is the re-evaluated GE value; the raw JSON’s solver-objective field should not be cited directly. The sign of every existence witness survives re-evaluation (all four probes remain positive at true GE), so this paragraph’s qualitative conclusion is unaffected, and the direction of the solver’s distortion cannot manufacture a false positive here: since the true argmax can only exceed what a distorted-objective optimizer finds, the achievable unilateral gain is, if anything, understated by this exercise, not overstated. A companion single-country placebo confirms every member with a negotiating lever (US, Japan, Australia, South Korea) can likewise achieve a positive own-gain in isolation, Australia’s margin thin ($+0.0085$) but positive; Canada has no negotiated lines and so no lever to test (`p43_pair_ir_probe.md` §B). The R&R fix-path ranking is unaffected by this probe’s result specifically: an architecture-preserving fix (a better solver, or damped Gauss–Seidel updates) remains the better-supported route to a convergent $F_{\text{pairs}}^{eq,IR}$ for reasons independent of this probe — a fix that merges shared members into a joint sub-system would itself change the coalition structure being tested (axis (d)), which is why it is de-prioritized, not because this probe demonstrated joint feasibility.

What actually drives the boundary result? (Revised under the corrected D -convention — the paragraph banked here through Round 21 is retracted, not revised: see the discharged scope note above.) Under the corrected economy the distributional picture is materially different, and offers no single dominant driver. Only two of six members — Japan ($g_i(F_{\text{pairs}}^{eq}) - g_i^{HW} = +0.0381$) and South Korea ($+0.0329$) — are **strictly better off** under F_{pairs}^{eq} than under BSY’s actual realized outcome. The remaining four are worse off, and two of them **flip sign** relative to the pre-fix reading: the United States ($g_{\text{US}} = 0.0182$ vs. $g_{\text{US}}^{HW} = 0.0314$, a -0.0132 shortfall, reversing a previously reported $+0.080$ gain) and Canada ($g_{\text{Canada}} = 0.0024$ vs. $g_{\text{Canada}}^{HW} = 0.0039$, a -0.0015 shortfall, reversing a previously reported $+0.001$ gain). The EU — previously reported as this result’s sole dominant driver, via a -19.2% -of- $v(N)$ individual-rationality violation — in fact *satisfies* its own 1990 floor at the corrected fixed point ($g_{\text{EU}} = +0.0155$, a shortfall of -0.0492 relative to $g_{\text{EU}}^{HW} = 0.0647$, not a floor violation): this **reverses** the paper’s previously banked “primary mechanism” finding outright, it does not merely revise its magnitude. The member now closest to its own individual-rationality floor is Australia, at $g_{\text{AUS}} = -0.0013$ (-0.175% of $v(N)$) — three orders

of magnitude smaller than the retracted EU figure, and plausibly within solver-tolerance noise rather than a genuine, decisive violation (`p40_dconvention_rerun_result.md`). With no single member’s own infeasibility dominating the aggregate shortfall at this scale, the paper no longer reports a $21\times/30.9\times$ -style violation-magnitude ratio: there is no floor violation of that size left to measure a multiple of. (A structurally analogous, but conceptually distinct, ratio can be computed on the corrected numbers — the EU’s own shortfall relative to its realized $\hat{\zeta}$ outcome, divided by the aggregate boundary margin, $0.0492/0.001077 = 45.7\times$ — but this measures the EU’s distance from its own bargained outcome, not an individual-rationality floor violation, and is reported here only to forestall a mechanical recomputation of the retracted ratio under a new label.) The pair-by-pair construction still shows individual members sacrificed at several pairs’ own isolated optima (a definitional consequence of a TU welfare-sum objective with no per-member floor, ECON-115); the specific pair-by-pair pushdown figures cited in the pre-fix draft (EU’s isolated US–EU/EU–Japan trajectory, the externality signs on outside countries) have **not** been recomputed against the corrected fixed point and are marked **pre- D -fix figures, recomputation pending**, not silently retained. Ranking the relative contribution of pair-level coordination externality against any one member’s own shortfall is therefore left open, not resolved in either direction, pending that recomputation.

Joining a pair is not obviously better or worse than not joining one, at these corrected numbers. Canada, which negotiates in *zero* of the seven pairs, gains $g_{\text{Canada}} = +0.0024$ at F_{pairs}^{eq} as a free-riding outsider, slightly *below* its full BSY outcome ($g_{\text{Canada}}^{HW} = +0.0039$) rather than capturing it in full as previously reported. Australia, which negotiates in *two* pairs, ends barely negative ($g_{\text{AUS}} = -0.0013$), close to, not decisively worse than, Canada’s zero-pair outcome. The qualitative point that a pair-decomposed, TU welfare-sum construction offers no individual-rationality guarantee to any member survives this correction — Australia’s outcome is still the worst of the six, and still non-positive — but the sharp “negotiating is not guaranteed to pay, and a small member is worse off in the room than out of it” framing is weakened: the gap between Canada’s outside outcome and Australia’s in-pair outcome shrank from roughly 0.005 to roughly 0.004, both now close to zero, and neither is any longer a decisive, order-of-magnitude departure from the other.

A regime-matched comparator, $F_{\text{pairs}}^{eq,IR}$, imposing $g_i \geq 0$ on every pair’s own sub-problem, was attempted to correct for this and did not converge: five of the six $\mathcal{N}_6$ members participate in two to four pairs simultaneously, so no single pair’s own-floor sub-problem is well posed in isolation while a pair sharing one of its members has itself not yet converged, and the naive sequential update has no mechanism to prevent the resulting cascade (`f_pairs_eq_ir_ATTEMPT_member_aggregate.md`). We tested directly whether the EU’s own individual-rationality floor is unsatisfiable within either of its two pairs, considered unilaterally: it is not — both admit a strictly positive, GE-re-evaluated witness ($+0.130$, $+0.043$, above). This rules out the simplest failure mode, EU unilateral infeasibility, but it does **not** test whether both members’ floors can be satisfied *simultaneously* within a pair — exactly what the $F_{\text{pairs}}^{eq,IR}$ sub-problem itself requires — so an existence failure at the joint, two-sided constraint is not ruled out, only left untested (`p43_pair_ir_probe.md`). This remains a diagnosed coordination/scheduling problem as the leading account, not a demonstrated non-existence result, deferred to a revision with either a si-

multaneous (not sequential) sub-system solve or a damped-update re-implementation. The one iterate this failed attempt did produce before diverging — round 1’s assembled value, 0.323202 — is not a bankable estimate (four of seven sub-solves failed to converge and the point is itself IR-infeasible, $\min_i g_i = -0.331857$), but it is the only directional information this paper has on the regime-matched leg, and it points *above* $\Sigma_k g_k^{HW} = 0.312441$, not below it: disclosed here, alongside the boundary verdict, rather than only in the registry, per this paper’s own adverse-diagnostic-disclosure discipline (as with P-37’s placebo rider above). Pending a converged $F_{\text{pairs}}^{eq,IR}$, the ranking of explanations for why F_{pairs}^{eq} falls short of $\Sigma_k g_k^{HW}$ is revised under the *D*-convention fix: the previously banked “a single member’s own infeasibility (the EU’s IR violation) dominates” reading is **retracted** (above); genuine pair-level coordination externality and the diffuse aggregate margin itself are left as open, unranked candidate explanations pending the pre-fix pair-by-pair diagnostics’ own recomputation; what *is* settled is that a coverage/mechanism mismatch is **ruled out**: BSY’s own cached $\hat{\zeta}$ equilibrium moves zero tariff lines outside the 151-line negotiated mask, 0 of 251 off-mask cells across all six members, Canada’s full 67-line row included (`check_hw_offmask_displacement.md`) — there is no “other rounds” or MFN-spillover mechanism inside this cached, one-shot equilibrium object. This withdraws the coverage explanation offered in an earlier draft of this passage and, as a free consequence, upgrades *H2*’s within-agenda framing (above) from presumed to verified: $\Sigma_k g_k^{HW}$ and $F_{\text{pairs}}^{IR,corr}/F_{\text{pairs}}^{eq}$ are confirmed measured on the identical 151-line agenda.

Was the pair-decomposed welfare-sum objective itself the problem? (Caveat, added with the *D*-convention fix.) The motivating premise below — an EU individual-rationality violation to be explained away by solution concept — no longer holds under the corrected economy (above); `run_p41_f_pairs_np.py` is itself on the confirmed-affected ten-script list and has not been re-run, so its own numbers below are **likely themselves confounded by the same bug, re-run deferred to R&R**, independent of the question it originally motivated. ECON-115’s own diagnosis is that a TU welfare-sum planner, without a disagreement point or an individual-rationality guarantee, is the wrong object to be reading an IR violation off of in the first place; a genuine Nash-product reconstruction of F_{pairs} — each pair maximizing $(g_n - g_n^0)^{\hat{\mu}}(g_m - g_m^0)^{1-\hat{\mu}}$ against its own 1990 disagreement point, using BSY’s own estimated pair weights $\hat{\mu}$, which endogenously rules out any negative individual gain by construction — was attempted (P-41) to test this directly. The attempt stalled after ≈ 4 hours at round 1 of its own Gauss–Seidel sweep (five of seven pairs reached, two never started), with zero log output for the full stall duration, and was killed by an unstructured manual operator call ($\approx 14,421$ s) rather than a triggered, pre-set automatic cap — the run’s own 2400s nominal time-budget check fires only at the top of the outer loop and never re-fired mid-round (`f_pairs_np_member_aggregate.md`); **no F_{pairs}^{NP} point estimate is banked**, and this remains an open R&R item, not a substitute result for F_{pairs}^{eq} or $F_{\text{pairs}}^{eq,IR}$ (`f_pairs_np_member_aggregate.md`). What the stalled round did recover is nonetheless suggestive: four of the five pairs reached failed the domain-protection pre-check outright at the starting configuration — one member’s endogenous gain-over-disagreement was already negative before optimization began — landing at a *no-deal* (revert to 1990) outcome for that pair rather than at any negative-gain point; the fifth pair’s optimizer nominally succeeded but its

solution still fell outside the positive-gains domain post-solve. Round 1’s no-deal outcomes are a pure artifact of the domain-protection pre-check evaluated at the starting configuration — the optimizer was never called for four of the five pairs reached — and do not constitute evidence for or against the existence of a mutually beneficial agreement at any of those pairs; this paper’s own P-43 direct feasibility probe, run on the identical F_{pairs}^{eq} fixed point, points the opposite direction: every EU-containing pair, and every member with a negotiating lever, admits a strictly positive unilateral IR-feasible gain in isolation (above). The mechanism this paragraph might otherwise be read as illustrating — that sequential bilateral negotiation can fail to internalize gains a joint, multilateral solve would capture — has a better-anchored precedent in BSY’s own replication package: their Table VIII reports that an isolated US–Japan negotiation, evaluated on their own estimated model, fails to reach any deal at all, the literature’s own anchor for a genuine no-deal outcome, and the same table reports total cross-pair interaction effects of approximately 20% of the Uruguay Round’s realized world-welfare gain (Bagwell et al. 2021, Table VIII, p. 1620, col. 3), comparable in order of magnitude to this paper’s own $\Delta_d^{TU} = 17.42\%$ figure below (re-optimized under the corrected D -convention, superseding the pre-fix 18.19%; both denominated relative to a realized-or-assembled multilateral total, not identical objects, so the two should be read as similarly scaled rather than as cross-validating one another); this is a materially different object from the +0.081% **true** cross-pair interaction measured on the pair-decomposed welfare-sum construction below (superseding the pre-fix +0.20%), and the two should not be conflated — BSY’s $\approx 20\%$ measures interaction across sequential, multi-round bilateral negotiation dynamics over the full Uruguay Round, while this paper’s +0.081% measures the non-additivity of six simultaneous, one-shot two-country welfare-sum optima assembled into a single joint point; the former is a multilateral-versus-sequential-bargaining gap, the latter is an assembly-nonlinearity gap, and the near-two-orders-of- magnitude difference in level between them is itself informative about which mechanism the pair-decomposed construction actually isolates. The suspected proximate cause of the stall is **not** numerical ill-conditioning at extreme pair weights: by $\min(\hat{\mu}, 1 - \hat{\mu})$, the two pairs where the sweep actually stalled (Japan–South Korea $\hat{\mu} = 0.845$, Australia–South Korea $\hat{\mu} = 0.830$; $\min(\hat{\mu}, 1 - \hat{\mu}) = 0.155/0.171$) are the *least* extreme of the seven pair weights, while the one pair whose optimizer ran to termination, US–South Korea, has the *most* extreme weight ($\hat{\mu} = 0.005$) and terminated normally — the reverse of what an extreme-weight-conditioning account predicts. Objective rescaling is also not a viable fix on its own terms and is **withdrawn** as a proposed remedy: rescaling an objective by a constant leaves its condition number unchanged ($\kappa(cH) = \kappa(H)$). A third hypothesis, the domain-protection implementation itself, was proposed — the constraint is enforced as $\max(g - g^0 + 10^{-10}, 10^{-16})$ inside the log, creating a numerically flat, zero-gradient plateau at the domain boundary, not the steeply curved log-barrier the Nash-product objective in fact has there (the true curvature, $(1 - \hat{\mu})/g^2$, grows steep, not flat, as $g \rightarrow 0^+$) — with a proposed test: re-running the two stalled pairs alone with `maxiter=200` (ten times the family default) would distinguish a genuinely slow optimization from a stuck plateau at zero additional cost. **This test has since run and rules the plateau account out as stated:** both pairs still fail, at `status=8` (“positive directional derivative for linesearch,” a genuine line-search failure, not an iteration-limit exhaustion) well short of the raised budget (Japan–South Korea 117/200 iterations, Australia–South Ko-

rea 14/200), with the market-clearing equality constraint left unsatisfied by 2.83×10^{-2} and 2.87×10^{-2} respectively at the failed terminal point (`p41_maxiter_probe.md`). This residual regime is distinct from, and should not be conflated with, MATH-124’s *D*-convention finding below: there, terminations reported `success=True` with a machine-precision (10^{-16} – 10^{-14}) constraint residual whose *reported welfare number* was nonetheless computed against the wrong economy; here, the optimizer genuinely fails to converge at all, and the $\sim 2.8 \times 10^{-2}$ residual is the ordinary signature of that failure, not a convention mismatch. Raising `maxiter` is therefore conclusively ruled out as a fix, and the plateau account, while not directly confirmed by this test, survives it: the observed line-search failure is consistent with a flat, uninformative gradient at the domain boundary rather than with a genuinely slow but eventually convergent search. **This diagnosis remains possibly contention-confounded:** this batch ran P-41, P-43, and an isolated GATE-1 retry concurrently on the same twelve-core machine, so all three of this round’s negative or null results — P-41’s stall, the disclosed solver-fidelity gap above, and GATE-1’s own FAIL — share an undischarged CPU-contention confound, and none should be read as dispositive evidence about the object it nominally tests until an isolated re-run is available. A full re-attempt — with iteration-level heartbeat logging, the two stalled pairs isolated and diagnosed in a standalone sub-problem first, a different warm start for these two pairs specifically, and the domain-protection implementation replaced with a smooth interior-point barrier rather than rescaled — is deferred to R&R.

Interaction diagnostics and the coalition-structure cost (numbers re-optimized under the corrected *D*-convention; the pre-fix values are superseded, not merely re-verified). The pair-own objectives, summed naively across all seven pairs (each counting only its own two members’ welfare), give 0.228362 (superseding the pre-fix 0.210338), against the assembled joint value $F_{\text{pairs}}^{BR1} = 0.310033$ (superseding 0.302243) — a -11.34% -of- $v(N)$ gap (superseding -12.76%) that an initial pass, pre-fix, read as “cross-pair interaction” exceeding the pre-registered materiality gate ($0.25 \cdot H2 = 0.028908$) *roughly 2.8-fold* ($0.081671/0.028908 = 2.83$, superseding the pre-fix “roughly threefold,” $0.091906/0.028908 = 3.18$). This reading is corrected: 99.3% of the gap (superseding 98.4%) is not interaction but a **coverage mismatch** between the pair-own objective (each pair’s own 2 members) and the six-country welfare its isolated optimum actually generates once re-evaluated at full general equilibrium (0.309448, superseding 0.300775, using this run’s own `g_by_country_full16_isolated` field, zero new compute). Decomposed: a **pair externality** of $+0.081086$ ($+11.26\%$ of $v(N)$, superseding $+0.090437/+12.56\%$) — welfare each pair’s own optimum creates for the four countries outside that pair, which the pair’s own objective never counts — and a **true cross-pair interaction** of only $+0.000585$ ($+0.081\%$ of $v(N)$, superseding $+0.001468/+0.20\%$, Σ (six-country isolated optima) vs. the single joint assembled solve), which does *not* clear the materiality gate (reversing the original “gate fires” reading, and by a wider margin than pre-fix). The pair externality is 26.2% of the six-country surplus each pair creates when it optimizes only its own two members’ welfare ($0.081086/0.309448$, superseding $30.1\%/0.090437/0.300775$): this is, with zero new compute, a direct measurement, on this paper’s own fully-matched object, of the mechanism this section has argued for throughout — bilateral bargaining pairs cannot price the welfare their

own concessions confer on, or extract from, parties outside the pair. This is the scale of the *uninternalized spillover*, not the scale of a *loss*: the welfare the 26.2% figure describes (superseding the pre-fix 30.1%) is actually realized by the four outside countries, not destroyed; whatever efficiency cost this mechanism carries shows up instead as a distortion in each pair’s own incentive to cut deeply, not as forgone world welfare per se. This corroborates, rather than duplicates, BSY’s own Table VII decomposition of the Uruguay Round’s actual realized world-welfare gain, in which “free-rider gains” — non-participants’ uncompensated benefit from MFN — account for 35.38% of the total (Bagwell et al. 2021); the two figures are not the same object (ours is a share of each pair’s own *potential* surplus at an idealized bilateral optimum; theirs is a share of the *realized* gain from the actual multilateral round), so the two are a **structurally analogous ratio, not a comparable estimate** of the same underlying parameter, and should not be treated as a replication of one another; both nonetheless put the uninternalized-externality share of bilateral tariff bargaining in the same broad range — mid-twenties to mid-thirties percent (this paper’s corrected 26.2%, BSY’s own 35.38%) — from two independent constructions. Institutionally, **MFN is the transmission mechanism that creates this externality, not the device that internalizes it**: because every bilaterally negotiated concession is automatically extended to all GATT/WTO members regardless of their own concessions, a pair’s own tariff cut mechanically generates the very free-rider gains quantified above and in BSY’s own Table VII (Bagwell and Staiger 1999). The reciprocity norm operates on a narrower margin: it neutralizes the terms-of-trade externality *within* a negotiating pair — disciplining each side to match concessions so that neither can unilaterally shift the bilateral terms of trade in its own favor — but has no mechanism to price the externality a pair’s concession confers on countries *outside* that pair. The principal-supplier rule this paper’s 151-line mask restricts to is GATT’s own second-best device for narrowing that residual gap without a fully multilateral negotiation, by routing each product’s negotiation through the pair with the largest stake in it, which pushes a pair’s private bargaining incentive closer to, though not equal to, the multilateral incentive a planner internalizing all six countries’ welfare would have; the 26–35% residual (superseding the pre-fix 30–35%) documented here and in BSY’s own Table VII is a direct measure of how much of that gap the principal-supplier device still leaves unpriced. Separately, one further quantity holds coverage, granularity, and the transfer-unconstrained regime fixed while varying only coalition structure: $\Delta_d^{TU} = F_{\text{line}}^{v2} - F_{\text{pairs}}^{eq} = 0.436840 - 0.311364 = 0.125476 = \mathbf{17.42\%}$ of $v(N)$ (superseding the pre-fix $0.130973 = 18.19\%$). This figure is **no longer numerically coincident** with the $H2(0.5) = 18.19\%$ figure earlier in this section — the pre-fix “coincident by rounding, not by construction” remark above is now moot, since the corrected Δ_d^{TU} and $H2(0.5)$ no longer round to the same value; $H2(0.5)$ compares $F^{\text{IR,corr}}$ against the $\zeta = 0.5$ symmetric-weight placebo holding coalition structure at the grand coalition, while Δ_d^{TU} compares F_{line}^{v2} against F_{pairs}^{eq} holding ζ at its point estimate and varying only coalition structure — the two were never the same object, and the correction removes what had been a distracting numerical coincidence. We report Δ_d^{TU} as a **descriptive**, not hypothesis-test, finding: replacing a joint 151-line planner with BSY’s own seven-pair bilateral architecture, holding coverage and instrument fixed, moves the pair-own objective’s own dollar total by 17.42% of $v(N)$ on a transfer-unconstrained basis. This number also embeds a **solution-concept substitutive**

tion: F_{pairs}^{eq} 's pair-decomposed welfare-sum objective replaces BSY's own estimated bargaining weights $\hat{\mu}$ with each pair's income-based welfare weight $\tilde{P}_i$, so part of 17.42% is the price of that substitution, not of bilateralism per se (ECON-115); a genuine Nash-product reconstruction that would isolate the two effects was attempted and deferred to R&R (P-41, above), so this component cannot yet be sized. Δ_d^{TU} differences two transfer-unconstrained objects (the joint planner F_{line}^{v2} against the pair-decomposed equilibrium F_{pairs}^{eq}), so it does *not* carry $H2$'s asymmetric TU-vs-actual-protocol bias, and its own regime-correction direction is **not established**: on the numerator side, the IR-consistent joint planner is known to sit below its TU counterpart ($F_{\text{line}}^{IR} = F_{\text{line}}^{v2} - 0.008321$, above); on the pair-decomposed side, the one IR-floored iterate this paper has computed (0.323202, non-convergent, disclosed above — itself a pre-fix figure, recomputation pending) sits *above* F_{pairs}^{eq} , not below it. An earlier draft's "lower bound" characterization of 18.19% is therefore **retracted**: we report 17.42% (superseding the pre-fix 18.19%) as a TU-regime point estimate, with the sign of its eventual IR-regime correction left open pending a convergent $F_{\text{pairs}}^{eq,IR}$ or the Nash-product reconstruction (P-41, above). See Appendix E for the full P-40 disposition.

Does the shortfall survive a stress test against BSY's actual bargain, not just against 1990? (P-36). F_{line}^{IR} 's individual-rationality floor is the 1990 status quo, not what the six members actually achieved at $\hat{\zeta}$; at F_{line}^{IR} 's optimum, four of six members (US, EU, Australia, South Korea) are worse off than at BSY's own actual outcome (§4.1.2 above, ECON-79), so a no-transfer institution starting from the real 1990 $\rightarrow \hat{\zeta}$ agreement could not implement it, and citing $H2$ as surplus "genuinely on the table" presupposes exactly the question this raises. We test the stronger, correctly-targeted claim directly: does a no-transfer, line-by-line improvement exist that leaves every member at least as well off as they *actually* ended up, not merely as well off as 1990? Re-solving the identical weight-corrected MPEC with $g_i(t, w) \geq g_i^{HW}$ replacing the 1990-relative IR constraint for all six members gives $F^{PD} = 0.410870$ (fully converged at all three planned starting points, cross-start F difference 9.8×10^{-14} , argmax also unique across starts (per-line difference $\leq 1.8 \times 10^{-7}$), IR-binding to machine precision, `p36_pareto_dominant_frontier.md`), which nests correctly below the corrected frontier $F^{IR,corr} = 0.428073$ and $F_{\text{line}}^{IR} = 0.428519$ in turn ($F^{PD} \leq F^{IR,corr} \leq F_{\text{line}}^{IR}$, confirmed to 6+ significant figures — a useful independent cross-check of both re-runs), as required since $g_i \geq g_i^{HW}$ is a strictly tighter constraint than $g_i \geq 0$ for **six of six members** (BSY's own bargained outcome is strictly individually rational at every member; minimum $g_i^{HW} = 0.003873$, Canada). $F^{PD} - \sum_k g_k^{HW} = 0.098429 = \mathbf{13.67\%}$ of $v(\mathcal{N}_6)$ — 85.1% of the 1990-IR-constrained point estimate $H2 = 16.06\%$ — clears the pre-registered 5% materiality band by a wide margin.

This is a weak, not a strict, Pareto improvement: at F^{PD} 's optimum, the US, EU, Australia, and South Korea are each pinned exactly at their own g_i^{HW} floor (binding to machine precision), i.e. indifferent between F^{PD} and BSY's actual outcome; the entire net gain accrues to Japan (0.263027 vs. 0.173316 at $\hat{\zeta}$) and the zero-negotiated-lines Canada (0.012592 vs. 0.003873). Under any strictly positive negotiation or ratification cost, the four indifferent members would have no material incentive to move from the status quo to F^{PD} ; this sharpens, rather than undercuts, the architectural reading discussed above, since it is precisely the bilateral

structure’s inability to price Canada’s free-riding and Japan’s concentrated gains that leaves four of six members exactly indifferent.

Reconciling this with the fairness-gap finding. Japan receiving the entire net gain here sits in apparent tension with $\Gamma_{\text{Japan}} = -12.54\text{pp}$, the most negative fairness gap in the panel (Table 2 above): the Shapley benchmark says Japan already holds *more* of the pie than its marginal contribution warrants, while F^{PD} ’s own optimum directs the entire no-transfer net gain over BSY’s actual outcome *to* Japan. These are not contradictory findings; they answer different questions under different constraints. Γ is a transfer-feasible, fairness-and-efficiency object: it asks how a fixed, efficient pie ought to be split given each member’s marginal contribution, with transfers available to reach any point on the frontier. F^{PD} is a no-transfer, efficiency-only object: it asks which feasible point Pareto-dominates the status quo without transfers, and because four of six members are pinned at their own indifference floor by construction, any further no-transfer surplus can only flow to whichever member has the highest remaining marginal product and has not yet hit its own floor — Japan, on this coalition and line set. That the two objects send opposite distributive signals for the same country is not a defect of either; it is the direct empirical content of this paper’s own 13.51%-of-surplus gross-transfer finding (§4.1 above): a no-transfer institution and a transfer-feasible fairness benchmark generically disagree about who should receive the next dollar of surplus, and Japan is exactly where that disagreement is starkest in this panel.

A genuine, no-transfer-feasible, weakly Pareto-dominant improvement over BSY’s actual 1990 $\rightarrow \hat{\zeta}$ agreement exists and is material: the “genuine bargaining shortfall” reading of $H2$ survives this stress test at roughly 85% of its magnitude, with the remaining 15% attributable specifically to the gap between the 1990 and actual-outcome IR floors, not to the shortfall evaporating once BSY’s actual outcome, rather than 1990, is used as the comparison point. The multi-start confirmation has since landed: all three starting points converge to $F^{\text{PD}} = 0.410870$ (cross-start difference 9.8×10^{-14}), consistent with every other completed variant in this MPEC family, which has converged to 12+ significant figures across every tested starting point with zero exceptions.

The masked point’s own decomposition: a third instance of the paper’s participation-constraint story. Although the masked TW-max point does not resolve P-30, its country-by-country decomposition is independently informative and worth reporting on its own terms. World-welfare maximization — what TW-max computes, not a *member-group* optimum for the six negotiators — imposes a net loss of -3.220370 on the six-country group, against their own true constrained best, $F_{\text{line}}^{v2} = +0.436840$ (P-31, weight-corrected, above; P-29’s $v^{\text{negotiated}} = 0.303512$ is a looser lower bound on the same quantity): a gap of 3.657210 , roughly $5.1 \times v(\mathcal{N}_6) = 0.720163$. This net loss is *not* a broad transfer of surplus away from negotiators generally: five of the six negotiators are *better off* at the masked world optimum than at their 1990 baseline — US $+1.77$, Japan $+2.32$, South Korea $+0.49$, Canada $+0.11$ (unaffected, since it has zero negotiated lines) — and only Australia is marginally negative (-0.03), while the EU alone loses -7.87 at the world optimum, about $10.9 \times v(\mathcal{N}_6)$, and drives essentially the entire six-country net loss on its own (p30_twmax_member_aggregate.md §2). Every one of the five

non-strategic (NES) rest-of-world regions gains (+1.6 to +2.3%, above). **World efficiency here is blocked by a single member’s participation constraint — the EU — not by a mechanism that systematically favors outsiders over negotiators** (ECON-76): the same single-veto-player logic that blocks the member-group optimum at Australia’s participation constraint elsewhere in this section, here operating at world scale and with a different binding country. This is a third instance, and the largest in dollar terms, of the participation-constraint mechanism documented elsewhere in this section — Australia’s participation constraint at the member-group optimum, and the core’s dependence on feasible transfers (§4.1): a world-efficient policy is not *member-group* efficient for the six GATT principal-supplier negotiators, and it takes only one binding participation constraint, not a broad transfer away from the negotiating group, to produce a large net loss for that group. We flag this as a magnitude only, not as a claim that this paper’s participation-constraint or core machinery has been extended to the eleven-region world game; a γ -response core check on the full world panel is future work (§5.3).

A country-level pattern the mismatch does not erase (P-17). One pattern survives the agenda mismatch above, and it is the paper’s only positive evidence that the efficiency benchmark carries descriptive content beyond the mechanical headroom relabeling documented in Section 4.2: on the 151 lines both objects can be compared on directly, the *country-level* order of liberalization depth matches. D1 cuts the five negotiating countries with negotiated lines in the same rank order the Uruguay Round actually did — deepest for Australia, then Korea/Japan, shallowest for the US and EU — with by-country mean-cut correlation +0.916 ($n = 5$; Canada holds no negotiated lines and is excluded, consistent with its absence from every bilateral pair, `p17_tariff_fit.md`). This is an $n = 5$ descriptive pattern, not a test statistic, and this paper’s own pre-registration discipline requires it be read together with its placebo, which is adverse: at the individual-line level, an atheoretical uniform-proportional-cut baseline outperforms *both* D1 (+0.229) and BSY’s own fitted model (+0.375) on correlation with the actual 2000 line-level tariff *level*, not the 1990–2000 tariff *change* (uniform-*s* placebo +0.593), so the line-level fit carries essentially no information beyond tariff persistence for either model (`p17_tariff_fit.md`). On the *cuts* metric itself (correlation with the actual 1990–2000 tariff change), only D1 (+0.710) and the placebo (+0.720) have been computed here; BSY’s own fitted model has not been evaluated on cuts. Reported together, as both the placebo design and prediction P-17 require: the efficiency benchmark’s country-level concentration pattern tracks the Uruguay Round’s actual concentration; its line-level schedule does not fit better than a naive persistence rule, and neither does BSY’s own estimated model.

The efficient benchmark against external data, with a placebo-adverse rider (P-37, corrected 2026-08-19). Every other comparison in this section is model-internal — D1 versus D1, or D1 versus BSY’s own cached $\hat{\zeta}$ estimate. We report one comparison against data neither model targets: the actual 1990 $\rightarrow$ 2000 MFN applied-tariff trajectory, on the identical 151-line negotiated mask. For each line, the model cut is $\tau^{90} - s^*$ at F_{line}^{IR} ’s argmax (primary) or F_{line}^{v2} ’s argmax (robustness); the actual cut is $\tau^{90} - \tau^{2000}$. The pooled Spearman rank correlation across all 151 lines and six countries is $\rho = +0.6405$ (primary) / +0.6665 (robustness), both above the pre-registered $\rho \geq 0.3$ falsification threshold, so the literal decision rule fires: **D1’s**

efficient-frontier cuts are directionally, not unrelatedly, associated with where 1990s trade liberalization actually went. Every country with negotiated lines shows a positive within-country correlation (+0.49 to +0.78), so this is not driven by a single country. We report this as the paper’s first *external-data* comparison — data neither D1 nor BSY’s own estimator targets, as opposed to every other comparison in this paper, which is model-versus-model or model-versus-BSY’s-own-estimate.

This is not, however, an unqualified validation, and reporting it as one would repeat an error this paper’s own P-17 already corrected. A model-free uniform- s placebo — any constant-scaling cut is a positive scalar multiple of τ^{90} , so its Spearman correlation with the actual cut equals $\text{Spearman}(\tau^{90}, \text{actual cut})$ exactly, for every valid scaling, no GE solve required — scores $\rho = +0.7136$ on the identical 151 lines, *beating both D1 legs* ($\Delta\rho_{IR} = -0.073$, $\Delta\rho_{v2} = -0.047$; p37_placebo_addendum.md). Per-country, the pattern is not uniform: the US is the sole exception where D1 beats the placebo ($\Delta\rho_{IR} = +0.046$), while every other country with negotiated lines underperforms it, most severely South Korea ($\Delta\rho_{v2} = -0.126$) — descriptively, one positive and four negative among the five country-level comparisons, range +0.046 to -0.126 ; the US reading is the maximum of five, and selection among them is not priced. The correct terminal reading, following the same dual-face discipline P-17 already applies to its own line-level finding: **PASS on the literal $\rho \geq 0.3$ band, with a placebo-adverse rider that must always travel with it** — D1’s cuts and actual liberalization move in the same direction, well above zero, but this directional agreement carries no informational content beyond simple tariff persistence (higher-1990-tariff lines mechanically got cut more), and on this metric D1 does slightly *worse* than persistence alone for four of the five countries with negotiated lines (Canada has none). Accordingly, “the paper’s first genuine model-versus-data validation” is downgraded throughout this paper to “the paper’s first external-data comparison,” and all significance/ p -value language is removed: the original permutation test’s free reshuffling assumes line exchangeability, which this artifact’s own caveats already document as violated (the 151 lines share country-level dependency), so a p -value computed under a violated assumption is not reported.

Does D1 track the data better than BSY’s own fitted model does? (P-42). Every comparison above is D1 against D1, D1 against data, or D1 against BSY’s own *cached point estimate* of the bargained outcome; none scores BSY’s own *fitted bargaining model* against the same 1990–2000 actual-cut target. We add that comparator directly, with zero new GE solves: for each of the 151 negotiated lines, BSY’s model-implied cut is $\tau^{90} - t^{HW}$, read off the identical cached $\hat{\zeta}$ schedule used throughout this paper.

Five readings, each of which must travel with the others; reporting only the first would repeat the selective-reporting error this paper’s own P-17/P-37 discipline exists to prevent. (i) **D1 vs. the placebo:** unchanged from above — both D1 legs trail the model-free uniform-cut placebo on Spearman, so neither model is shown to add directional content beyond simple persistence; a paired-inference check (below) confirms this is not merely a point-estimate reading. (ii) **D1 vs. BSY’s own fitted model, on BSY’s cached $\hat{\zeta}$ schedule, no longer merely tie-handling-conditional, though not fully resolved either:** the mid-rank recomputation

Table 5: Correlation with the actual 1990–2000 MFN cut, 151 negotiated lines, six countries. D1 legs and the placebo are the same objects as the text above; the BSY row is new (P-42). Ordinal-rank Spearman uses an index-order tie-break and is not a well-defined rank statistic under BSY’s 57% tie block; mid-rank Spearman (`scipy`’s average-rank convention) is the methodologically correct, primary statistic and is reported alongside it.

Row	Spearman ρ (ordinal)	Spearman ρ (mid-rank)	Pearson r
Uniform- s placebo	+0.7136	+0.7137	+0.7200
D1-line-IR (primary)	+0.6405	+0.6407	+0.6831
D1-line-v2 (robustness)	+0.6665	+0.6668	+0.6933
BSY’s own fitted model	+0.3318	+0.3168	+0.6785

this paragraph previously flagged as pending has landed (`p42_tie_census_addendum.json`) and *strengthens*, rather than narrows, the gap: under `scipy`’s standard average-rank tie convention, ρ_{BSY} moves *against* BSY (+0.3318 $\rightarrow$ +0.3168), while both D1 legs move marginally in D1’s favor (+0.6405 $\rightarrow$ +0.6407, +0.6665 $\rightarrow$ +0.6668); the original ordinal tie-break was, if anything, slightly flattering to BSY. Two further checks rule out the two most natural mechanical explanations for the gap. **The tie count itself is not the explanation:** pinning a *random* 86-of-151 line subset of D1’s own argmax to BSY’s exact pin value, repeated 5,000 times, produces a mean $\rho = +0.1315$ (std 0.084) — BSY’s actual $\rho_{BSY} = +0.3168$ sits at the 98.7th percentile of this distribution, far above what pure tie-counting predicts; BSY’s cached schedule carries real information about the data’s ranking well beyond its tie rate. **Coverage, however, is where the pooled advantage does not survive:** restricted to the 65 lines where BSY’s schedule is *not* pinned — the only comparison that does not structurally penalize BSY for lines it was mechanically incapable of losing ties on, and the caliber this paper’s own registry treats as the fair, terminal comparison — D1’s point estimate still leads ($\rho_{\text{mid}} = +0.633$ IR / +0.625 v2 against BSY’s +0.573; on this restricted sample BSY’s own schedule also falls below the placebo, +0.600), but a formal paired-bootstrap test (P-45, below) does not resolve any of the three pairwise differences at this sample size, so the pooled-sample gap does not, by itself, establish a robust D1-over-BSY finding. What the gap *does* substantially reflect is a genuine, reportable mechanism: $\rho(\text{D1 cut, placebo}) = +0.8815$ against $\rho(\text{BSY cut, placebo}) = +0.5016$ — D1’s argmax resembles naive τ^{90} -persistence far more closely than BSY’s country-heterogeneous-weight schedule does, and since the placebo already correlates strongly with the actual data ($\rho = +0.714$), a substantial share of D1’s advantage over BSY is attributable to D1 tracking persistence more closely, not (only) to D1 resolving the data’s true determinants better. Per-country, on the mid-rank statistic, D1 exceeds BSY at every country with negotiated lines except South Korea (near tie). **(iii) The extensive-margin contrast, tie-robust by construction:** D1-line-IR cuts 118 of 151 lines (78.1%; 111/151, 73.5%, for D1-line-v2) carrying a positive 1990 tariff, against BSY’s own cached schedule, which leaves 86 of 151 (57%) essentially uncut at its own solver’s tolerance. D1 cuts a substantially larger share of the negotiated agenda than BSY’s cached outcome does, not the entire agenda. **(iv) Partial-rank increment, guarding against over-reading (ii):** rank-residualizing both the model cut and the actual cut on τ^{1990} and correlating the residuals gives $\rho_{\text{partial}} = +0.035$ for D1-line-IR and -0.046 for BSY’s model — both inside the $|\rho_{\text{partial}}| < 0.2$ no-incremental-content

band. This is not evidence that D1 contains directional information the data does not already contain through own-tariff level; it is evidence that **BSY’s own fitted model contains less rank information than D1 does**, not that D1 contains rank information beyond τ^{1990} . Read together with the levels face reported earlier in this section, where BSY’s own model correlates more closely with the actual 2000 tariff *level* than D1 does (+0.375 against +0.229, both trailing the placebo’s +0.593), the full picture across five faces — levels, cuts-Pearson, cuts-Spearman (no longer tie-confounded, but coverage-confounded on the pooled sample), the extensive-margin cut/no-cut contrast, and the placebo-proximity mechanism — is: BSY’s model fits levels better, D1 and BSY are close on cut magnitude, D1 beats BSY on cuts-rank on the pooled sample (a margin that survives the tie-count confound but not a formal coverage-matched test), D1 cuts a much larger share of negotiated lines than BSY’s cached schedule leaves uncut, and part of that rank advantage is D1’s own closeness to naive persistence rather than superior tracking of the data’s true determinants; none of the five faces should be cited alone. **(v) Statistical inference on the head-to-head comparisons (P-45):** point estimates alone understate how thin some of these margins are. A line-paired bootstrap (10,000 draws) on the pooled 151-line sample gives 95% CIs that exclude zero for all three pairwise comparisons: D1 beats BSY ($\Delta\rho = +0.324$, CI [+0.204, +0.459]), but the placebo beats *both* models — placebo beats D1 ($\Delta\rho = -0.073$, CI [-0.148, -0.005]) and placebo beats BSY by a wide margin ($\Delta\rho = -0.397$, CI [-0.563, -0.247]). On the same-coverage 65-line sample, by contrast, none of the three CIs excludes zero (D1 vs. BSY: [-0.014, +0.147]; D1 vs. placebo: [-0.071, +0.152]; BSY vs. placebo: [-0.143, +0.096]) — nothing is resolved once the coverage confound is removed. The Williams–Steiger paired-correlation test agrees in sign and CI-exclusion status with the bootstrap at every entry (p45_paired_inference.json). **The three-layer conclusion this paper draws from P-42/P-45, stated precisely because each layer answers a different question, and because the registry’s own controlling ruling (identification critic, P-42) sets the terminal disposition at INTERMEDIATE, not at the pooled sample’s literal trigger:** (a) *D1 vs. BSY* — on the pooled 151-line sample D1 beats BSY’s own fitted model, resolved (CI excludes zero), but that comparison is coverage-confounded, since BSY structurally cannot lose rank ties on the 86 lines it never touched; on the same-coverage 65-line sample — the caliber the registry’s controlling ruling treats as the fair, terminal comparison — the difference is *not* resolved (P-45 CI includes zero), so the row’s official disposition is **INTERMEDIATE**, not a settled D1 victory; (b) *D1 vs. the zero-parameter placebo* — D1 does *not* beat a naive constant-cut rule; on the pooled sample the placebo is resolved to beat D1, and BSY loses to the placebo by an even wider, resolved margin; (c) *mechanism* — both models’ apparent skill is substantially mediated by closeness to τ^{90} -persistence ($\rho(\text{model}, \text{placebo}) \times \rho(\text{placebo}, \text{actual})$) reproduces the measured model-vs-actual correlation to within 2–8% for both D1 and BSY), so neither model is shown to add rank information the naive persistence rule does not already capture. **This paper does not claim, and this evidence does not support, that D1 beats a zero-parameter or persistence-only benchmark;** the defensible claim is narrower and, on BSY, officially INTERMEDIATE rather than a settled D1 victory — D1 outperforms BSY’s own estimated bargaining model on the pooled, coverage-confounded sample, but the two are statistically indistinguishable on the coverage-matched sample that this paper’s own

registry treats as the terminal caliber, while BSY’s own advantage, and D1’s own advantage, over naive persistence are each limited or absent.

Why does BSY’s cached schedule score lower: tie count or tie placement? (MATH-125.) The random-pin placebo above isolates the pure *tie-count* mechanism (losing rank information on 86 lines) from BSY’s specific choice of *which* 86 lines to leave essentially uncut. Pinning D1’s own argmax at BSY’s *specific* 86 line indices, rather than a random subset, gives $\rho = +0.4948$ — the 100th percentile of the random-pin distribution (mean $+0.1315$). BSY’s choice of which lines to leave uncut is itself strongly informative about the data’s true ranking, not an arbitrary or interchangeable subset. Decomposing ρ_{BSY} relative to D1’s own $\rho = +0.6407$: the *tie-count* effect (random-pin mean vs. D1’s own ρ) is -0.509 ; the *tie-placement* effect (BSY-specific pin vs. random-pin mean) is $+0.363$; the residual — BSY’s own continuous cut choices on the 65 unpinned lines, relative to the BSY-specific-pin figure — is -0.178 , pulling BSY’s final $\rho = +0.3168$ back down from the tie-placement-only figure. Read together: BSY’s tie *placement* carries genuine information about the data (a favorable finding for BSY), but BSY’s *continuous* cut choices on the lines it did negotiate track the data less well than D1’s argmax does (an unfavorable one) — two distinct, additive-in-spirit components with opposite signs, neither of which alone characterizes BSY’s overall performance on this metric (p45_paired_inference.json).

4.1.3 How rule-dependent is the fairness gap?

Table 6: Fairness gap Γ_i (pp) under four efficient, individually-rational allocation rules. “Sign-robust” countries have the same sign under all four rules. DESCRIPTIVE (P-27): computed in-sample from the same $v(S)$ table as Table 2; never cite as a predictive success. Source: p27_descriptive_gamma_sensitivity.json.

Rule	USA	EU	Japan	Canada	Australia	S. Korea
Shapley	+3.29	+4.30	−12.54	+2.12	+0.69	+2.14
Equal split (1/6)	+6.63	−4.05	−38.81	+15.43	+14.45	+6.34
1990 revenue share	+20.97	+15.95	−39.46	+2.36	+2.41	−2.25
1990 income share	+23.21	+15.91	−32.12	+1.56	−0.79	−7.77
Sign-robust?	yes	no	yes	yes	no	no

$\Gamma_{US} > 0$, $\Gamma_{Japan} < 0$, and $\Gamma_{Canada} > 0$ are sign-robust across all four efficient, individually-rational rules tested; Γ_{EU} , $\Gamma_{Australia}$, and $\Gamma_{South Korea}$ change sign across rules. Magnitudes are strongly rule-dependent: Γ_{US} ranges from $+3.3\text{pp}$ (Shapley) to $+23.2\text{pp}$ (1990 income share), a factor of seven. The correct reading is that the *sign* of the US, Japan, and Canada gaps is a robust, rule-independent finding; the *magnitude* of any of them is specific to the allocation rule and should not be quoted without naming the rule that produced it. This is the evidentiary basis for the C3 restatement adopted below, and it is deliberately an empirical, not an axiomatic, robustness check.

How much of Γ is a rescaling of the efficient policy’s own incidence? $\phi_i(v)/v(\mathcal{N}_6)$ (Table 2) and $\Delta_i = g_i(s^*)/v(\mathcal{N}_6)$ (Table 4) are strongly correlated: Pearson = 0.985 ($R^2 = 0.970$), OLS fit $\phi_i/v(\mathcal{N}_6) = 4.13 + 0.752 \cdot \Delta_i$ (percentage points), largest residual 3.47pp (South Korea). **This is descriptive (P-27), computed in-sample, $n = 6$ with a two-parameter fit — four degrees of freedom — and no significance language is permitted.** It is reported because it is informative in both directions. Favorably for the framework: 97% of the cross-country variance in the Shapley benchmark is already present in the efficient policy’s own physical incidence, and Shapley’s independent contribution is a single, describable operation — it compresses the dispersion of $g_i(s^*)$ by roughly a quarter (OLS slope $0.752 < 1$), in the direction the symmetry and dummy axioms are designed to produce: protecting players with positive marginal contribution but small or negative physical incidence (Australia, South Korea). Adversely for any claim that Γ_{US} is independent evidence of a genuine gap: a referee can reproduce 97% of the headline fairness-gap pattern with a one-parameter shrinkage rule applied to $g_i(s^*)$ alone, and the residual where Shapley’s axioms add information beyond that linear rescaling (3.47pp, at South Korea) is the same order of magnitude as the headline $\Gamma_{US} = +3.29\text{pp}$ — i.e. the US finding sits inside the part of Γ that a simple shrinkage rule already explains, not in the part that is distinctively Shapley’s. Source: `p27_descriptive_gamma_sensitivity.json`.

4.1.4 A symmetric-weight placebo

Table 7: Fairness gap at BSY’s point estimate $\hat{\zeta}$ versus the symmetric placebo $\zeta = 0.5$. Four of six countries flip sign.

Country	$\Gamma_i^{\text{raw}}(\hat{\zeta})$ (pp)	$\Gamma_i^{\text{raw}}(0.5)$ (pp)	Sign flips?
USA	+3.29	−0.06	yes
EU	+4.30	−0.76	yes
Japan	−12.54	+6.79	yes
South Korea	+2.14	−8.41	yes
Canada	+2.12	+1.97	no
Australia	+0.69	+0.47	no

Neither $\hat{\zeta}$ nor 0.5 is privileged by the model: $\hat{\zeta}$ is BSY’s own GMM point estimate, and 0.5 is the natural symmetric, no-information comparator. That two-thirds of the panel’s fairness-gap signs are not robust to this comparison is a direct sensitivity of the *comparator* side of the bridge, $s_i^{HW}(\zeta)$ — it is not a sensitivity of the Shapley benchmark $\phi_i(v)/v(\mathcal{N}_6)$, which does not depend on ζ at all (§4.1). It would be a mistake to read Table 6’s four-rule check below as resolving this sensitivity: that check varies the *allocation rule* applied to $v(S)$ while holding $\zeta = \hat{\zeta}$ fixed on the comparator side, so it cannot by construction speak to sensitivity in ζ — the two robustness checks answer different questions and should not be conflated. What this table actually shows is sharper: at $\zeta = 0.5$, Shapley and BSY’s own Horn–Wolinsky bargain assign the United States almost exactly the same share ($\Gamma_{US}^{\text{raw}}(0.5) = -0.06\text{pp}$). The entire $+3.29\text{pp}$ gap at $\hat{\zeta}$ is therefore not evidence of a structural distance between the Shapley benchmark and bargaining outcomes in general; it is a direct readout of the size of the asymmetry BSY’s own GMM estimation happened to find at their published point estimate. $\Gamma_i^{\text{raw}}(\hat{\zeta})$ should be read

accordingly: not as “how far is Shapley from bargaining,” but as “how large is the asymmetry BSY estimated.” Table 6 remains useful for its own, narrower purpose — establishing that the *sign* of Γ_{US} , Γ_{Japan} , and Γ_{Canada} is unchanged across four efficient, individually-rational allocation rules, holding $\zeta = \hat{\zeta}$ fixed — and that is the evidentiary basis this paper uses for any sign claim about Γ_i ; it is robustness to the rule, not to ζ , and the ζ -sensitivity documented in this table is disclosed, not resolved, by that check.

4.1.5 A registered search for the Shapley-consistent weight set (ζ^*)

A complementary question to the $\zeta = 0.5$ placebo above is whether *any* Horn–Wolinsky bargaining-weight vector $\zeta \in [0.005, 0.995]^7$ (one per bilateral pair) rationalizes the Shapley benchmark, i.e. whether the Shapley-consistent weight set is non-empty for an economically small tolerance — the object predictions P-07', P-08, and P-09 register (`paper/theory/predictions.md`). We search 8 starting points — BSY’s own $\hat{\zeta}$, the symmetric point $\zeta = 0.5^7$, and 6 Sobol/random draws — minimizing $L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(\mathcal{N}_6)\|_2^2$ with a box-constrained local optimizer. **All 8/8 starts stall** (`zstar_run.json`) before a single accepted optimizer step, but the proximate cause is compute scheduling, not equilibrium non-convergence: the run’s own per-evaluation log (`zstar_evals.jsonl`) shows all 44 of its “non-finite residual” entries carry the batch-level 1800-second wall-clock-timeout sentinel, not the genuine-solver-failure sentinel, which fires zero times across all 52 attempted evaluations. Successful evaluations take 504–1757 seconds (median ≈ 900 s), so assembling a forward-difference Jacobian (7 perturbation tasks split across 3 workers) needs roughly three worker-rounds (≈ 2700 s), structurally longer than the 1800-second per-batch budget: the first Jacobian batch at *any* starting point is scheduled to time out. This is confirmed directly in the log — at one identical starting point, one evaluation timed out and a second, later evaluation of the same point succeeded in 587s. We therefore withdraw the earlier reading of this run’s two **stalled** reason strings (`nonfinite_residual_fd`, `nonfinite_residual_at_start`) as a *third* independent observation of finite-difference-through-bargaining non-convergence: both are aliases for “killed by the wall clock” here, not a property of the objective, and 8-start coverage together with the design-sampled coverage sweep — which ran to completion, all 10 requested Sobol/random draws submitted, at a 2/10 per-draw success rate (the other 8 hit the identical per-batch timeout, not a 40-hour budget cutoff) — never reached a point where the objective’s differentiability could actually be assessed.

What this run’s evaluations *do* establish, without new computation, is a sharper and independently verifiable diagnosis of why this objective resists finite differencing near the step size anyone has tried. BSY’s own Nash-in-Nash fixed point takes a discrete deal/no-deal branch on each negotiated line — a pair reaches a deal only if its minimum individual gain clears a -10^{-7} tolerance, otherwise the pair reverts to its 1990 baseline tariff (`nash_in_nash.py`). At the symmetric placebo $\zeta = 0.5^7$, two independent re-solves of the *identical* point — one during this run’s own optimization pass, one during its noise-floor probe — return loss values that differ by $\Delta L \approx 4.1 \times 10^{-4}$ (`zstar_run.md` §§1, 3–4), direct evidence that at least one line’s deal/no-deal branch is decided differently across nominally identical re-solves of the same ζ . A later isolated, fully-converged re-run of this same $\zeta = 0.5^7$ point supplies a decisive, independent

confirmation of the same mechanism at a coarser grain: run to machine precision rather than compared at two partial iterates, the fixed point still lands on a branch 10.0%/5.08pp away from BSY’s cached comparator (below, `zstar_gate1_half_isolated.md`). The forward-difference step used to assemble the Jacobian, $h = 0.005$ — chosen to match BSY’s own `DiffMinChange` — is comparable in scale to this branch-flip noise, so a branch flip inside the perturbation window can dominate the finite-difference signal: at this resolution, $L(\zeta)$ is better described as a path-dependent step function than as a smooth objective. This mechanism, not a generic “non-finite residual,” is the correct reading of why BSY’s own estimation needed a coarse `DiffMinChange` to begin with (`phase5_verdict.md` §4) and why this project’s own Phase 5 re-estimation attempt diverged inside the same Jacobian-assembly step (`phase5_verdict.md` §3) — two independent observations of the same branch-selection fragility, each standing on its own evidence; this run’s stalls do not add a third, for the scheduling reasons above, but are consistent with, and sharpen, the same diagnosis. The registered test of P-08/P-09 is therefore **optimizer-layer INCONCLUSIVE**: no start reached a first-order- stationary point, for reasons now correctly attributed to scheduling rather than to the objective’s non-differentiability, though the branch-selection mechanism just identified independently explains why any finite-difference search near $h = 0.005$ is likely to remain fragile regardless of scheduling (`zstar_run.json`, P08 block).

Per the 2026-08-14/15 author rulings, we report as a summary quantity the smallest loss value *evaluated* across all starts and diagnostic sweeps (not only at the 8 stalling points), $L_{\min} = 0.0034$ — equivalently an ℓ_2 distance of 5.8 percentage points in HW-weight/Shapley-share space. **This quantity, and any comparison built on it, is provisional:** the author-authorized A0 comparator-gate re-verification has since run and aborted at its own acceptance gate (reported below), so the qualifier below is strengthened rather than resolved. $L(\hat{\zeta}) = 0.0196$ (14.0pp), the value against which $L_{\min}$ is compared, is computed from BSY’s own *cached* equilibrium (§3.1); $L_{\min}$ ’s best-evaluated point, by contrast, comes from this run’s own re-solved Nash-in-Nash pipeline, which returns a different value at the identical $\hat{\zeta}$ point, $L(\hat{\zeta})_{\text{resolved}} = 0.0381$ (19.5pp, `zstar_run.md` §§1, 3), a discrepancy consistent with the branch-level disagreement documented in §3.1 (this project’s own re-solve lands 26 of the 151 negotiated lines at the no-deal corner where BSY’s cached solution records a deal). The A0 gate — re-solving $\hat{\zeta}$ and $\zeta = 0.5^7$ and confirming agreement with the cached comparator to a pre-registered tolerance — has since been attempted at both anchors and, in each case, failed to confirm agreement (below): by budget timeout at $\hat{\zeta}$, by a determinate comparator-match failure at $\zeta = 0.5^7$; absent that confirmation the headline “2.4× closer” comparison mixes two pipelines and should be read as suggestive rather than certified; on a same-pipeline reading (both numbers from this run’s own re-solves, $L_{\min} = 0.0034$ against $L(\hat{\zeta})_{\text{resolved}} = 0.0381$), the ratio is 3.3×, not 2.4×. Both readings are reported here; neither is treated as established. **The A0 comparator-gate re-run has since been attempted at both anchors, reaching two different determinate outcomes.** A three-gate protocol (comparator-reproduction gate, measured-noise-floor δ , per-task timeouts) warm-started from BSY’s own cached equilibrium at $\hat{\zeta}$ (cache distance 7×10^{-5}) and required the re-solve to reproduce their cached aggregate, $\Sigma_k g_k^{HW}(\hat{\zeta}) = 0.312441$, within a fixed compute budget before any downstream ζ^* search could proceed. At $\hat{\zeta}$, five of seven bilateral pairs classified as “no deal,” and the first Gauss–Seidel sweep alone consumed

1627s of the 1800s chunk, leaving only 173s for a second sweep that a discrete deal/no-deal fixed point routinely needs; the gate aborted on a budget-scale mismatch, not a specific pathology of the probe search. At the symmetric placebo $\zeta = 0.5^7$, a first 1800s-budget attempt was likewise killed by the same clock while *visibly converging* (sweep 1, 1451s, all seven pairs reaching a deal; sweep 2, 335s more, $\max |\Delta_{\text{mfn}}|$ falling from 0.1469 to 0.001656, cumulative 1786s breaching the 1800s cap on the next step, `zstar_run.md`, GATE 1 addendum), which at the time left open whether a larger budget would resolve the gate. A dedicated single-anchor re-run with a 3600s budget and no concurrent load has since answered this: the Gauss–Seidel fixed point converges fully, in 1018s, to machine precision ($\max |\Delta_{\text{mfn}}| = 5.085 \times 10^{-7}$ across three sweeps, with over 2500s of headroom to spare) — but the converged point misses BSY’s cached comparator by 10.0% on $\Sigma_k g_k$ (against a 1% tolerance) and 5.08 percentage points on the resolved share vector’s ℓ_2 distance (against a 0.5pp tolerance), failing both gate criteria by roughly an order of magnitude (isolated single-anchor re-run, `zstar_gate1_half_isolated.md`). This anchor’s failure is therefore *determinate*, not a timeout: the fixed point is stable and fully converged, and it simply is not BSY’s cached branch. GATE 1 now fails at both anchors, by two distinct and independently diagnosed mechanisms — insufficient budget at $\hat{\zeta}$, sufficient-budget convergence to the wrong branch at $\zeta = 0.5^7$ — and no further re-run is indicated at either anchor. Independently of these two mechanisms, three of the seven bilateral pairs at $\hat{\zeta}$ have a minimum gain within $\sim 2 \times 10^{-6}$ of the -10^{-7} deal/no-deal cutoff — two no-deal pairs, (0, 2) and (0, 5), and one deal pair, (4, 5) — numerically indistinguishable from the boundary; a further no-deal pair, (2, 4), is looser by an order of magnitude (-5.88×10^{-5}). This near-boundary clustering stands on its own as evidence for the branch-instability diagnosis documented above, and $\zeta = 0.5^7$ ’s own determinate wrong-branch convergence is a second, independent piece of the same evidence obtained by a different route (a fully converged solve landing on the wrong branch, rather than boundary-proximity in the deal/no-deal classification). Per the gate’s own design, a failed comparator reproduction aborts the batch rather than proceeding on a possibly-mismatched branch. The ζ^* -layer objects (P-07’, P-08, P-09) remain **INCONCLUSIVE** on this evidence, and the solver-branch stability fix needed to make this comparison reliable is deferred to future work. Independent of this caveat, $L_{\min}$ is, by construction, an **upper bound** on the true minimum distance: it certifies that some evaluated HW weight vector gets this close to the Shapley benchmark, not that no weight vector gets closer, since the search that would certify a minimum did not converge at any start. Predictions P-07’ (the sign of the lower bound of the Γ_{US}^{raw} confidence-set interval over $\hat{\Sigma}$), P-08 (whether the Shapley-consistent set Z_δ^* hugs the boundary of the feasible weight box, the registered CORNER null), and P-09 (a Wald test against the registered set) are therefore all recorded **INCONCLUSIVE** on this evidence, not FAIL — the registered null could not be tested, not that it was tested and rejected (full detail and status bookkeeping: `paper/theory/predictions.md`).

4.1.6 The corrected C3 reading

Putting the four results in this subsection together supersedes an earlier draft’s framing of C3 as “BSY’s bargaining outcome is unfair to the United States.” The corrected reading, adopted after author review of Round 12 remediation: the observed Horn–Wolinsky allocation largely

tracks the liberalization’s own physical incidence (Japan 55.5% of the HW surplus against 49.8% of the efficient policy’s own incidence, Table 4); the Shapley benchmark deviates from *both* the bargained outcome and the physical incidence, redistributing toward members whose marginal contribution is positive but whose physical incidence is small or negative (Australia +7.0% of $v(\mathcal{N}_6)$, South Korea +6.0%, Table 4). $\Gamma_{US} > 0$, $\Gamma_{\text{Japan}} < 0$, and $\Gamma_{\text{Canada}} > 0$ are sign-robust across all four efficient, individually-rational allocation rules tested (Table 6); magnitudes are strongly rule-dependent (Γ_{US} spans +3.3 to +23.2pp); Γ_{EU} , $\Gamma_{\text{Australia}}$, and $\Gamma_{\text{South Korea}}$ are sign-unstable. This is evidence that the Shapley benchmark redistributes along a dimension — marginal contribution — that BSY’s observed bargaining outcome does not track, in a direction that happens to favor the US and Canada and disfavor Japan robustly and several other countries unstably; it is not evidence that BSY’s bargaining outcome is specifically unfair to the United States, and the ECON-19 agenda-truncation caveat above means even the robust part of the US finding should be read as a sign claim, not a magnitude claim.

4.1.7 Pre-registered predictions: failures and retractions

Four pre-registered Track A predictions failed against their frozen claims. This paper’s own pre-registration discipline (`paper/theory/predictions.md`) requires each to be reported here, in this section, not only in the registry.

P-07'' registered that the fairness gap Γ_{US} would clear a frozen +5pp band; the certified value is +3.29pp (Table 2), below the frozen threshold. **FAIL.** This verdict holds on the primary, cached comparator branch this paper uses throughout; it is not robust to the comparator-side branch instability disclosed above (Table 3), where the alternative branch’s $\Gamma_{US} = +5.81\text{pp}$ clears the same threshold. We report the FAIL as certified-but-branch-conditional, not as a branch-independent finding.

P-20 registered that the United States’ passive-outsider welfare gain would be negative on a majority of the 31 coalitions not containing the US; the certified count is 13/31 (41.9%), short of a majority. **FAIL.** The registered null — mirroring Track B’s 83.7%-positive pattern and requiring $\geq 80\%$ positive — also fails, since Track A is only 58.1% positive; the US’s passive-outsider exposure on this economy is genuinely mixed-sign, neither the systematically adverse pattern conjectured nor Track B’s near-uniform free-riding.

P-19’s strong form registered that the grand-coalition optimum is strictly interior at every player, $s_i^* \in (0, 1)$ for all i ; South Korea’s certified optimum is exactly $s_{\text{Korea}}^* = 0.000$, a boundary point. **FAIL.** A post-hoc weak form — five of six coordinates strictly interior, the certified point 17.4% above the best vertex — **passes**, but may only be cited with the post-hoc label, since it was not the claim as registered.

P-26 — an earlier draft’s headline claim that implementing the Shapley benchmark requires a transfer of $\sim 19\%$ of $v(\mathcal{N}_6)$ to the United States, on the premise that the US is an absolute loser at the point attaining $v(\mathcal{N}_6)$ — is **retracted**. At the certified optimum the US is not the loser; Australia is ($g_{\text{AUS}}(s^*) < v(\{\text{AUS}\}) = 0$, Table 4), and the correct budget-balanced transfer has the US *paying* into, not receiving from, the reallocation. The retraction and its corrected replacement (gross transfer 13.51% of $v(\mathcal{N}_6)$, Australia the binding recipient) are carried in full in the core/transfer discussion above (§4.1); it is repeated here only so that the failure appears

in this section’s own inventory, per the registry’s disclosure rule.

4.2 Robustness: A Second Economy (Track B, Caliendo–Parro)

We report, in condensed form, the parallel exercise on a second, independently calibrated economy: the $N = 3$ (NAFTA) through $N = 7$ (NAFTA + China, India, Brazil, Korea) coalitions built on the Caliendo and Parro (2015) 30-country $\times$ 40-sector 1993 panel (“Track B”). This panel was the empirical core of an earlier draft of this paper; it now serves as a robustness check that the D1/Shapley machinery of Section 2 produces a qualitatively similar pattern — and some informative differences — on a structurally distinct economy. Full tables, figures, and legacy diagnostics are in Appendix C; this subsection reports headline numbers only.

Pipeline sanity check. As a check on the underlying Caliendo and Parro (2015) equilibrium solver, independent of the D1/Shapley layer, we replicate their Table 2 NAFTA welfare-change estimates: Mexico +1.174%, Canada −0.041%, USA +0.085%, against CP’s reported +1.17%, −0.04%, +0.09% (all errors < 0.01pp), and their Table 5 bilateral trade-flow changes to within 5pp on flows of $\sim 100\%$ magnitude (`verify_cp2015_nafta.py`).

Raw coalition values and near-dummy players. As on Track A, the raw Shapley imputation on Track B tilts toward the highest-tariff, highest-headroom economies: China’s raw share reaches 89.6% at $N = 4$ against a GDP share of 7.3% (Appendix Table 13). Low-applied-tariff members are near-dummy at every N : Canada’s raw Shapley share is under 1% of grand-coalition surplus at $N = 7$ ($\phi_{\text{Canada}}(v)/v(N) = 0.877/178.354$, `ir_verification.md`), echoing the Track A dummy-player pattern for Australia and Canada (Section 4.1). The legacy per- N enumeration table and figures are in Appendix C (Table 10); they predate the current D1 recompute, are superseded in *level* by the $v(N)$ figures cited in this subsection, and are retained only because the *qualitative* ranking they document (high-tariff dominance, low-tariff near-dummy) is unchanged.

Unilateral vs. cooperative decomposition. With $v(\{n\})$ the unilateral (own-tariff) gain and the cooperative game $w(S) = v(S) - \sum_{n \in S} v(\{n\})$ (eq. (9)), Shapley linearity splits the raw imputation exactly, $\phi_n(v) = v(\{n\}) + \phi_n(w)$. Once a high-tariff outsider enters the coalition ($N \geq 4$), only 13.1–16.1% of grand-coalition surplus is genuinely cooperative rather than unilateral, against 75.2% at $N = 3$ (NAFTA alone, no high-tariff outsider); the United States is the largest cooperative contributor at every $N \geq 4$ (35.5–45.2% of $w(N)$) — not the raw-share leader China — once each country’s own unilateral gain is netted out (`ir_verification.md`). Individual rationality holds at the correct baseline, $\phi_n(v) \geq v(\{n\})$ (equivalently $\phi_n(w) \geq 0$), for every player at every $N \in \{3, \dots, 7\}$, with global binding cushion \$0.266B (Canada, $N = 3$) — roughly three orders of magnitude above the model’s measured noise floor (`ir_verification.md`, `noise_floor_eps_B.md`). The full legacy decomposition table (Appendix C) predates this `ir_verification.md` pass and reports different, superseded percentages; the figures in this paragraph are the current ones and should be quoted in their place.

Horse races. Raw Shapley share is close to uncorrelated with GDP share ($|\rho| \leq 0.28$ across $N = 3\text{--}7$) but, once a high-tariff outsider is present, close to empirically indistinguishable from base-tariff-revenue share (Pearson 0.84–0.87 at $N = 4\text{--}7$; markedly weaker at $N = 3$, +0.29) — an adverse fact for any claim that the Shapley structure performs discriminating work beyond a largely mechanical headroom calculation (`horse_race_v3.md`, pre-committed reporting rule; full tables and per-country premia in Appendix C, Tables 13–14).

Convexity and the core. The cooperative game is convex to within numerical resolution (pooled supermodularity violation rate 0.56% across $N = 3\text{--}7$, worst at $N = 6, 7$; `superadditivity_sweep_D1.r`) and we verify core membership directly rather than merely inferring it from convexity: on the full $N = 7$ panel all 126 proper-coalition constraints hold (0/126 violations), minimum slack 0.37% of $v(N)$, binding at a genuinely out-of-sample six-country coalition (`{USA, Mexico, Canada, China, India, Korea}`; `p28_trackB_core.json`) — the same direct-verification strategy used on Track A (Section 4.1). Both core results are checks on the status-quo-outsider (α -style) core maintained throughout; the γ -response outsider convention (§2.6) is untested on either track. Full detail, including the per-triple supermodularity table, is in Appendix C (Table 15).

Registered but not executed. Two Track B rows, **P-14'** (a two-tier anchor/non-anchor Shapley-share stability band across N) and **P-15** (an alternative threat point at $N = 4$), remain registered but untested (status `DEFERRED-rescope` in `paper/theory/predictions.md`): the claims stand as written, but their execution path was orphaned by the 2026-08-12 Track-A-recentering decision, which redirected compute priority to Track A, and no compute is currently queued. They are disclosed here, on the track they actually belong to, rather than left silently absent from this paper or misfiled under Track A.

Further robustness (Appendix C). Three additional checks are reported in full in the appendix rather than here, because their headline magnitudes predate the current D1 recompute and are flagged superseded at their source (Section 2): a two-step (BSY-style) vs. single-step MPEC benchmark on the NAFTA grand coalition, illustrating that separating the trade equilibrium from the outer optimization can both leave surplus on the table and silently violate individual rationality; a threat-point sensitivity exercise on NAFTA showing the Shapley ranking is not invariant to the choice of disagreement point; and the numerical evidence on existence and uniqueness of the σ^* fixed point from a retired (ADR-1) specification of the characteristic function, retained purely as an archival diagnostic alongside Appendix D.

5 Conclusion

5.1 Summary

This paper replaces the exogenous bilateral bargaining weight of Bagwell et al. (2021) (BSY) with a characteristic function $v(S)$ that carries no bargaining-weight parameter of any kind (D1, the utilitarian-optimal coalition surplus), and divides the resulting surplus using the Shapley value. We apply this framework to BSY’s own six-country 1990 negotiating economy (“Track

A”), the paper’s primary application, after first replicating their equilibrium map and key counterfactuals (their Table III to a maximum deviation of 0.024pp; their Nash-in-Nash bargaining fixed point to 0.40pp mean tariff deviation) but not their $\hat{\zeta}$ estimator, whose re-estimation fails to converge — a disclosed negative result that does not affect any downstream number, since $v(S)$ under D1 carries no bargaining weight of any kind and every $\hat{\zeta}$ -dependent object we report is evaluated at BSY’s own published estimate.

Comparing the Shapley benchmark to BSY’s own estimated bargaining outcome through a fairness-gap statistic $\Gamma_i = \phi_i(v)/v(N) - s_i^{HW}(\hat{\zeta})$, $\Gamma_{US} > 0$, $\Gamma_{Japan} < 0$, and $\Gamma_{Canada} > 0$ are sign-robust across four efficient, individually-rational allocation rules we test (Shapley, equal split, 1990 revenue share, 1990 income share), while magnitudes are strongly rule-dependent (Γ_{US} spans +3.3 to +23.2pp) and Γ_{EU} , $\Gamma_{Australia}$, Γ_{Korea} are sign-unstable (§4). We read this as evidence that the Shapley benchmark redistributes along a dimension — marginal contribution — that BSY’s observed bargaining outcome does not track, in a direction that favors the US and Canada and disfavors Japan robustly, not as evidence that BSY’s outcome is specifically unfair to the United States.

The bridge comparison also shows that the observed HW allocation largely tracks the liberalization’s physical incidence (Japan receives 55.5% of the HW surplus against 49.8% of the efficient policy’s own physical incidence): the Shapley benchmark deviates from *both*, redistributing toward members whose marginal contribution is positive but whose physical incidence is small or negative (Australia +7.0% of $v(\mathcal{N}_6)$, South Korea +6.0%, `phase7_zero_ge_batch.md` §3).

A companion recomputation that restricts the coalition-value lever to BSY’s own 151-line negotiated set, solved on a genuine, weight-corrected line-by-line lever ($F_{line}^{v2} = 0.436840$, IR-constrained $F_{line}^{IR} = 0.428519$, predictions P-31/P-33, re-purposing the same MPEC machinery BSY’s own Table III counterfactual and P-30’s inconclusive first attempt used), together with the identical instrument extended to the full 294-line schedule ($D = 0.885046$, prediction P-34), closes an *exact* identity against the raw gap between $v(\mathcal{N}_6)$ and BSY’s aggregate bargained surplus ($\Sigma_k g_k^{HW} = 0.312441$): $D - \Sigma_k g_k^{HW} = 79.51\%$ of $v(\mathcal{N}_6)$, splitting additively into three matched-instrument terms. The first two are non-negative by construction of the nested optimization problems that produce them; the third is not — it would require BSY’s own cached tariff schedule to be itself a feasible point of the constraint set that generates it, and a direct check finds it is not (5 of 151 masked lines, 3.3%, all below zero and legal under BSY’s own solver bound $lb = -0.25$, `check_hw_box_constraint.md`), so its non-negativity is confirmed empirically in this run’s numbers, not guaranteed by construction.

The three terms, each measured consistently against the TI-5-corrected frontier $F^{IR,corr} = 0.428073$ (§4): extending coverage — to lines outside BSY’s own bilateral negotiating structure, an estimation-sample construct rather than the Uruguay Round’s actual agenda — from these 151 lines to the full 294-line schedule, at an identical line-by-line instrument, is worth 62.24% of $v(\mathcal{N}_6)$; the transferable-utility/individually-rational wedge on the 151-line set itself is worth 1.22%; and a within-agenda bargaining shortfall restricted to points individually rational for every member (relative to the 1990 threat point) is worth 16.06% (the TI-5-correct point estimate, using each member’s own line-by-line unilateral floor rather than a uniform zero; a

uniform-zero floor instead gives a 16.12% upper bound, only 0.06 percentage points higher; the three two-decimal components sum to 79.52% by independent rounding, exact total 79.51% from the unrounded values above), essentially unchanged from the unconstrained reading, so the feasibility axis is not the source of the shortfall — though four of six members are worse off at that IR-feasible point than at BSY’s own cached bargain, so this is a transferable-utility aggregate efficiency gap, not a demonstrated Pareto improvement at that point.

A stronger stress test, using BSY’s own actual outcome rather than 1990 as every member’s floor, finds a materially large no-transfer *weak* Pareto improvement survives regardless (four of six members exactly indifferent; the net gain accrues to Japan and Canada): 85% of the 1990-IR-constrained shortfall remains. Coverage, the TU–NTU wedge, and within-agenda bargaining performance are jointly, not separately, responsible for BSY’s distance from full efficiency, and in that order of magnitude — widening what is negotiable is worth 3.88 times more than the within-agenda shortfall on a regime-matched instrument, now a fully floor-corrected point estimate on both operands (the 294-line per-country IR-floor correction on the coverage side has been executed, moving the ratio down slightly from a previously reported $3.91\times$, §4): the coverage term specifically reflects what is on the table, not how skillfully BSY’s negotiators bargained over it, while the within-agenda shortfall is consistent with an architectural gap in what a bilateral bargaining protocol can price, though cut-depth (axis (a)) and coalition-structure effects (axis (d)) remain jointly embedded in that reading and have not been separately measured.

A dedicated test, P-40, was designed to separate them directly by constructing F_{pairs} , a **pair-decomposed welfare-sum equilibrium** of BSY’s own seven bilateral principal-supplier pairs — each pair maximizing its own two members’ dollar-welfare sum, with no disagreement point and no Horn–Wolinsky-style weight, so not itself a Nash bargain (ECON-115) — spot-checked, not merely locally verified, against a second start at every pair and against reversed update order and an independent starting configuration across three trajectories, all converging to the identical fixed point to within 7.2×10^{-8} (§4); this equilibrium ($F_{\text{pairs}}^{\text{eq}} = 0.311364$, re-optimized under the corrected D -convention, superseding the pre-fix 0.305867) falls short of what BSY’s negotiators actually achieved ($\Sigma_k g_k^{\text{HW}} = 0.312441$, on the identical 151-line mask), triggering a pre-registered boundary contingency rather than either dispositive branch, so the axes (a)/(d) entanglement is diagnosed but not resolved. This entire F_{pairs} family — the level number just cited, the boundary disposition, and the pair-externality/ Δ_d^{TU} diagnostics reported below — has been **re-optimized under the corrected D -convention and independently verified to machine precision**; the provisional flag it previously carried is discharged (p40_dconvention_rerun_result.md, §4).

The re-optimization reverses, not merely revises, what the paper had identified as this shortfall’s driver. Previously reported: a single member’s individual-rationality violation — the EU, $21\times$ the aggregate boundary margin in absolute-level terms, $30.9\times$ measured as a like-for-like deviation from its own actual outcome — with four of six members better off at this equilibrium than at BSY’s actual outcome. **Under the corrected economy the EU does not violate its own individual-rationality floor at all** ($g_{\text{EU}} = +0.0155$, a shortfall of -0.0492 relative to its actual outcome, not a floor violation); the $21\times/30.9\times$ readings and the “four of six better off” distributional claim are **retracted**. Only two members (Japan,

South Korea) are strictly better off at the corrected fixed point than at BSY’s actual outcome, and two that were previously reported better off (the United States, Canada) **flip sign** to worse off. The member closest to its own individual-rationality floor is now Australia, by three orders of magnitude less than the retracted EU figure. With no single member’s own shortfall dominating, the boundary result’s driver is not currently attributable to any one country; the boundary result itself remains comparator-sensitive, landing inside the “architectural constraint ESTABLISHED” branch at the paper’s own $\zeta = 0.5$ robustness point even though it does not at $\hat{\zeta}$ — disclosed, not banked, since $\hat{\zeta}$ remains the paper’s primary comparator and the margin’s own thinness (now 0.149% of $v(N)$, an 83.6% relative contraction from the pre-fix 0.913%) is reason enough on its own not to treat switching anchors as decisive.

(Caveat: with the EU no longer IR-violating under the corrected economy, this probe’s motivating question is largely moot, and its own script is on the confirmed- D -convention-affected list, not yet re-run — likely confounded by the same bug, re-run deferred to R&R.) A direct feasibility probe closes the one pre-registered route to an emptiness proof: every EU-containing pair admits a strictly positive unilateral IR-feasible gain in isolation, so EU-only infeasibility cannot explain the non-convergence. It does *not*, however, test *joint* two-member feasibility, which is what the regime-matched $F_{\text{pairs}}^{\text{eq,IR}}$ construction actually requires: at the US–EU probe’s own argmax, the partner’s own gain is sharply negative ($g_{US} = -0.2568$), the EU’s unilateral gain achieved substantially at the US’s expense, so satisfying one member’s floor is no evidence the other’s is anywhere close to being jointly satisfiable at the same point. The regime-matched comparator’s own non-convergence remains attributed to Gauss–Seidel coordination failure across shared members, with existence at the joint constraint neither established nor ruled out (§4).

(Caveat: this attempt’s own script is likewise on the confirmed- D -convention-affected list and has not been re-run; deferred to R&R independent of the question below.) A genuine Nash-product reconstruction of F_{pairs} , using BSY’s own estimated pair weights and an endogenous disagreement point, was attempted to isolate the solution-concept component of this result and stalled after ≈ 4 hours of Gauss–Seidel iteration with no point estimate banked; the partial round it did recover shows most reached pairs failing a pre-optimization domain check and reverting to no-deal — a pure artifact of that check being evaluated at the starting configuration, since the optimizer was never called for those pairs, and not evidence for or against the existence of a mutually beneficial agreement (this paper’s own direct feasibility probe on the same fixed point finds the opposite, §4) — deferred to R&R.

The same exercise nonetheless establishes, with zero additional compute, that replacing a joint 151-line planner with BSY’s own seven-pair bilateral architecture moves the pair-own objective’s total by 17.42% of $v(\mathcal{N}_6)$ on a transfer-unconstrained basis (re-optimized under the corrected D -convention, superseding the pre-fix 18.19%; direction of its own IR-regime correction not established, since the two available data points on a regime-matched correction move in opposite directions), and that roughly a quarter (26.2%, superseding the pre-fix 30.1%) of the surplus any one pair creates when optimizing only its own two members’ welfare spills, uninternalized, to the other four countries — corroborating, on this paper’s own fully-matched object, BSY’s own Table VII finding that free-rider gains account for 35.38% of the Uruguay

Round’s realized world-welfare gain (Bagwell et al. 2021; the two figures are a structurally analogous ratio, not a comparable estimate of the same underlying parameter, and measure realized, uninternalized spillover rather than deadweight loss), reported descriptively rather than as a verdict on axes (a)/(d).

D1’s own optimal line-by-line cuts on this agenda correlate positively with the actual observed 1990–2000 tariff change (Spearman $\rho = +0.64$, clearing the pre-registered $\rho \geq 0.3$ band) — the paper’s first external-data comparison, though a paired bootstrap shows a model-free uniform-cut placebo scores higher on the identical pooled metric, resolved ($\rho = +0.71$; the 95% CI on the difference excludes zero and favors the placebo).

On BSY’s own cached $\hat{\zeta}$ schedule, 86 of the 151 negotiated lines (57%) are left at exactly BSY’s own 10^{-6} solver tolerance — effectively uncut, though the data show these same lines were, on net, cut over 1990–2000 — while D1’s efficient frontier cuts 118 of the 151 lines (78%), including all but 27 of the lines that carried a positive 1990 tariff; a mid-rank-corrected rank comparison on the same metric favors D1 over BSY’s own fitted model on the pooled sample ($\rho_{BSY} = +0.32$ against $\rho_{D1} \in [+0.64, +0.67]$), a margin that survives a direct test against the tie-count mechanism but not against coverage: BSY structurally cannot lose rank ties on lines it never negotiated, and on the same-coverage 65-line subsample — the caliber this paper’s own registry treats as the fair, terminal comparison — none of the three pairwise comparisons is resolved, so the official disposition is **intermediate**, not a settled D1 victory; part of the pooled-sample gap reflects D1’s own closer resemblance to naive 1990-tariff persistence than BSY’s fitted schedule shows. The two are close on the level (Pearson) face, and neither carries incremental content beyond own-tariff persistence once rank is partialled on τ^{1990} ($|\rho_{\text{partial}}| < 0.05$ for both).

The extensive-margin contrast — BSY’s own model leaving a majority of negotiated lines essentially untouched while D1 cuts a large majority of them — is the more defensible reading of this comparison; on BSY’s own cached $\hat{\zeta}$ schedule, it establishes directional agreement, a pooled-sample rank advantage over BSY’s own estimated model that does not survive a coverage-matched test, and a documented coverage difference from it, but not incremental agreement with the data beyond persistence, nor an advantage over a naive persistence rule (§4).

We verify directly, rather than merely infer from convexity, that the Shapley imputation lies in the core on Track A despite superadditivity failing outright (20/301 disjoint-pair violations, worst case the US joining {EU, Canada}): all 62 proper-coalition constraints hold (0/62 violations, minimum slack 2.59% of $v(\mathcal{N}_6)$, status-quo-outsider convention, `phase7_zero_ge_batch.md` §1). Core membership is a property of the *transferable-utility* game: it establishes that the fair allocation is self-enforcing *given* feasible transfers, not that the underlying efficient policy is implementable without them. On Track A, absent a transfer instrument, the efficiency-maximizing policy itself is blocked by Australia’s participation constraint ($g_{\text{AUS}}(s^*) < v(\{\text{AUS}\}) = 0$); reaching the Shapley allocation requires a budget-balanced gross transfer of 13.5% of $v(\mathcal{N}_6)$ (the US pays 2.49%, Australia receives 7.0%).

As a robustness check on a second, structurally distinct economy — coalitions of size $N = 3$ (NAFTA) through $N = 7$ (NAFTA + China, India, Brazil, Korea) built on the Caliendo and Parro (2015) dataset (“Track B”) — the identical D1/Shapley machinery produces a qualitatively similar pattern together with informative differences (item (x) below elaborates). Raw

Shapley shares tilt toward high-tariff, high-headroom economies (China, India); splitting each country’s raw share into a unilateral (own-tariff) and a genuinely cooperative component reverses this story. Once high-tariff economies enter ($N \geq 4$), only 13.1–16.1% of grand-coalition surplus is genuinely cooperative rather than unilateral (versus 75.2% at $N = 3$, where NAFTA has no high-tariff outsider), and this cooperative surplus accrues to the large, import-diversified United States (top cooperative contributor at every $N \geq 4$, 35.5–45.2%) rather than to the high-tariff economies that dominate the raw share (`ir_verification.md`). Both the magnitude of the cooperative share and who leads it diverge sharply from Track A: only 13.1–16.1% of surplus is genuinely cooperative on Track B against 94.05% on Track A ($\kappa_A^{s \leq 1}$, §4.1), and while the United States leads Track B’s cooperative component, it ranks third on Track A’s (Japan 42.9%, EU 25.0%, US 13.3%); the raw-versus-cooperative reversal documented here has no analog on Track A, where $\phi_i(v) = \phi_i(w)$ for four of six players by construction ($v(\{i\}) = 0$, §4.1). Two horse races show the raw imputation is not a GDP relabel (Pearson ranging from -0.28 to $+0.07$ across $N = 3$ –7) but is, on Track B at $N \geq 4$, close to empirically indistinguishable from a tariff-revenue relabel (Pearson 0.84–0.87; markedly weaker at $N = 3$, $+0.29$, where no high-tariff outsider is present) — an adverse finding reported honestly rather than as something the exercise establishes (`horse_race_v3.md`, pre-committed reporting rule). We also verify $\text{Shapley} \in \text{core}$ directly on Track B (all 126 proper-coalition constraints hold at $N = 7$, 0/126 violations, minimum slack 0.37% of $v(N)$, binding at a genuinely out-of-sample six-country coalition, `p28_trackB_core.json`). Full detail is in Section 4.2 and Appendix C.

5.2 Policy Implications

Three implications follow directly from the model. First, the asymmetry of *raw* Shapley shares toward high-distortion economies should not be read as a leverage or bargaining-power finding. On Track B, where the unilateral-cooperative decomposition is informative (§4.2), the raw tilt is dominated by $v(\{i\})$ — the surplus a high-tariff country captures acting alone, an outside-option effect on *that country’s own* payoff, not evidence about its influence over the coalition. The genuinely cooperative component, $\phi_i(w)$, instead accrues to the large, import-diversified United States ($\arg\max_i \phi_i(w)$ at every $N \geq 4$), not to the high-tariff economies the raw ranking favors. Simple GDP- or trade-volume-based bargaining-weight assignments would miss this distinction entirely, and so would reading the raw Shapley ranking on its own terms.

Second, the direct core-verification results above have a policy reading that only holds *conditionally*. Core membership shows the Shapley-fair allocation is self-enforcing once transfers are feasible, on both economies we study; it does not show that the underlying policy is implementable without them. GATT/WTO is, by design, a no-transfer institution, operating through issue linkage and reciprocal concessions rather than side payments. On Track A this is not a hypothetical concern: absent transfers, the efficiency-maximizing policy itself violates Australia’s participation constraint outright, so no transfer-free institution can implement either the surplus-maximizing benchmark or the Shapley reallocation built on top of it there, and the transfer volume required (13.5% of $v(\mathcal{N}_6)$) is far outside anything GATT’s reciprocity mechanism can replicate. Both core results reported here are checks on the status-quo-outsider (α -style) core; the γ -response outsider convention of Chander (2007) is unchecked on either

track and is a natural next robustness step (see Future Work). Third, the inclusion of Rest-of-World as a non-strategic outsider (rather than as a coalition member) avoids the spurious size-jumps that would arise from accounting aggregation.

5.3 Limitations and Scope

The contribution of this paper is a *normative imputation rule* for cooperative tariff bargaining surplus, not an identification of real-world bargaining power from data. Within that scope, six limitations deserve explicit acknowledgment.

(i) Solution-concept choice. The Shapley value is one of several imputation rules consistent with efficiency, symmetry, the dummy axiom, and additivity (Shapley, 1953). Alternative cooperative solution concepts—the nucleolus, the kernel, the γ -core (Chander, 2007), or the coalitional Nash bargaining of Ray and Vohra (2024)—would generate different per-country imputations from the same coalitional values $v(S)$. Without a moment condition tying $v(S)$ to data, the choice of solution concept is normative, not identified (Manski, 1995). We adopt Shapley for its axiomatic parsimony and for direct nesting of symmetric Nash bargaining at $N = 2$ (see Proposition 1), but do not claim it is uniquely correct. The Track A sensitivity exercise summarized above (§4) quantifies how much this choice matters in dollar terms: swapping Shapley for an equal-split, revenue-share, or income-share rule leaves two countries’ fairness-gap signs unchanged but moves a third country’s magnitude by an order of magnitude.

(ii) Scalar policy lever. Each importer n controls a single scalar $s_n \in [0, 1]$ that scales *all* of its applied tariffs proportionally. This is a substantial simplification of WTO and PTA practice, where sectoral schedules, Swiss-formula nonlinear cuts, and sensitive-product carve-outs are first-order features. The scalar lever preserves a clean cooperative game structure and mechanically links Shapley shares to base-tariff revenue (Section C.7). Its effect on convexity is track-dependent: on Track B it produces a near-convex game (pooled supermodularity violation rate 0.56% across $N = 3$ –7, Section 4.2) in which tariff complementarity, not strategic substitutability, dominates at every tested N — not the free-riding mechanism that might otherwise be expected to drive non-convexity. On *Track A — our main panel — it does not*: superadditivity fails outright on 20/301 disjoint pairs (Section 4), the worst case the US joining {EU, Canada}. Both economies’ core results nonetheless hold, because both are verified directly rather than inferred from convexity (Section 4, Section 4.2). Whether a richer sectoral or Swiss-formula lever would change either track’s convexity property is untested. The price of the scalar simplification is directly quantifiable on Track A: replacing the country-scalar lever with a line-by-line one on the identical policy set raises attainable member-sum welfare by 18.51% of $v(\mathcal{N}_6)$ restricted to BSY’s own 151 negotiated lines, or 22.90% of $v(\mathcal{N}_6)$ at full (294-line) coverage (§4, TI-13(vi)); a richer instrument is therefore not a marginal robustness concern but a first-order one. A full sectoral extension is the natural next step but requires both an MPEC reformulation and a substantially larger inner solve.

(iii) Threat-point endogeneity. The applied tariff used as the disagreement point is itself the outcome of decades of prior GATT bargaining (1947–1993 for Track B’s 1993 baseline, 1947–1990 for Track A’s own baseline); treating it as exogenous is a real identification compromise on both tracks. The robustness exercise in Section C.9 illustrates the sensitivity on NAFTA. We do not claim to fully resolve threat-point endogeneity; we report the result, and recommend that a Nash-tariff disagreement point in the spirit of Ossa (2014) be used as the principled alternative in future work.

(iv) Non-member individual rationality. The cooperative program is solved over coalition S with non-members held at status-quo tariffs. Because each importer’s lever s_n scales its MFN schedule uniformly across all trading partners (Section 2), this is *not* a discriminatory-PTA interpretation: non-members receive the same liberalized access as coalition members by construction, since MFN treatment cannot legally discriminate among origins. The correct reading is *plurilateral agreement with free-riding*: coalition S negotiates and implements a joint tariff cut, and non-members free-ride on the resulting MFN externality without reciprocating — this is also not the GATT Article XXIV reciprocity-with-MFN setup, which requires substantially-all-trade coverage between the parties to a preferential arrangement. We have directly verified, on both tracks, whether sub-coalition optima ever reduce non-member welfare, and the answer is track-dependent. On Track B, non-member welfare is overwhelmingly but not universally positive: $369/441 = 83.7\%$ of (coalition, non-member) pairs show a strictly positive welfare change, clearing our pre-registered $\geq 80\%$ threshold (`outsider_welfare_N7.md`); the remaining 16.3% are genuine negative externalities. On Track A the analogous registered prediction **fails**: only 58.1% of (coalition, non-member) pairs are positive (13 of 31 strictly negative), so a majority-positive reading does not transfer to an economy of six large, mutually exposed negotiators (`phase7_full_primary.md` §3). Both core results reported in this paper (§4.1, §4.2) are checks on the status-quo-outsider, α -style core (assumption A4): non-members are held at their 1990 applied tariffs, not at a best response, and the core constraint is evaluated against that fixed outside option. Imposing non-member IR ($\widehat{W}_n \geq 1$ for $n \notin S$) inside the coalition-value program itself, in the spirit of the γ -core of Chander (2007), is therefore a natural next test, and Track A’s negative-externality share suggests it would bind there; a companion result sharpens the stakes of this scope-out rather than closing it. The masked TW-max decomposition reported in §4.1 shows a world-welfare-maximizing policy imposes a net loss of 3.52 (roughly $4.9 \times v(\mathcal{N}_6)$) on the six negotiators as a group relative to their own constrained optimum, with the EU alone losing $10.9 \times v(\mathcal{N}_6)$ while every rest-of-world region gains — a non-member externality an order of magnitude larger than anything documented on the α -core convention above. This is suggestive, not dispositive: it is computed on the world game, not the six-country coalition game the α/γ -core distinction applies to, so it motivates rather than substitutes for a genuine γ -core recomputation, which we leave, explicitly, to an R&R rather than treat as implicit in the α -core results above.

(v) No data moment on $v(S)$ for $S \neq N$. The coalitional values $v(S)$ for proper sub-coalitions are not anchored to observed welfare-change data; they are GE-model outputs under stated modeling restrictions. Following the partial-identification posture of Manski (1995), this

is a deliberate choice: identifying $v(S)$ from data would require linking the cooperative value of historical RTAs (NAFTA 1994, EU enlargements, MERCOSUR, ASEAN, USMCA, CPTPP) to observed welfare changes—a substantial empirical project beyond this paper’s scope.

(vi) Free-riding at large N . The cooperative game does not internalize the public-goods nature of liberalization. [Olson \(1965\)](#) classical free-riding logic predicts that as N grows, coalitional stability weakens. Our $N \leq 7$ exercises do not explore this scaling, and the extension to $N = 31$ via the Monte-Carlo Shapley sampler validated in [Section C.4](#) would not, by itself, address strategic incentives to defect. Free-riding analysis at large N is a substantive extension, not a robustness check.

5.4 Future Work

Several directions remain. (i) Disaggregate the policy lever to the sectoral or HS-4 product level (with Swiss-formula or sensitive-product carve-outs), to decouple Shapley shares from τ^0 -revenue; (ii) implement a true MPEC formulation in Pyomo+IPOPT (or GAMS+CONOPT) in which all equilibrium variables are simultaneous decision variables, enabling both efficient sectoral solution and access to multi-start / global solvers (e.g., BARON) for verification of local optima; (iii) replace the base-year disagreement point with a Nash-tariff disagreement point in the spirit of [Ossa \(2014\)](#); (iv) extend the α -style core checks reported above ([§5.3](#)) to the γ -response outsider convention of [Chander \(2007\)](#) on both tracks, and impose non-member IR ($\widehat{W}_n \geq 1$ for $n \notin S$) directly inside the coalition-value program; (v) run the Monte-Carlo Shapley sampler at the full $N = 31$ panel and confront the resulting imputation with cross-RTA welfare-change moments à la [Manski \(1995\)](#); (vi) add a substantive comparison to the coalitional Nash bargaining of [Ray and Vohra \(2024\)](#) to clarify how the choice of cooperative solution concept moves the imputation; (vii) for the working-paper extension, engage the PTA-formation literature ([Limão \(2007\)](#); [Saggi and Yildiz \(2010\)](#)) on free-riding and stumbling-block effects; (viii) the agenda mismatch in the Track A bridge comparison ([§4](#)) runs in two directions. The coverage direction running through our own $v(S)$ lever — agriculture and tobacco outside BSY’s own 151-line bilateral negotiating structure, itself their estimation-sample screen (Canada negotiates zero lines under it) rather than the Uruguay Round’s actual agenda — is resolved, not merely bounded: predictions P-31/P-33/P-34 — re-purposing the same MPEC machinery P-30’s inconclusive first attempt used, weight-corrected to match this paper’s own dollar objective (MATH-83) and solved at four points (the 151-line and 294-line agendas, each at both a country-scalar and a line-by-line instrument) — close an *exact* additive identity anchored at the full-coverage frontier $D = 0.885046$: $D - \sum_k g_k^{HW} = 79.51\%$ of $v(\mathcal{N}_6)$, splitting into 62.24% from extending coverage from 151 to 294 lines at a matched line-by-line instrument (lines outside BSY’s own bilateral negotiating structure, an estimation-sample construct rather than the Uruguay Round’s actual agenda); 1.22% from the transferable-utility/ individually-rational wedge on the 151-line set, both measured against the TI-5-corrected $F^{\text{IR,corr}} = 0.428073$; and 16.06% (the TI-5-correct point estimate, using each member’s own line-by-line unilateral floor rather than a uniform zero; a uniform-zero floor instead gives a 16.12% upper bound, only 0.06 percentage points higher, [§4](#)) from BSY’s bargaining performance *within* its own nego-

tiated agenda falling short of the true, IR-constrained line-by-line efficient frontier (the three two-decimal components sum to 79.52% by independent rounding; exact total 79.51% from the unrounded values), $3\times$ the pre-registered materiality band and essentially unchanged by the individual-rationality restriction — though four of six members are worse off at that IR-feasible point than at BSY’s own cached bargain, so this is a TU-aggregate efficiency gap, not a demonstrated Pareto improvement at that point; a stronger stress test against BSY’s own actual outcome, rather than 1990, finds a *weak* Pareto improvement (four of six members exactly indifferent) survives regardless: 85% of the 1990-IR-constrained shortfall remains (§4). Coverage is worth 3.88 times the within-agenda shortfall on a regime-matched instrument, now a fully floor-corrected point estimate on both operands (the 294-line per-country IR-floor correction on the coverage side has been executed, moving the ratio down slightly from a previously reported $3.91\times$). Prediction P-29’s coarser country-scalar coverage figure, 57.9% of $v(\mathcal{N}_6)$, is *not* conflated with or superseded by this decomposition: it answers a well-posed, internally consistent question at the coarser instrument and remains on file alongside the finer, 62.2% line-instrument figure. The opposite direction — BSY’s $\hat{\zeta}$ is estimated from tariff concessions alone and cannot see TRIPS/GATS side payments, and this project’s own applied-tariff data shows agricultural protection did *not* fall over 1990–2000 in this sample (within-country median change positive for five of six countries, -0.31pp to $+11.77\text{pp}$, pooled-cell median $+1.45\text{pp}$ across all 54 country-sector cells; the earlier-cited $+7.35\text{pp}$ -to- $+35.91\text{pp}$ range is the country-level *mean*, sensitive to a small number of capped tariff lines, §4), so whatever the Agreement on Agriculture’s formula-cut track achieved on bound rates, it is invisible to, and not commensurable with, the applied-MFN channel either s_i^{HW} or our own lever uses (Ingco, 1996; Hoda, 2001) — remains unquantified and would need an external estimate of Uruguay Round non-tariff (TRIPS/GATS) concession magnitude; (ix) replace the forward-difference Jacobian used both in the $\hat{\zeta}$ re-estimation and in the Shapley-consistent weight-set search (§4.1.5) with an implicit-function-theorem analytic gradient through the Nash-in-Nash fixed point. The registered test of P-07’/P-08/P-09 is optimizer-layer INCONCLUSIVE (0/8 starts survive a single accepted step), and the reported closest-approach distance ($L_{\min} = 0.0034$, 5.8pp in share space) is an upper bound, not a certified minimum, and remains provisional: the author-authorized A0 comparator-gate re-verification against BSY’s cached equilibrium (§3.1) has since run under a three-gate protocol and aborted at its own first gate on its *first sweep* — the solver cannot reproduce BSY’s cached aggregate $\Sigma_k g_k^{HW}(\hat{\zeta}) = 0.312441$ within budget, for a reason that is a budget-scale mismatch rather than solely a probe-search pathology: at $\hat{\zeta}$, five of seven bilateral pairs classified as no deal and a single Gauss–Seidel sweep alone consumed 1627s of the 1800s chunk. An earlier 1800s-budget attempt at the symmetric placebo $\zeta = 0.5^7$ was killed on its second sweep while visibly converging ($\max|\Delta\text{mfn}|$ falling to 0.0017), which at the time we read only as evidence against a stalled solve, not as evidence of what it would converge to. An isolated single-anchor re-run with a 3600s budget and dedicated resources has since settled this: the $\zeta = 0.5^7$ Gauss–Seidel fixed point converges fully, in 1018s, to machine precision ($\max|\Delta\text{mfn}| = 5.085 \times 10^{-7}$ across three sweeps, with headroom to spare) — but the converged point lands on a *different* equilibrium branch than BSY’s cached comparator ($\Sigma_k g_k = 0.267337$ vs. the frozen 0.297081, a 10.0% relative gap against a 1% tolerance; share vector L_2 distance 0.0508 vs. a 0.005 toler-

ance), failing both gate criteria by roughly an order of magnitude past their tolerances (isolated single-anchor re-run, `zstar_gate1_half_isolated.md`). This anchor’s failure is therefore *determinate*, not a timeout artifact: convergence and comparator-match are logically distinct, and this re-run supplies the missing evidence that the inner solve converges cleanly to a stable point that simply is not BSY’s.

Three of the seven bilateral pairs at $\hat{\zeta}$ do sit within $\sim 2 \times 10^{-6}$ of the deal/no-deal threshold, evidence for the branch-instability diagnosis below on its own terms. GATE 1 now fails at both anchors by two independently diagnosed and distinct mechanisms: $\hat{\zeta}$ by insufficient budget (a genuine timeout, not yet distinguished from branch instability), and $\zeta = 0.5^7$ by a sufficient-budget, determinate convergence to the wrong branch. The two mechanisms are not the same finding restated twice, and no further GATE 1 re-run is indicated at either anchor.

Two hypotheses remain open and are not distinguished by this evidence: (H-A) BSY’s own Nash-in-Nash Gauss–Seidel sweep is genuinely branch-unstable near the deal/no-deal boundary, or (H-B) this project’s own re-implementation carries a systematic bias (a leading suspect, documented in the script itself, is that the wage guess resets to a cold baseline rather than warm-starting from the previous iterate). We pre-register, for R&R, a discriminating test: evaluate BSY’s own MATLAB `gains` routine directly on their cached tariffs at $\hat{\zeta}$ and compare it to this project’s own `gains_at` function at the identical point. Agreement in sign and magnitude would support (H-A) and strengthen the branch-instability reading used elsewhere in this paper; a systematic discrepancy would instead point to (H-B) and require a code-level fix before any further ζ^* -search attempt.

On a same-pipeline reading against this run’s own re-solve of $\hat{\zeta}$ ($L = 0.0381$, 19.5pp) the closest-approach ratio is $3.3\times$, not the cross-pipeline $2.4\times$ reported against BSY’s cached point (§4).

The obstacle is not merely finite-difference imprecision: the Nash-in-Nash fixed point takes a discrete deal/no-deal branch on each of the 151 negotiated lines, and two independent re-solves of the identical symmetric-placebo point return loss values differing by $\Delta L \approx 4.1 \times 10^{-4}$ — noise on the same order as the $h = 0.005$ forward-difference step used to assemble the Jacobian (matching BSY’s own `DiffMinChange`), so a branch flip inside the perturbation window can dominate the finite-difference signal, making $L(\zeta)$ a path-dependent step function at this resolution rather than a smooth objective (§4.1.5). This is consistent with the class of difficulty documented independently in BSY’s own replication package (which needed the same coarse `DiffMinChange` to begin with) and in this project’s own $\hat{\zeta}$ re-estimation; an analytic IFT gradient — or a step size large enough to average over branch flips — is the natural next step before this object can support a registered test.

(x) The two-economy contrast as a companion paper. Running the identical D1/Shapley machinery on two structurally distinct economies surfaced a cluster of findings that this paper reports as robustness checks but does not fully explain: the optimal policy is bang-bang (corner) on Track B but genuinely interior for two of Track A’s six players (Japan, Canada); the non-member welfare externality is overwhelmingly positive on Track B (83.7% of pairs) but only bare-majority positive on Track A (58.1% of pairs; 13 of 31 strictly negative, more than

two-and-a-half times Track B’s negative-externality rate); and the cooperative game is near-convex on Track B but fails superadditivity outright on Track A (20/301 disjoint pairs). All three contrasts point in the same direction — toward the negotiating coalition’s share of world economic activity as the underlying mechanism, since Track A’s six players are collectively far larger, relative to the rest of the world economy, than Track B’s — but we have not established this formally. We regard this cluster as the seed of a companion paper studying the contrast directly (varying coalition size and world-economy share parametrically, rather than across two fixed, differently-sized panels), rather than as an incidental robustness footnote to the present one.

Items (i) and (ii) require additional computational infrastructure, as does (ix) (an implicit-function-theorem gradient through the Nash-in-Nash fixed point); items (iii)–(iv), (vi), and (x) are within scope of the existing Python pipeline.

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A Proofs

This appendix collects the formal statements and proofs of every result used in the main text, together with an explicit ledger of what is *proved*, what is *proved under a maintained hypothesis that we do not establish*, and what remains *conjectural*. The ledger is in [Section A.11](#) at the end.

Notation reconciliation with Section 2. Notation follows the main text except for three declarations, each of which resolves a genuine collision and none of which may be dropped when this appendix is read alongside Section 2. (i) *Player set and its cardinality.* Where the main text writes N for both the player set and its cardinality, we write $\mathcal{N}$ for the set and $n \equiv |\mathcal{N}|$ for its cardinality. (ii) *The symbol n .* Section 2 uses n as a *country index* (as in $\widehat{W}_n, \tau_{ni}^j, s_n$); in this appendix n denotes *only* the cardinality $|\mathcal{N}|$, and country indices are i, j, k, m, h throughout. Where an expression inherited from the main text carries a country subscript n — for instance $\widehat{\kappa}_{ni}^j$ in (10) — the reading is the main text’s, and it is flagged at the point of use. (iii) *The symbol κ .* Section 2 uses $\kappa_{ni}^j = (1 + \tau_{ni}^j)d_{ni}^j$ for the bilateral trade wedge and $\widehat{\kappa}_{ni}^j$ for its hat-change. In this appendix the *unsubscripted* $\kappa \equiv w(\mathcal{N})/v(\mathcal{N})$ is the *cooperative ratio* of (A.11) — a different object, appearing only from [Section A.7](#) onward, and never carrying country–sector subscripts. To keep the two apart we also depart from the theory note `derivations/03` by renaming its corner-rate constant κ_{ij} to ρ_{ij} ([Proposition A.38](#)); κ_{ij} is not used anywhere in this appendix.

A.1 Primitives, assumptions, and the maintained hypothesis

The characteristic function (D1). For a coalition $S \subseteq \mathcal{N}$ write $\Sigma_S \equiv [0, 1]^{|S|}$ for the coalition’s lever box and $s_S = (s_m)_{m \in S} \in \Sigma_S$ for a lever profile; outsiders are held at the status quo, $s_m = 1$ for $m \notin S$, which we abbreviate $\mathbf{1}_{-S}$. Define the *coalitional objective*

$$f_S(s_S) \equiv \sum_{k \in S} I_k^{\text{base}} [\widehat{W}_k(s_S, \mathbf{1}_{-S}) - 1], \quad s_S \in \Sigma_S, \quad (\text{A.1})$$

and the canonical (D1) characteristic function

$$v(S) = \max_{s_S \in \Sigma_S} f_S(s_S), \quad v(\emptyset) = 0, \quad (\text{A.2})$$

which is eq. (7) of the main text. There is no bargaining weight, no Nash product, and no individual-rationality constraint inside (A.2) (ADR-1). The *cooperative game* is the zero-

normalization

$$w(S) \equiv v(S) - \sum_{k \in S} v(\{k\}), \quad w(\{k\}) = 0 \text{ by construction}, \quad (\text{A.3})$$

and the Shapley value of a TU game u on $\mathcal{N}$ is

$$\phi_i(u) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [u(S \cup \{i\}) - u(S)] = \mathbb{E}_\pi [u(P_\pi^{<i} \cup \{i\}) - u(P_\pi^{<i})], \quad (\text{A.4})$$

the second expression being the expectation over a uniformly random arrival order π , with $P_\pi^{<i}$ the set of players preceding i .

Assumptions invoked. The canonical registry is `paper/theory/00_assumptions.md`; we restate here only what the proofs actually use, with the labels of that registry.

(A1) GE primitives. The world economy is the multi-country, multi-sector Eaton–Kortum model with input–output linkages of [Caliendo and Parro \(2015\)](#), with welfare $W_k = (w_k L_k + R_k + D_k)/P_k$ and hat-algebra system (1)–(5). The D1 objective (A.1) sums *member* welfare only; the welfare of non-players is a reported diagnostic, not part of the maximand.

(A2) Selected-equilibrium continuity, *as corrected on 2026-08-09*. For each S and each $s_S \in \Sigma_S$ the hat-algebra system admits a solution on **the branch selected by the damped fixed-point iteration cold-started from the base-year data** (`cp2015_value_solve_welfare_at_scaling`, damping `ufactor` $\in (0.2, 0.05)$ retry schedule), and the induced map $s_S \mapsto \widehat{W}_k(s_S, \mathbf{1}_{-S})$ is continuous on Σ_S for every $k \in \mathcal{N}$ *along that selection*. This is a stipulation, not a theorem: [Álvarez and Lucas \(2007\)](#) is single-sector, with a tariff wedge but without intermediates or deficits (**corrected 2026-08-16, literature-curator Round 4, #7**: an earlier draft of this sentence stated Álvarez–Lucas (2007) as “without tariffs” — this reversed the paper’s own scope; its abstract establishes existence and uniqueness of an equilibrium with balanced trade *where each country imposes an import tariff*, so the tariff wedge is a feature their proof covers, not one it excludes; the overall argument this sentence supports — that the nearest single-sector precedent with a uniqueness proof does not extend to this paper’s multi-sector CP2015 framework — is unaffected, only the single-sector precedent’s feature list changes from “excludes tariffs” to “includes tariffs”), and [Caliendo and Parro \(2015\)](#) do not prove multi-sector uniqueness. Branch selection is in principle damping-dependent; the measured cross-damping discrepancy is 2.5×10^{-5} relative and the independent re-solve agreement is bitwise on all 127 coalitions (`paper/replication/results/noise_floor_eps_B.md`, strands 1–2). Every result below that depends on A2 is stated *along the selected branch*.

(A3) Policy space. $s_m \in [0, 1]$ for members (no tariff-raising threat, no import subsidies); outsiders at $s_m = 1$ by the A4 convention. Hence Σ_S is a non-empty compact box.

(A4) Outsider convention. $v(\cdot)$ is a projection of a partition function; the projection chosen is the status quo, outsiders at $s_m = 1$.

(A5) Transferable utility within a coalition. $v(S)$ is dollar-denominated and intra-coalition

transfers are feasible. Under D1 this is a *precondition for the characteristic function to have meaning*, not a convenience: if surplus were non-transferable the utilitarian sum would not represent any single allocation the coalition can reach.

(A6) Player set. The bridge’s player set is the full six BSY negotiating countries $\mathcal{N}_6$; NES/ROW aggregates are non-player GE filler.

(A7) Threat point. The disagreement point is the 1990 applied tariff vector — the same convention as Bagwell et al. (2021).

(A8) Share-language preconditions. (a) *raw*: $v(\mathcal{N}) > 0$ and $\phi_i(v) \geq 0$ for all i ; (b) *co-operative*: $w(\mathcal{N}) > 0$ and $\phi_i(w) \geq 0$ for all i (equivalently imputation IR, $\phi_i(v) \geq v(\{i\})$); (c) *comparator side*: $\sum_k g_k > 0$ and $g_i \geq 0$ for all i . A8-b $\Rightarrow$ A8-a. If a tier fails, the corresponding object is reported in dollar-gap form via Proposition A.36 rather than as a share.

(A9) γ -equilibrium existence. In the γ -response robustness branch only, a Nash equilibrium among the outsiders exists on $[0, 1]^{|\mathcal{N} \setminus S|}$. Not proved; a stipulation confined to the robustness fork.

The one hypothesis we cannot discharge — and, on Track A, have refuted. Several results below — imputation IR, strict positivity of cooperative surplus for non-null players, and the upper half of the fairness-gap interval — require that adding a single player never destroys cooperative surplus. We isolate it once, name it, and carry it explicitly in the statement of every result that uses it. Its empirical status is *track-split*: maintained (supported at 100%) on Track B, **refuted by counterexample on Track A** — see Proposition A.2(iii’); Track A results that would otherwise rest on it are instead established by direct verification.

Maintained Hypothesis A.1 (Single-player w -monotonicity). For every $S \subseteq \mathcal{N}$ and every $i \notin S$,

$$w(S \cup \{i\}) \geq w(S), \quad \text{equivalently} \quad \Delta_i^v(S) \equiv v(S \cup \{i\}) - v(S) \geq v(\{i\}). \quad (\text{A.5})$$

Remark A.2 (Status of Proposition A.1: empirically supported, not proved). Four points must travel with every use of Proposition A.1 (count updated 2026-08-12 on the addition of (iii’)). (i) *It is not general superadditivity.* (A.5) is the restriction of $v(S \cup T) \geq v(S) + v(T)$ to singleton $T = \{i\}$; it is implied by, but strictly weaker than, superadditivity, and the two must not be joined by “ $\equiv$ ”. (ii) *It is not implied by any result in this paper.* In particular the replication lower bound of Proposition A.12 does not deliver it: its w -analogue is generically a large negative number for high-tariff countries (Proposition A.14). (iii) *It is empirically supported at 100% on the current D1 value tables* — 2059/2059 single-player additions at $N = 7$ and 3019/3019 pooled over $N = 3, \dots, 7$, with the companion general superadditivity check passing 966/966 at $N = 7$ (paper/replication/results/superadditivity_sweep_D1.md). The binding margins exceed the measured numerical noise floor $\varepsilon_B \approx 2.5 \times 10^{-5}$ relative — converting to dollars via $\varepsilon_B \cdot |v(\mathcal{N})|$, e.g. $\approx \$4.46\text{M}$ at $N = 7$ ($v(\mathcal{N}) = \$178.4\text{B}$) — by **two to three orders of magnitude**, not five: the $N = 7$ cushion of $\$0.683\text{B}$ clears it by $\approx 153\times$ and the

global binding $N = 3$ cushion of \$0.266B clears its own ($N = 3$ -scaled) floor of $\approx \$0.195M$ by $\approx 1363\times$ (`noise_floor_eps_B.md`, corrected 2026-08-09) — both multiples clear the $100\times$ structural-evidence threshold declared in [Proposition A.22](#) (MATH-57(a), added 2026-08-14; the two remarks apply one standard, not two). This is strong evidence and is not a proof; [Proposition A.1](#) is therefore a *maintained hypothesis* throughout, and every result conditioned on it says so in its own statement. (iii') *Track-A refutation* — added 2026-08-11 (Round 11, MATH-46; *supersedes any reading of (iii) as track-free*). Point (iii) is a **Track B** statement. On **Track A** (the BSY 1990 six-country economy, cuts-only $s \in [0, 1]^6$ per TI-13(d)), the full 63-coalition continuous-optimization sweep *refutes* [Proposition A.1](#): 17/192 single-player additions violate (A.5) (min margin -0.026338 , i.e. 3.7% of $v(\mathcal{N}_6)$) and general superadditivity fails 20/301, with the failure already present at the pairwise level, $w(\{US, CAN\}) = -0.005591$. **Retracted claim (2026-08-18, TEST-17-11; matching the parallel correction in theory-invariants.md TI-12):** an earlier version of this remark asserted “every violation has US and Canada in the same coalition without Japan or South Korea.” That was the initial, qualitative signature only, and it is refuted as a universal claim by the full 192-addition census: 8 of the 17 violations fire *without* it (e.g. Canada joining {South Korea, US}). The tighter, single-country characterization that survives is that **Canada is present in 16 of the 17 violations** — only US joining {Australia, EU} has no Canada anywhere in the coalition (`paper/replication/results/bsy\_replication/p35\_coalition\_panel.md`) (`paper/replication/results/bsy\_replication/phase7\_full\_primary.md` §2.1). Consequently *no Track A result may be argued via [Proposition A.1](#)*; the results that would otherwise rest on it survive there by *direct verification* — $\phi_i(v) \geq v(\{i\})$ at every player with minimum slack 0.018632 (2.59% of $v(\mathcal{N}_6)$) and $w(\mathcal{N}_6) = 0.677 > 0$ — which suffices because the hypothesis is sufficient, not necessary, for those conclusions (point (i)). The theoretical status therefore sharpens from “unproven” to “**false in general for this class of games**”: Track A furnishes counterexamples, and Track B’s 100% support is a property of that economy (a small bloc facing passive outsiders), not of the model class.

A.2 Well-definedness of $v(\cdot)$

Proposition A.3 (Existence, finiteness, sign, and the empty-coalition convention). *Assume A1–A4 and fix the equilibrium selection of A2. Then for every non-empty $S \subseteq \mathcal{N}$:*

- (a) (Existence) *the program (A.2) attains its maximum: there exists $s_S^* \in \Sigma_S$ with $f_S(s_S^*) = v(S)$, so $v(S)$ is a maximum and not merely a supremum;*
- (b) (Finiteness and sign) *there is $M_S < \infty$ with $0 \leq v(S) \leq (\sum_{k \in S} I_k^{base}) M_S < \infty$; the lower bound is constructive and exact;*
- (c) (Convention) $v(\emptyset) = 0$, *matching the characteristic-function-form convention of [Shapley \(1953\)](#).*

All statements hold along the branch selected in A2.

Proof of [Proposition A.3](#). Step 1 (the feasible set is compact). By A3 the lever box is $\Sigma_S = [0, 1]^{|S|}$, a non-empty closed bounded subset of $\mathbb{R}^{|S|}$, hence compact by Heine–Borel. Note that no individual-rationality constraint carves a subset out of Σ_S : under D1 the whole box is

feasible, which is what removes the ϵ -buffer construction that the retired Nash-product definition required.

Step 2 (the objective is continuous). By A2, for each $k \in S$ the map $s_S \mapsto \widehat{W}_k(s_S, \mathbf{1}_{-S})$ is continuous on Σ_S along the selected branch. The objective f_S in (A.1) is a finite linear combination of these maps with constant coefficients I_k^{base} , hence continuous on Σ_S . Two features of (A.1) matter here and are worth stating because they are exactly what the retired Nash-product objective lacked: f_S contains no logarithm, so there is no singularity at a boundary of a feasible set, and f_S is additive rather than multiplicative, so it does not vanish identically on a face.

Step 3 (existence). A continuous real function on a non-empty compact set attains its maximum (Weierstrass). Applying this to f_S on Σ_S gives (a).

Step 4 (bounds). By Step 2 and compactness, each $\widehat{W}_k(\cdot)$ is bounded on Σ_S : there is $M_S < \infty$ with $|\widehat{W}_k(s_S) - 1| \leq M_S$ for all $s_S \in \Sigma_S$ and $k \in S$. Hence $v(S) = f_S(s_S^*) \leq \sum_{k \in S} I_k^{\text{base}} M_S < \infty$. For the lower bound, the status-quo lever $s_S = \mathbf{1}$ lies in Σ_S by A3 and reproduces the base-year tariff vector exactly; the identity counterfactual returns $\widehat{W}_k = 1$ for every k , so $f_S(\mathbf{1}) = 0$ exactly. Since s_S^* maximizes f_S , $v(S) = f_S(s_S^*) \geq f_S(\mathbf{1}) = 0$. This is a constructive non-negativity argument: it uses a named feasible point, not a sign restriction on the primitives. Its exactness — that $f_S(\mathbf{1})$ is 0 and not merely ≈ 0 — is enforced numerically by assertion (invariant TI-15; `paper/replication/results/welfare_baseline_bug.md`), the identity counterfactual returning $\max |\widehat{W} - 1| = 0.00\text{e}+00$ (`noise_floor_eps_B.md`, strand 3). This proves (b).

Step 5 (empty coalition). For $S = \emptyset$ there are no decision variables and (A.2) is vacuous; we adopt $v(\emptyset) = 0$ as a definitional convention, which is the convention under which the Shapley formula (A.4) is stated. $\square$

Corollary A.4 (The value is selection-free; argmax-valued objects are not). Under the hypotheses of Proposition A.3, the argmax correspondence $S \mapsto \arg \max_{\Sigma_S} f_S$ may be multi-valued, but every maximizer attains the same value $v(S)$; hence $v(\cdot)$, and therefore $w(\cdot)$, $\phi(v)$, $\phi(w)$ and every object built from them, are independent of which maximizer a solver returns. Objects that are functions of the maximizing *point* — the outsider welfare table $\{\widehat{W}_m(s_S^*)\}_{m \notin S}$, the non-player terms-of-trade diagnostic, and the right-hand side of Proposition A.12 — inherit an ordinary numerical branch dependence, which must be reported with the solver protocol.

Proof of Proposition A.4. Immediate from the definition of $\arg \max$: if s^*, s^{**} both maximize f_S then $f_S(s^*) = f_S(s^{**}) = v(S)$. The contrast with the retired construction is the substantive content: there the reported number was the dollar sum f_S evaluated at the maximizer of a *different* (Nash-product) objective, so two maximizers tying on the Nash product could report different dollar values. Under D1 objective and payoff coincide, and the indeterminacy disappears. $\square$

Remark A.5 (A benefit of the corrected A2). The corrected (cold-start) form of A2 makes $\widehat{W}$ a function of the *world lever vector alone*: the solver is re-initialized from the base-year data at every s and never warm-started along a path. Two proofs below rely on precisely this. First, $v(S)$ is well defined as a function of S : had the branch been selected by homotopy *within* Σ_S , the selected equilibrium at a given world tariff vector could in principle have depended on the

path, hence on which coalition was optimizing. Second, the replication step in [Proposition A.12](#) compares $\widehat{W}_k$ at the *same* world tariff vector reached from two different programs; equality there is an identity under the corrected A2, whereas under a homotopy reading it would have required an additional path-independence argument. The cost is that branch selection is damping-dependent in principle; the live monitors are multi-damping agreement (2.5×10^{-5} relative) and independent re-solve agreement (127/127 bitwise), both in `noise_floor_eps_B.md`.

A.3 Bang-bang structure and exactness of the grid

The D1 program has no interior optimum anywhere in the current evidence: the persisted census records 127/127 coalitions with every coordinate at a bound (448/448 coordinates, 409 at $s = 0$ and 39 at $s = 1$, zero interior, at tolerance 10^{-6} ; `paper/replication/results/outsider_welfare_N7.json`, `san01_census`). The following proposition states the structural condition under which that pattern is the *necessary* form of the solution, and derives the payoff that matters computationally: grid search over a grid containing the vertices is then exhaustive and exact.

Definition 4 (Coordinatewise strict monotonicity). Coalition S satisfies condition **BB** if for every member $m \in S$ and every fixed profile $s_{-m} \in [0, 1]^{|S|-1}$ of the other members' levers, the section $s_m \mapsto f_S(s_m, s_{-m})$ is strictly monotone on $[0, 1]$. The direction of monotonicity is allowed to depend on m and on s_{-m} . Condition **BB**⁺ strengthens BB by requiring that, for each m , the *direction* of monotonicity does not depend on s_{-m} .

Proposition A.6 (Bang-bang optima and exactness of an endpoint-containing grid). *Assume A1–A4, fix the A2 selection, and let $V_S \equiv \{0, 1\}^{|S|}$ denote the vertex set of Σ_S . If S satisfies BB, then:*

- (a) *every maximizer of (A.2) lies in V_S ; no interior or mixed optimum exists;*
- (b) *$v(S) = \max_{s \in V_S} f_S(s)$; consequently, for any finite grid $G \subseteq \Sigma_S$ with $V_S \subseteq G$, the grid-search value $\max_{s \in G} f_S(s)$ equals $v(S)$ exactly, so grid search is exhaustive and exact and no local polish step can improve on it;*
- (c) *if in addition S satisfies **BB**⁺, the maximizer is unique and is the vertex $s^* = (s_m^*)_{m \in S}$ with $s_m^* = 0$ when f_S is decreasing in s_m and $s_m^* = 1$ when it is increasing — i.e. the optimum is determined coordinatewise, and the model's content is participation (which members liberalize) rather than concession depth.*

Proof of Proposition A.6. Step 1 (existence). A maximizer exists by [Proposition A.3\(a\)](#); let s^* be one.

Step 2 (no interior coordinate). Suppose, for contradiction, that $s_m^* \in (0, 1)$ for some $m \in S$. By BB the section $g(\cdot) \equiv f_S(\cdot, s_{-m}^*)$ is strictly monotone on $[0, 1]$. If g is strictly increasing then $g(1) > g(s_m^*)$; if strictly decreasing then $g(0) > g(s_m^*)$. Either way there is a feasible point — namely s^* with coordinate m replaced by the corresponding endpoint — with strictly higher objective value, contradicting optimality of s^* . Hence $s_m^* \in \{0, 1\}$ for every $m \in S$, i.e. $s^* \in V_S$, proving (a).

Step 3 (grid exactness). By (a) the maximum over Σ_S is attained on V_S , so $v(S) = \max_{\Sigma_S} f_S = \max_{V_S} f_S$. If $V_S \subseteq G \subseteq \Sigma_S$ then $\max_{V_S} f_S \leq \max_G f_S \leq \max_{\Sigma_S} f_S$, and the two ends coincide, giving $\max_G f_S = v(S)$. This proves (b). Note that (b) converts what is otherwise a discretization-error concern (invariant TI-10) into a closure argument: under BB the discretization is not an approximation at all.

Step 4 (uniqueness under BB^+). Under BB^+ the direction of monotonicity in coordinate m is a property of m alone. Write $e_m = 1$ if f_S is increasing in s_m and $e_m = 0$ if decreasing. Step 2 applied at any maximizer forces $s_m^* = e_m$ for every m , so the maximizer is the single vertex $(e_m)_{m \in S}$, proving (c). $\square$

Remark A.7 (Status of condition BB: swept for singletons, inferred for $|S| \geq 2$ — **Track B only; refuted on Track A**). We do *not* claim BB is verified for $|S| \geq 2$. What is on record is the following, and the distinction is deliberate. **Scope correction (2026-08-09, MATH-22, model-refiner batch fix item 1):** everything in this remark below the mechanism paragraph is evidence about *Track B* (the seven-country Caliendo–Parro economy). On *Track A* (the BSY six-country 1990 economy) condition BB is **refuted**, not merely unswept, and the mechanism argument below does not survive as a general argument — see the Track A paragraph after the mechanism paragraph.

Mechanism. The economic reason to expect BB is that the utilitarian aggregate internalizes intra-coalition terms-of-trade transfers: member m ’s cut moves rents between m and its coalition partners, and those movements cancel inside $\sum_{k \in S} I_k^{\text{base}}(\widehat{W}_k - 1)$, leaving a net coalition-level effect whose sign is plausibly constant over the box. A member therefore either liberalizes fully or not at all. This mechanism is refuted as a general argument by the Track A evidence below; it survives only as a fact about *Track B’s own calibration*, not as a structural property of D1 objectives in general. The economic reason the two tracks differ is itself instructive (Bagwell and Staiger, 1999; Ossa, 2014): Track B’s seven players are a small minority of the 31-region modelled world, so a member’s terms-of-trade losses from full liberalization spill mostly onto non-player regions, leaving the coalition-level objective monotone in each member’s own lever (bang-bang); Track A’s six players are themselves the bulk of the modelled 1990 world economy, so intra-coalition terms-of-trade transfers largely cancel and what remains is the coalition’s market power over a small non-player fringe — the classical result that a bloc with market power retains a strictly positive optimal external tariff, i.e. an interior optimum. This is a genuine two-economy contrast turning on the coalition’s share of world economic size, not an inconsistency between the two tracks.

Track A counterexamples (refutation, not merely non-confirmation). On Track A the refutation is now carried by the *certified continuous optimum* (evidence upgraded 2026-08-11, Phase 7, per MATH minor 9): the grand-coalition optimum is the strongly asymmetric interior point on five of six coordinates — **Korea sits exactly at the zero boundary**, $s_{\text{Korea}}^* = 0$; **the registered strong form of prediction P-19 failed on that coordinate**, and “interior” here is the post-hoc weak form (corrected 2026-08-14, TEST-13-02) — $s^* = [0.533, 0.472, 0.027, 0.326, 0.111, 0.000]$ with $v(\mathcal{N}_6) = +0.720163$, beating the best vertex of $\{0, 1\}^6$ (+0.613241) by 17.4% (paper/replication/results/bsy_replication/phase7_full_primary.md, 63/63 coalitions converged, L-BFGS-B with exact IFT gradients); the

continuous singleton optima confirm Japan ($s^* = 0.514$) and Canada ($s^* = 0.652$) interior. The original detection — the uniform probe $s = 0.25^6$ at $+0.644756$ already beating every vertex, with all 144 solves converged (`paper/replication/results/bsy\_replication/d33c\_screen.md`, git 4dc362b, since superseded on the values though not on the finding) — is retained as the historical first evidence; the continuous optimum strengthens it strictly. Condition BB’s mechanism paragraph above is therefore refuted as a *general* argument for D1 objectives: it does not predict Track A, where two of six singletons are strictly interior on a resolution (Canada’s $s^* \approx 0.65$) that a 5-node grid would have missed entirely. **Any future Track A computation of $v(S)$ (Phase 7 and beyond) must use continuous optimization, not vertex census or grid search**; a vertex-based method would silently understate $v(S)$ on Track A.

Direct evidence, singletons (Track B, upgraded 2026-08-09 — an 11-node recheck, not a corner-sign check). An earlier version of this remark reported only that six of seven Track B singletons were “signed at $s = 0$ ” at a coarse resolution. That evidence is superseded by a stronger result: a dedicated 11-node (0.1-resolution) recheck of every Track B singleton and three grand-coalition axes finds **VERTEX_CENSUS_SURVIVES** — all seven singletons endpoint-optimal (USA and Canada and Korea at $s = 1$; Mexico, China, India and Brazil at $s = 0$) and non-monotone-interior-free, and all three grand-coalition axes (USA, China, India) monotone non-increasing with $\arg \max s = 0$, 110/110 GE solves converged, reading rules pre-registered in the script docstring *before* the run (`paper/replication/results/bb\_recheck\_trackB.md`, git 4dc362b). Per that file’s own pre-registered scope note: a clean pass at 11 nodes does not *prove* BB (fine structure between nodes could still be non-monotone; an interior win at any node would have refuted it outright, and none did) — so [Proposition A.6](#)’s conditionality on the current Track B evidence still stands, but the evidentiary base is now a resolution an order of magnitude finer than the corner-sign check it replaces, at exactly the resolution (0.1) that was sufficient to expose Track A’s Canada counterexample. The dollar values at the coarser corners agree with the finer sweep — positive at $s = 0$ for Mexico (\$1.94B), China (\$119.68B) and Brazil (\$1.67B), negative at $s = 0$ for the USA (−\$2.21B), Canada (−\$0.13B) and Korea (−\$0.38B), whence $v(\{\text{USA}\}) = v(\{\text{Canada}\}) = v(\{\text{Korea}\}) = 0$ (`paper/replication/results/india\_existence\_boundary.md` §2, the six-player row; these six values agree exactly with the v3 tables). The seventh singleton, India, was swept over nine nodes in the same table, but that sweep predates the damping fix: three of its nine nodes failed to converge and its recorded values belong to the quarantined chain (the sweep’s \$17.071B and \$5.927B against the post-fix $v(\{\text{India}\}) = \$26.411\text{B}$). It is therefore cited here as **pre-fix directional evidence only** — it indicates a monotone decreasing profile but cannot be used as a converged sweep. A damped re-sweep of the India singleton (≈ 9 GE solves) is queued; the 11-node recheck above does not cover India (it targets the six singletons and axes that were already corner-consistent at coarse resolution), so India remains the one Track B singleton without either coarse- or fine-resolution converged sweep evidence.

Direct evidence, $|S| \geq 2$: what the executed grand-coalition sweep does and does not establish. A one-dimensional sweep at the $N = 7$ recorded optimum has been run (`paper/replication/results/rsan04_grand_coalition_sweep.json`): two axes (USA and China), 11 nodes each

on $\{0, 0.1, \dots, 1\}$, the other six members held at $s = 0$, 22 GE solves, both axes strictly monotone non-increasing with zero violations and $\arg \max s = 0$ on both (f_N falls from \$178.354B at $s_{USA} = 0$ to \$169.873B at $s_{USA} = 1$, and to \$46.784B at $s_{China} = 1$). What this establishes is precise and worth having: *the recorded vertex $s^* = \mathbf{0}$ is a global maximum along two of the seven grand-coalition axes*, so the optimizer’s $\arg \max$ is not a resolution artifact on those axes. What it does *not* establish is condition BB for $|S| \geq 2$. [Definition 4](#) quantifies over *every* member $m \in S$ and *every* fixed profile $s_{-m} \in [0, 1]^{|S|-1}$; the sweep fixes one profile ($s_{-m} = \mathbf{0}$), for two members, in one coalition out of 127 — two of 448 recorded coordinates, each at a single point of a continuum. BB for $|S| \geq 2$ therefore remains **inferred from the mechanism, not swept**, which is finding RSAN-04 of `quality_reports/model/shapley-tariff-v2/sanity_report_rerun.md` in its correct scope.

Indirect evidence. The full census is consistent with BB at every size: 127/127 coalitions are at vertices, including 35 of size 3, 35 of size 4, and the grand coalition, and the interior grid node $s = 0.5$ — which was on the grid and was selected 22 times before the solver fix — is selected zero times afterwards. Consistency is not implication: a vertex optimum can also arise without strict monotonicity (for instance if f_S is quasi-convex in a coordinate), so the census supports BB but does not establish it.

What would close it, and what follows meanwhile. No finite sweep can discharge BB, because the condition quantifies over a continuum of profiles s_{-m} for each of $|S|$ members: sweeps can only accumulate axis-wise counterexample-free evidence, and a proof would require signing $\partial f_S / \partial s_m$ over the whole box, which A2’s bare continuity does not support (see [Proposition A.8](#)). Two things follow, and the draft must respect both. First, [Proposition A.6](#) is used throughout as a *conditional* structural statement; the census and the two swept axes are its empirical counterparts, never its proof. Second — and this is the operative consequence — [Proposition A.6\(b\)](#)’s grid-exactness corollary, which is what would dissolve invariant TI-10 and license reporting grid-3 values as exact rather than approximate, inherits that conditionality in full. Until BB is established for $|S| \geq 2$, the correct manuscript wording is that the grid contains every vertex and therefore returns the exact value *under BB*, with the cross-grid agreement (grid 3 versus grid 5, identical on all non-India coalitions) reported as corroborating evidence. The unconditional sentence “the discretization is not an approximation at all” is not available on the current evidence.

Remark A.8 (No interior first-order condition exists). Because all 448 recorded lever coordinates sit at bounds, the D1 program has no interior stationary point anywhere in the evidence, and there is consequently no interior first-order condition on which to hang an implicit-function-theorem comparative-statics apparatus at the lever level. We therefore do not derive $\partial s^* / \partial \theta$ for any primitive θ : the object does not exist. The comparative statics that remain well defined, and that the paper uses, are of four kinds: value-level statements obtained from the maximum theorem and the envelope property of [\(A.2\)](#); compositional statements comparing ϕ across player sets (cross- N); threat-point statements (the alternative-threat-point exercise); and statements in the bargaining-weight ζ over the level set of [Proposition A.43](#). The cooperative ratio κ of [Proposition A.23](#) is itself the sharpest comparative-static object in the paper, since by [Proposition A.27](#) it measures how much any conclusion can depend on the choice of allocation

rule.

A.4 Normalization and the exact unilateral-cooperative decomposition

Proposition A.9 (Singleton values and the cooperative normalization). *Assume A1–A4 and fix the A2 selection. Then:*

- (a) $v(\{k\}) \geq 0$ for every k , with $v(\{k\})$ equal to k 's welfare gain from its own unilateral optimal liberalization;
- (b) $v(\{k\}) = 0$ exactly if and only if $s_k = 1$ (no cut) is a maximizer of the singleton program; in particular a country whose applied schedule is already at or below its unilateral optimum has $v(\{k\}) = 0$ exactly, not approximately;
- (c) the cooperative game w in (A.3) is well defined, satisfies $w(\emptyset) = w(\{k\}) = 0$ for all k , and delivers the exact level decomposition $v(\mathcal{N}) = \sum_{k \in \mathcal{N}} v(\{k\}) + w(\mathcal{N})$.

Proof of Proposition A.9. (a) is Proposition A.3(b) applied to $S = \{k\}$, together with the reading of the singleton program: with $S = \{k\}$ the only lever is k 's own, all other countries are at the status quo, and the objective is $I_k^{\text{base}}(\widehat{W}_k - 1)$, i.e. k 's own welfare gain.

(b) The “if” direction: if $s_k = 1$ is a maximizer, then $v(\{k\}) = f_{\{k\}}(1) = 0$ exactly by Step 4 of the proof of Proposition A.3. The “only if” direction: if $v(\{k\}) = 0$ then $s_k = 1$ attains the maximum, since $f_{\{k\}}(1) = 0 = v(\{k\})$, so it is a maximizer. The exactness in both directions is the content of TI-15 — the identity counterfactual must return exactly zero welfare change, an invariant now enforced by assertion. On the current value tables $v(\{\text{USA}\}) = v(\{\text{Canada}\}) = v(\{\text{Korea}\}) = \0.000B exactly, while $v(\{\text{Mexico}\}) = \$1.941\text{B}$, $v(\{\text{Brazil}\}) = \$1.669\text{B}$, $v(\{\text{India}\}) = \$26.411\text{B}$ and $v(\{\text{China}\}) = \$119.681\text{B}$ (superadditivity_sweep_D1.md, §3, $N = 7$ panel).

(c) w is a finite difference of finite numbers by Proposition A.3(b), hence well defined; $w(\{k\}) = v(\{k\}) - v(\{k\}) = 0$ by construction and $w(\emptyset) = 0$; evaluating (A.3) at $S = \mathcal{N}$ and rearranging gives the level decomposition. $\square$

Proposition A.10 (Exact unilateral-cooperative decomposition of the Shapley value). *Let a denote the additive game $a(S) = \sum_{k \in S} v(\{k\})$, so that $w = v - a$. Then for every $i \in \mathcal{N}$*

$$\phi_i(v) = v(\{i\}) + \phi_i(w), \quad \sum_{i \in \mathcal{N}} \phi_i(w) = w(\mathcal{N}), \quad \sum_{i \in \mathcal{N}} \phi_i(v) = v(\mathcal{N}). \quad (\text{A.6})$$

The decomposition is exact, not approximate, and holds regardless of whether v is monotone, superadditive or convex.

Proof of Proposition A.10. Step 1 (Shapley value of an additive game). For the additive game a and any $S \not\ni i$, $a(S \cup \{i\}) - a(S) = \sum_{k \in S \cup \{i\}} v(\{k\}) - \sum_{k \in S} v(\{k\}) = v(\{i\})$, a constant independent of S . Substituting into (A.4),

$$\phi_i(a) = v(\{i\}) \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} = v(\{i\}),$$

because the Shapley weights are the probabilities of the events $\{P_\pi^{<i} = S\}$ under a uniform random order and therefore sum to one.

Step 2 (linearity). The Shapley value is linear in the game: from (A.4), $\phi_i(\cdot)$ is a fixed finite linear combination of the game's values, so $\phi_i(v) = \phi_i(a + w) = \phi_i(a) + \phi_i(w)$. Combining with Step 1 gives the first identity in (A.6). (Linearity here is the algebraic property of the formula; the additivity *axiom* is the normative statement that jointly with efficiency, symmetry and the dummy property characterizes ϕ uniquely (Shapley, 1953).)

Step 3 (efficiency). Summing (A.4) over i and using the random-order representation, $\sum_i \phi_i(u) = u(\mathcal{N})$ for any game u with $u(\emptyset) = 0$, because for each fixed order π the marginal contributions telescope to $u(\mathcal{N}) - u(\emptyset)$. Applying this to $u = w$ and to $u = v$ gives the two remaining identities; the two are mutually consistent by Proposition A.9(c). $\square$

Remark A.11 (Numerical check of efficiency). Invariant TI-1 ($\sum_i \phi_i = v(\mathcal{N})$) is verified directly on the computed imputation: the efficiency gap at $N = 7$ is 3.05×10^{-5} dollars against $v(\mathcal{N}) = \$178.354\text{B}$ (paper/replication/results/extended_shapley/N7_D1.json, field `efficiency_gap_abs`), i.e. at the level of double-precision accumulation and far below the numerical noise floor.

A.5 Marginal contributions: the replication bound and the mechanical near-dummy

Lemma A.12 (R6 replication lower bound). Assume A1–A4 and fix the A2 selection. For every coalition S and every $i \notin S$, let s_S^* be any maximizer of the S -program and define the *outsider gain*

$$\text{og}_i(S) \equiv I_i^{\text{base}}[\widehat{W}_i(s_S^*, \mathbf{1}_{-S}) - 1], \quad (\text{A.7})$$

country i 's passive welfare gain when S liberalizes optimally and i does nothing. Then

$$\Delta_i^v(S) = v(S \cup \{i\}) - v(S) \geq \text{og}_i(S). \quad (\text{A.8})$$

The inequality holds *for every* choice of maximizer s_S^* , and hence for the infimum of og_i over the argmax set; the notation $\text{og}_i(S)$ suppresses a selection dependence that Proposition A.4 flags and that Proposition A.16 handles explicitly.

Proof of Proposition A.12. Step 1 (a feasible point for the larger program). The profile $(s_S^*, s_i = 1) \in \Sigma_{S \cup \{i\}}$ is feasible for the $S \cup \{i\}$ -program: it replicates S 's optimal levers and adds i passively at the status quo. Hence

$$v(S \cup \{i\}) \geq \sum_{k \in S} I_k^{\text{base}}[\widehat{W}_k(s_S^*, s_i=1, \mathbf{1}_{-S \cup \{i\}}) - 1] + I_i^{\text{base}}[\widehat{W}_i(s_S^*, s_i=1, \mathbf{1}_{-S \cup \{i\}}) - 1].$$

Step 2 (the two world tariff vectors coincide). Under the S -program the world lever vector is $(s_S^*, \mathbf{1}_{-S})$, in which i — an outsider to S — carries $s_i = 1$. Under the point of Step 1 the world lever vector is $(s_S^*, s_i=1, \mathbf{1}_{-S \cup \{i\}})$. These are the *same* vector. By A2 in its corrected (cold-start) form, the selected equilibrium is a function of the world lever vector alone, so $\widehat{W}_k$ takes the same value in both, for every $k \in \mathcal{N}$. (See Proposition A.5: this step is an identity

under the corrected A2, whereas a homotopy-based selection would have required a separate path-independence argument.)

Step 3 (conclusion). By Step 2 the first sum in Step 1 equals $f_S(s_S^*) = v(S)$ and the second term equals $\text{og}_i(S)$. Hence $v(S \cup \{i\}) \geq v(S) + \text{og}_i(S)$, which is (A.8). $\square$

Corollary A.13 (Sufficient condition for monotonicity of v ; the converse fails). If $\text{og}_i(S) \geq 0$ for every pair (S, i) with $i \notin S$, then v is monotone: $\Delta_i^v(S) \geq 0$ for all such pairs. The converse does *not* hold — v can be monotone at pairs where the outsider gain is negative, because (A.8) is a lower bound and not an equality. (An earlier draft stated this as “if and only if”; that was an error.)

Proof of Proposition A.13. Immediate from (A.8). For the failure of the converse it suffices to observe that (A.8) is an inequality derived from a single feasible point, so $\Delta_i^v(S)$ may exceed $\text{og}_i(S)$ by any amount, in particular enough to be non-negative when $\text{og}_i(S) < 0$. $\square$

Corollary A.14 (The w -analogue and why it has no force). Subtracting $v(\{i\})$ from both sides of (A.8),

$$\Delta_i^w(S) = \Delta_i^v(S) - v(\{i\}) \geq \text{og}_i(S) - v(\{i\}). \quad (\text{A.9})$$

For a high-tariff country the right-hand side is generically a large negative number: $v(\{i\})$ reaches \$26.4B (India) and \$119.7B (China) while $\text{og}_i(S)$ is a passive spillover an order of magnitude or more smaller. Hence Proposition A.12 *cannot* establish Proposition A.1, and every downstream result that needs single-player w -monotonicity must carry it as a maintained hypothesis rather than derive it here.

Proof of Proposition A.14. The identity $\Delta_i^w(S) = \Delta_i^v(S) - v(\{i\})$ follows from (A.3): $w(S \cup \{i\}) - w(S) = v(S \cup \{i\}) - v(S) - v(\{i\})$. Substituting (A.8) gives (A.9). The magnitude claim is a statement about the computed primitives, not a theorem (`superadditivity_sweep_D1.md`, §3; the outsider gains themselves are tabulated in `outsider_welfare_N7.json`, where 369 of 441 outsider-coalition pairs have a positive gain and the R6 bound registers 0 violations out of 441). $\square$

We now state the near-dummy result. The version in the theory notes bounded a country’s marginal contribution by a term proportional to its base tariff alone; formalizing it revealed that the bound must also carry a *passive* free-riding term, which does not vanish with the country’s own tariff. The corrected statement is a two-sided sandwich, and it is economically sharper: as a country’s tariff headroom vanishes, its marginal contribution does not go to zero but collapses to the free-riding band.

Maintained Hypothesis A.15 (Lipschitz welfare response). There exists $L < \infty$ such that for every coalition T , every $k \in \mathcal{N}$, and all $s, s' \in \Sigma_T$, $|\widehat{W}_k(s, \mathbf{1}_{-T}) - \widehat{W}_k(s', \mathbf{1}_{-T})| \leq L d(s, s')$, where $d(s, s') \equiv \max_{m \in T} \max_{h,j} |\log \widehat{\kappa}_{mh}^j(s_m) - \log \widehat{\kappa}_{mh}^j(s'_m)|$ measures the policy perturbation in log tariff wedges. This strengthens A2’s continuity to a uniform modulus; it holds automatically if $\widehat{W}$ is continuously differentiable along the selected branch on the compact box.

Proposition A.16 (Mechanical near-dummy: a sandwich, not a one-sided bound). Assume A1–A4, fix the A2 selection, and assume Proposition A.15. For country i let $\bar{\tau}_i \equiv \max_{h,j} \tau_{ih}^{j,0}$

be its base applied-tariff scale and let $C \equiv \sum_{k \in \mathcal{N}} I_k^{\text{base}}$. Write $M_S \equiv \arg \max_{\Sigma_S} f_S$ for the (non-empty, compact) set of maximizers of the S -program and define the free-riding band

$$\omega_i \equiv \min_{S \subseteq \mathcal{N} \setminus \{i\}} \min_{s_S^* \in M_S} I_i^{\text{base}} [\widehat{W}_i(s_S^*, \mathbf{1}_{-S}) - 1], \quad \Omega_i \equiv \max_{S \subseteq \mathcal{N} \setminus \{i\}} \max_{s_S \in \Sigma_S} I_i^{\text{base}} [\widehat{W}_i(s_S, \mathbf{1}_{-S}) - 1].$$

Selection qualifier (required by [Proposition A.4](#)): $\text{og}_i(S)$ is a function of the maximizing point, not of $v(S)$, so it is selection-dependent whenever M_S is not a singleton; ω_i is therefore defined as the infimum over the whole argmax set, which makes it a valid lower bound under every selection. The inner minimum is attained because M_S is compact ($M_S = f_S^{-1}(\{v(S)\})$ is closed in the compact Σ_S by continuity of f_S) and $\widehat{W}_i$ is continuous on it. Ω_i needs no such qualifier: it already maximizes over all of Σ_S , not only over maximizers. Then for every $S \subseteq \mathcal{N} \setminus \{i\}$,

$$\text{og}_i(S) \leq \Delta_i^v(S) \leq \Omega_i + C L \log(1 + \bar{\tau}_i), \quad (\text{A.10})$$

and therefore $\omega_i \leq \phi_i(v) \leq \Omega_i + C L \log(1 + \bar{\tau}_i)$. Consequently, as $\bar{\tau}_i \rightarrow 0$ the policy channel of i 's marginal contribution vanishes at rate $O(\bar{\tau}_i)$ and i 's Shapley value is squeezed into the free-riding band $[\omega_i, \Omega_i]$. In the exact limit $\bar{\tau}_i = 0$: i 's lever is inert ($\widehat{\kappa}_{ih}^j(s_i) \equiv 1$ for all s_i), $v(\{i\}) = 0$ exactly, and $\omega_i \leq \Delta_i^v(S) \leq \Omega_i$ for every S ; if in addition i has no general-equilibrium linkage to any coalition (so $\widehat{W}_i \equiv 1$), then $\omega_i = \Omega_i = 0$ and i is an exact dummy with $\phi_i(v) = 0$.

Proof of Proposition A.16. Step 1 (lower bound). This is [Proposition A.12](#).

Step 2 (the lever's reach is $O(\bar{\tau}_i)$). By the lever definition (10), $\widehat{\kappa}_{ih}^j(s_i) = (1 + s_i \tau_{ih}^{j,0}) / (1 + \tau_{ih}^{j,0})$, which is increasing in s_i with $\widehat{\kappa}_{ih}^j(1) = 1$ and $\widehat{\kappa}_{ih}^j(0) = 1 / (1 + \tau_{ih}^{j,0})$. Hence for any $s_i, s'_i \in [0, 1]$, $|\log \widehat{\kappa}_{ih}^j(s_i) - \log \widehat{\kappa}_{ih}^j(s'_i)| \leq \log(1 + \tau_{ih}^{j,0}) \leq \log(1 + \bar{\tau}_i)$. A country with no applied tariffs ($\bar{\tau}_i = 0$) therefore has a literally inert lever.

Step 3 (upper bound). Let $(s_S^\dagger, s_i^\dagger)$ maximize the $S \cup \{i\}$ -program. Since $s_S^\dagger$ is feasible for the S -program, $v(S) \geq f_S(s_S^\dagger)$, whence

$$\begin{aligned} \Delta_i^v(S) &\leq \sum_{k \in S} I_k^{\text{base}} [\widehat{W}_k(s_S^\dagger, s_i^\dagger) - \widehat{W}_k(s_S^\dagger, s_i=1)] \\ &\quad + I_i^{\text{base}} [\widehat{W}_i(s_S^\dagger, s_i^\dagger) - \widehat{W}_i(s_S^\dagger, s_i=1)] + I_i^{\text{base}} [\widehat{W}_i(s_S^\dagger, s_i=1) - 1], \end{aligned}$$

where we used Step 2 of the proof of [Proposition A.12](#) to identify $f_S(s_S^\dagger)$ with the terms evaluated at $s_i = 1$. The first two brackets compare two world tariff vectors differing only in i 's lever, so by [Proposition A.15](#) and Step 2 each is bounded in absolute value by $L \log(1 + \bar{\tau}_i)$; multiplying by the income weights and summing gives at most $C L \log(1 + \bar{\tau}_i)$. The third term is i 's passive gain at $s_S^\dagger$, bounded above by Ω_i by definition. This yields (A.10).

Step 4 (from marginals to the Shapley value). By (A.4), $\phi_i(v)$ is a convex combination of the $\Delta_i^v(S)$, hence lies between their minimum and maximum, giving the displayed bounds on $\phi_i(v)$.

Step 5 (the exact limit). If $\bar{\tau}_i = 0$ then by Step 2 the map $s_i \mapsto \widehat{\kappa}_{ih}^j(s_i)$ is constant at 1, so the $S \cup \{i\}$ -objective does not depend on s_i at all and equals $f_S(s_S) + I_i^{\text{base}} [\widehat{W}_i(s_S, \mathbf{1}_{-S}) - 1]$. Taking $S = \emptyset$ gives $v(\{i\}) = 0$ exactly (consistent with [Proposition A.9\(b\)](#)): $s_i = 1$ is a maximizer, trivially, since every s_i is). For general S , maximizing the displayed sum gives $v(S \cup \{i\}) \leq$

$v(S) + \Omega_i$ and, by [Proposition A.12](#), $v(S \cup \{i\}) \geq v(S) + \text{og}_i(S) \geq v(S) + \omega_i$. If moreover $\widehat{W}_i \equiv 1$ then both bounds are zero, so $v(S \cup \{i\}) = v(S)$ for all S and $\phi_i(v) = 0$ by the dummy property (invariant TI-3). $\square$

Remark A.17 (What [Proposition A.16](#) does and does not license). The proposition licenses the statement that a low-tariff country’s small Shapley share is *mechanical* in the policy channel: it follows from the scalar lever $s_i \in [0, 1]$ acting on a small τ_i^0 , not from any finding about strategic position. It does *not* license the stronger claim that such a country’s share must be near zero: the free-riding band $[\omega_i, \Omega_i]$ is not bounded by $\bar{\tau}_i$ and is an empirical magnitude. On the current tables that band is small — Canada’s raw share is 0.45–0.49% of $v(\mathcal{N})$ across $N = 4$ –7 and Korea’s is 1.37% at $N = 7$, both countries with $\bar{\tau} < 5\%$ (prediction P-13, `extended_shapley/N{4..7}_D1.json`) — but this is measured, not derived. Two further caveats survive unchanged: the ranking implication is confined to the low end (the Shapley ordering and a GDP-weighted ordering disagree at the top), and τ^0 is itself the cumulative outcome of 1947–1990 GATT bargaining, so the near-dummy mechanism partly encodes historical bargaining outcomes rather than exogenous position.

A.6 Imputation individual rationality and strictly positive cooperative surplus

Proposition A.18 (Imputation IR under [Proposition A.1](#)). *Assume A1–A4, fix the A2 selection, and assume [Proposition A.1](#) (single-player w -monotonicity; not proved — see [Proposition A.2](#)). Then*

$$\phi_i(w) \geq 0 \quad \text{for every } i \in \mathcal{N}, \quad \text{equivalently} \quad \phi_i(v) \geq v(\{i\}),$$

which is invariant TI-5 and the second half of precondition A8-b. If in addition $w(\mathcal{N}) > 0$, then A8-b holds in full and therefore A8-a holds, so both cooperative and raw share language are well posed.

Proof of [Proposition A.18](#). Step 1. By the random-order representation in (A.4), $\phi_i(w) = \mathbb{E}_\pi[w(P_\pi^{< i} \cup \{i\}) - w(P_\pi^{< i})]$, an average of single-player marginal contributions of w with strictly positive weights summing to one.

Step 2. By [Proposition A.1](#) every such marginal contribution is non-negative. An average of non-negative numbers is non-negative, so $\phi_i(w) \geq 0$.

Step 3. By [Proposition A.10](#), $\phi_i(v) = v(\{i\}) + \phi_i(w)$, so $\phi_i(w) \geq 0$ is equivalent to $\phi_i(v) \geq v(\{i\})$. Together with $w(\mathcal{N}) > 0$ this is exactly A8-b; and A8-b implies A8-a because $\phi_i(v) = v(\{i\}) + \phi_i(w) \geq 0$ by [Proposition A.9\(a\)](#), while $v(\mathcal{N}) = \sum_k v(\{k\}) + w(\mathcal{N}) > 0$. $\square$

Remark A.19 (Empirical status of A8-a/A8-b). A8-a and A8-b are checkable numerical propositions, and they are checked. On the D1 tables all five audited value tables pass TI-5, A8-a and A8-b, with minimum cooperative cushion $\min_i \phi_i(w)$ equal to +\$0.266B ($N = 3$, Canada), +\$0.659B ($N = 4$, Canada), +\$0.771B ($N = 5$, Canada), +\$0.614B ($N = 6$, Brazil) and +\$0.683B ($N = 7$, Brazil) (`paper/replication/results/ir_verification.md`). The **global** binding cushion across all N is the $N=3$ value, +\$0.266B (Canada) — not the $N=6$ figure above,

which is a local minimum only (MATH-31 fix). These cushions exceed the measured noise floor by **two to three orders of magnitude**, not five (**correction, MATH-24, 2026-08-09**: ε_B is a relative quantity; converting via $\varepsilon_B \cdot |v(\mathcal{N})|$ gives $\approx \$4.46\text{M}$ at $N = 7$ and $\approx \$0.195\text{M}$ at $N = 3$, so the $N = 7$ cushion clears by $\approx 153\times$ and the global $N = 3$ cushion by $\approx 1363\times$) (`noise_floor_eps_B.md`, corrected 2026-08-09). This does not convert [Proposition A.1](#) into a theorem; it means the hypothesis and its consequence are jointly consistent with every number the project has produced under D1. Note also that the earlier `verify_ir.py` threshold $\phi_i \geq 0$ tested the wrong inequality: under D1 the singletons are large and positive, so $\phi_i \geq 0$ can pass while true IR fails by tens of billions. The correct threshold is $\phi_i(v) \geq v(\{i\})$ and it is what the audit above reports.

Proposition A.20 (Non-null players receive strictly positive cooperative surplus). *Assume [Proposition A.1](#) (not proved). If player i is non-null in the cooperative game — that is, there exists $S \subseteq \mathcal{N} \setminus \{i\}$ with $w(S \cup \{i\}) > w(S)$ — then*

$$\phi_i(w) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [w(S \cup \{i\}) - w(S)] > 0.$$

Proof of [Proposition A.20](#). Every Shapley weight $|S|!(n - |S| - 1)!/n!$ is strictly positive. Under [Proposition A.1](#) every bracket is non-negative, and by the non-null assumption at least one bracket is strictly positive. A weighted sum with strictly positive weights of non-negative terms, at least one of which is strictly positive, is strictly positive. $\square$

Corollary A.21 (Zero surplus for a non-null player is impossible under the Shapley axioms but permitted by the generalized-Nash family). Assume [Proposition A.1](#). Then no non-null player receives zero cooperative surplus under the Shapley imputation. By contrast, in the Horn–Wolinsky/generalized-Nash bilateral family the corner weight $\zeta \rightarrow 0^+$ drives the weak party’s surplus to exactly zero at rate $O(\zeta)$ ([Proposition A.38](#)), and corner weights lie inside the admissible parameter space. The correct statement of the contrast is therefore one about *permissiveness of a solution-concept family*, not about a parameter value: the generalized-Nash family admits, at admissible parameters, an allocation that the Shapley axioms exclude for *every* game.

Proof of [Proposition A.21](#). The first sentence is [Proposition A.20](#). The second is [Proposition A.38](#). The third is the conjunction: the two solution concepts differ not merely in the value of a free parameter but in whether a zero-surplus allocation to a non-null player is attainable at all. $\square$

Remark A.22 (Convexity: what the sweep does and does not show — **Track B only**). **Track qualifier (added 2026-08-14, MATH-59)**. This remark is Track B (the seven-country Caliendo–Parro economy) throughout; Track A’s supermodularity is not swept here, and Track A’s general superadditivity outright fails 20/301 times, a fact this remark’s “convex to within numerical resolution” conclusion describes on a different track and must not be read to contradict (§2.5, `02_model.tex`). Convexity (supermodularity) is a different property from [Proposition A.1](#) and is not needed for anything above. On the D1 tables the supermodularity failure rate is

0.6% (31 of 5103 valid triples at $N = 7$; 40 of 7101 pooled; `superadditivity_sweep_D1.md`). **Evidence-strength threshold (added 2026-08-14, MATH-57(a); stated once here and cross-cited by Proposition A.2(iii) so the two remarks apply one standard, not two).** We judge a violation-magnitude comparison against its noise floor as follows: a binding gap or margin below $20\times$ the relevant noise floor is read as a numerical artifact, not structural evidence; above $100\times$ it is read as structural evidence; strictly in between, we report it as inconclusive rather than asserting either reading. The correct reading under this threshold, and the only one the draft may use, is: *the game is convex to within numerical resolution; the residual 0.61% of violating triples have gaps of $\$0.016$ – $\$0.046B$ against a cross-configuration sensitivity of about $\$0.003$ – $\$0.006B$, a ratio of roughly $3\times$ – $15\times$ — below the $20\times$ artifact threshold just stated — and is not evidence of genuine non-convexity.* Reporting 0.61% as a measured non-convexity finding would be a “significant inside the error band” claim in mirror image. Substantively this is a pre-registered FAIL of the non-convexity half of prediction P-12 (threshold: failure rate $\geq 10\%$), and the pre-written branch was taken and reported as a failure rather than reframed.

A.7 The fairness gap is locked into an interval of width κ

This subsection contains the paper’s sharpest formal result about the positive–normative bridge. It answers a question left open in earlier rounds — whether imputation IR pins down the *sign* of the fairness gap — with a negative answer and a more useful positive one: IR does not sign Γ_i , but it confines Γ_i to an interval whose width is exactly the cooperative share of the surplus, uniformly across players and across allocation rules.

Notation. Fix the bridge player set $\mathcal{N}_6$ (A6) and assume A8-a, $v(\mathcal{N}) > 0$. Write

$$u_i \equiv \frac{v(\{i\})}{v(\mathcal{N})} \quad (\text{player } i\text{'s unilateral share}), \quad \kappa \equiv \frac{w(\mathcal{N})}{v(\mathcal{N})} \quad (\text{the cooperative ratio}), \quad (\text{A.11})$$

so that $\sum_i u_i = 1 - \kappa$ by Proposition A.9(c). Let $s_i^{HW}(\zeta, b)$ denote the comparator share of Definition 2. For an arbitrary allocation rule ψ applied to the cooperative game, write $\Gamma_i^{\text{raw}}(\psi) \equiv [v(\{i\}) + \psi_i(w)]/v(\mathcal{N}) - s_i^{HW}$, which specializes to (14) when $\psi = \phi$ by Proposition A.10.

Proposition A.23 (IR interval for the fairness gap). *Assume A8-a ($v(\mathcal{N}) > 0$) and $w(\mathcal{N}) \geq 0$ (added 2026-08-13, MATH-50: efficiency (i) forces $\sum_j \psi_j(w) = w(\mathcal{N})$ while IR (ii) forces every summand ≥ 0 ; if $w(\mathcal{N}) < 0$ no allocation can satisfy both, so the class of allocations the proposition quantifies over is empty and the interval is vacuous rather than proved — the hypothesis is therefore necessary, not merely convenient, and must be checked before invoking this proposition on any track. On Track B $w(\mathcal{N}) > 0$ at every $N \in \{3, \dots, 7\}$ and on Track A $w(\mathcal{N}_6) = 0.677 > 0$ (paper/replication/results/bsy_replication/phase7_full_primary.md §2); both are verified, not assumed, when the proposition is actually invoked in this paper). Let ψ be any allocation of the cooperative game satisfying*

- (i) efficiency: $\sum_{j \in \mathcal{N}} \psi_j(w) = w(\mathcal{N})$;
- (ii) imputation IR: $\psi_j(w) \geq 0$ for every $j \in \mathcal{N}$.

Then for every i ,

$$u_i - s_i^{HW} \leq \Gamma_i^{\text{raw}}(\psi) \leq u_i + \kappa - s_i^{HW}, \quad (\text{A.12})$$

an interval of width exactly κ — independent of i , independent of ψ , and independent of the comparator s^{HW} .

Proof of Proposition A.23. Step 1 (lower bound on ψ_i). By (ii) applied to $j = i$, $\psi_i(w) \geq 0$.

Step 2 (upper bound on ψ_i). By (i), $\psi_i(w) = w(\mathcal{N}) - \sum_{j \neq i} \psi_j(w)$, and by (ii) applied to each $j \neq i$ the subtracted sum is non-negative, so $\psi_i(w) \leq w(\mathcal{N})$.

Step 3 (rescale). Steps 1–2 give $\psi_i(w) \in [0, w(\mathcal{N})]$. By Proposition A.10's decomposition applied to ψ (which here is a definition, $\psi_i(v) \equiv v(\{i\}) + \psi_i(w)$, and is the Shapley identity when $\psi = \phi$), dividing by $v(\mathcal{N}) > 0$ gives $\psi_i(v)/v(\mathcal{N}) \in [u_i, u_i + \kappa]$.

Step 4 (subtract the comparator). Subtracting s_i^{HW} from each side of Step 3's interval gives (A.12). The width is $(u_i + \kappa) - u_i = \kappa$, which involves neither i nor ψ nor s^{HW} . $\square$

Proposition A.24 (Both endpoints are attained: the interval cannot be narrowed). *Let $n \geq 2$ and $w(\mathcal{N}) > 0$. Then within the class of allocations satisfying (i)–(ii) of Proposition A.23, both bounds in (A.12) are attained, so the interval is the exact range of $\Gamma_i^{\text{raw}}(\psi)$ over that class and no strengthening is possible without an axiom beyond efficiency and IR.*

Proof of Proposition A.24. Step 1 (lower endpoint). Fix $j_0 \neq i$ (possible since $n \geq 2$) and let ψ^ℓ assign $\psi_{j_0}^\ell(w) = w(\mathcal{N})$ and $\psi_j^\ell(w) = 0$ for all $j \neq j_0$. Then $\sum_j \psi_j^\ell = w(\mathcal{N})$, satisfying (i), and every component is non-negative since $w(\mathcal{N}) > 0$, satisfying (ii). Its i -component is 0, so $\Gamma_i^{\text{raw}}(\psi^\ell) = u_i - s_i^{HW}$, the lower bound.

Step 2 (upper endpoint). Let ψ^u assign $\psi_i^u(w) = w(\mathcal{N})$ and $\psi_j^u(w) = 0$ for $j \neq i$. The same two checks give (i)–(ii), and $\Gamma_i^{\text{raw}}(\psi^u) = u_i + \kappa - s_i^{HW}$, the upper bound.

Step 3 (range). By Proposition A.23 the map $\psi \mapsto \Gamma_i^{\text{raw}}(\psi)$ takes values in $[u_i - s_i^{HW}, u_i + \kappa - s_i^{HW}]$; the class (i)–(ii) is convex (it is the scaled simplex $\{\psi \geq 0 : \sum_j \psi_j = w(\mathcal{N})\}$) and Γ_i^{raw} is affine in ψ , so its image is the closed interval between the two attained endpoints, i.e. the whole interval. The Shapley value is one interior point of that set; it is not at either endpoint except in degenerate games. $\square$

Proposition A.25 (Consistency with $\sum_i \Gamma_i = 0$). *Under efficiency of ψ and of the comparator shares ($\sum_i s_i^{HW} = 1$), $\sum_i \Gamma_i^{\text{raw}}(\psi) = 0$, and this is compatible with (A.12): summing the lower bounds gives $-\kappa \leq 0$ and summing the upper bounds gives $(n-1)\kappa \geq 0$.*

Proof of Proposition A.25. $\sum_i \Gamma_i^{\text{raw}}(\psi) = \frac{1}{v(\mathcal{N})} [\sum_i v(\{i\}) + \sum_i \psi_i(w)] - 1 = \frac{1}{v(\mathcal{N})} [\sum_i v(\{i\}) + w(\mathcal{N})] - 1 = \frac{v(\mathcal{N})}{v(\mathcal{N})} - 1 = 0$, using Proposition A.9(c). For the bounds, $\sum_i (u_i - s_i^{HW}) = (1 - \kappa) - 1 = -\kappa$ and $\sum_i (u_i + \kappa - s_i^{HW}) = (1 - \kappa) + n\kappa - 1 = (n-1)\kappa$, and $-\kappa \leq 0 \leq (n-1)\kappa$ whenever $\kappa \geq 0$. $\square$

Corollary A.26 (Shapley-free sign conditions). Under the hypotheses of Proposition A.23: $u_i > s_i^{HW} \Rightarrow \Gamma_i^{\text{raw}} > 0$, and $u_i + \kappa < s_i^{HW} \Rightarrow \Gamma_i^{\text{raw}} < 0$. Neither condition requires computing any Shapley value: they use only the singleton value $v(\{i\})$, the grand-coalition value $v(\mathcal{N})$, and the comparator share.

Proof of Proposition A.26. Immediate from the two ends of (A.12). $\square$

Corollary A.27 (Small κ makes the conclusion independent of the allocation rule). As $\kappa \rightarrow 0$ the interval (A.12) collapses to the single point $u_i - s_i^{HW}$: *every* efficient, IR allocation rule then delivers the same fairness gap. Consequently κ is a direct, reportable measure of how sensitive the paper’s distributional conclusion is to the choice of solution concept, and should be reported alongside Γ as a robustness statistic rather than treated as a descriptive aside.

Proof of Proposition A.27. The interval has width κ by Proposition A.23 and contains $u_i - s_i^{HW}$ as its left endpoint; as $\kappa \downarrow 0$ the right endpoint converges to the left. By Proposition A.24 the interval is exactly the range over the class, so the collapse is not merely of an outer bound but of the range itself. $\square$

Corollary A.28 (Computation-free screen for the pre-registered Γ_{US} claim). Suppose $v(\{US\}) = 0$, so $u_{US} = 0$. Then $\Gamma_{US}^{\text{raw}} \in [-s_{US}^{HW}, \kappa - s_{US}^{HW}]$ for every efficient, IR allocation. Hence:

(a) $\Gamma_{US}^{\text{raw}} > 0$ under *some* such allocation requires $\kappa > s_{US}^{HW}$;

(b) $\Gamma_{US}^{\text{raw}} \geq +5\text{pp}$ requires $\kappa \geq s_{US}^{HW} + 0.05$.

With the measured comparator value $s_{US}^{HW}(\hat{\zeta}) = 10.04\%$ on the six-country denominator (`paper/replication/results/bsy_replication/sanity_assertions.md`), these read $\kappa > 10.04\%$ and $\kappa \geq 15.04\%$ respectively.

Proof of Proposition A.28. Set $u_{US} = 0$ in (A.12) and read off the two endpoints; (a) and (b) are the statements that the relevant threshold lies below the upper endpoint. Necessity is exactly Proposition A.23; the endpoint is attainable by Proposition A.24, so the conditions are necessary and, at the level of the whole allocation class, also sufficient for attainability. $\square$

Remark A.29 (Scope of Proposition A.23 and its corollaries). Four qualifications travel with this block and none may be dropped. (1) Both bounds use hypothesis (ii), but asymmetrically: the lower bound uses i ’s own IR and the upper bound uses the IR of the *others*. If IR fails for a set of players, the upper bound relaxes to $\psi_i(w) \leq w(\mathcal{N}) - \sum_{j \neq i} \min\{\psi_j(w), 0\}$ and the lower bound to $\psi_i(w) \geq \min\{\psi_i(w), 0\}$; the width is then κ plus the total IR violation, divided by $v(\mathcal{N})$. Hypothesis (ii) for the Shapley value is Proposition A.18, whose own hypothesis Proposition A.1 is not proved; but Proposition A.23 itself is stated for an arbitrary efficient, IR allocation and therefore does not depend on the Shapley value at all. (2) κ and u_i must be measured on the same calibration track as each other; the paper’s two-track protocol forbids importing a κ computed on the CP-1993 economy (Track B) into a statement about the BSY-1990 economy (Track A). The Track B values are $\kappa = 75.2\%$ ($N = 3$), 13.4% ($N = 4$), 13.1% ($N = 5$), 13.6% ($N = 6$) and 16.1% ($N = 7$) (`ir_verification.md`); these are a *prior* for Track A, not a substitute computation, and Proposition A.28’s screen must be run with Track A’s own eight solves (seven singletons plus the grand coalition). (3) The result is about the raw gap Γ^{raw} . The analogous bound for the cooperative gap Γ^{coop} requires re-deriving with the aligned comparator of eq. (15) and has not been done. (4) Both sides of Γ^{raw} must use the same threat point (A7, TI-13(a)), the same denominator (TI-13(b)), and the same player set (TI-13(c)); in particular the six-country denominator is primary and the five-country in-pair renormalization is a robustness

variant whose systematic inflation must be disclosed. *Two further TI-13 clauses were added after this remark was first drafted and are completed here (2026-08-13, MATH-50): (4a) same policy space (TI-13(d))* — the comparator $s_i^{HW}(\hat{\zeta})$ is the outcome of BSY’s cuts-only bargaining problem ($s \in [0, 1]^n$), so u_i , κ , and any ψ invoked in [Proposition A.28](#) must also be computed on the cuts-only policy space; mixing a cuts-only comparator against an extended-space ($\bar{s} > 1$) $v(\mathcal{N})$ or $v(\{i\})$ is forbidden as a primary object and is reported only as the separate policy-space placebo of TI-13(d)(iii). *(4b) same transferability regime, or an explicit decomposition of the difference (TI-13(d'))* — ψ is a TU allocation of w , while s^{HW} is the outcome of BSY’s NTU bilateral bargain (transfer variables fixed at zero, `est_barg_power_all_lsnonlin.m` L221–223); the TU feasible set contains the NTU one, so Γ_i^{raw} ’s sign readings inherit a known-direction bias (TU is weakly more favorable to any given i than NTU) that is not corrected by this proposition and must be disclosed alongside any Γ reported from it.

Remark A.30 (A8 checks do not automatically transfer across policy-space placebo rows). (Added 2026-08-13, MATH-51.) The policy-space placebo table (TI-13(d)(iii), $\bar{s} \in \{1, 1.5, 2, \infty\}$) recomputes $v(S)$ and $w(S)$ over a *different* feasible set at every row, so A8-a ($v(\mathcal{N}) > 0$), A8-b ($w(\mathcal{N}) \geq 0$, the hypothesis added to [Proposition A.23](#) above), and A8-c (comparator-side positivity, $\sum_k g_k > 0$ and $g_i \geq 0$) are properties of the row’s own game and are *not* inherited from the $\bar{s} \leq 1$ primary row by default. Two consequences, one cautionary and one exculpatory: (i) the deep-sweep row’s originally reported $\kappa_A^{\text{deep}} = -2.17\%$ would, taken at face value, have meant $w(\mathcal{N}) < 0$ at that row — an A8-b violation that voids any invocation of [Proposition A.23](#) there, not merely a cosmetic sign flip; **the corrected value satisfies $\kappa_A^{\text{deep}} \leq +8.53\%$ as a point bound (corrected 2026-08-14, MATH-58 — the text previously here wrote $\geq$, reversing the direction MATH-48 established elsewhere in the repo: the numerator $\sum_i v(\{i\})^{\text{deep}}$ is a 10-node grid lower bound on the true singleton values, so the ratio $\kappa = 1 - \sum_i v(\{i\})/v(\mathcal{N})$ it enters is an upper bound, not a lower one), and $\gtrsim +8.4\%$ under the MATH-48 robustness reconstruction (Japan/Canada continuous re-values plus a quadratic fit at the US/EU peaks) — which is the lower bound A8-b actually needs to be restored at that row; the -2.17% was a stale-denominator artifact (MATH-48), not a genuine subadditivity finding at that policy space, and the row was re-checked against its own $w(\mathcal{N})$, never because A8-b was assumed to survive from $\bar{s} \leq 1$. Every future placebo row must state its own A8-a/b/c check explicitly; a row that has not been re-checked must be reported as A8-status UNVERIFIED, not silently treated as PASS. (ii) A8-c is the one exception to the no-inheritance rule: the comparator $s_i^{HW}(\hat{\zeta})$ is computed once, on BSY’s own (cuts-only) bargaining problem, which the $v(S)$ -side placebo sweep never touches — so A8-c’s PASS at the primary row (`sanity\_assertions.md` MC-15) is unaffected by which $\bar{s}$ row is being read on the $v(S)$ side, and does not need re-verification per row.**

A.8 Two players: the collapse propositions and the threat-point equivalence

Throughout this subsection $n = 2$ with players $\{i, j\}$, utility is transferable within the pair (A5), and the feasible set is the TU frontier $\{(x_i, x_j) : x_i + x_j \leq V\}$ with $V \equiv v(\{i, j\})$. The non-negativity of the pair’s cooperative surplus, $w(\{i, j\}) = V - v(\{i\}) - v(\{j\}) \geq 0$ — the two-player instance of [Proposition A.1](#), and therefore *not* established — is needed by exactly

one result below and is carried in that result's own statement (Proposition A.33), not assumed silently here. Proposition A.31 and Proposition A.32 do not require it; Proposition A.34 inherits it through Proposition A.33 and says so.

Proposition A.31 (Shapley on the cooperative game splits evenly). $\phi_i(w) = \phi_j(w) = \frac{1}{2}w(\{i, j\})$, and hence $\phi_i(v) = v(\{i\}) + \frac{1}{2}w(\{i, j\})$.

Proof of Proposition A.31. With $n = 2$ the Shapley formula (A.4) gives $\phi_i(w) = \frac{1}{2}[w(\{i\}) - w(\emptyset)] + \frac{1}{2}[w(\{i, j\}) - w(\{j\})]$. By Proposition A.9(c), $w(\emptyset) = w(\{i\}) = w(\{j\}) = 0$, so $\phi_i(w) = \frac{1}{2}w(\{i, j\})$, symmetrically for j . Adding $v(\{i\})$ via Proposition A.10 gives the raw statement. Note that this is a property of the *cooperative* game: the corresponding statement for the raw game, $\phi_i(v) = \frac{1}{2}V$, is false whenever $v(\{i\}) \neq v(\{j\})$ — see Proposition A.32. $\square$

Proposition A.32 (B-1a: raw side, BSY's threat point). *Let the disagreement point be the 1990 status quo, $d = (0, 0)$, matching A7 and Bagwell et al. (2021). The symmetric Nash bargaining solution on the TU frontier is $x_i = x_j = \frac{1}{2}V$, and this coincides with the raw-game Shapley value $\phi_i(v)$ if and only if $v(\{i\}) = v(\{j\})$.*

Proof of Proposition A.32. Step 1. The symmetric Nash solution maximizes $x_i x_j$ subject to $x_i + x_j \leq V$, $x \geq 0$. The constraint binds at any solution (increasing either coordinate increases the product), so substituting $x_j = V - x_i$ and maximizing $x_i(V - x_i)$ gives the unique interior maximizer $x_i = V/2$, hence $x_i = x_j = V/2$.

Step 2. By Proposition A.31, $\phi_i(v) = v(\{i\}) + \frac{1}{2}[V - v(\{i\}) - v(\{j\})] = \frac{1}{2}V + \frac{1}{2}[v(\{i\}) - v(\{j\})]$.

Step 3. Therefore $\phi_i(v) = \frac{1}{2}V$ if and only if $v(\{i\}) = v(\{j\})$. The condition is not innocuous in this application: the computed singletons range from exactly \$0 (USA, Canada, Korea) to \$119.7B (China), so the coincidence generically fails. $\square$

Proposition A.33 (B-1b: cooperative side, unilateral-optimum threat point). *Let the disagreement point be each country's own unilateral optimum, $d_i = v(\{i\})$, $d_j = v(\{j\})$, and assume the pair's cooperative surplus is non-negative, $w(\{i, j\}) \geq 0$ (the two-player instance of Proposition A.1, which is not established — see Proposition A.2; without it the bargaining set $\{x \geq d : x_i + x_j \leq V\}$ is empty and the solution is undefined). Then the symmetric Nash bargaining solution on the TU frontier is $x_i = v(\{i\}) + \frac{1}{2}w(\{i, j\}) = \phi_i(v)$, with no symmetry condition on the singletons required — “unconditional” here means unconditional in the singletons, in contrast with Proposition A.32's iff, not free of the non-emptiness hypothesis just stated.*

Proof of Proposition A.33. The symmetric Nash solution maximizes $(x_i - d_i)(x_j - d_j)$ subject to $x_i + x_j \leq V$ and $x \geq d$. Writing $y_k \equiv x_k - d_k \geq 0$, the constraint becomes $y_i + y_j \leq V - d_i - d_j = w(\{i, j\})$, a non-negative budget by the stated hypothesis. By the argument of Step 1 of Proposition A.32 applied in y -coordinates, the solution is $y_i = y_j = \frac{1}{2}w(\{i, j\})$, i.e. $x_i = v(\{i\}) + \frac{1}{2}w(\{i, j\})$, which equals $\phi_i(v)$ by Proposition A.31. $\square$

Proposition A.34 (R8/B-2: zero-normalization is a threat-point redefinition). *Comparing Proposition A.32 and Proposition A.33 (and inheriting the latter's hypothesis $w(\{i, j\}) \geq 0$):*

at $n = 2$ with transferable utility and a linear frontier, computing the Shapley value on the zero-normalized game w instead of on v is exactly equivalent to moving the disagreement point from “1990 tariffs frozen” to “every country at its own unilateral optimum”. For general n the corresponding statement is the identity of [Proposition A.10](#), $\phi_i(v) = v(\{i\}) + \phi_i(w)$, read as the axiomatic expression of a two-stage procedure: each country first takes its unilateral dividend, then the residual cooperative surplus is divided by the Shapley axioms.

Proof of [Proposition A.34](#). Step 1 ($n = 2$). [Proposition A.33](#) shows that the symmetric Nash solution at $d = (v(\{i\}), v(\{j\}))$ equals $\phi_i(v) = v(\{i\}) + \phi_i(w)$, which is the w -Shapley allocation translated back to v -space. [Proposition A.32](#) shows that the symmetric Nash solution at $d = (0, 0)$ equals the raw-Shapley allocation only under the singleton-symmetry condition. Hence the operation “zero-normalize, then apply Shapley” is the same operation as “move the disagreement point to the unilateral optima, then apply the symmetric Nash bargaining solution”. *Step 2 (general n).* [Proposition A.10](#) is an identity, so the two-stage reading is available for any n ; but the equivalence with a bargaining solution is established here only for $n = 2$ with a linear frontier, and we do not claim it beyond that. $\square$

Corollary A.35 (Why the raw game is the bridge’s primary operand). Three independent reasons, each established above. (1) *Threat-point comparability (A7)*. The comparator gains $g_i(\zeta, b)$ are measured from the 1990 status quo with nothing subtracted; $\phi_i(w)/w(\mathcal{N})$ has already netted out $v(\{i\})$. By [Proposition A.34](#) the cooperative object implicitly uses a different disagreement point, so only $\phi_i(v)/v(\mathcal{N})$ and s_i^{HW} share a threat point (invariant TI-13(a)). (2) *Behavioral credibility*. The disagreement point of [Proposition A.33](#) is a counterfactual — if China could unilaterally capture \$119.7B, the question of why it did not is unanswered by the model — whereas the 1990 applied schedule is an observed historical fact. (3) *Weaker preconditions*. Raw share language needs only A8-a, and A8-b $\Rightarrow$ A8-a strictly ([Proposition A.18](#)), so the raw bridge survives a failure of the maintained hypothesis that would invalidate the cooperative one. The cooperative gap is not thereby deprecated: it answers a well-posed and independently interesting question, and is reported as a secondary object with its differing threat point stated.

Proof of [Proposition A.35](#). (1) is [Proposition A.34](#) plus the definition of g_i ; (3) is the last sentence of [Proposition A.18](#); (2) is an interpretive statement about the primitives, recorded here because it is load-bearing for the choice of primary object and must not be relegated to a footnote. $\square$

A.9 The dollar identity and the inverse map

Proposition A.36 ($D4'.0$: exact dollar identity). *Fix ζ and an equilibrium branch b , let $g_i(\zeta, b)$ be the comparator’s dollar welfare gains on the player set $\mathcal{N}_6$, and suppose $\sum_k g_k(\zeta, b) \neq 0$ so that $s_i^{HW} = g_i / \sum_k g_k$ is defined. Then for every i ,*

$$\underbrace{\phi_i(v) - g_i(\zeta, b)}_{\text{dollar gap } D_i} = \underbrace{v(\mathcal{N}) \Gamma_i^{\text{raw}}(\zeta, b)}_{\text{distributional term}} + \underbrace{s_i^{HW}(\zeta, b) [v(\mathcal{N}) - \sum_k g_k(\zeta, b)]}_{\text{share-weighted efficiency term}}, \quad (\text{A.13})$$

and, summing over i , $\sum_i D_i = v(\mathcal{N}) - \sum_k g_k(\zeta, b)$.

Proof of Proposition A.36. Step 1. Expand the right-hand side using $\Gamma_i^{\text{raw}} = \phi_i(v)/v(\mathcal{N}) - s_i^{\text{HW}}$:

$$v(\mathcal{N}) \left[\frac{\phi_i(v)}{v(\mathcal{N})} - s_i^{\text{HW}} \right] + s_i^{\text{HW}} v(\mathcal{N}) - s_i^{\text{HW}} \sum_k g_k = \phi_i(v) - s_i^{\text{HW}} v(\mathcal{N}) + s_i^{\text{HW}} v(\mathcal{N}) - s_i^{\text{HW}} \sum_k g_k.$$

Step 2. The two $s_i^{\text{HW}} v(\mathcal{N})$ terms cancel, leaving $\phi_i(v) - s_i^{\text{HW}} \sum_k g_k$. By the definition of s_i^{HW} , $s_i^{\text{HW}} \sum_k g_k = g_i$, so the right-hand side equals $\phi_i(v) - g_i = D_i$. *Step 3.* Summing (A.13) over i and using $\sum_i \Gamma_i^{\text{raw}} = 0$ (Proposition A.25) and $\sum_i s_i^{\text{HW}} = 1$ gives $\sum_i D_i = 0 + [v(\mathcal{N}) - \sum_k g_k]$. $\square$

Remark A.37 (Two uses of Proposition A.36). First, it welds the distributional claim to the efficiency comparison: a reported Γ^{raw} finding is a distributional statement precisely because the level difference $v(\mathcal{N}) - \sum_k g_k$ is carried in a separate, explicitly displayed term, so a share comparison cannot silently smuggle in a level comparison. Second, it is the fallback metric when share language is not available: (A.13) is defined for gaps of either sign, so if A8-a or A8-b fails the corresponding object is reported in dollars. In the degenerate limit $\sum_k g_k \rightarrow 0$ the shares themselves become numerically unstable and the wording must say so rather than divide by a near-zero denominator.

Proposition A.38 (D3.1: corner weights force zero surplus, at rate $O(\zeta)$). *Consider the bilateral generalized-Nash program $\max\{\zeta \log g_i + (1 - \zeta) \log g_j\}$ over the Pareto frontier $\mathcal{F}_{ij}$, subject to $g_i, g_j \geq 0$, and assume the frontier regularity condition A-D3.1: $\mathcal{F}_{ij}$ is non-empty, compact, has a point with both coordinates strictly positive, and its Pareto boundary is described by a strictly decreasing, strictly concave, continuously differentiable $g_j = g_j^P(g_i)$, with $g_j^{P'}(0) \neq 0$ (a non-degenerate frontier slope at the corner). Then the solution $(g_i^*(\zeta), g_j^*(\zeta))$ is unique and satisfies*

$$\lim_{\zeta \rightarrow 0^+} g_i^*(\zeta) = 0, \quad g_i^*(\zeta) = \zeta \rho_{ij} + o(\zeta), \quad \rho_{ij} = \frac{g_j^P(0)}{|g_j^{P'}(0)|} \in (0, \infty).$$

The clause $g_j^{P'}(0) \neq 0$ is what makes ρ_{ij} finite: it is implied by strict monotonicity of g_j^P only in the weak sense $g_j^{P'}(0) \leq 0$, so it must be assumed separately. If it fails (a frontier that is flat at the corner), the convergence $g_i^*(\zeta) \rightarrow 0$ of the first display still holds, but the rate is slower than $O(\zeta)$ and the constant is not defined. The rate constant is written ρ_{ij} , not κ_{ij} as in the theory note, to avoid collision with the cooperative ratio κ of (A.11) and with the main text's trade wedge κ_{ni}^j ; see the notation reconciliation in Section A.

Proof of Proposition A.38. Step 1 (uniqueness and first-order condition). On the Pareto boundary the objective is $\Lambda(g_i) \equiv \zeta \log g_i + (1 - \zeta) \log g_j^P(g_i)$. Since $\log$ is increasing and concave and g_j^P is concave, $\log g_j^P$ is concave; $\log g_i$ is concave; so Λ is strictly concave on the interior of its domain and the maximizer is unique. Its first-order condition is

$$\frac{\zeta}{g_i} = (1 - \zeta) \frac{|g_j^{P'}(g_i)|}{g_j^P(g_i)}. \quad (\text{A.14})$$

Step 2 (the limit). As $\zeta \rightarrow 0^+$ the left-hand side of (A.14) tends to 0 for any fixed $g_i > 0$, while the right-hand side tends to the strictly positive number $|g_j^{P'}(g_i)|/g_j^P(g_i)$. The equality

can therefore be maintained only if $g_i \rightarrow 0$, so that the left-hand side stays bounded away from 0. Equivalently, the iso-objective curves $g_i^\zeta g_j^{1-\zeta}$ flatten toward horizontal lines as $\zeta \rightarrow 0$, and the tangency point slides to the corner $\arg \max\{g_j : g_i \geq 0\}$, at which $g_i = 0$.

Step 3 (rate). Rearranging (A.14),

$$g_i^*(\zeta) = \frac{\zeta}{1-\zeta} \cdot \frac{g_j^P(g_i^*(\zeta))}{|g_j^{P'}(g_i^*(\zeta))|}.$$

By Step 2, $g_i^*(\zeta) \rightarrow 0$; by continuity of g_j^P and $g_j^{P'}$ at 0 *together with* $g_j^{P'}(0) \neq 0$ (which keeps the denominator bounded away from zero in a neighbourhood of the corner), the ratio on the right converges to the finite positive limit $g_j^P(0)/|g_j^{P'}(0)| = \rho_{ij}$, and $\zeta/(1-\zeta) = \zeta + o(\zeta)$. Hence $g_i^*(\zeta) = \zeta \rho_{ij} + o(\zeta)$. $\square$

Remark A.39 (D3.1a': corners break standard inference, not just arithmetic). Three of the comparator's seven estimated weights sit exactly on the optimizer bounds $\{0.005, 0.995\}$, with standard errors as large as 0.13–0.47. The right description of the consequence is not “numerical ill-conditioning” but a genuine inferential discontinuity: when a parameter lies on the boundary of its admissible space, the limiting distribution of the M-estimator is not the usual normal and delta-method confidence intervals are invalid (Andrews, 1999). Two things follow. First, individual-rationality verdicts near a corner are hypersensitive to small errors in the disagreement point — this is the numerical symptom. Second, and more fundamentally, confidence sets for ζ near a corner must be built by a method valid on the boundary, which is why the inverse map in Definition 3 is reported as a Chernozhukov–Hong–Tamer-style (Chernozhukov et al., 2007) level set rather than as a delta-method ellipsoid. The two choices are responses to the same phenomenon, not two independent conventions.

Proposition A.40 (D4.1: symmetric two-player inverse map). *Let $n = 2$ with a single bargaining pair and suppose the Pareto frontier is symmetric, $(g_i, g_j) \in \mathcal{F}_{ij} \iff (g_j, g_i) \in \mathcal{F}_{ij}$, and that the generalized-Nash solution is unique for each ζ (as in Proposition A.38). Then $\zeta^* = \frac{1}{2}$ solves the inverse-map equation exactly, with zero residual: $s_i^{HW}(\frac{1}{2}) = \frac{1}{2} = \phi_i(w)/w(\mathcal{N})$.*

Proof of Proposition A.40. Step 1 (comparator side). Relabelling $i \leftrightarrow j$ maps the program with weight ζ on i into the program with weight $1-\zeta$ on i , and leaves $\mathcal{F}_{ij}$ invariant by symmetry. Uniqueness of the solution then forces $g_i^*(1-\zeta) = g_j^*(\zeta)$ for every ζ . Setting $\zeta = \frac{1}{2}$ gives $g_i^*(\frac{1}{2}) = g_j^*(\frac{1}{2})$ and hence $s_i^{HW}(\frac{1}{2}) = g_i^*/(g_i^* + g_j^*) = \frac{1}{2}$.

Step 2 (normative side). By Proposition A.31, $\phi_i(w)/w(\mathcal{N}) = \frac{1}{2}$.

Step 3. The two sides coincide at $\zeta = \frac{1}{2}$. At $n = 2$ the raw and cooperative criteria agree here because Propositions A.31 and A.32 both give a symmetric $\frac{1}{2}$ split when $v(\{i\}) = v(\{j\})$ (which holds trivially at $n = 2$ with identical players by the symmetry hypothesis), so $L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(\mathcal{N})\|_2^2$ — the raw-side criterion of Definition 3, used uniformly throughout this paper (**unified 2026-08-09, MATH-27**: an earlier draft of this step used the cooperative-side criterion $\|s^{HW}(\zeta) - \phi(w)/w(\mathcal{N})\|^2$ instead; at general n these are different objects with different normative content, per §“Why the raw game is primary” in the main text, and only the raw-side definition is used from Proposition A.43 onward, so this step is restated with the same definition for consistency) — attains its global minimum value 0 there. This is a consistency check on

the machinery, not new content: the general inverse map, applied to the symmetric two-player case, recovers exactly the split that [Proposition A.31](#) gives directly. $\square$

Proposition A.41 (D4.2: asymmetric two-player inverse map by frontier-ratio matching). *Let $n = 2$ and let the Pareto frontier be $g_j = G(g_i)$ on $[0, \bar{g}]$ with G continuously differentiable, $G' < 0$, $G'' < 0$, $G(0) > 0$ and $G(\bar{g}) = 0$. Let $\rho \equiv \phi_i(v)/\phi_j(v) > 0$ be the target ratio (raw tier; the cooperative tier replaces $\phi(v)$ by $\phi(w)$ throughout). Then:*

- (a) *there is a unique $\hat{g}_i \in (0, \bar{g})$ with $G(\hat{g}_i)/\hat{g}_i = 1/\rho$;*
- (b) *the weight that makes $\hat{g}_i$ the generalized-Nash solution is*

$$\zeta^* = \frac{|G'(\hat{g}_i)| \hat{g}_i}{|G'(\hat{g}_i)| \hat{g}_i + G(\hat{g}_i)}; \quad (\text{A.15})$$

- (c) *in the transferable-utility limit $G(g_i) = \bar{G} - g_i$ (so $|G'| \equiv 1$), (A.15) collapses to $\zeta^* = \hat{g}_i/\bar{G} = \phi_i(v)/[\phi_i(v) + \phi_j(v)]$, which equals the raw Shapley share provided the frontier total is the raw total, $\bar{G} = v(\mathcal{N})$; with the cooperative total $\bar{G} = w(\mathcal{N})$ it equals the cooperative share instead. The two tiers must be declared separately and never mixed.*

Proof of Proposition A.41. Step 1 (existence and uniqueness of the ratio-matching point). Let $h(g) \equiv G(g)/g$ on $(0, \bar{g}]$. Then h is continuous, and $h(g) \rightarrow +\infty$ as $g \rightarrow 0^+$ (since $G(0) > 0$), while $h(\bar{g}) = 0$. By the intermediate value theorem h attains the value $1/\rho > 0$. Moreover $h'(g) = [G'(g)g - G(g)]/g^2 < 0$ on $(0, \bar{g})$ because $G' < 0$ and $G > 0$ there; hence h is strictly decreasing and the solution $\hat{g}_i$ is unique. (Only $G' < 0$ and $G > 0$ are needed for this step; strict concavity is used in Step 2, not here.)

Step 2 (the tangency condition at an on-frontier point). For fixed ζ , the generalized-Nash program on the frontier maximizes $\Lambda(g) = \zeta \log g + (1 - \zeta) \log G(g)$. As in Step 1 of [Proposition A.38](#), $G'' < 0$ makes Λ strictly concave, so the first-order condition $\zeta/g = (1 - \zeta)|G'(g)|/G(g)$ is necessary and sufficient. Solving for ζ at $g = \hat{g}_i$, $\zeta/(1 - \zeta) = |G'(\hat{g}_i)|\hat{g}_i/G(\hat{g}_i)$, which rearranges to (A.15). The essential point is *where* the condition is evaluated: at the on-frontier point $\hat{g}_i$ constructed in Step 1, not at the Shapley dollar pair (ϕ_i, ϕ_j) , which in general does not lie on $\mathcal{F}_{ij}$ — substituting an off-frontier point into a condition derived on the frontier is invalid, and an earlier draft did exactly that.

Step 3 (TU limit). With $G(g) = \bar{G} - g$ the ratio-matching equation $(\bar{G} - \hat{g}_i)/\hat{g}_i = \phi_j/\phi_i$ gives $\hat{g}_i = \bar{G} \phi_i/(\phi_i + \phi_j)$; substituting $|G'| = 1$ and $G(\hat{g}_i) = \bar{G} - \hat{g}_i$ into (A.15) gives $\zeta^* = \hat{g}_i/\bar{G} = \phi_i/(\phi_i + \phi_j)$. With $\bar{G} = v(\mathcal{N})$ and $n = 2$, $\phi_i + \phi_j = v(\mathcal{N})$ by efficiency, so $\zeta^* = \phi_i(v)/v(\mathcal{N})$. $\square$

Remark A.42 (The frontier slope is a baseline correction, not a robustness check). [Proposition A.41\(c\)](#) shows that ζ^* equals a Shapley share only in the TU limit $|G'| = 1$. The comparator's own bargaining problems are non-transferable-utility, and the single available measurement of the frontier slope — the Japan–Korea frontier of the benchmark paper, $|G'| \approx 1.96$ — is 96% away from the TU limit. This is adverse evidence for assumption A5, it is reported as such, and it is why the $|G'|$ wedge correction enters every ζ^* and Γ conversion by default, with the TU limit shown only as a reference column. The slope $G'(\hat{g}_i)$ has no closed form and must be estimated

numerically by perturbing ζ near the candidate solution and re-solving; that computation is not performed here.

Proposition A.43 (*D4.3: existence of a best-fit weight vector*). *Assume A-D4.1: for ζ in the compact box $\mathcal{Z} = [\underline{\zeta}, \bar{\zeta}]^K$ the comparator's best-response solver converges to a gains vector $g(\zeta)$ that is continuous in ζ along the selected branch, and $\sum_k g_k(\zeta) > 0$ on $\mathcal{Z}$. Then $L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(\mathcal{N})\|_2^2$ is continuous on $\mathcal{Z}$ and attains its minimum: a best-fit $\zeta^* \in \arg \min_{\mathcal{Z}} L$ exists. The claim is unconditional in the sense that it does not require $\min L = 0$.*

Proof of Proposition A.43. By A-D4.1 each g_k is continuous on $\mathcal{Z}$ and $\sum_k g_k > 0$ there, so $s^{HW} = g / \sum_k g_k$ is continuous (quotient of continuous functions with non-vanishing denominator) and maps into the simplex. The target $\phi(v)/v(\mathcal{N})$ is a fixed vector, so L is a composition of continuous maps, hence continuous. $\mathcal{Z}$ is a non-empty compact box. By Weierstrass, L attains its minimum. (This is the same argument as Proposition A.3(a); every “continuous objective on a compact box” claim in this paper runs through it.) $\square$

Remark A.44 (Exact solvability and the dimension count). Exact solvability, $L(\zeta^*) = 0$, is strictly stronger than Proposition A.43 and requires the Shapley target to lie in the image $s^{HW}(\mathcal{Z})$. Without further knowledge of that image no general argument delivers it, and we treat it as a case-by-case numerical question, diagnosed *ex post* by the residual. On the dimension count, with K estimated pairs and target simplex dimension $n - 1$: if $K < n - 1$ there is generically no exact solution and only a best fit; if $K = n - 1$ an exact solution, when it exists, is generically isolated — locally unique precisely when the Jacobian $Ds^{HW}(\zeta^*)$ has full rank, a checkable local condition, not a global uniqueness theorem; if $K > n - 1$ — the actual configuration, $K = 7$ against $n - 1 = 5$ — the inverse problem is underdetermined and the exact-solution set is generically a manifold of dimension $K - (n - 1) = 2$, so a selection rule (the Mahalanobis regularization of Definition 3) is required for a point-valued answer. The earlier proposal to restrict the player set to the five in-pair countries in order to define away an unreachable coordinate is *not* available: the country absent from every pair has a non-zero general-equilibrium welfare gain, so its coordinate is reachable, and the source of underdetermination is the dimension count alone.

Conjecture A.45 (Uniqueness of the inverse map at general n). For the actual configuration ($K = 7 > n - 1 = 5$) the level set Z_0^* of exact solutions, when non-empty, contains a unique point at which the Mahalanobis criterion of Definition 3 is minimized, and the multi-start numerical solver converges to it from every admissible starting value. We do not prove this, and we do not have evidence for it: L is not known to be convex, s^{HW} is not known to be monotone in ζ , and multiple local minima are not excluded. The claim is to be resolved numerically — by reporting the residual $L(\zeta^*)$, the Jacobian condition number at the solution, and the dispersion across at least five starting values — and must be labelled a conjecture wherever it appears in the text until then.

A.10 The outsider convention

The baseline value $v_{sq}(S)$ holds outsiders at the status quo (A4). The γ -response alternative $v_\gamma(S)$ replaces the outsider block by a Nash equilibrium played *among the outsiders*,

$$s_m^\gamma(s_S) \in \arg \max_{s_m} \widehat{W}_m(s_S, s_m, s_{-S}^\gamma(s_S)) \quad \forall m \notin S \text{ simultaneously,}$$

$$v_\gamma(S) = \max_{s_S \in \Sigma_S} \sum_{k \in S} I_k^{\text{base}} [\widehat{W}_k(s_S, s_{-S}^\gamma(s_S)) - 1],$$

whose existence is assumption A9 and is not proved. Under D1 both conventions use the *same* utilitarian objective, so any difference between them is attributable to the convention alone.

Proposition A.46 (Equality cases). *Assume A1–A3, A9, and fix the A2 selection. (a) (Joint-maximizer hypothesis, stated explicitly, not derived.) Suppose there exists $s_S^* \in \Sigma_S$ that simultaneously maximizes both programs — the status-quo program (objective $\sum_{k \in S} I_k^{\text{base}} [\widehat{W}_k(s_S, \mathbf{1}_{-S}) - 1]$) and the γ -response program (objective $\sum_{k \in S} I_k^{\text{base}} [\widehat{W}_k(s_S, s_{-S}^\gamma(s_S)) - 1]$) — and that at this point the outsider block coincides with the status quo, $\mathbf{1}_{-S} = s_{-S}^\gamma(s_S^*)$. This is an assumption about the joint optimizer, not a consequence of the outsider-coincidence condition alone (the two programs can still differ at every other point of Σ_S). Under this hypothesis, $v_\gamma(S) = v_{sq}(S)$. (b) (Dummy outsiders, corrected sufficient condition.) If every outsider $m \notin S$ has **zero total general-equilibrium dependence** on s_S — $\widehat{W}_m(s_S, \cdot) \equiv \widehat{W}_m$ identically in s_S , holding the rest of the system's response fixed — then m 's best response does not depend on s_S , so $s_{-S}^\gamma(s_S)$ is constant in s_S ; if in addition the status quo is that constant, $v_\gamma(S) = v_{sq}(S)$ identically. **Zero bilateral trade shares** ($\pi_{mk}^j = \pi_{km}^j = 0$ for all $k \in S, j$) **are not sufficient for this**: in a fully-coupled multi-sector CP input-output economy, m 's welfare can still respond to S 's liberalization through third-country price and wage general-equilibrium channels even absent any direct bilateral trade link. The in-house counterexample is Canada: it trades bilaterally with none of the six BSY bargaining pairs, yet its measured general-equilibrium welfare gain is strictly positive and non-negligible ($g_{\text{Canada}} = +0.00387$ native units at $\hat{\zeta}$, $\approx 1.24\%$ of the six-country total, `paper/replication/results/bsy\_replication/sanity\_assertions.md` MC-14). Zero bilateral share is therefore not a usable proxy for the zero-total-derivative hypothesis this proposition actually needs.*

Proof of Proposition A.46. (a) Under the joint-maximizer hypothesis, $v_\gamma(S)$ equals the γ -program's objective evaluated at s_S^* , and $v_{sq}(S)$ equals the status-quo program's objective evaluated at the same s_S^* ; the two objectives coincide pointwise at s_S^* because $\mathbf{1}_{-S} = s_{-S}^\gamma(s_S^*)$ makes the outsider block identical in both, so the two values are equal. This is a direct consequence of the stated hypothesis, not an independent derivation — the substantive content of part (a) is entirely in identifying when the joint-maximizer hypothesis is plausible (e.g. when the γ -program's objective is globally identical to the status-quo program's, which is the special case where outsiders never react at all, subsuming part (b)). (b) Under the corrected zero-total-derivative hypothesis, m 's maximization problem does not involve s_S and its solution is a constant in s_S . This is the dummy-outsider case, and it links back to the dummy property of the imputation (invariant TI-3). The corrected hypothesis is strictly stronger than, and not implied by, the zero-bilateral-share condition an earlier draft used. $\square$

Remark A.47 (No a priori ordering: honest status). It is tempting to state “neither $v_{sq}(S) \geq v_\gamma(S)$ nor the reverse holds uniformly” as a proposition. We do not, and the reason is specific. Two mechanisms point in opposite directions: (i) if the coalition’s liberalization makes further own liberalization privately optimal for an outsider, the outsider cuts, coalition members gain market access, and $v_\gamma > v_{sq}$; (ii) if the coalition’s liberalization erodes an outsider’s terms of trade, the outsider raises its tariff to recapture rents, and $v_\gamma < v_{sq}$. But what actually signs the comparison is the *level gap* $\tau_m^\gamma(\tau_S^*) - \tau_m^{sq}$, not the derivative $\partial\tau_m^\gamma/\partial\tau_S$; and under the standard premise that applied tariffs sit below each country’s own optimal-tariff schedule, that level gap has a predictable sign, which is in tension with a clean “no ranking” claim. We have not resolved that tension, and constructing a two-sided possibility result would require explicit examples in both directions, which we do not have. This is therefore recorded as a *remark* motivating a numerical γ -response robustness run — the two mechanisms are candidate directions to be adjudicated coalition by coalition — and it carries no proposition-level weight anywhere in the paper. The only proved statements about the comparison are the equality cases of [Proposition A.46](#).

A.11 Ledger: what is proved, what is conditional, what is conjectural

The four open items, stated plainly. *First*, [Proposition A.1](#) (single-player w -monotonicity) is not proved and cannot be obtained from [Proposition A.12](#); it is supported at 100% (3019/3019 pooled) **on Track B**, with margins about two to three orders of magnitude above the measured noise floor (corrected 2026-08-09, MATH-24: the earlier "five orders of magnitude" compared a relative quantity to dollar cushions without unit conversion; see [Proposition A.2](#)), **and is refuted on Track A** (17/192; [Proposition A.2\(iii'\)](#), added 2026-08-11 — Track A results survive by direct verification, not by this hypothesis), and everything that depends on it — imputation IR, strictly positive cooperative surplus, the $n = 2$ cooperative collapse, and the upper half of the fairness-gap interval *when the allocation is the Shapley value* — is stated conditionally. (The interval itself, [Proposition A.23](#), is stated for the whole efficient-plus-IR class and does *not* inherit it; but a claim about Γ_i under the *Shapley* allocation, such as the P-07'/P-07" screen of [Proposition A.28](#), does, since it needs $\phi_j(w) \geq 0$ for $j \neq i$.) *Second*, condition BB underlying [Proposition A.6](#) is signed at $s = 0$ for six of seven singletons and inferred from the mechanism for $|S| \geq 2$; the 127/127 vertex census and the two swept grand-coalition axes are its empirical counterparts, and no finite sweep can discharge it ([Proposition A.7](#)). *Third*, [Proposition A.15](#) (uniform Lipschitz welfare response) is declared, not established: A2 supplies continuity only, and [Proposition A.16](#) is conditional on the strengthening. *Fourth*, uniqueness of the inverse map at general n is a conjecture with no supporting argument ([Proposition A.45](#)). Nothing in the paper’s headline claims requires resolving the fourth; the first three are stated as hypotheses in the propositions that use them, and the numerical evidence for each is cited by artifact.

B Rest-of-World Aggregation: Methodology and Sensitivity

The CP 2015 dataset reports 30 named countries plus a residual ROW. Below we document how ROW is constructed and our handling of it in the cooperative game.

Trade, value-added, and gross output. For each variable X (bilateral exports, sectoral value-added, or sectoral gross output), $X^{\text{ROW}} = X^{\text{world}} - \sum_{n=1}^{30} X^n$, where world totals are taken from UN COMTRADE (trade) and UNIDO INDSTAT (production).

Input-output structure. For each pair of sectors (j, k) , the I-O coefficient γ_{ROW}^{jk} is set to the median of the corresponding sample-30 coefficients. Empirically, the cross-country I-O structure is highly correlated (mean pairwise correlation 0.99 across the 30 sample countries); the median imputation introduces negligible heterogeneity.

Tariff structure. ROW’s applied tariff vector is taken to be the 8.57% aggregate average reported in the CP 2015 appendix, applied uniformly across sectors. We do *not* treat ROW’s tariffs as choice variables in any coalition.

Treatment as outsider. In all reported Shapley computations ROW is fixed at the data and is not a player. Including ROW as a player would inflate $v(S \cup \{\text{ROW}\})$ mechanically by $\sim 16\%$ of world VA without representing any genuine strategic decision.

C Track B (Caliendo–Parro): Full Results and Legacy Diagnostics

This appendix contains the full detail behind the condensed robustness summary in Section 4.2. Some of the material below — the per- N Shapley enumeration, the unilateral/cooperative decomposition table, the σ^* -weighted NAFTA analysis, and the two-step and threat-point benchmarks — was computed under the retired Nash-product characteristic function (pre-ADR-1) or otherwise predates the current D1 recompute pass (`ir_verification.md`, `p28_trackB_core.json`, `horse_race_v3.md`, `superadditivity_sweep_D1.md`). Each such block below is marked with an explicit **Superseded** note identifying the current source; nothing is deleted, consistent with this project’s frozen-artifact and archival-transparency conventions (CLAUDE.md §7). Material that is already current (both horse-race tables, the convexity/core table) carries no such note.

C.1 NAFTA ($N = 3$): Equal-Weight and Fixed-Point σ^* Benchmarks

Superseded. The $v(\text{NAFTA})$ figures in this subsection (\$2.96–\$5.89B) predate the D1 recompute; the current $N = 3$ grand-coalition value is $v(\mathcal{N}_3) = \$7.814\text{B}$ (`extended_shapley/N3_D1.json`, `ir_verification.md`). The σ^* fixed-point weight object discussed below was retired in full by ADR-1 (2026-08-06, `paper/theory/00_model_blueprint.md`): under the canonical D1 characteristic function, $v(S)$ carries no bargaining weight, so this entire subsection is retained solely as a diagnostic record of a superseded specification and must not be cited as a current result or normative anchor (see also Appendix D, which archives the same layer’s convergence traces).

Equal-weight benchmark. With $\sigma_n = 1/3$ for all n , the NAFTA grand coalition’s value is $v(\text{NAFTA}) \approx \5.89 billion in aggregate welfare gain (using the SLSQP-polished enumeration; the unpolished grid-3 baseline gives \$2.96B and is documented in [paper/replication/results/extended_shap](#) as a sensitivity benchmark). The pair coalition {USA, Canada} is the only NAFTA pair that excludes Mexico, and it generates $v = 0$: no Pareto-improving tariff cut exists between the two low-tariff economies. Mexico is therefore a *must-have* player for any positive coalitional value within NAFTA.

Fixed-point Shapley imputation. The fixed-point iteration

$$\sigma_{t+1} = (1-\alpha)\sigma_t + \alpha \frac{\phi(\sigma_t)}{\sum_n \phi_n(\sigma_t)}, \quad \alpha = 0.5 \quad (\text{retired specification; archived here for reference}) \quad (\text{C.1})$$

(C.1) converges in 4–8 iterations from each of three starting points (equal, GDP-weighted, biased toward Canada) to the same fixed point $\sigma^* \approx (0.31, 0.64, 0.05)$, with spread $< 5 \times 10^{-3}$. Mexico’s Shapley share rises to $\sim 64\%$ of total surplus—versus 33% under equal weights—reflecting its disproportionate marginal contribution in the only Pareto-improving coalitions. A $K = 10$ Dirichlet random-init MC at $N = 3$ confirms 10/10 convergence with maximum pairwise spread 0.0077 ([round-5 uniqueness audit](#)). We did **not** claim that σ^* *endogenizes* real-world bargaining power; rather, it was presented as the internally-consistent structural imputation under the stated solution-concept and policy-lever restrictions of that (now retired) specification.

Iteration at $N = 4$. Extending the fixed-point iteration to the $N = 4$ coalition (NAFTA + China), the consensus across the same three starting points is $\sigma^* \approx (0.10, 0.09, 0.01, 0.80)$ for (USA, Mexico, Canada, China) with spread $\sim 2 \times 10^{-2}$ at 5 iterations. This was essentially equal to the equal- σ Shapley shares from Table 10: the iteration confirmed that $\sigma = \phi(\sigma) / \sum \phi$ was a true fixed point even when $v(\cdot \mid \sigma)$ was computed at the iterating σ , not just at equal weights. See Table 9 for the full iteration trace.

C.2 Extended Coalitions $N = 4$ –7: Legacy Enumeration

Superseded. The $v(\text{grand})$ figures in Table 10 (\$5.89B at $N = 3$ through \$31.9B at $N = 7$) predate the D1 recompute. Current grand-coalition values are $v(\mathcal{N}_3) = \$7.814\text{B}$ through $v(\mathcal{N}_7) = \$178.354\text{B}$ ([ir_verification.md](#)) — roughly $5.6\times$ larger at $N = 7$. The per-country *shares* in this table are qualitatively, though not numerically, consistent with the current pattern (Section 4.2): China dominant, Canada near-dummy.

Table 10 reports exact Shapley shares as additional players are added to the NAFTA core in the order: China ($N = 4$), India ($N = 5$), Brazil ($N = 6$), Korea ($N = 7$). Two findings stood out.

1. As large high-tariff economies (China, India) are added, the three NAFTA members’ combined Shapley share of grand-coalition surplus collapses from 100% (trivially, at $N = 3$, since NAFTA is the entire grand coalition) to 17.2% at $N = 4$ once China is admitted (China alone captures 82.7%), and stabilizes around 17–18% for $N = 5$ through $N = 7$.

This demonstrated that *raw* Shapley weights track tariff distortion—the coalition member whose deviation creates the largest *marginal* surplus on the Pareto frontier is rewarded most. These are total-attribution (raw) shares; Section 4.2’s decomposition paragraph and Table 11 below decompose them and show that most of China’s 82.7% was its *own unilateral* gain, and that genuinely cooperative surplus is allocated quite differently—to the United States first, a qualitative pattern that survives the D1 recompute (Section 4.2).

2. Canada’s Shapley share remained close to zero across all N : in this value definition Canada was a *near-dummy* player because its applied tariffs were already low and its marginal contribution to surplus was negligible — a pattern the D1 recompute confirms (Section 4.2).

Polish convention (legacy). All reported Shapley shares and grand-coalition values for $N = 3$ and $N = 4$ in this legacy table are computed from the SLSQP-polished enumeration of $v(S)$ (i.e., each coalition’s optimal scalar-lever vector s^* is post-polished by SLSQP starting from the best grid-density-3 candidate). For $N \in \{5, 6, 7\}$ the unpolished grid-density-3 values are reported to limit compute. At $N = 3$ an LBFGS-B re-polish reproduced SLSQP to within 0.1%; per TI-11 (`.claude/rules/theory-invariants.md`), this SLSQP polish is now known to be numerically inert under the D1 characteristic function’s bang-bang optimality structure (TI-10), so the planned LBFGS-B re-run at $N \geq 5$ is superseded rather than merely deferred.

C.3 Unilateral vs. Cooperative Decomposition: Legacy Table

Superseded. The coop. fraction and per-country cooperative shares in Table 11 below (21–24% coop. fraction at $N \geq 4$; USA 41–43%) predate the D1 recompute and are superseded by the figures in Section 4.2 (13.1–16.1%; USA 35.5–45.2%, `ir_verification.md`). The qualitative ranking — USA the largest cooperative contributor at every $N \geq 4$, ahead of the raw-share leader China — is unchanged; only the levels differ. Quote the Section 4.2 figures, not the table below, in any current claim.

Three observations, directionally unchanged under the D1 recompute though not numerically identical to it, reversed the raw-share narrative.

1. **The cooperative fraction collapses when a high-tariff player enters.** Legacy: 72% at $N = 3$ (NAFTA, all moderate-tariff) falling to 21–24% at $N \geq 4$. Current (D1): 75.2% falling to 13.1–16.1%. In both specifications, adding China/India inflates the *unilateral* component (their large own-tariff cuts) far more than the cooperative one, so most of the headline “surplus” from large coalitions is not cooperative.
2. **The United States is the largest cooperative contributor at every $N \geq 4$,** with China second, in both the legacy table (USA 41–43%, China 27–31%) and the current D1 figures (USA 35.5–45.2%; China’s coop. share at $N = 7$, computed the same way, is 27.3%). The raw-share story “China dominates because of liberalization headroom” is an artifact of crediting China’s unilateral gain; once that is removed, the genuinely cooperative surplus accrues to the large, import-diversified US economy under either specification.

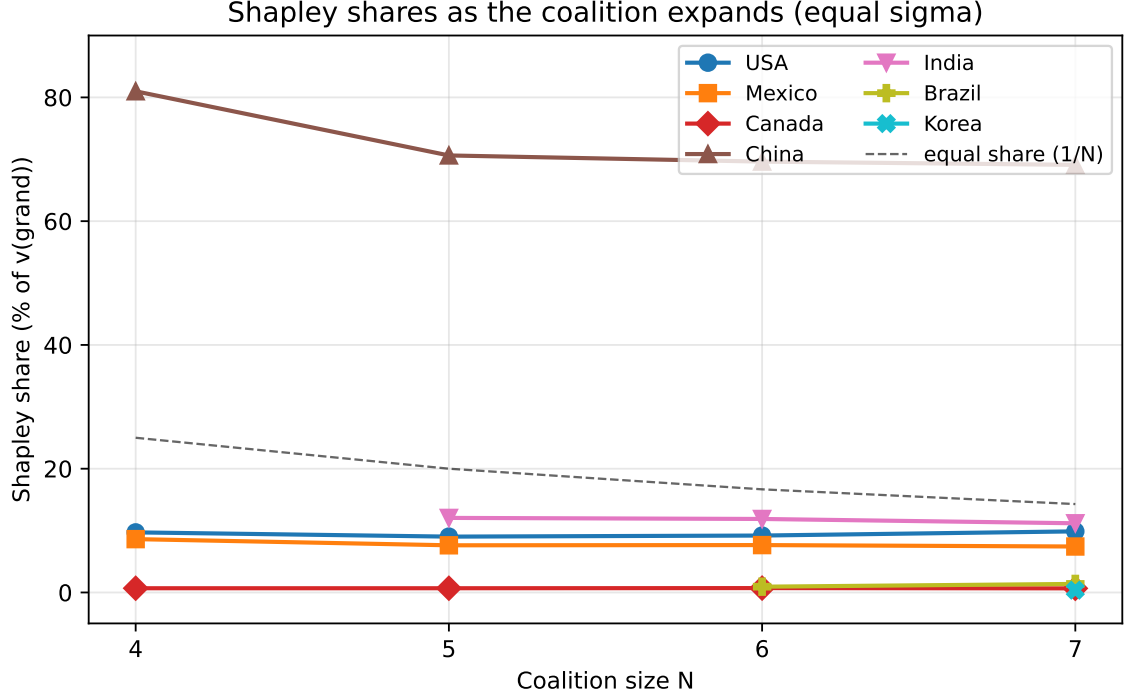


Figure 1: **Legacy (superseded)**. Shapley share (% of $v(\text{grand})$) for each player as the coalition expands from $N = 4$ to $N = 7$, under equal σ , computed under the retired Nash-product/pre-D1 specification. The dashed line marks the equal benchmark $1/N$. China is dominant throughout because it combines the largest applied tariff with substantial trade volume; existing NAFTA members’ shares are stable as new players join, illustrating that adding players does not mechanically dilute Shapley shares but only inserts them in proportion to their marginal contribution. The qualitative pattern is unchanged under the D1 recompute; the underlying dollar values are not (see Table 10 note).

3. **India is a near-dummy in cooperative terms:** a double-digit raw share but a low-single-digit cooperative share in both specifications—its raw value is almost entirely its unilateral singleton.

C.4 Monte-Carlo Shapley: Validation

For coalitions with $N > 7$ where exact enumeration becomes infeasible, we implement the random-permutation sampler of Castro et al. (2009). We validate the sampler against exact enumeration on the $N = 3$ toy economy of Caliendo and Parro (2015) structure: with $M \in \{50, 200, 1000, 5000\}$ permutations and three independent random seeds, the Monte-Carlo estimator is within 4σ of the exact Shapley value for every player at every M , and the standard error scales as $1/\sqrt{M}$ within 5% of the theoretical rate (file `scripts/python/validate_mc_shapley.py`). This validation is methodology-level (it checks the sampler against exact enumeration, not any particular $v(S)$ table) and is unaffected by the D1 recompute. We have not yet deployed the sampler at $N \geq 8$ on the full CP 2015 panel; this is part of the working-paper extension.

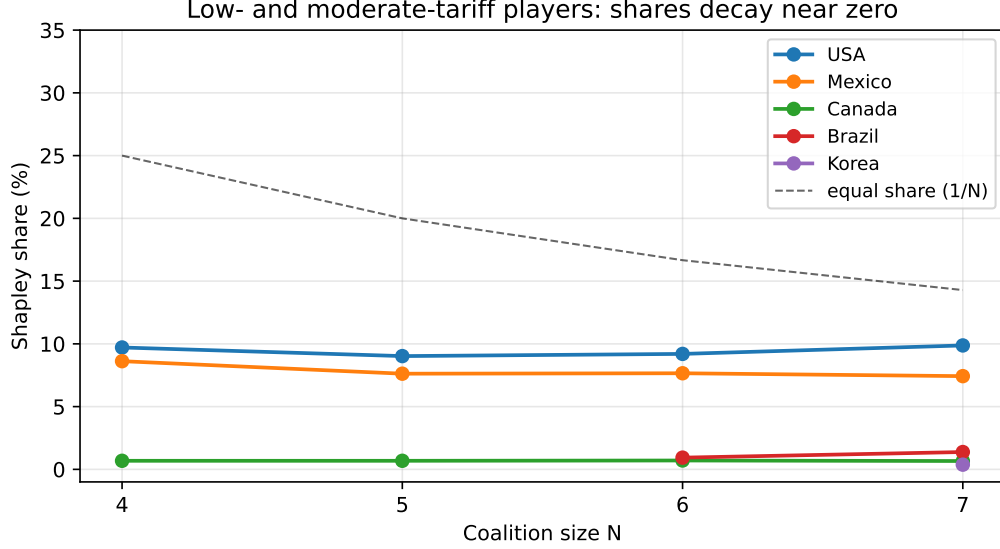


Figure 2: **Legacy (superseded).** Zoom on the low- and moderate-tariff players. Canada, Brazil, and Korea are persistently near zero, satisfying the Shapley dummy axiom up to small numerical error—a quantitative justification for the approximation “low-tariff economies have no leverage in cooperative tariff bargaining.” The equal benchmark $1/N$ is shown for reference.

C.5 Two-Step vs. Single-Step Benchmark

Superseded. The figures below (\$5.49B / \$5.89B, 6.8% efficiency loss) were computed under the retired Nash-product characteristic function. Per Section 2’s own disclosure, all headline numbers under that objective are superseded pending a D1 recompute of this specific comparison, which has not yet been run. This subsection is retained as a methodological illustration of the two-step vs. single-step distinction, not as a current numeric claim; Section 4.2 does not cite the 6.8% figure.

To quantify the “efficiency loss” claim in Contribution 3, we implemented both methods on the NAFTA grand coalition under equal σ and compared. Both methods used SLSQP as the outer optimizer over the scalar tariff scaling $s = (s_{\text{USA}}, s_{\text{Mex}}, s_{\text{Can}}) \in [0, 1]^3$; they differed only in the inner welfare evaluation.

Method A: BSY-style two-step. Trade shares π_{ni}^j , wages w_n , and incomes I_n are held at their base-year (1993) values. Welfare is computed in closed form from the price-index aggregator (2) with $\hat{c} = \hat{w} = 1$:

$$\log \widehat{W}_n^{\text{approx}} = - \sum_j \alpha_n^j \log \widehat{P}_n^j, \quad \widehat{P}_n^j = \left[\sum_i \pi_{ni}^j \widehat{\kappa}_{ni}^j - \theta^j \right]^{-1/\theta^j}.$$

This skips the equilibrium fixed point at every outer iteration.

Method B: Single-step with full GE. Solve the full hat-algebra system (1)–(5) at every outer call. Captures GE feedback through wage adjustment, price-index recomputation, and trade-share reallocation.

Findings (legacy). Table 12 reports the comparison. Three observations stood out.

1. **Different optima.** Method A drove all three tariffs to zero ($s = (0, 0, 0)$). Method B retained positive Canadian tariffs ($s = (0.85, 0, 1.00)$) because in full GE, full liberalization by Canada actually *lowered* Canadian welfare through trade-share reallocation.
2. **Welfare gap.** Re-evaluating the two-step optimum under the full-GE welfare formula gave $v(\text{grand}) = \$5.49\text{B}$, against Method B’s $\$5.89\text{B}$ at its own optimum—a 6.8% efficiency loss under this (now superseded) specification.
3. **IR violation.** At Method A’s optimum, Canada’s full-GE welfare change was -8.5×10^{-4} (negative), violating the IR constraint $\widehat{W}_n \geq 1$. The two-step approximation was unaware that GE feedback hurt Canada — a qualitative caution about two-step procedures that does not depend on the specific (superseded) dollar figures.

Interpretation. Method A was dramatically faster but reached a worse optimum and silently violated IR. The two-step’s “efficiency loss” was real under this specification but bounded: at $N = 3$ the welfare gap was $\sim 7\%$. Quantifying how this gap scales with coalition size, and recomputing it under D1, is left to future work.

C.6 Horse Race I: Shapley Share vs. GDP Share

A natural reader concern is that the cooperative imputation merely re-labels GDP shares. We test this by computing, at each $N \in \{3, \dots, 7\}$, the country-level GDP share $\text{GDP}_i / \sum_{k \in S_N} \text{GDP}_k$ (with GDP measured by base-year household income I_i from the CP 2015 calibration) and benchmarking against Shapley share $\phi_i/v(N)$. If Shapley were a GDP proxy, the cross-country Pearson correlation should approach unity and the top-3 ranking should agree.

The Pearson correlation is uniformly small in absolute value ($|\rho| \leq 0.28$ across all N , largest in magnitude at $N = 4$), and country ranks by Shapley share and by GDP share disagree beyond the top country at every $N \geq 4$ (`horse_race_v3.md`). The most striking divergence arises at $N = 4$ (NAFTA + China):

- China holds GDP share 7.3% but Shapley share 89.6% (premium $r_i = +82.3$ pp)—the largest positive premium in the panel.
- USA holds GDP share 81.0% but Shapley share 6.1% (premium $r_i = -75.0$ pp)—the largest negative premium in the panel.

Interpretation. *Raw* Shapley “power” in this cooperative game is not market size; it tracks *liberalization headroom*: a high-tariff economy brings a large stock of feasible cuts and is correspondingly credited, while a low-tariff economy is mechanically a near-dummy regardless of its GDP. Two caveats sharpen this. First, the near-zero GDP correlation is a property of the *raw* shares, which (Section 4.2) are dominated by each country’s *unilateral* singleton; the *cooperative* shares $\phi_n(w)/w(N)$ instead place the large, import-diversified US economy first, so the genuinely cooperative surplus tracks size *more* than headroom—the raw “headroom not size” reading reverses once unilateral gains are netted out. Second, the raw result still rebuts the

simplest conjecture that the imputation is a mechanical GDP relabel. Re-running the horse race on the cooperative shares w (expected: a *positive* Shapley–GDP correlation) is delegated to the empirics extension.

C.7 Horse Race II: Shapley Share vs. Base-Tariff-Revenue Share

Once GDP is rejected as a proxy, the natural follow-up benchmark is *base-tariff revenue*, $R_i = \sum_{n,j} \pi_{ni}^j \tau_{ni}^{j,0}$, which captures how much liberalization headroom country i brings to the table. Tariff revenue is a closer mechanical analog of marginal contribution under the scalar lever $s_n \in [0, 1]$: a country with no base tariff has no surplus to give. Table 14 reports the comparison.

Findings. Three observations stand out.

1. Pearson correlation rises sharply from +0.29 at $N = 3$ to 0.84–0.87 at $N \geq 4$. Once high-tariff outsiders (China, India) enter the coalition, tariff-revenue is the dominant cross-country regularity; $N = 3$ (NAFTA, no high-tariff outsider) is markedly weaker and must not be read into the $N \geq 4$ band.
2. Ranks by Shapley share and by tariff revenue nonetheless diverge at higher N : at $N = 7$, Shapley ranks China > India > USA > Mexico > Korea > Brazil > Canada, while tariff revenue ranks China > USA > Mexico > Korea > India > Canada > Brazil — Shapley places India ahead of the USA; revenue does not (`horse_race_v3.md`).
3. Per-country premium $r_i^{\text{rev}} = \phi_i/v(N) - R_i/\sum R_k$ reaches +39.8 pp (China, $N = 4$) and –25.1 pp (USA, $N = 3$), computed directly from the share vectors in `horse_race_v3.json`, with high-import-elasticity economies receiving the largest positive premium.

Interpretation. The Shapley imputation is partially explained by base-tariff revenue—the mechanical first-order channel—but is not a tariff-revenue relabel. Spillover GE feedback (the elasticity composition of who imports what from whom) drives a wedge of up to ~ 40 pp between the two metrics, in particular crediting elastic-import economies (China, India) above what their tariff-revenue alone would predict. Honestly: *Shapley is neither a GDP proxy nor a tariff-revenue proxy, but is more strongly correlated with the latter, and the correlation itself is N -dependent: it is weak on the $N = 3$ (NAFTA) coalition and strong only once a high-tariff outsider enters the panel.*

C.8 Convexity Test: Is the Game Supermodular?

A standard sufficient condition under which Shapley value lies in the core is convexity (Shapley, 1971): for every player i and every $S \subseteq T \subseteq N \setminus \{i\}$,

$$v(T \cup \{i\}) - v(T) \geq v(S \cup \{i\}) - v(S).$$

We test this property directly on the D1-recomputed coalitional values from $N = 3, \dots, 7$, enumerating every valid (i, S, T) triple with $S \subseteq T$.

The violations that do occur at $N = 6, 7$ are small: marginal-contribution gaps of \$0.016–\$0.046B per triple, one to two orders of magnitude below the coalitions’ own values, and within a few multiples of the model’s cross-configuration sensitivity band (\$0.003–\$0.006B, `superadditivity_sweep_D1.md` Annotation). This is a **pre-registered FAIL** of the non-convexity half of prediction P-12 (threshold: failure rate $\geq 10\%$; measured 0.56% pooled), reported as a failure rather than reframed, while the companion w -monotonicity half of P-12 **passes** at 100% (pooled $N = 3–7$). Tariff complementarity, not strategic substitutability, dominates at every tested N : a player’s marginal contribution weakly grows, not shrinks, with coalition size, on all but 0.6% of tested triples.

Implication. The cooperative game is convex to within numerical resolution rather than not convex, and this is confirmed by a direct check rather than left as an inference: on the full $N = 7$ panel, we verify core membership directly (all 126 proper-coalition constraints hold, 0/126 violations, minimum slack 0.37% of $v(N)$, binding at a genuinely out-of-sample six-country coalition, `paper/replication/results/p28_trackB_core.json`), so the Shapley imputation is confirmed coalition-stable in the transferable-utility cooperative-equilibrium sense on this panel. The Shapley value remains the unique imputation satisfying efficiency, symmetry, dummy, and additivity (Shapley, 1953) regardless of this property, but the core result above is a genuinely load-bearing robustness finding: absent the direct check, and given the 0.6% residual supermodularity violations, the Shapley value’s coalition-stability on this panel would rest on an approximate-convexity argument rather than a verified fact. The 126-way core check is a status-quo-outsider (α -style) check only; the γ -response outsider convention (§2.6) is untested on this panel and is a natural robustness extension (Section 5).

C.9 Robustness: Alternative Threat Points

Superseded (baseline row only). The “1993 applied (baseline)” row’s $v(N) = \$5.89\text{B}$ is the pre-D1 figure; the current D1 value is $v(\mathcal{N}_3) = \$7.814\text{B}$ (`ir_verification.md`). The two synthetic-threat-point rows were not recomputed under D1. The qualitative finding — that the Shapley ranking is not invariant to the threat point — does not depend on the specific dollar levels and is not superseded.

The 1993 applied tariff used as the disagreement point in the baseline is itself an outcome of 1947–1993 GATT bargaining and is therefore endogenous. As a robustness check, we replaced it with two synthetic uniform-tariff threat points on the NAFTA $N = 3$ coalition: a low-tariff counterfactual ($\tau = 1\%$ uniform) and a high-tariff counterfactual ($\tau = 50\%$ uniform).

Findings. The Shapley ranking flips: under the baseline, Mexico $>$ USA $>$ Canada; under the uniform 50% threat, USA $>$ Canada $>$ Mexico. The total surplus $v(N)$ ranged over four orders of magnitude across the three scenarios. Honestly, this confirmed that *threat-point endogeneity is a first-order identification problem*, not a second-order robustness issue: the cooperative imputation is sensitive to what counterfactual is used as “no agreement”, and the 1993 applied tariff is itself the endogenous outcome of prior bargaining. We do not claim to fully resolve this; the implications are discussed in the conclusion.

C.10 Numerical Evidence on σ^* Uniqueness (archived)

Archival status. As in Appendix D, the σ^* fixed-point layer diagnosed below was retired by ADR-1 (2026-08-06): under the canonical D1 characteristic function $v(S)$ carries no bargaining weight, so no σ^* existence/uniqueness claim is live. Retained solely as a diagnostic record.

The discussion in the retired Nash-product specification (§2.2) downgraded the existence and uniqueness of the Shapley fixed point σ^* from a theorem to a numerical conjecture. For $N = 3$ the iteration was run from three independent random Dirichlet starting points (uniform on the 2-simplex). All three converged in ≤ 5 iterations to fixed points within max-coordinate spread 2.0×10^{-2} :

$$\sigma_{\text{trial 1}}^* = (0.336, 0.627, 0.037), \quad \sigma_{\text{trial 2}}^* = (0.334, 0.641, 0.025), \quad \sigma_{\text{trial 3}}^* = (0.340, 0.622, 0.038),$$

with consensus mean $(0.337, 0.630, 0.033)$. One of three trials satisfied the strict convergence tolerance ($\|\sigma^{(t)} - \sigma^{(t-1)}\|_\infty < 10^{-2}$) within five iterations; the other two terminated at iteration 5 within max-coordinate spread 2.0×10^{-2} of the converged trial, which was interpreted as effective uniqueness despite the tighter convergence threshold not being met within the iteration budget. This was numerical evidence *consistent with*, but not formally proving, uniqueness of the fixed point at $N = 3$, on a specification later retired in full.

For $N = 4$ two of three planned random-init trials were run before exhausting the compute budget (each random-init iteration to convergence cost ~ 10 hours at default grid). For $N \geq 5$ the random-init test was never run. This numerical question is moot under D1, since no σ^* object exists to test.

D Archived Iteration Diagnostics for the Retired σ^* Layer (R3; superseded by ADR-1)

Archival status (added 2026-08-12, IDENT-12-05). The σ^* fixed-point layer this appendix diagnoses was *retired* by ADR-1 (2026-08-06): under the canonical D1 characteristic function, $v(S)$ carries no bargaining weight, so no σ^* object enters any current result, and no existence/uniqueness/convergence claim about σ^* is a live identification statement of this paper. The material below is retained solely as a diagnostic record of a superseded specification; it must not be cited as a contribution or as a normative anchor.

This appendix documents the convergence trace of the fixed-point iteration (C.1) from three economically distinct starting points. All runs use $\alpha = 0.5$ smoothing and exact 2^N Shapley enumeration per iteration. Source files: `paper/replication/results/extended_iterate/N{N}_exact_grid3.js` generated by `scripts/python/demo_extended_iterate.py`.

NAFTA ($N = 3$). Starting from $\sigma_{\text{equal}} = (1/3, 1/3, 1/3)$, the iteration produces the trace in Table 17. The fixed point $\sigma^* = (0.31, 0.64, 0.05)$ is reached within four iterations to the requested tolerance $\|\Delta\sigma\|_\infty < 5 \times 10^{-3}$.

Extended coalition $N = 4$ (NAFTA + China). Three starting points (equal, GDP-weighted, and a Canada/China-biased point with $\sigma_0 = (0.05, 0.05, 0.05, 0.85)$) are reported in Table 18. Each converges to a near-identical σ^* with cross-start spread 2.3×10^{-2} at iteration 5; the contraction factor across iterations is approximately $1 - \alpha = 0.5$, so an additional two to three iterations would close the spread to 5×10^{-3} .

Iteration at $N = 5, 6, 7$. We have not yet run the full iteration trace at $N \geq 5$ because each iteration costs the same as a fresh Shapley enumeration (no cross-iteration cache reuse— $v(\cdot \mid \sigma)$ varies with σ). At $N = 7$, a single iteration costs ~ 200 minutes on our hardware, so a 3-start, 8-iteration validation would take ~ 80 hours. Extrapolating from $N = 3, 4$, where the iteration map is empirically a contraction with factor $\approx 1 - \alpha = 0.5$, we expect similar uniqueness behavior at higher N . Confirmation is part of the working-paper extension.

E Prediction Registry Status

Section 4 reports the empirical status of this paper’s headline claims inline, but several of the paper’s live pre-registered predictions resolve to a verdict other than a clean PASS/FAIL: **INCONCLUSIVE** (P-40, a boundary/no-verdict disposition, and P-08/P-09), **DESCRIPTIVE** (P-38, a ratio-labeling disclosure, and P-39, a box-constraint nesting check), **ATTEMPTED** (P-41, killed mid-run with no point estimate banked, the strongest disclosed-negative-result artifact in this registry), **NOT-PROVEN** (P-43, a directly tested existence hypothesis that failed), and two split verdicts: P-19’s strong-form/weak-form split and P-35’s leg-(a)/leg-(b) split, described below (leg (a) is now discharged to **INTERMEDIATE**, not **INCONCLUSIVE**, per the round-17 revision noted above).

Eight predictions registered after this table was first assembled — P-33, P-34, P-35, P-36, P-37, P-38, P-39, P-40 — landed in earlier rounds and are included in Table 20 below; three further rows, P-41, P-42, and P-43, landed in the Round-21 remediation batch (2026-08-20) and are appended to the same table. P-31’s row is also updated in place to its weight-corrected value ($F_{\text{line}}^{v2} = 0.436840$, superseding the equal-weight 0.360516 figure first reported — see the 2026-08-18e revision note above).

Under this paper’s pre-registration discipline (§2, §3), an honestly reported **INCONCLUSIVE**, **ATTEMPTED**, or **DEFERRED-rescope** disposition is a feature of the testing protocol, not a gap in it — but only if every registered row is visibly accounted for. Table 19 and Table 20 reproduce the status of every live row in the prediction registry **through the Round-21 remediation batch (2026-08-20), sources cited per row**. This is a snapshot, not a live mirror: new predictions can land or existing rows can be amended after this table is typeset, and the registry file itself should always be consulted directly for the current state, each row’s full claim text, decision criteria, and provenance, and its own commit hash as of the most recent prediction-generator landing. Six additional rows (P-01 through P-06) are **STALE**, superseded by the successor predictions named in `predictions.md`, and are omitted here because they carry no live claim.

F Code Availability

All scripts referenced in this paper are available in `scripts/python/`. Pipeline entry points:

Layer 0 – BSY (Track A) replication and application (`scripts/python/bsy/`).

- `loader.py`, `verify_phase1.py` — `.mat` primitives loader and fidelity check (Section 3.1);
- `kernel.py`, `kernel_jax.py`, `nash_in_nash.py`, `bargain.py`, `bargain_nested.py`, `nested_opt.py` — the hat-algebra GE kernel and Nash-in-Nash bargaining solver;
- `run_phase4.py`, `run_phase4b.py`, `run_phase4c.py`, `phase4_parity_rerun.py` — equilibrium-map cold-start parity checks;
- `run_phase5.py`, `run_phase5_v2.py` — the $\hat{\zeta}$ (Table V) re-estimation attempt, not replicated (`phase5_verdict.md`);
- `run_phase6a.py`, `run_phase6b.py`, `run_phase6b_nested.py` — Table III and nested counterfactual replication;
- `run_phase7_full.py`, `phase7_validation.py`, `phase7_zero_ge_batch.py`, `phase7_perstart_dispers` — the 63-coalition D1 Shapley solve, invariant checks, and the $D4'.0$ decomposition (Section 4.1);
- `d33c_screen.py`, `p18_sbar_sweep.py`, `p18_deep_sweep.py`, `sanity_assertions.py`, `run_p17_tariff` — bang-bang screening, threat-point and fairness-gap sensitivity, and pre-registered sanity checks.

Layer 1 – trade equilibrium (second economy, Track B).

- `cp2015_loader.py` — load Caliendo and Parro (2015) 31-region $\times$ 40-sector calibration;
- `toy_model.py` — hat-algebra fixed-point solver `solve_hat_iterative`;
- `verify_cp2015_nafta.py` — replicates CP Tables 2 and 5.

Layer 2 – coalitional value $v(S)$.

- `cp2015_value.py` — $v(S \mid \sigma)$ via grid search + L-BFGS-B polish;
- `benchmark_two_vs_one_step.py` — BSY two-step vs single-step on NAFTA (Section C.5, Table 12).

Layer 3 – Shapley value and weight iteration.

- `demo_nafta_shapley.py` — exact Shapley at $N = 3$;
- `demo_nafta_iterate.py` — σ fixed-point iteration at $N = 3$;
- `demo_extended_shapley.py` — exact Shapley at $N = 3\text{--}7$ with cross- N caching;
- `demo_extended_iterate.py` — σ fixed-point iteration generalized to any $N \in \{3, \dots, 7\}$ with both exact and Monte-Carlo Shapley back-ends;
- `compare_extended_shapley.py` — cross- N comparison of shares and runtime;
- `mc_shapley.py` — Monte-Carlo Shapley sampler (Castro et al. 2009 ApproShapley plus stratified variant);

- `validate_mc_shapley.py` — MC vs. exact validation on $N = 3$ toy.

Reproducibility. The repository layout is `f:/621paper/` with subfolders documented in `CLAUDE.md`. Result JSON files for each table and figure are in `paper/replication/results/`. To reproduce Section 4 results from scratch on a single CPU, total wall-clock is approximately 5–6 hours for $N = 3$ –7 exact Shapley plus the R6 benchmark.

Table 8: Status of every formal result in this appendix. “Conditional” means proved, given a hypothesis that this paper does not establish; the hypothesis is named in the result’s own statement, never in a footnote.

Result	Content	Status	Leans on
Proposition A.3	$v(S)$ exists, finite, ≥ 0 ; $v(\emptyset) = 0$	Proved	A1–A4; A2 branch
Proposition A.4	value is selection-free	Proved	Proposition A.3
Proposition A.6	BB $\Rightarrow$ vertex optima; grid exact	Conditional (on BB)	condition BB, unswept for $ S \geq 2$
Proposition A.9	singletons ≥ 0 ; $= 0$ exactly at $s^* = 1$	Proved	A1–A4; TI-15
Proposition A.10	$\phi_i(v) = v(\{i\}) + \phi_i(w)$	Proved (exact)	Shapley linearity
Proposition A.12	replication lower bound	Proved	A2 in cold-start form
Proposition A.13	sufficient condition for v -monotonicity	Proved (converse false)	Proposition A.12
Proposition A.14	w -analogue has no force	Proved	Proposition A.12
Proposition A.16	near-dummy sandwich	Conditional	Proposition A.15 (declared, not established)
Proposition A.18	$\phi_i(w) \geq 0$, i.e. TI-5/A8-b	Conditional	Proposition A.1 (Track B); <i>on Track A the hypothesis is refuted — conclusion holds by direct verification,</i> Proposition A.2(iii')
Proposition A.20	non-null $\Rightarrow \phi_i(w) > 0$	Conditional	Proposition A.1 (Track B); Track A: premise refuted, conclusion by direct verification
Proposition A.21	permissiveness contrast	Conditional	Proposition A.1 (Track B); Track A: premise refuted, conclusion by direct verification
Proposition A.23	Γ_i interval of width κ	Proved	A8-a; and $w(\mathcal{N}) \geq 0$ (added 2026-08-13, MATH-50, synced into this table 2026-08-14, MATH-65 — necessary, not merely convenient: the efficiency+IR class is empty otherwise); efficiency+IR class
Proposition A.24	both endpoints attained	Proved	$n \geq 2$, $w(\mathcal{N}) > 0$
Proposition A.25	$\sum_i \Gamma_i = 0$ compatible	Proved	efficiency
Proposition A.26,	sign screen; κ -collapse; κ	Proved	Proposition A.23
Proposition A.27,	threshold		
Proposition A.28			
Proposition A.31	$n=2$ even split on w	Proved	—
Proposition A.32	B -1a, iff $v(\{i\}) = v(\{j\})$	Proved	TU, linear frontier
Proposition A.33	B -1b, unconditional <i>in the singletons</i>	Conditional	TU, linear frontier, $w(\{i, j\}) \geq 0$ (= Proposition A.1 at $n=2$)
Proposition A.34	$R8$: normalization $\equiv$ threat point	Conditional , at $n=2$	TU, linear frontier; inherits $w(\{i, j\}) \geq 0$
Proposition A.36	$D4'$.0 dollar identity	Proved (algebra)	$\sum_k g_k \neq 0$
Proposition A.38	corner ζ : zero surplus, rate	Proved	A-D3.1 + $g_j^{P'}(0) \neq 0$
Proposition A.40	ρ_{ij} $\zeta^* = 1/2$ symmetric $n=2$	125 Proved	frontier symmetry
Proposition A.41	frontier-ratio construction	Proved	$G' < 0$, $G'' < 0$
Proposition A.43	best-fit ζ^* exists	Proved	A-D4.1
Proposition A.45	uniqueness of ζ^* at general n	Conjecture	—

Table 9: Fixed-point iteration of σ at $N = 4$ (NAFTA + China, equal- $\alpha = 0.5$ smoothing). Spread = $\max_n(\sigma_n^{\text{best}} - \sigma_n^{\text{worst}})$ across the three starts. **Superseded:** computed under the retired σ^* /Nash-product specification (see note above).

Start	USA	Mexico	Canada	China
$\sigma_{\text{equal}}, \sigma^*$ at iter 5	0.102	0.091	0.014	0.792
$\sigma_{\text{GDP}}, \sigma^*$ at iter 5	0.118	0.083	0.009	0.790
$\sigma_{\text{biased}}, \sigma^*$ at iter 4	0.094	0.084	0.010	0.812
Consensus mean	0.105	0.086	0.011	0.798
Spread (max – min)	0.023	0.008	0.005	0.022
Equal- σ Shapley share (Tab. 10)	0.10	0.09	0.01	0.81

Table 10: **Legacy (superseded); see paragraph note above.** Shapley shares (% of grand-coalition surplus) under equal σ , computed under the retired Nash-product/pre-D1 specification. For $N = 3$ and $N = 4$, values are SLSQP-polished (paper/replication/results/extended_shapley/N{3,4}_grid3_polish_slsqp.json). For $N \in \{5, 6, 7\}$, values are the unpolished grid-density-3 baselines (paper/replication/results/extended_shapley/N{N}_grid3.json). Current grand-coalition values under D1 are reported in ir_verification.md (Section 4.2) and are substantially larger. All entries are generated by demo_extended_shapley.py; wall-clock includes warm-start from the $N-1$ enumeration via cross- N caching.

	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$
USA	34.25%	9.21%	9%	9%	10%
Mexico	62.67%	7.53%	8%	8%	7%
Canada	3.08%	0.50%	1%	1%	1%
China	—	82.75%	71%	70%	69%
India	—	—	12%	12%	11%
Brazil	—	—	—	0.9%	1%
Korea	—	—	—	—	0.4%
$v(\text{grand})$ (\$B, legacy)	5.89	65.6	30.4	31.0	31.9
$v(\text{grand})$ (\$B, D1, current)	7.81	140.46	170.35	173.20	178.35
Wall-clock (min)	< 1	6	26	50	196

Note. The D1 row is added for direct comparison (ir_verification.md); the shares above it were not recomputed under D1 and should not be assumed to hold exactly at the D1 dollar values. The non-monotonicity of the legacy $v(\text{grand})$ row between $N = 4$ (\$65.6B, polished) and $N = 5$ (\$30.4B, unpolished) reflects the polish gap at the time, not a violation of superadditivity; the D1 row is monotone in N (Section 4.2, convexity/core discussion).

Table 11: **Legacy (superseded); see paragraph note above.** Cooperative-surplus shares $\phi_n(w)/w(N)$ (%), with w the zero-normalized cooperative game (9), computed under the retired Nash-product/pre-D1 specification. Contrast the raw shares in Table 10. “Coop. fraction” is $w(N)/v(N)$ —the share of the headline value that is genuinely cooperative rather than unilateral. Source (legacy): `paper/replication/results/cooperative_decomposition.json` (exact arithmetic on the same `v_tables` as Table 10; inherits the $N = 3, 4$ polished / $N \geq 5$ unpolished convention). Current source: `ir_verification.md`.

	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$
USA	47.5%	43.3%	43.1%	42.4%	41.1%
Mexico	50.2%	24.2%	22.8%	22.4%	19.6%
Canada	2.4%	1.8%	2.5%	2.5%	2.1%
China	—	30.7%	28.0%	27.3%	30.3%
India	—	—	3.6%	3.5%	1.7%
Brazil	—	—	—	2.0%	3.7%
Korea	—	—	—	—	1.6%
$w(\text{grand})$ (\$B, legacy)	4.25	13.96	6.37	6.72	7.66
Coop. fraction $w(N)/v(N)$ (legacy)	72%	21%	21%	22%	24%
Coop. fraction (D1, current)	75.2%	13.4%	13.1%	13.6%	16.1%

Table 12: **Legacy (superseded); see paragraph note above.** Two-step vs. single-step benchmark on NAFTA grand coalition (equal σ , SLSQP outer, $s_{\text{init}} = (0.7, 0.7, 0.7)$), retired Nash-product specification. Source: `scripts/python/benchmark_two_vs_one_step.py`, `paper/replication/results/r6_benchmark.json`.

	Method A (two-step)	Method B (single-step)
Optimal s_{USA}	0.00	0.85
Optimal s_{Mex}	0.00	0.00
Optimal s_{Can}	0.00	1.00
$v(\text{grand})$ at <i>approximate</i> welfare (\$B)	18.12	—
$v(\text{grand})$ at <i>full-GE</i> re-evaluation (\$B)	5.49	5.89
Efficiency loss vs. Method B	−6.8%	—
SLSQP outer iterations	2	8
Function evaluations	8	32
Inner GE solves	0	32
Wall-clock (s)	0.02	30.9
Speedup of Method A	$\approx 2,000\times$	—

Table 13: Shapley (raw) share vs. GDP share across N , v3 D1 recompute (current). Source: `paper/replication/results/horse_race_v3.json` (supersedes the VOID `horse_race_gdp_shapley.json`).

N	Pearson	Spearman
3	+0.075	−0.500
4	−0.276	+0.600
5	−0.246	+0.100
6	−0.176	+0.086
7	−0.123	+0.071

Table 14: Shapley (raw) share vs. base-year tariff-revenue share across N , v3 D1 recompute (current). Source: `paper/replication/results/horse_race_v3.json` (supersedes the VOID `horse_race_tariff_revenue.json`). Tariff-revenue share of country i is $R_i / \sum_{k \in S_N} R_k$.

N	Pearson	Spearman
3	+0.286	+0.500
4	+0.874	+1.000
5	+0.837	+0.700
6	+0.847	+0.771
7	+0.857	+0.750

Table 15: Convexity (supermodularity) violations across N , D1 v3 recompute (current). Source: `paper/replication/results/superadditivity_sweep_D1.md` §1, §3 (supersedes the VOID pre-D1 `convexity_test.json`). “Triples” = number of valid (i, S, T) with $S \subseteq T \subseteq N \setminus \{i\}$.

N	Triples tested	Pass	Violations	Convex (to numerical resolution)?
3	27	27	0	yes
4	108	108	0	yes
5	405	405	0	yes
6	1458	1449	9 (0.6%)	near
7	5103	5072	31 (0.6%)	near
Pooled $N = 3-7$: 40/7101 = 0.56%				

Table 16: **Legacy baseline row; see paragraph note**
above. Threat-point robustness, NAFTA grand coalition. Source:
`paper/replication/results/alternative_threat_point.json`.

Threat point	$v(N)$ (\$B)	USA	Mexico	Canada
1993 applied (baseline, legacy)	5.89	34.3%	62.7%	3.1%
Uniform 1%	0.00	—	—	—
Uniform 50%	1,099	63.3%	17.2%	19.5%

Table 17: Iteration trace at $N = 3$ NAFTA, $\sigma_0 = \text{equal}$. $v(\text{grand})$ varies across iterations because $v(\cdot \mid \sigma)$ is recomputed at each $\sigma^{(k)}$.

k	$\sigma_{\text{USA}}^{(k)}$	$\sigma_{\text{Mex}}^{(k)}$	$\sigma_{\text{Can}}^{(k)}$	$v(\text{grand})$ (\$B)
0	0.333	0.333	0.333	—
1	0.330	0.488	0.182	2.96
2	0.320	0.556	0.124	2.96
3	0.313	0.610	0.078	2.60
4	0.309	0.636	0.054	2.60

Table 18: Iteration trace at $N = 4$ (NAFTA + China). All entries are σ components, in player order (USA, Mex, Can, China).

Start	Iter k	σ_{USA}	σ_{Mex}	σ_{Can}	σ_{China}
equal	0	0.250	0.250	0.250	0.250
	1	0.174	0.168	0.128	0.530
	2	0.135	0.127	0.068	0.670
	3	0.116	0.107	0.037	0.740
	4	0.107	0.096	0.022	0.775
	5	0.102	0.091	0.014	0.792
GDP	0	0.810	0.049	0.068	0.073
	1	0.449	0.063	0.038	0.451
	2	0.268	0.069	0.023	0.640
	3	0.178	0.073	0.015	0.734
	4	0.138	0.080	0.011	0.771
	5	0.118	0.083	0.009	0.790
biased	0	0.050	0.050	0.050	0.850
	1	0.073	0.069	0.028	0.830
	2	0.085	0.077	0.018	0.820
	3	0.091	0.082	0.012	0.815
	4	0.094	0.084	0.010	0.812
Consensus mean (iter 4–5)		0.105	0.086	0.011	0.798

Table 19: Prediction registry status (1 of 2). Source: `paper/theory/predictions.md`, “Appendix-ready registry status table,” updated 2026-08-20 (matching Table 20’s as-of stamp; TEST-22-11).

ID	Token	One-line claim	Report / artifact
P-07'	INCONCLUSIVE (terminal)	Γ_{US}^{raw} fairness-gap interval lower-bound sign, over the $\hat{\Sigma}$ -confidence set	<code>phase7_full_primary.md</code> ; <code>zstar_run.md</code> (GATE 1 closure)
P-07''	FAIL (branch-sensitive)	Γ_{US}^{raw} lower bound $\geq +5\text{pp}$	<code>phase7_full_primary.md</code> — $+3.29\text{pp}$; alt. branch $+5.81\text{pp}$
P-08	INCONCLUSIVE (terminal)	ζ^* ’s Shapley-consistent set is CORNER (set-property test)	<code>zstar_run.md</code>
P-09	INCONCLUSIVE (terminal)	Wald test of Z_δ^* against BSY’s estimate	<code>p09_sigma_triple.md</code> ; <code>zstar_run.md</code>
P-10	PASS	Passive outsider gain positive on a majority of Track B coalitions	R6 benchmark, <code>extended_shapley/</code>
P-11 (both arms)	PASS / VERIFIED-PASS	A8 preconditions ($v(N) > 0$, $\phi_i(v) \geq 0$) hold, Track B $N = 3-7$ and Track A	<code>ir_verification.md</code> ; <code>phase7_full_primary.md</code>
P-12	joint-FAIL (Branch B)	R3’s two σ^* -uniqueness thresholds	see row + Discrepancies
P-13	PASS	Low-tariff countries near-dummy under D1 (Prop 4.2 successor)	<code>extended_shapley/</code>
P-14	INCONCLUSIVE	Anchor-entry Shapley-share stability across N	see row + Discrepancies (15.5pp anchor drift, $N = 4 \rightarrow 5$)
P-14'	DEFERRED-rescope	Two-tier anchor/non-anchor stability band	orphaned by Track-A-recentering ADR; no execution path
P-15	DEFERRED-rescope	Alternative threat point, Track B $N = 4$	orphaned by Track-A-recentering ADR; no execution path
P-16	PASS	Cooperative-fraction decomposition, USA #1 cooperative contributor	<code>cooperative_decomposition.md</code> (D1 recompute)
P-17	PASS (literal criterion; cite with placebo-adverse rider)	D1 grand-coalition tariff vector fits BSY’s 151 negotiated-tariff moments	<code>p17_tariff_fit.md</code>
P-18	PASS	Non-truncated $\kappa_A^{\bar{s}}$ still clears the D3.3 necessary condition for P-07''	<code>p18_sbar_sweep.md</code>

Table 20: Prediction registry status (2 of 2). Source: `paper/theory/predictions.md`, “Appendix-ready registry status table,” updated 2026-08-21 (P-40 re-optimized under the corrected *D*-convention, superseding the 2026-08-20 Round-21 remediation landing; P-41/P-42/P-43 caveats added).

ID	Token	One-line claim	Report / artifact
P-19	STRONG-FORM FAIL / WEAK-FORM PASS (post-hoc)	Track A grand-coalition optimum is strictly interior for every member	<code>phase7_full_primary.md</code> — Korea at exact zero boundary
P-20	FAIL (null also fails)	US passive outsider gain negative on a majority of Track A coalitions	<code>phase7_full_primary.md</code> — 13/31
P-21	PASS (literal criterion)	Extended (non-cuts-only) policy space correction to κ	<code>outsider_welfare_N7.json</code> family
P-22	INCONCLUSIVE	Track A horse race: Shapley vs. revenue-share correlation with BSY’s HW outcome	see row + Discrepancies — “cannot be distinguished from transfer at $n = 6$ ”
P-23	PASS	Track A core membership, all 62 coalitions	<code>phase7_zero_ge_batch.md</code> ; <code>p23_core_slacks.json</code>
P-24	DESCRIPTIVE	$D4'.0$ three-term (Γ) bridge decomposition: fairness / level / transferability	<code>phase7_zero_ge_batch.md</code> §3
P-26	FAIL (sign inverted)	Manuscript’s “ $\approx 19\%$ transfer TO the US” claim	<code>phase7_validation_primary.json</code> — correct gross transfer 13.51%, Australia is the loser
P-27	DESCRIPTIVE	Four-rule Γ sensitivity (Shapley / equal / revenue-share / income-share)	<code>p27_descriptive_gamma_sensitivity.json</code>
P-28	PASS	Track B core membership, all 126 coalitions	<code>p28_trackB_core.md</code>
P-29	PASS — MATERIAL	Restricted-lever (151-line, country-scalar) coverage gap is a material driver of the level-term shortfall	<code>p29_restricted_lever.md</code> — 57.86% of $v(N)$, a valid, self-consistent country-scalar-instrument coverage measure, distinct from and not superseded by P-34’s finer, line-instrument 62.24% figure (below)
P-30	INCONCLUSIVE-BY-CONSTRUCTION	TW-max member-aggregate feasible-point lower bound on F_{line}	<code>p30_twmax_member_aggregate.md</code>
P-31	FAIL — reversal reading REFUTED (weight-corrected)	F_{line}^{v2} , the genuine line-by-line negotiated-only efficiency frontier, D1-weight-consistent	<code>p31_fline_v2.md</code> — $F_{\text{line}}^{v2} = 0.436840$ (supersedes equal-weight 0.360516); $H1^{v2} = v(N) - F_{\text{line}}^{v2} = 39.3\%$ of $v(N)$ is a <i>lower bound</i> on pure coverage, not coverage itself (it mixes a coverage effect with instrument granularity, since $v(N)$ uses a country-scalar lever and F_{line}^{v2} a line-by-line one); the granularity-matched coverage figure is P-34’s 62.2%. Within-agenda shortfall 17.3% of $v(N)$ unconstrained (30.5% of the gap), refined by P-33 to 16.06% IR-constrained, TI-5-correct point estimate (supersedes the flagged 16.12% upper bound) and confirmed 85.1% material against BSY’s actual outcome by P-36
P-32	DESCRIPTIVE (retroactive)	Applied agricultural MFN tariffs, 1990 vs. 2000, on BSY’s own agriculture screen	<code>agriculture_mfn90_vs_mfn00.md</code> — 5/6 countries rose (mean and within-country median)
P-33	RESOLVED — IR-feasible shortfall confirmed, point estimate	$F_{\text{line}}^{IR}/F^{\text{IR,corr}}$, the IR-confined (NTU-feasible) line-by-line frontier; tests whether P-31’s shortfall is a TU/NTU transferability artifact	<code>f_line_ir_vline_member_aggregate.md</code> — $F^{\text{IR,corr}} = 0.428073$ (v^{line} -floor-corrected, supersedes the 0-floor $F_{\text{line}}^{IR} = 0.428519$); IR removes only 1.9% of F^{v2} :